JORDAN CONTROLS ACTUATORS

Jordan Controls electric and electro-hydraulic actuators, in their stock configurations, service a variety of needs within the Power Generation, Process Control and Industrial Automation markets. With options and custom configurations, Jordan products can meet any need, no matter how precise, no matter how demanding.

Each actuator section opens with general information on the torque or thrust ranges available, and follows with tables that detail the performance and options for each model. By using the information in these tables, the actuator model that most closely matches the application requirements can be determined.

Once the proper model has been identified, go to that model's section of the catalog. Each model section contains a selection chart that enables the model number and all applicable configurations to be identified including options for that model.

A complete listing of options as well as a section on typical wiring diagrams is available toward the back of the catalog.

On the final page of the catalog you will find information about our Service Solutions. Jordan Controls offers a preventative maintenance program, on-site and in-house repair, start-up assistance and on-site and in-house training.

TABLE OF CONTENTS

HOW TO SELECT AN ACTUATOR 2

1. ROTARY ELECTRIC ACTUATORS 3

SM-1000 Series - *Up to 350 in. lbs. (40 Nm)* 4

SM-1100/1500/1600 Series - *Up to 1,000 in. lbs. (113 Nm)* 11

J Series - *Up to 13,000 in. lbs. (1470 Nm)* 18

SM-1700 Series - *Up to 200 ft. lbs. (271 Nm)* 29

SM-5000 Series - *Up to 12,500 ft. lbs. (16950 Nm)* 35

SM-6000 Series - *Up to 26,000 ft. lbs. (35251 Nm)* 53

ROTARY APPLICATIONS 60

2. LINEAR ELECTRIC ACTUATORS 61

LA-1000 Series - *Up to 150 lbf. (667 N)* 62

LA-1100 Series - *Up to 200 lbf. (890 N)* 66

LA-2000 Series - *Up to 12,000 lbf. (53379 N)* 70

LINEAR APPLICATIONS 86

3. LINEAR ELECTRIC VALVE ACTUATORS 87

MV-1000 Series - *Up to 200 lbf. (890 N)* 88

VA-1000 Series - *Up to 900 lbf. (4003 N)* 93

MV-1100/MV-1500 Series - *Up to 2,000 lbf. (8896 N)* 97

VA-1600 Series - *Up to 6,000 lbf. (26689 N)* 103

LA-5000 Series - *Up to 6,500 lbf. (28912 N)* 106

LINEAR VALVE APPLICATIONS 114

4. ELECTRO-HYDRAULIC ACTUATORS 115

EH-2000 Rotary Series - *Up to 5,400 ft. lbs. (7321 Nm)* 116

EH-2000 Linear Series - *Up to 16,000 lbf. (70,400 N)* 130

AD-2000 Amplifier for EH-2000 Series 139

ELECTRO-HYDRAULIC APPLICATIONS 142

5. SPECIALTY ACTUATORS 143

TA-1200 Throttle Actuator - *Up to 45 in. lbs. (5 Nm)* 143

MC-1100 Electric-to-Pneumatic Converter - *Up to 200 psi* 148

SM-3330 Rotary Spring Return - *Up to 1000 in. lbs. (113 Nm)* .. 153

LA-3330 Linear Spring Return - *Up to 800 lbf. (3559 N)* 153

6. POSITION & SIGNAL TRANSMITTERS 159

7. REMOTE SERVO AMPLIFIERS 169

AD-7500 Series Servo Amplifiers 170

AD-8240 Series Servo Amplifiers 173

AD-8900 Series Servo Amplifiers 177

AD-7830 Series Servo Amplifiers 180

AD-9120 Digital Servo Amplifier 184

8. MISCELLANEOUS PRODUCTS 189

RP-4000 Series - Reserve Power Control 189

MT-6400 Series - Digital Meter 193

CS-7200 Series - Control Stations 195

8. COMPLETE LISTING OF OPTIONS 197

9. APPROVALS & CERTIFICATIONS 207

10. APPLICATION CONVERSIONS 210

11. TYPICAL WIRING DIAGRAMS 216

12. GLOSSARY OF TERMS 221

13. SERVICE SOLUTIONS 224

HOW TO SELECT AN ACTUATOR

Selecting a Jordan Controls actuator is a simple and logical process. Selection charts are located in each section to assist you in choosing the correct actuator for your application. Each actuator has a series of cells which allow you to choose the basic model. All actuators have five cells to choose the basic model except the SM-6000, MV-1000, MV-1500 and LA-3300 Spring Return actuators, which have six. After determining the basic model, you can then add options. Each actuator has its own unique list of options. An example of model configuration is illustrated below, though actual actuator selection must be completed using the Selection Chart located in the product section of the catalog.

Cell Numbers from Model Selection Charts

1 — **2** — **3** — **4** — **5**

Basic Model
with Amplifier Type
N = None
A = Analog
D = Digital

Input Voltage *
1 = 115 or 120 Vac, 1 Ph.
2 = 230 or 240 Vac, 1 Ph.
3 = 380 Vac, 3 Ph.
4 = 480 Vac, 3 Ph.
5 = 240 Vac, 3 Ph.
6 = 24-36 Vdc
7 = 90 Vdc
8 = 180 Vdc
9 = 208 Vac, 1 Ph.
10 = 208 Vac, 3 Ph.
11 = 28 Vdc
12 = 12-36 Vdc
13 = 24 Vac
14 = 380 Vac, 3 Ph., 60 Hz
16 = 110 Vac, 1 Ph.
17 = 415 Vac, 3 Ph., 50 Hz
18 = 575 Vac, 3 Ph., 50-60 Hz
19 = 220 Vac, 1 Ph., 50 Hz
20 = 115 Vac, 3 Ph., 50-60 Hz
21 = 20-30 Vdc
22 = 400 Vac, 3 Ph., 50 Hz

Speed
___ sec. for full rotation, stroke, or RPM.

Torque
___ in. lbf. or ft. lbf. (Nm)
or
Thrust
___ lbs. (N)

Rotation
___ degrees or turns
or
Linear Travel
___ inches (mm)

Options
List all desired options from the list of available options for each model.

A001
F002
P001
S001

** All input voltages are +/- 10%*

EXAMPLE:

If you wanted a 120 Vac SM-1720 rotary actuator with a digital amplifier that required a speed of 60 sec. for 90° rotation and a torque of 1,800 in. lbs.

Options needed include Auto/Manual toggle switches, a fixed drive arm, position indicator and two adjustable position switches.

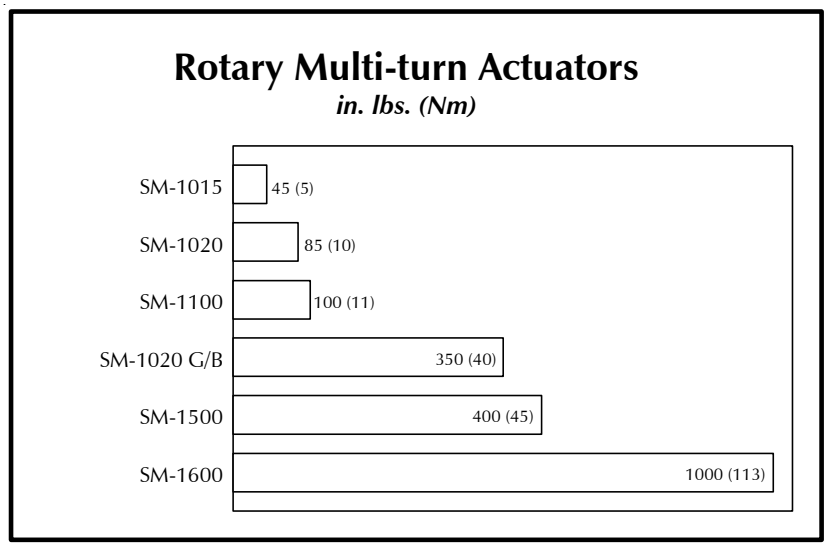
Your selection would look like this:

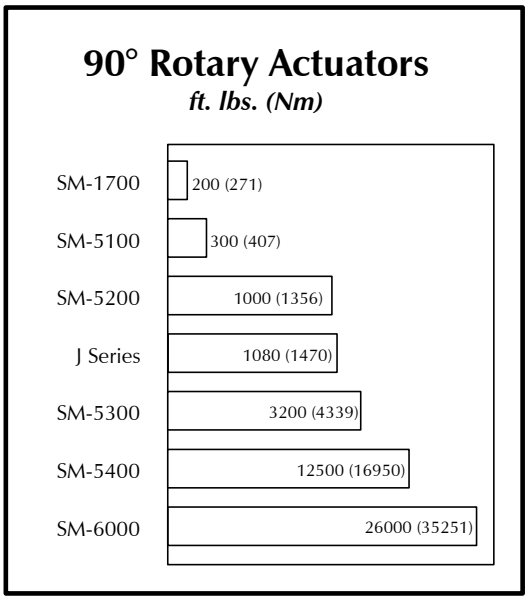
SM-1720-D — 1 — 60 — 1800 — 90°

A001
D001
P001
S001

Jordan CONTROLS

TORQUE RANGES





MODEL SERIES SELECTION CHART

Model Series	Performance			Input Voltage **			Catalog Page
	Maximum Torque in. lbs./ft. lbs. (Nm)	Speed (Sec./90°)		Single Phase 50/60 Hz	DC	Three Phase 50/60 Hz	
		Minimum	Maximum				
SM-1015	45/3.75 (5)	150	1.2	115/230	24-36	—	4
SM-1020	350/29 (40)	210	7	115/230	12-36	—	4
SM-1100	100/8.3 (11)	231	2	120/240	24	—	11
SM-1500	400/33 (45)	38	2	120/240	24	—	11
SM-1600	1000/83 (113)	7.5	2	120/240	24	—	11
J Series	13000/1080 (1470)	7	22	120/240	12, 24	220/380/440	18
SM-1700	2400/200 (271)	60	12	120/240	24, 90	240/480	27
SM-5100	3600/300 (407)	48	13	120/240	24, 90	208/240/380/480	33
SM-5200	12000/1000 (1356)	50	20	120/240	90	208/240/380/480	33
SM-5300	38400/3200 (4339)	47	23	120/240	90	208/240/380/480	33
SM-5400	150000/12500 (16950)	73	40	240*	180	208/240/380/480	33
SM-6000	312000/26000 (35251)	42	10	120/240/208	—	208/240/480	51

** Requires Jordan Controls Amplifier for AC Input*
*** +/- 10%*

SM-1000 SERIES ROTARY ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The SM-1000 Series are multi-turn, rotary actuators, designed to meet the exacting requirements for closed-loop modulating positioning control. Designed for low to medium torque rotary applications, these actuators are capable of accepting analog current and voltage command signals. The brushless stepper motor design provides smooth, highly accurate positioning, with positive position-lock when not in motion and powered. The SM-1000 Series is ideally suited for regulators, pilot valves, small quarter-turn valves, choke valves and dampers.

All SM-1000 Series actuators come with a standard digital internal amplifier. These amplifiers are all full-featured devices designed to work seamlessly with the actuator for closed loop control.



FEATURES

- Permanently lubricated for any position mounting
- Amplifier supplies current to hold last position and prevent backdriving, up to torque rating under power
- AC or DC input power versions
- 4 to 20 mA position, loop-powered, feedback signal
- Field selectable adjustments for:
  - speed
  - deadband
  - zero and span
  - command signal type
  - standard or reverse acting
  - manual-auto operation
  - output shaft position on loss of signal
- Wide ambient temperature range
- Full one year warranty

BASE MODEL INCLUDES:

- Motor (stepper)
- Manual override
- Drive shaft
- Split range command input
- Amplifier
- Electronic torque limiting
- Cast aluminum NEMA 4 (IP65) & dust and hazardous gas ignition-proof enclosure
- Four adjustable position switches (40 mA at 40 Vdc)
- Internal spur gear train
- 4-20 mA transmitter for customer use

POPULAR OPTIONS *(See Pages 197-206)*

- Signal Conversion Module: Convert 40 mA, 40 Vdc low level to 5 Amp, 120/240 Vac current rated position limit switches
- Low current at null (24 Vdc only)
- Switching input powered ac or dc positioning for positioning actuator using ac or dc remote voltage control
- Process Variable Controller to control one process variable - 120/240 Vac only
- Custom mounting and interface hardware
- Local Auto/Manual and INC/OFF/DEC toggle switches (Close-coupled, NEMA 4)
- Various enclosure coatings
- Exd IIB (Hydrogen) enclosure (up to 85 in. lbs.)
- Battery backup to position actuator on loss of ac power
- ATEX Approval

SPECIFICATIONS

Torque:
SM-1015: Up to 45 in. lbs. (5 Nm)
SM-1020: Up to 85 in. lbs. (10 Nm)
SM-1020 with optional gearbox:
Up to 350 in. lbs. (40 Nm). Only available up to 4.25 turns rotation.

Speed: See selection chart on next page.

Input Voltage: See selection chart on next page.

Rotation: 10° to 20 turns standard; 90° to 4.25 turns with gearbox

Environmental Ratings: Explosion-proof for Class I, Division 1, Groups C and D; dust-ignition-proof for Class II, Division 1, Groups E, F, G. Also rated NEMA 4 (IP65) indoor/outdoor. (Optional Exd IIB enclosure)

Command Input:

Full Range: 4-20 mA into 200 ohms
Split Range: 4-12 and 12-20 mA into 200 ohms
Voltage: 0-5 Vdc or 0-10 Vdc into 100,000 ohms
Switch: Dry contact closures

Temperature Limits: -40°F to 150°F (-40°C to 65°C)
For greater temperature ranges, consult factory.

Field Wiring: Barrier terminal blocks

Torque Limiting: Current sensing PWM

Positioning Accuracy: 1.5% of full range

Feedback: 4-20 mA, customer supplied (loop power)

Loss of Signal: Lock-in-place or run to preset.

Loss of Power: Lock-in-place or run to preset with battery backup option.

Direction: CW or CCW

Modulation Rate: Unrestricted modulating duty

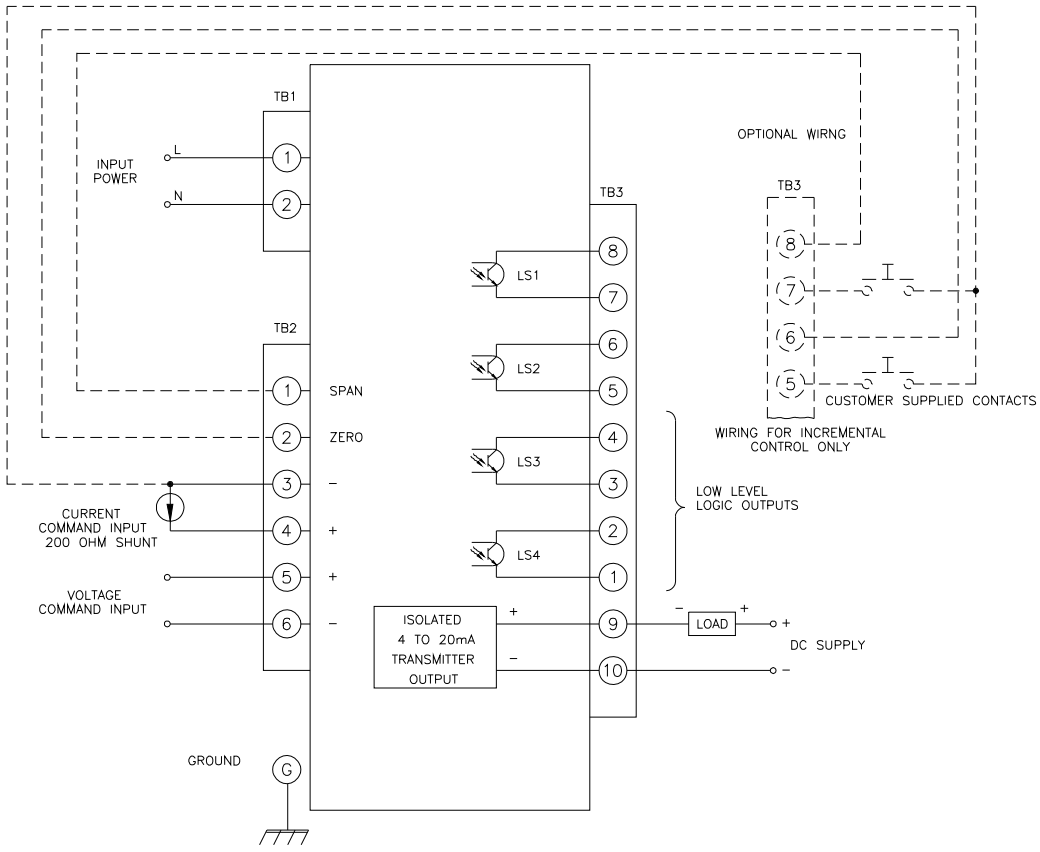
Backdrive: Self-locking up to torque rating

Approximate Weight:
SM-1015: 12 lbs. (5.4 kg)
SM-1020: 16 lbs. (7.3 kg)
SM-1020 with gearbox: 19 lbs. (8.6 kg)

Approvals: See page 207

Position Switch Rating: 40 mA at 40 Vdc

TYPICAL WIRING DIAGRAM



These dimensions and specifications are subject to change without notice. Current drawings and specifications are available upon request.

SM-1000 SERIES SELECTION CHART

						Selection
1	Basic Model	**SM-1015-D**	With built-in digital amplifier			
		SM-1020-D	With built-in digital amplifier			
2	Input Voltage	**1**: 115 Vac, 1 Phase, 50/60 Hz (All Models)				
		2: 230 Vac, 1 Phase, 50/60 Hz (All Models)				
		6: 24-36 Vdc (All Models)				
		12: 12-36 Vdc (SM-1020-D), maximum output torque is limited - consult factory				
		13: 24 Vac (All Models)				
3	Speed, RPM / Torque, in. lbs. (Nm)	SM-1015		SM-1020		
		12.5/15 (1.7)	**7/25** (2.8)	**10/20** (2.3)	**8/40** (4.5)	
4		**3/37** (4.2)	**1/45** (5.1)	**5.5/60** (6.7)	**1/85** (9.6)	
				2.1/100 (11.2)	**1.5/200** (22.6)	
				1/300 (33.9)	**0.7/350** (39.5)	
5	Rotation	**Select rotation between 10° to 20** turns (Up to 4.25 turns for units exceeding 85 in. lbs. torque)				

To convert RPM to sec./90°:

X = *RPM*
Y = *Sec. for 90°*

$$\frac{1}{X} \times 15 = Y$$

SM-1000 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure) NEMA 4	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure) NEMA 4	
Enclosure		
E005	Group B enclosure for use in Hydrogen atmosphere (SM-1020, AC input only)	
Identification/Certifications/Warranties		
J001	CE Marking	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
J006	ATEX Approval (Input voltages 1, 2 & 6) Consult factory for option compatibility.	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	

Code	Description	Selection
Special Electrical Options		
X012	Signal conversion module, installed in a close-coupled, NEMA 4 enclosure (not availabe with X015)	
X013	2 wire incremental dc input	
X014	3 wire incremental 120/240 Vac input	
X015	Battery Backup, only for 120/240 Vac, installed in a close-coupled, NEMA 4 enclosure (not availabe with X012)	
X017	Process variable (SM-1020 only)	
X018	Low Current (24 Vdc only)	
Special Mechanical Options		
Y001	Increased rotation to 33 turns	
Y005	Metric mounting holes, 6mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

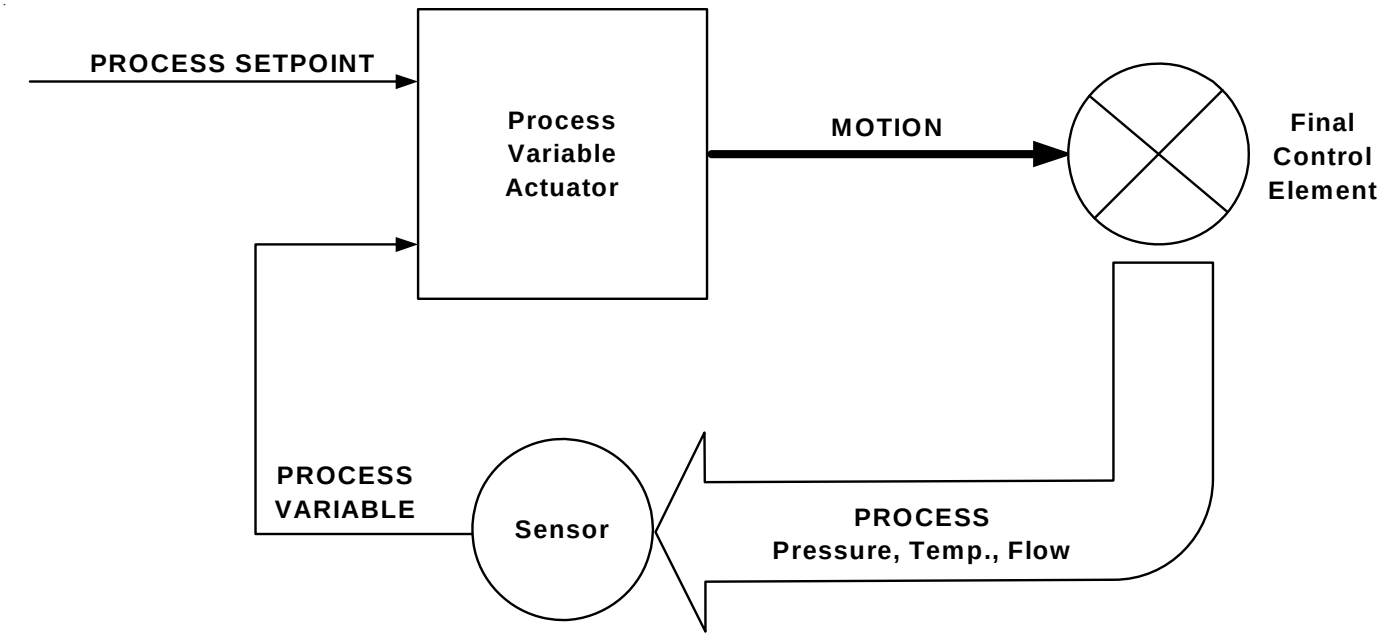
PROCESS VARIABLE OPTION

(Option X017)

The Process Variable option (X017) combines an SM-1020 and a proportional loop controller into one package.

With this option, the SM-1000 series actuator can function as a self-contained process controller for many applications. By eliminating an external process controller and by simplifying control wiring, actuators with the Process Variable option allow users to include control in the final element.

This rugged actuator/process controller unit may be mounted in any position and will withstand the most adverse environmental conditions.



LOW CURRENT AT NULL OPTION

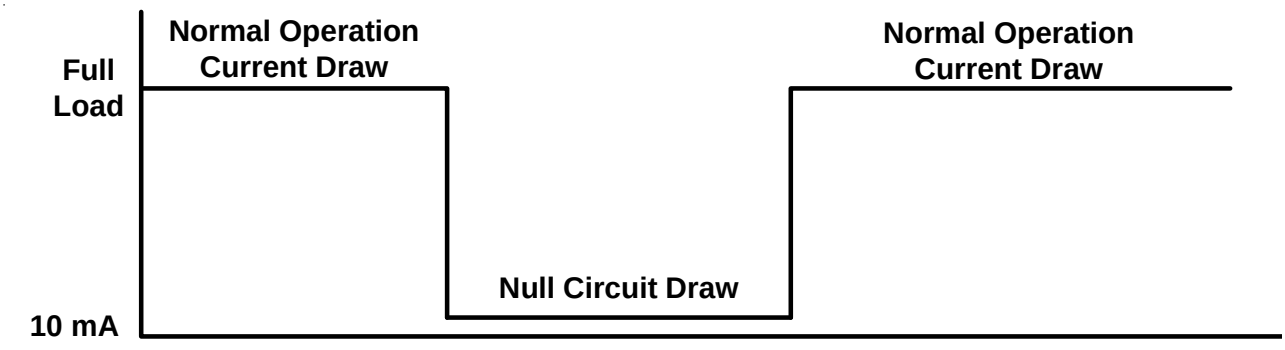
(Option X018)

Jordan Controls has developed a Low Current at Null Circuit (Sleeper Circuit) for the 1000 Series (24 Vdc power input).

The specially-designed circuit allows for very low power consumption while at null. While at null, the amplifier will not draw more than 10 mA.

This option is ideal for remote locations where actuators are operated by batteries or solar panels.

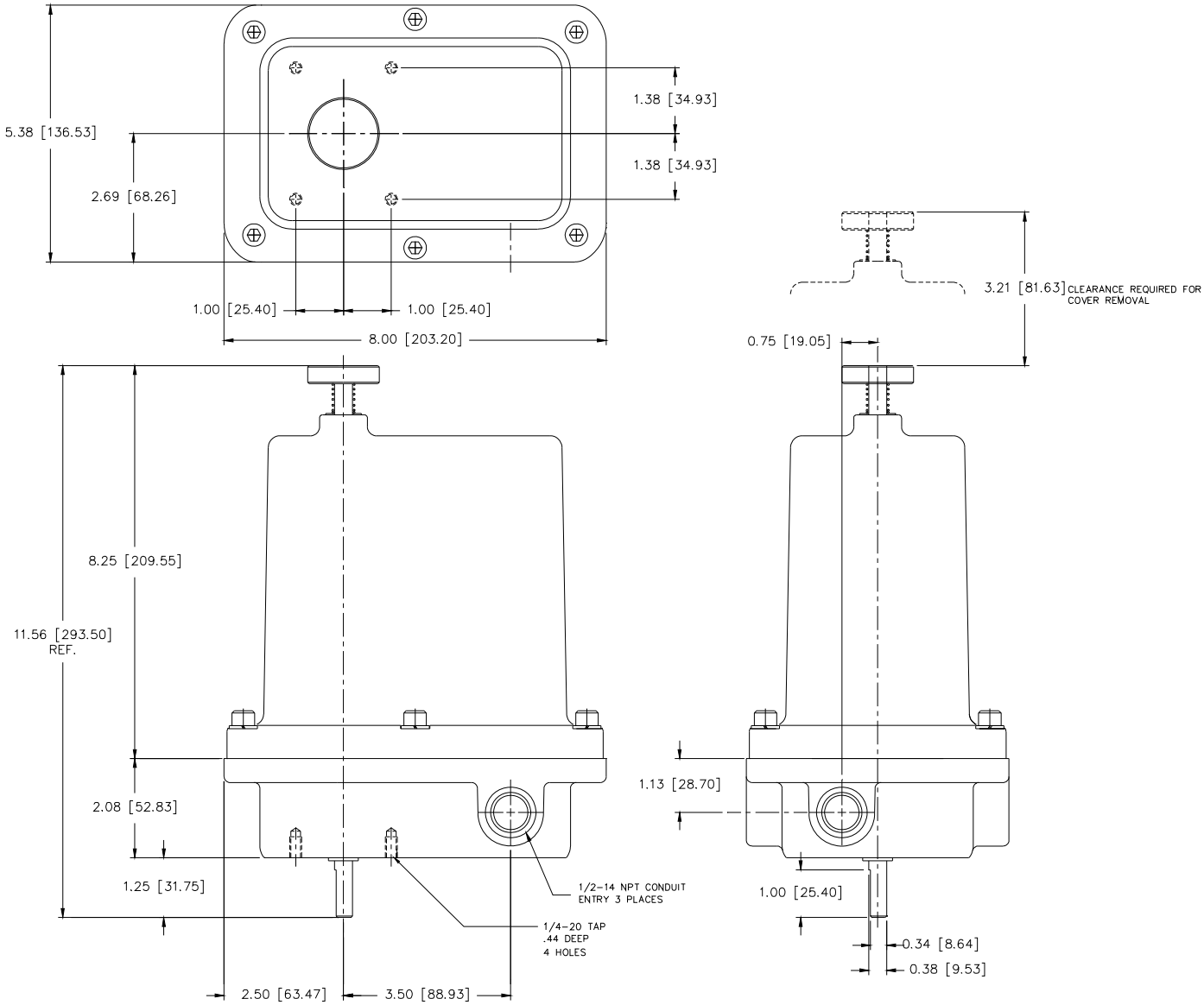
It is also ideal for critical applications which require the actuator to operate on very little power. The sensitivity is switch selectable in the field from 0.05 mA to 1.5 mA.



SM-1000 MAJOR DIMENSIONS

(Up to 85 in. lbs. torque)

INCHES (MILLIMETERS)

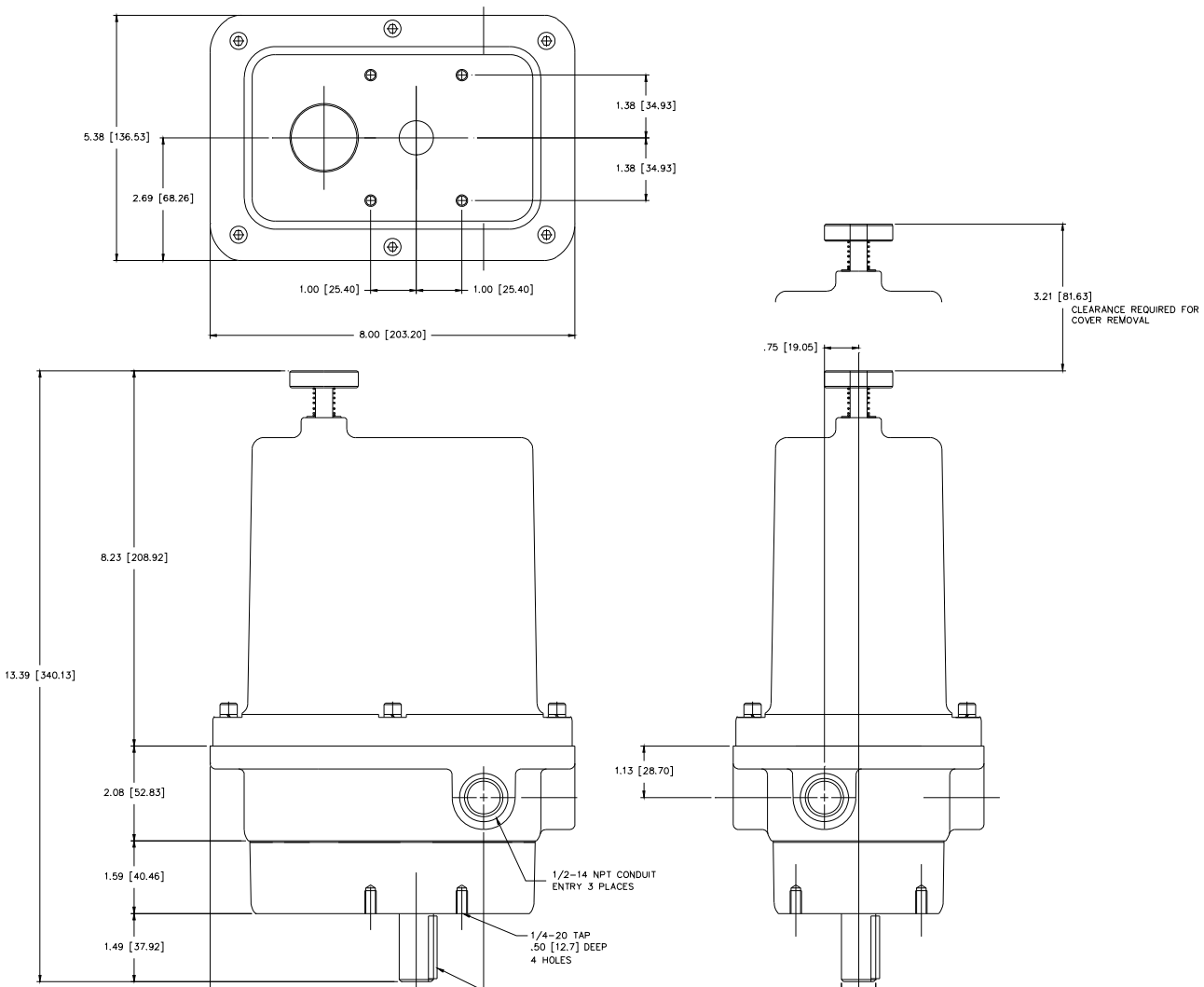


These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

SM-1000 MAJOR DIMENSIONS

(86 to 350 in. lbs. torque)

INCHES
(MILLIMETERS)

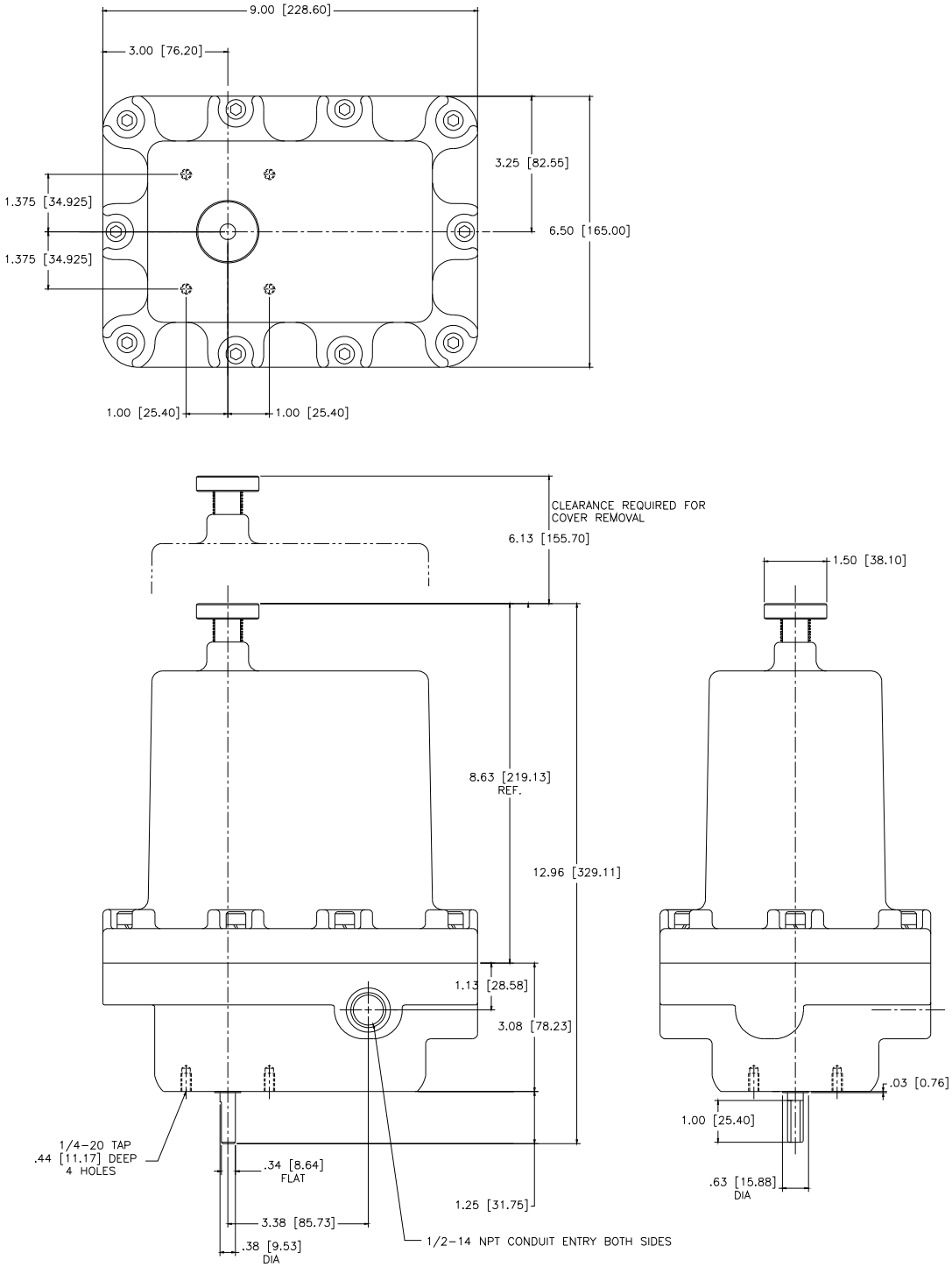


rication of
om.

SM-1000 MAJOR DIMENSIONS

Optional Group B Enclosure (E005)

INCHES (MILLIMETERS)



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

GENERAL DESCRIPTION

The SM-1100, SM-1500 and SM-1600 Series are multi-turn, rotary actuators, designed to meet the exacting requirements for closed-loop modulating positioning control. They can be applied to valves, metering pumps, hydraulic and pneumatic pressure regulators, dampers, and many more industrial/ process control applications where reliable remote positioning control is required.

All SM-1100, SM-1500 and SM-1600 Series actuators are also available with an internal amplifier (AC voltage only). These amplifiers are all full-featured ac switching devices designed to work seamlessly with the actuator for closed loop control.

FEATURES

- Permanently lubricated for any position mounting
- A wide variety of speed/torque combinations
- AC or DC input power versions
- Wide ambient temperature range
- Full one year warranty

BASE MODEL INCLUDES:

- Motor
- Enclosure: NEMA Type 4 (IP65) indoor/outdoor, and/or hazardous locations for Class I, Division 1, Groups C and D, or Class II, Division 1, Groups E, F and G.
- Internal spur gear train
- 4-20 mA transmitter for customer use (amplifier models only)
- End of travel limit switches, not for customer use

AMPLIFIER FEATURES

- Isolated, two-wire, 4-20mA feedback
- Customer-adjustable deadband
- Customer-adjustable zero and span
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place
  - Run to preset
- Field selectable dynamic brake
- Solid state switching of motor for increased life
- Easy pushbutton setup



POPULAR OPTIONS *(See Pages 197-206)*

- Potentiometer or 4 to 20 mA position feedback
- Custom mounting and interface hardware
- Anti-condensation heater and thermostat
- Motor brake (SM-1500/SM-1600 only)
- Manual Handcrank (SM-1500/SM-1600 only)
- Battery back-up, RP-4000

SPECIFICATIONS

ACTUATOR	SM-1100	SM-1500	SM-1600
Torque Range in. lbs. (Nm)	3.5 to 100 (0.4 to 11.3)	40 to 400 (4.5 to 45.2)	500 to 1000 (56.5 to 113)
Rotation	40° to 60 turns	72° to 324 turns	
Speed	0.065 to 70 rpm	0.60 to 73 rpm	2.5 to 5 rpm
Approx. Weight lbs. (kg)	12-14 (5.4) to (6.4)	15 to 22 (6.8 to 10)	

Temperature Limits: -40°F to 150°F (-40°C to 65°C) *For greater temperature ranges, consult factory.*

Field Wiring: Barrier terminal blocks

Approvals: See page 207

Position Switch Rating: 120/240 Vac, 10 Amp

The SM-1100, SM-1500 and SM-1600 Series do not have overtorque protection. Overtorquing the actuator will cause permanent damage.

SM-1100/SM-1500/SM-1600 SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Command Input	Input Impedance in ohms	Feedback	Modulation Rate*
		Run	Stall				
SM-1110-N	120/1/50/60	0.25	0.26	N/A	N/A	N/A	2000 starts/hr
SM-1110-D	120/1/50/60	0.25	0.26	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	2000 starts/hr
SM-1120-N	120/1/50/60	0.6	0.7	N/A	N/A	N/A	600 starts/hr
SM-1120-D	120/1/50/60	0.6	0.7	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	600 starts/hr
SM-1140-N	24 Vdc	1.2	4.8	N/A	N/A	N/A	4000 starts/hr
SM-1150-N	120/1/60	0.1	0.1	N/A	N/A	N/A	2000 starts/hr
SM-1150-D	120/1/60	0.1	0.1	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	2000 starts/hr
SM-1170-N	240/1/50/60	0.14	0.15	N/A	N/A	N/A	2000 starts/hr
SM-1170-D	240/1/50/60	0.14	0.15	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	2000 starts/hr
SM-1180-N	24 Vdc	1.9	5.2	N/A	N/A	N/A	4000 starts/hr
SM-1520-N	120/1/50/60	2.5	2.9	N/A	N/A	N/A	600 starts/hr
SM-1520-D	120/1/50/60	2.5	2.9	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	600 starts/hr
SM-1530-N	120/1/50/60	0.9	1.2	N/A	N/A	N/A	2000 starts/hr
SM-1530-D	120/1/50/60	0.9	1.2	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	2000 starts/hr
SM-1540-N	24 Vdc	1.7	1.9	N/A	N/A	N/A	4000 starts/hr
SM-1550-N	240/1/50/60	0.45	0.5	N/A	N/A	N/A	2000 starts/hr
SM-1550-D	240/1/50/60	0.45	0.5	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	2000 starts/hr
SM-1580-N	24 Vdc	5.2	6.25	N/A	N/A	N/A	4000 starts/hr
SM-1590-N	240/1/50/60	1.1	1.6	N/A	N/A	N/A	600 starts/hr
SM-1590-D	240/1/50/60	1.1	1.6	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	600 starts/hr
SM-1630-N	120/1/50/60	0.9	1.2	N/A	N/A	N/A	2000 starts/hr
SM-1630-D	120/1/50/60	0.9	1.2	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	2000 starts/hr
SM-1650-N	240/1/50/60	0.45	0.5	N/A	N/A	N/A	2000 starts/hr
SM-1650-D	240/1/50/60	0.45	0.5	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	1000 ohm pot or 4-20 mA	2000 starts/hr
SM-1680-N	24 Vdc	5.2	6.25	N/A	N/A	N/A	4000 starts/hr

**** Minimum Position Change of 1%.***

N/A = Not Applicable

REMOTE AMPLIFIER CHART

Actuator	Remote Mounted Amplifier	Catalog Page
SM-1110-N	AD-8240	173
SM-1120-N	AD-8240	173
SM-1140-N	AD-7530	170
SM-1150-N	AD-8240	173
SM-1170-N	AD-8240	173
SM-1180-N	AD-7530	170
SM-1520-N	AD-8240	173
SM-1530-N	AD-8240	173
SM-1540-N	AD-7530	170
SM-1550-N	AD-8240	173
SM-1580-N	AD-7540	170
SM-1590-N	AD-8240	173
SM-1630-N	AD-8240	173
SM-1650-N	AD-8240	173
SM-1680-N	AD-7540	170

ACTUATOR/AMPLIFIER SPECIFICATIONS

	SM-1110-D, SM-1120-D, SM-1150-D, SM-1170-D, SM-1520-D, SM-1530-D, SM-1550-D, SM-1590-D, SM-1630-D, SM-1650-D
Position Accuracy	1.0 % of full travel
Repeatability	0.5 % of full travel
Hysteresis	0.5 % of full travel
Linearity	+/-1 % of full travel
Deadband	0.75 % of full travel
Resolution	0.75 % of full travel

To convert RPM to sec./90°:

X = *RPM*
Y = *Sec. for 90°*

$$\frac{1}{X} \times 15 = Y$$

SM-1100 SERIES SELECTION CHART

							Selection
1	Basic Model	**SM-1110-N**	1 phase, 120 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1110-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1120-N	1 phase, 120 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1120-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1140-N	24 Vdc (If amplifier desired, see remote model AD-7530				
		SM-1150-N	1 phase, 120 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1150-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1170-N	1 phase, 240 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1170-D	1 phase, 240 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1180-N	24 Vdc (If amplifier desired, see remote model AD-7540				
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (SM-1110-N, SM-1110-D, SM-1120-N, SM-1120-D, SM-1150-N, SM-1150-D)					
		2: 240 Vac, 1 Phase, 50/60 Hz (SM-1170-N, SM-1170-D)					
		6: 24 Vdc (SM-1140-N, SM-1180-N)					
3	Speed, rpm	SM-1110 SM-1170	SM-1120	SM-1140	SM-1150	SM-1180	
		0.6/100 (11.3)	**0.6/100** (11.3)	**0.75/100** (11.3)	—	**1/100** (11.3)	
		1.2/100 (11.3)	**1.2/100** (11.3)	**1.5/100** (11.3)	—	**2/100** (11.3)	
		2.5/56 (6.3)	**2.5/100** (11.3)	**3/100** (11.3)	**0.065/100** (11.3)	**4/100** (11.3)	
4	Torque in. lbs. (Nm)	**5/28** (3.2)	**5/56** (6.3)	**6/100** (11.3)	**0.127/100** (11.3)	**8/100** (11.3)	
		10/14 (1.6)	**10/28** (3.2)	**12/70** (7.9)	**0.255/100** (11.3)	**17/90** (10.2)	
		20/7 (0.8)	**20/14** (1.6)	**25/35** (4.0)	**0.504/100** (11.3)	**35/45** (5.1)	
		40/3.5 (0.4)	**40/7** (0.8)	**50/19** (2.1)	**1.012/50** (5.6)	**70/25** (2.8)	
5	Rotation	**Select** rotation between 40° and 60 turns					

SM-1100 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual and INC/OFF/DEC (Close Coupled Enclosure)	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure)	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not available with amplifier)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA	
F006	One Potentiometer, 1000 ohms plus Transmitter, 4-20 mA (Not available with amplifier)	
F010	Anti Backlash Feedback Spring *(Up to 5.85 turns)*	
F011	Anti Backlash Feedback Gearing (5.85 turns or greater)	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certifications		
J001	CE Marking, AC input only (only available for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	

Code	Description	Selection
Shaft		
R001	Hollow Shaft,1/2" inner diameter with keyway	
R002	Round Shaft, .38" diameter, .38" long (MC Shaft)	
Auxiliary Switches		
S005	Aluminum Cam *(Up to 5.85 turns)*	
S006	Aluminum Feedback Frame (Up to 5.85 turns)	
S007	Aluminum Feedback Frame (5.85 turns or greater)	
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac, up to 5.85 turns	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac, up to 5.85 turns	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	
Special Mechanical Options		
Y006	Metric mounting holes, 6mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

SM-1500/SM-1600 SERIES SELECTION CHART

							Selection
1	Basic Model	**SM-1520-N**	1 phase, 120 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1520-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1530-N	1 phase, 120 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1530-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1540-N	24 Vdc (If amplifier desired, see remote model AD-7530				
		SM-1550-N	1 phase, 240 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1550-D	1 phase, 240 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1580-N	24 Vdc (If amplifier desired, see remote model AD-7540				
		SM-1590-N	1 phase, 240 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1590-D	1 phase, 240 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1630-N	1 phase, 120 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1630-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1650-N	1 phase, 240 Vac, 50/60 Hz, without built-in digital amplifier				
		SM-1650-D	1 phase, 240 Vac, 50/60 Hz, with built-in digital amplifier				
		SM-1680-N	24 Vdc (If amplifier desired, see remote model AD-7540				
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (SM-1520-N, SM-1520-D, SM-1530-N, SM-1530-D, SM-1630-N, SM-1630-D)					
		2: 240 Vac, 1 Phase, 50/60 Hz (SM-1550-N, SM-1550-D, SM-1590-N, SM-1590-D, SM-1650-N, SM-1650-D)					
		6: 24 Vdc (SM-1540-N, SM-1580-N, SM-1680-N)					
3 / 4	Speed, rpm / Torque in. lbs. (Nm)	SM-1520 SM-1590	SM-1530 SM-1550	SM-1540	SM-1580	SM-1630 SM-1650 SM-1680	
		0.6/400 (45)	**0.6/400** (45)	**0.8/400** (45)	**0.6/400** (45)	—	
		2/400 (45)	**2/400** (45)	**3/400** (45)	**2/400** (45)	**2.5/1000** (131)	
		4/400 (45)	**4/400** (45)	**6/400** (45)	**4/400** (45)	**5/500** (56)	
		7/400 (45)	**7/400** (45)	**10/300** (34)	**7/400** (45	—	
		15/400 (45)	**15/200** (23)	**20/140** (16)	**15/270** (31)	—	
		26/300 (34)	**26/100** (11)	**36/80** (9)	**26/160** (18)	—	
		41/190 (21)	**41/67** (8)	**56/52** (6)	**41/100** (11)	—	
		53/150 (17)	**53/50** (6)	**73/40** (4)	**53/75** (8)	—	
5	Rotation	**Select** rotation between 72° and 324 turns					

To convert RPM to sec./90°:

X = *RPM*
Y = *Sec. for 90°*

$$\frac{1}{X} \times 15 = Y$$

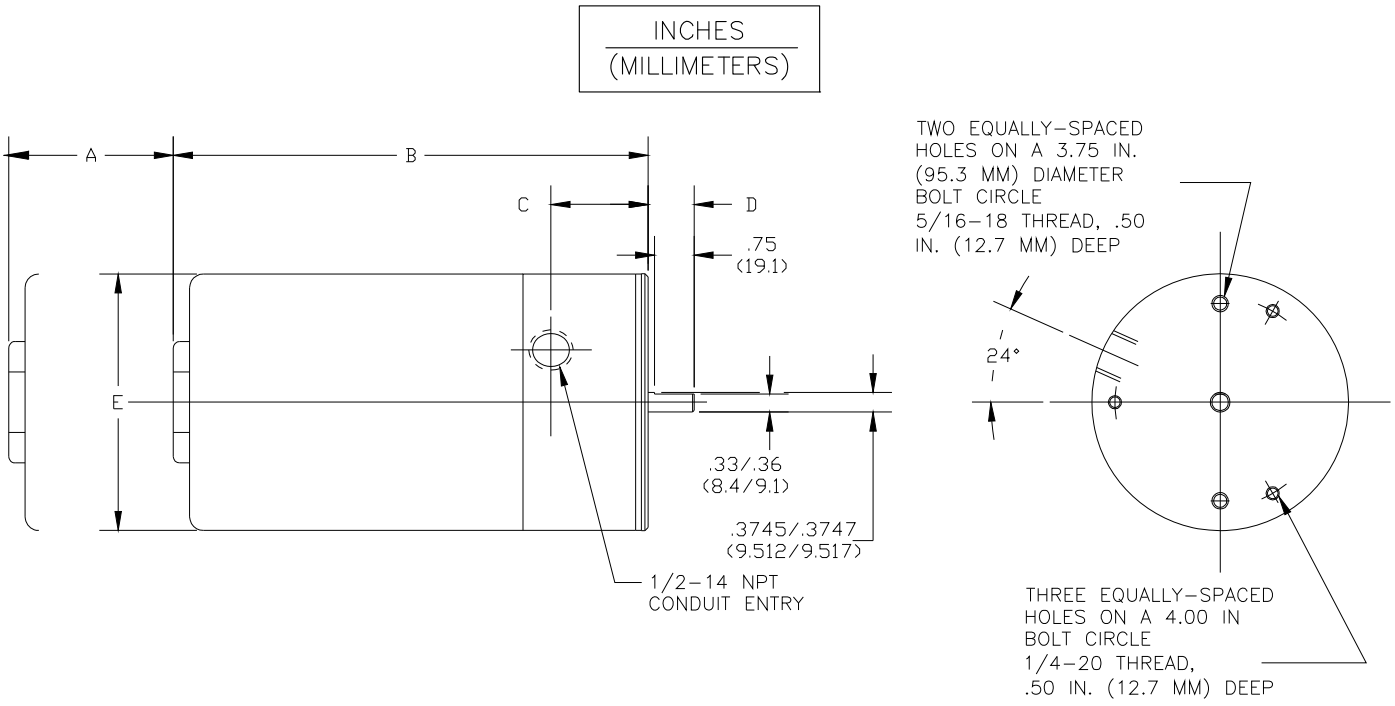
SM-1500/SM-1600 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual and INC/OFF/DEC (Close Coupled Enclosure)	
A015	ON/OFF Toggle Switch (Power) (Close Coupled Enclosure)	
Brake		
B001	Electrically Operated Brake 120 Vac	
B002	Electrically Operated Brake 240 Vac	
Drive Arm/Adapter Clevis/Driven Arm		
D041	Adjustable drive arm for standard shaft (SM-1500 only)	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not available with amplifier)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (Not available with amplifier)	
F010	Anti Backlash Feedback Spring *(Up to 27.26 turns)*	
F011	Anti Backlash Feedback Gearing (27.26 turns or greater)	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	

Code	Description	Selection
Identification/Certifications		
J001	CE Marking, AC input only (only available for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
J006	ATEX Approval (SM-1590-D only)	
Handcrank		
K001	Manual Handcrank (X-Proof)	
Shaft		
R003	Hollow Shaft, 1/2" diameter (SM-1500 only)	
Auxiliary Switches		
S005	Aluminum Cam *(Up to 27.26 turns)*	
S006	Aluminum Feedback Frame *(Up to 27.26 turns)*	
S007	Aluminum Feedback Frame (27.26 turns or greater)	
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac (Up to 27.26 turns)	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac (Up to 27.26 turns)	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	
Special Mechanical Options		
Y013	Metric mounting holes, 8mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

SM-1100 MAJOR DIMENSIONS



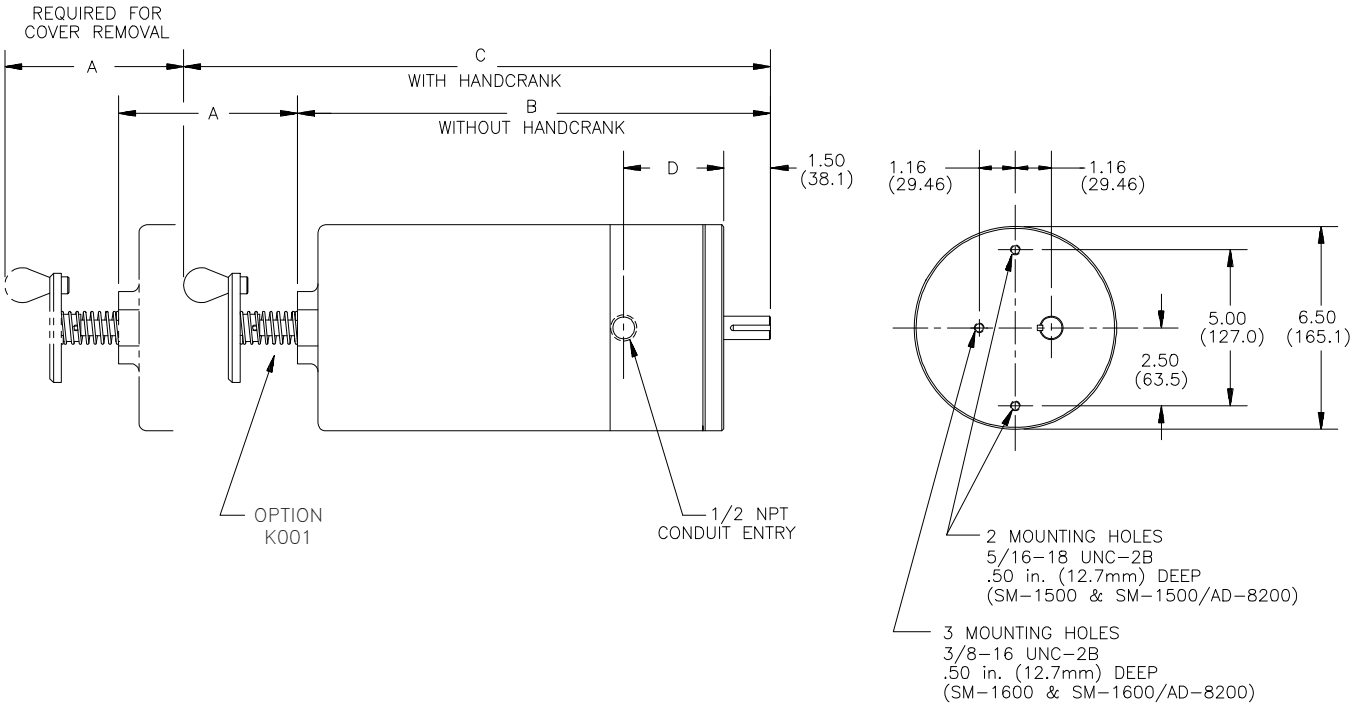
SM-1100-N	A	B	C	D	E (Dia.)
Nema Type 4 Watertight and X-proof	6.28 (159.5)	8.65 (219.7)	1.84 (46.7)	0.88 (22.4)	4.88 (124)

SM-1100-D	A	B	C	D	E (Dia.)
Nema Type 4 Watertight and X-proof	7.81 (198.4)	10.21 (259.3)	1.84 (46.7)	0.88 (22.4)	4.88 (124)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

SM-1500/SM-1600 MAJOR DIMENSIONS

INCHES
(MILLIMETERS)



SM-1500/SM-1600	A		B		C		D
	Without an internal amplifier	With an internal amplifier	Without an internal amplifier	With an internal amplifier	Handcrank without an internal amplifier	Handcrank with an internal amplifier	
Nema Type 4 Watertight and X-proof	7.00 (177.8)	10.13 (257.3)	12.20 (309.9)	15.33 (389.4)	15.76 (400.3)	18.89 (479.8)	3.20 (81.3)

Output Shaft Dimensions	SM-1500 Series	SM-1600 Series
Output Shaft Diameter	0.7490/0.7496 (19.025/19.040)	0.9990-0.9996 (25.375/25.390)
Keyway Dimensions	0.187 W x 0.105 D x 1.25 L	0.250 W x 0.145 D x 1.25 L
Key (supplied)	3/16 Sq. x 1.19 L	1/4 Sq. x 1.19 L

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

J Series Rotary Actuators

Jordan CONTROLS

GENERAL DESCRIPTION

The J Series are 90° actuators that produce up to 30,975 in. lbs. (3,500 Nm) of torque. This actuator series is designed to provide on/off, positioning and modulating control. These actuators provide quiet and reliable operation for small ball and butterfly valves, and dampers.

All J Series actuators are also available with an internal or remote mounted amplifier. These amplifiers are all full-featured ac switching devices designed to seamlessly work with the actuator for closed loop control.



FEATURES

- NEMA 4 & 4X enclosures (IP67); waterproof and dustproof
- Permanently lubricated for any position mounting
- Position indicator providing continuous output shaft position
- Manual override (except JOM & JBM 2 models)
- Corrosion-resistant enclosure
- Anti-condensation heater and thermostat
- Two auxiliary position limit switches
- Small, compact size
- Lightweight
- 30% duty cycle for AC models, 75% duty cycle for DC models
- Self-locking gear train
- On/Off control, modulating control is optional
- External, adjustable mechanical stops (Except JOM, JBM2 and JOM1)
- Single phase AC, three phase AC and DC versions
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Position feedback potentiometer, 1000 & 5000 ohm
- Valve mounting hardware kits or factory installation
- Torque switches (JOM2 - JOM12)
- Amplifier to provide modulating control and 4-20 mA feedback
- 75% duty cycle (1 Phase AC models)
- Auto/Manual, Increase, Decrease and Stop switches

SPECIFICATIONS

Power Input: AC and DC versions

Rotation: 90° maximum

Drive Train: High alloy steel gear trains provide self-locking function to avoid back drive

Motor Protection: Built-in thermal overload 275° F (135° C)
Class F Insulation: JBM2 - JOM12
Class H Insulation: JOMB, JOM & JOM1

Temperature Limits: -23° to 140°F (-30° to 60°C)

Environmental Ratings: Enclosures designed for NEMA 4, NEMA 4X (IP67)

Enclosure: Aluminum alloy, dry powder coated

Mounting Base: ISO 5211

Back Drive: Self-locking up to torque rating

Field Wiring: Screw terminal strip to accomodate 22 AWG on JOMB and JOM1 models, and 18 AWG for all other models.

Duty Cycle: 30% standard on all AC voltage models for ON/OFF and positioning applications, up to 300 starts per hour.

75% standard on all DC models for modulation up to 1,200 starts per hour. 75% optional on 1 Ph. AC models.

Approximate Weight:

JOMB - 4.5 lb. (2 kg)
JOM1 - 4.5 lb. (2 kg)
JOM - 7 lb. (3 kg)
JBM2 - 10 lb. (4.5 kg)
JOM2 - 20 lb. (9 kg)
JOM3 - 20 lb. (9 kg)
JOM4 - 39 lb. (17.5 kg)
JOM5 - 39 lb. (17.5 kg)
JOM6 - 39 lb. (17.5 kg)
JOM7 - 80 lb. (36 kg)
JOM8 - 80 lb. (36 kg)
JOM9 - 123 lb. (56 kg)
JOM10 - 123 lb. (56 kg)
JOM11 - 123 lb. (56 kg)
JOM12 - 123 lb. (56 kg)

J SERIES SPECIFICATION CHARTS

12 Vdc & 24 Vdc											
Model	**Maximum Torque in. lb. (Nm)**	**Speed sec./90°**	**Current (Amps) 12 Vdc**		**Current (Amps) 24 Vdc**		**Manual Override**	**Command Input**	**Feedback**	**Loss of Power**	**Loss of Command Signal**
			Run	***Stall***	***Run***	***Stall***					
JOM-N	440 (49.7)	20	0.5	3	0.27	1.8	NA	NA	NA	Lock-in-Place	NA
JOM-A	440 (49.7)	20	0.5	3	0.27	1.8	NA	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA or 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOMB-N	100 (11.5)	7	0.5	3	0.27	1.8	Lever	NA	NA	Lock-in-Place	NA
JOMB-A	100 (11.5)	7	0.5	3	0.27	1.8	Lever	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA or 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM1-N	300 (33.9)	12	0.5	3	0.27	1.8	Lever	NA	NA	Lock-in-Place	NA
JOM1-A	300 (33.9)	12	0.5	3	0.27	1.8	Lever	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA or 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM2-N	800 (90.3)	15	3.4	8.5	2	5	Handwheel	NA	NA	Lock-in-Place	NA
JOM2-A	800 (90.3)	15	3.4	8.5	2	5	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA or 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM3-N	1300 (146.9)	22	3.4	8.5	2	5	Handwheel	NA	NA	Lock-in-Place	NA
JOM3-A	1300 (146.9)	22	3.4	8.5	2	5	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA or 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM4-N	3500 (395.4)	16	5	10	3	6	Handwheel	NA	NA	Lock-in-Place	NA
JOM4-A	3500 (395.4)	16	5	10	3	6	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA or 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM5-N	4425 (500)	22	13	30	6.5	30	Handwheel	NA	NA	Lock-in-Place	NA
JOM5-A	4425 (500)	22	13	30	6.5	30	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA or 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM6-N	5700 (645)	28	14	30	7.5	30	Handwheel	NA	NA	Lock-in-Place	NA
JOM6-A	5700 (645)	28	14	30	7.5	30	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA or 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset

J SERIES SPECIFICATION CHARTS

120 & 240 Vac, Single Phase											
Model	Maximum Torque in. lb. (Nm)	Speed sec./90°	Current (Amps) 120 Vac		Current (Amps) 240 Vac		Manual Override	Command Input	Feedback	Loss of Power	Loss of Command Signal
			Run	*Stall*	*Run*	*Stall*					
JOM-N	440 (49.7)	20	0.5	0.6	0.3	0.5	NA	NA	NA	Lock-in-Place	NA
JOM-A	440 (49.7)	20	0.5	0.6	0.3	0.5	NA	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JBM2-N	1050 (118.6)	8	1	1.8	0.5	0.9	NA	NA	NA	Lock-in-Place	NA
JOMB-N	100 (11.5)	7	0.5	0.6	0.3	0.5	Lever	NA	NA	Lock-in-Place	NA
JOMB-A	100 (11.5)	7	0.5	0.6	0.3	0.5	Lever	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM1-N	300 (33.9)	12	0.5	0.6	0.3	0.5	Lever	NA	NA	Lock-in-Place	NA
JOM1-A	300 (33.9)	12	0.5	0.6	0.3	0.5	Lever	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM2-N	800 (90.3)	15	1	1.8	0.5	0.9	Handwheel	NA	NA	Lock-in-Place	NA
JOM2-A	800 (90.3)	15	1	1.8	0.5	0.9	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM3-N	1300 (146.9)	22	1	1.8	0.5	0.9	Handwheel	NA	NA	Lock-in-Place	NA
JOM3-A	1300 (146.9)	22	1	1.8	0.5	0.9	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM4-N	3500 (395.4)	16	1	3.6	0.5	1.8	Handwheel	NA	NA	Lock-in-Place	NA
JOM4-A	3500 (395.4)	16	1	3.6	0.5	1.8	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM5-N	4425 (500)	22	1.5	3.6	0.7	1.8	Handwheel	NA	NA	Lock-in-Place	NA
JOM5-A	4425 (500)	22	1.5	3.6	0.7	1.8	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM6-N	5700 (645)	28	1.8	3.6	0.8	1.8	Handwheel	NA	NA	Lock-in-Place	NA
JOM6-A	5700 (645)	28	1.8	3.6	0.8	1.8	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM7-N	8850 (1000)	46	3.2	10	1.6	4	Handwheel	NA	NA	Lock-in-Place	NA
JOM7-A	8850 (1000)	46	3.2	10	1.6	4	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM8-N	13275 (1500)	46	4	10	2	5	Handwheel	NA	NA	Lock-in-Place	NA
JOM8-A	13275 (1500)	46	4	10	2	5	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM9-N	17700 (2000)	58	3.2	6	1.6	4	Handwheel	NA	NA	Lock-in-Place	NA
JOM9-A	17700 (2000)	58	3.2	6	1.6	4	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM10-N	22125 (2500)	58	4	6	2	3	Handwheel	NA	NA	Lock-in-Place	NA
JOM10-A	22125 (2500)	58	4	6	2	3	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM11-N	26550 (3000)	58	3	5	1.6	3	Handwheel	NA	NA	Lock-in-Place	NA
JOM11-A	26550 (3000)	58	3	5	1.6	3	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM12-N	30975 (3500)	58	4	5	2.2	3	Handwheel	NA	NA	Lock-in-Place	NA
JOM12-A	30975 (3500)	58	4	5	2.2	3	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset

J SERIES SPECIFICATION CHARTS

220, 380 & 440 Vac, Three Phase													
Model	**Maximum Torque in. lb. (Nm)**	**Speed sec./90°**	**Current (Amps) 220 Vac**		**Current (Amps) 380 Vac**		**Current (Amps) 440 Vac**		**Manual Override**	**Command Input**	**Feedback**	**Loss of Power**	**Loss of Command Signal**
			Run	***Stall***	***Run***	***Stall***	***Run***	***Stall***					
JOM2-N	800 (90.3)	15	0.6	1.1	0.3	0.7	0.4	0.7	Handwheel	NA	NA	Lock-in-Place	NA
JOM2-A	800 (90.3)	15	0.6	1.1	0.3	0.7	0.4	0.7	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM3-N	1300 (146.9)	22	0.6	1.1	0.3	0.7	0.4	0.7	Handwheel	NA	NA	Lock-in-Place	NA
JOM3-A	1300 (146.9)	22	0.6	1.1	0.3	0.7	0.4	0.7	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM4-N	3500 (395.4)	16	1	3.5	0.7	2	0.8	2	Handwheel	NA	NA	Lock-in-Place	NA
JOM4-A	3500 (395.4)	16	1	3.5	0.7	2	0.8	2	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM5-N	4425 (500)	22	1	3.5	0.3	0.7	0.4	0.7	Handwheel	NA	NA	Lock-in-Place	NA
JOM5-A	4425 (500)	22	1	3.5	0.3	0.7	0.4	0.7	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM6-N	5700 (645)	28	1	3.5	0.7	2	0.8	2	Handwheel	NA	NA	Lock-in-Place	NA
JOM6-A	5700 (645)	28	1	3.5	0.7	2	0.8	2	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM7-N	8850 (1000)	46	0.6	1.8	0.4	1	0.4	1	Handwheel	NA	NA	Lock-in-Place	NA
JOM7-A	8850 (1000)	46	0.6	1.8	0.4	1	0.4	1	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM8-N	13275 (1500)	46	0.8	2.8	0.6	1.6	0.6	1.2	Handwheel	NA	NA	Lock-in-Place	NA
JOM8-A	13275 (1500)	46	0.8	2.8	0.6	1.6	0.6	1.2	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM9-N	17700 (2000)	58	0.4	2	0.4	1	0.4	1	Handwheel	NA	NA	Lock-in-Place	NA
JOM9-A	17700 (2000)	58	0.4	2	0.4	1	0.4	1	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM10-N	22125 (2500)	58	0.8	1.5	0.4	1	0.4	1	Handwheel	NA	NA	Lock-in-Place	NA
JOM10-A	22125 (2500)	58	0.8	1.5	0.4	1	0.4	1	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM11-N	26550 (3000)	58	1.2	3	0.6	1.5	0.6	1.5	Handwheel	NA	NA	Lock-in-Place	NA
JOM11-A	26550 (3000)	58	1.2	3	0.6	1.5	0.6	1.5	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset
JOM12-N	30975 (3500)	58	1.2	2.5	0.6	1.5	0.6	1.5	Handwheel	NA	NA	Lock-in-Place	NA
JOM12-A	30975 (3500)	58	1.2	2.5	0.6	1.5	0.6	1.5	Handwheel	4-20 mA, 1-5 Vdc or 2-10 Vdc	4-20 mA, 2-10 Vdc	Lock-in-Place	Lock-in-Place or Run to Preset

J SERIES SELECTION CHART

				Selection
1	Basic Model	**JOM-N**	440 in. lbs. (49.7 Nm)	
		JOM-A	440 in. lbs. (49.7 Nm)	
		JBM2-N	1050 in. lbs. (118.6 Nm)	
		JOMB-N	100 in. lbs. (11.2 Nm)	
		JOMB-A	100 in. lbs. (11.2 Nm)	
		JOM1-N	300 in. lbs. (33.9 Nm)	
		JOM1-A	300 in. lbs. (33.9 Nm)	
		JOM2-N	800 in. lbs. (90.3 Nm)	
		JOM2-A	800 in. lbs. (90.3 Nm) 1 Phase & Vdc	
		JOM2-A	800 in. lbs. (90.3 Nm) 3 Phase (Includes Auto/Manual Switches)	
		JOM3-N	1300 in. lbs. (146.9 Nm)	
		JOM3-A	1300 in. lbs. (146.9 Nm) 1 Phase & Vdc	
		JOM3-A	1300 in. lbs. (146.9 Nm) 3 Phase (Includes Auto/Manual Switches)	
		JOM4-N	3500 in. lbs. (395.4 Nm)	
		JOM4-A	3500 in. lbs. (395.4 Nm) 1 Phase & Vdc	
		JOM4-A	3500 in. lbs. (395.4 Nm) 3 Phase (Includes Auto/Manual Switches)	
		JOM5-N	4425 in. lbs. (500 Nm)	
		JOM5-A	4425 in. lbs. (500 Nm) 1 Phase & Vdc	
		JOM5-A	4425 in. lbs. (500 Nm) 3 Phase (Includes Auto/Manual Switches)	
		JOM6-N	5700 in. lbs. (645 Nm)	
		JOM6-A	5700 in. lbs. (645 Nm) 1 Phase & Vdc	
		JOM6-A	5700 in. lbs. (645 Nm) 3 Phase (Includes Auto/Manual Switches)	
		JOM7-N	8850 in. lbs. (1000 Nm)	
		JOM7-A	8850 in. lbs. (1000 Nm) 1 Phase	
		JOM7-A	8850 in. lbs. (1000 Nm) 3 Phase	
		JOM8-N	13275 in. lbs. (1500 Nm)	
		JOM8-A	13275 in. lbs. (1500 Nm) 1 Phase	
		JOM8-A	13275 in. lbs. (1500 Nm) 3 Phase	
		JOM9-N	17700 in. lbs. (2000 Nm)	
		JOM9-A	17700 in. lbs. (2000 Nm) 1 Phase	
		JOM9-A	17700 in. lbs. (2000 Nm) 3 Phase	
		JOM10-N	22125 in. lbs. (2500 Nm)	
		JOM10-A	22125 in. lbs. (2500 Nm) 1 Phase	
		JOM10-A	22125 in. lbs. (2500 Nm) 3 Phase	
		JOM11-N	26550 in. lbs. (3000 Nm)	
		JOM11-A	26550 in. lbs. (3000 Nm) 1 Phase	
		JOM11-A	26550 in. lbs. (3000 Nm) 3 Phase	
		JOM12-N	30975 in. lbs. (3500 Nm)	
		JOM12-A	30975 in. lbs. (3500 Nm) 1 Phase	
		JOM12-A	30975 in. lbs. (3500 Nm) 3 Phase	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz		
		2: 240 Vac, 1 Phase, 50/60 Hz		
		3: 380 Vac, 3 Phase, 50/60 Hz -- JOM2-N, JOM3-N, JOM4-N, JOM5-N, JOM6-N, JOM7-N, JOM8-N, JOM9-N, JOM10-N, JOM11-N, JOM12-N		
		4: 440 Vac, 3 Phase, 50/60 Hz -- JOM2-N, JOM3-N, JOM4-N, JOM5-N, JOM6-N, JOM7-N, JOM8-N, JOM9-N, JOM10-N, JOM11-N, JOM12-N		
		5: 220 Vac, 3 Phase, 50/60 Hz -- JOM2-N, JOM3-N, JOM4-N, JOM5-N, JOM6-N, JOM7-N, JOM8-N, JOM9-N, JOM10-N, JOM11-N, JOM12-N		
		6: 24 Vdc -- All models except JBM2, JOM-5, JOM-6, JOM-7, JOM-8, JOM9-N, JOM10-N, JOM11-N, JOM12-N		
		12: 12 Vdc -- All models except JBM2, JOM-5, JOM-6, JOM-7, JOM-8, JOM9-N, JOM10-N, JOM11-N, JOM12-N		
		13: 24 Vac -- All models except JBM2, JOM-5, JOM-6, JOM-7, JOM-8, JOM9-N, JOM10-N, JOM11-N, JOM12-N		
3	Speed, sec.	**20/440** (49.7 Nm) - JOM		
		8/1050 (118.6 Nm) - JBM2		
		7/100 (11.2 Nm) - JOMB		
		12/300 (33.9 Nm) - JOM1		
		15/800 (90.3 Nm) - JOM2		
		22/1300 (146.9 Nm) - JOM3		
		16/3500 (395.4 Nm) - JOM4		
		22/4425 (500 Nm) - JOM5		
4	Torque, in. lbs.	**28/5700** (645 Nm) - JOM6		
		46/8850 (1000 Nm) - JOM7		
		46/13275 (1500 Nm) - JOM8		
		58/17700 (2000 Nm) - JOM9		
		58/22125 (2500 Nm) - JOM10		
		58/26550 (3000 Nm) - JOM11		
		58/30975 (3500 Nm) - JOM12		
5	Rotation	**90°**		

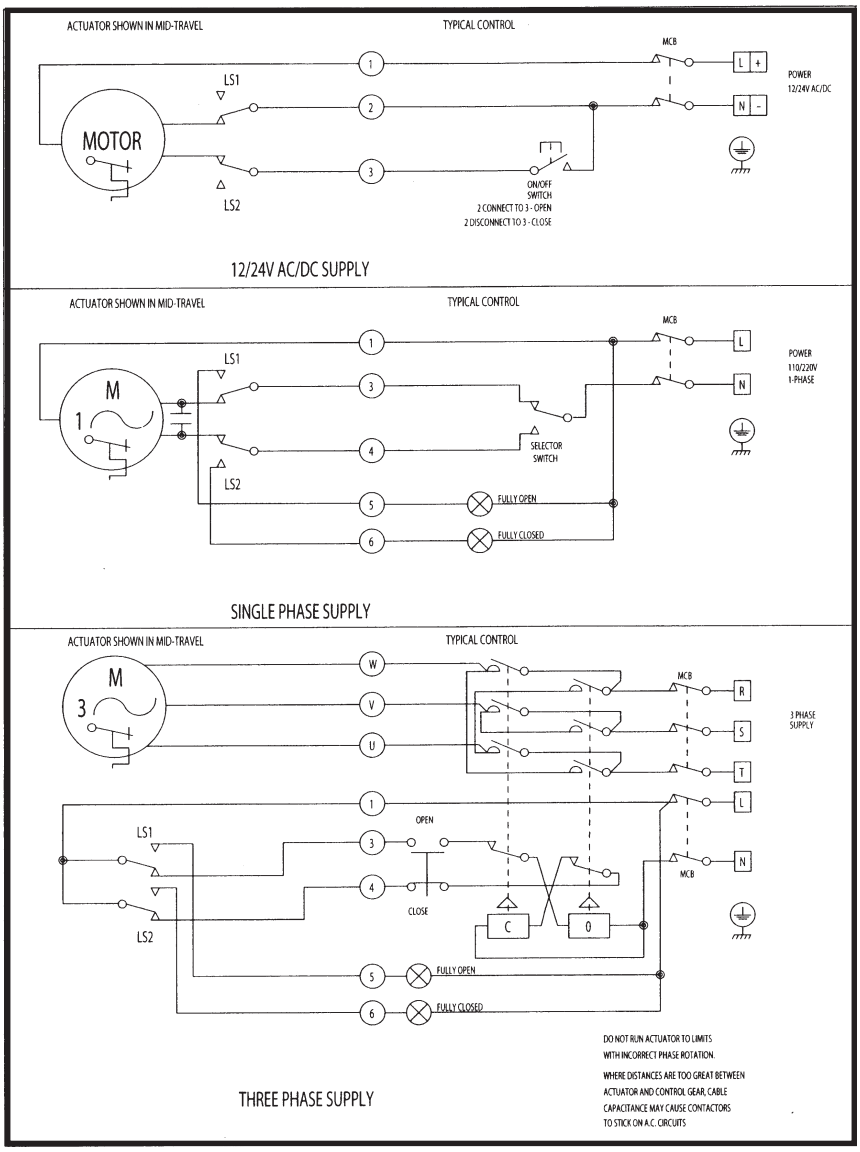
J SERIES OPTIONS

Code	Description	Selection
	Toggles/Lights	
A031*	Auto/Manual Switches (JOM2-N, JOM3-N), 1 Ph.	
A032*	Auto/Manual Switches (JOM4-N, JOM5-N, JOM6-N), 1 Ph.	
A033	Auto/Manual Switches (JOM7, JOM8), 1 Ph.	
A034*	Auto/Manual Switches (JOM2-N, JOM3-N), 3 Ph.	
A035*	Auto/Manual Switches (JOM4-N, JOM5-N, JOM6-N), 3 Ph.	
A036	Auto/Manual Switches (JOM7, JOM8), 3 Ph.	
A043	Auto/Manual Switches (JOM9, JOM10, JOM11, JOM12), 1 Ph.	
A044	Auto/Manual Switches (JOM9, JOM10, JOM11, JOM12), 3 Ph.	
	Feedback	
F001	Potentiometer, 1000 ohms	
F003	Potentiometer, 5000 ohms	
F036	4-20 mA Output Transmitter	
	Auxiliary Switches	
S042	Torque Switches (JOM2 & JOM3)	
S042	Torque Switches (JOM4 - JOM6)	
S057	Torque Switches (JOM7 - JOM12)	
	Special Electrical Options	
X088	75% Duty Cycle for JOM, JOM1 & JOMB (AC only)	
X089	75% Duty Cycle for JOM2 - JOM8 (1 Phase AC only)	
X095	75% Duty Cycle (All Vdc, 12 Vac & 24 Vac)	
	Special Mechanical Options	
Y033	M20 Conduit Entries	

** Standard on JOM2-A, JOM3-A, JOM4-A, JOM5-A, JOM6-A units with 3 Phase voltage*

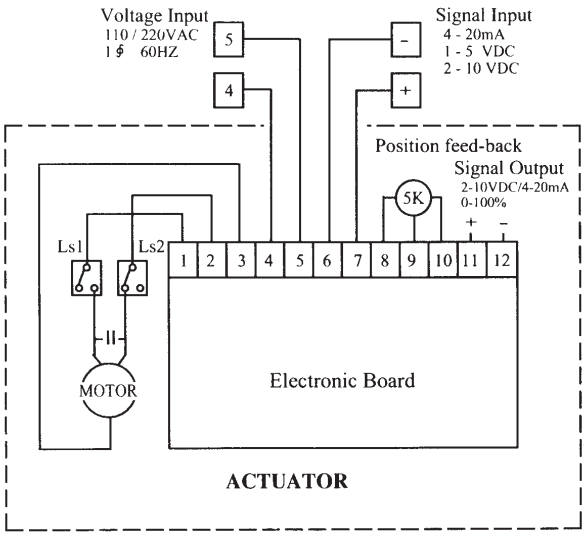
TYPICAL WIRING DIAGRAMS

For models without amplifier



For models with amplifier

110 /220 Vac
12/24 DC, AC



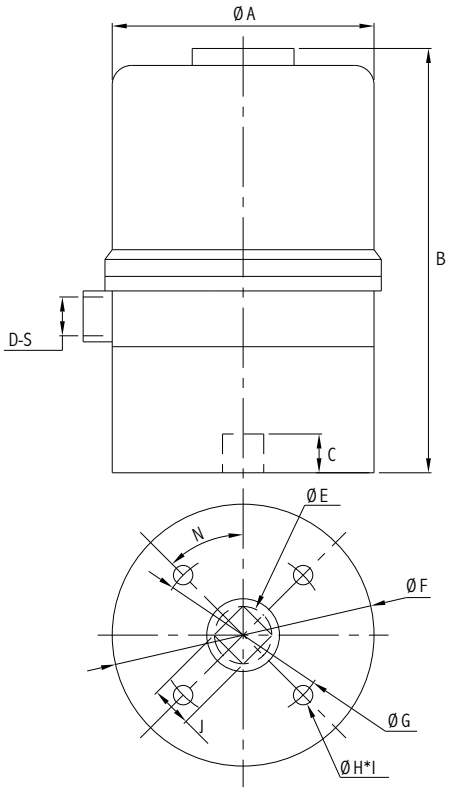
Consult Factory for Three Phase wiring.

MAJOR DIMENSIONS

JOM

	A	B	C	D	E	F	G
inch (mm)	4.25 (108)	7 (178)	0.63 (16)	0.04 (1)	.94 (24)	4.17 (106)	2.75 (70)

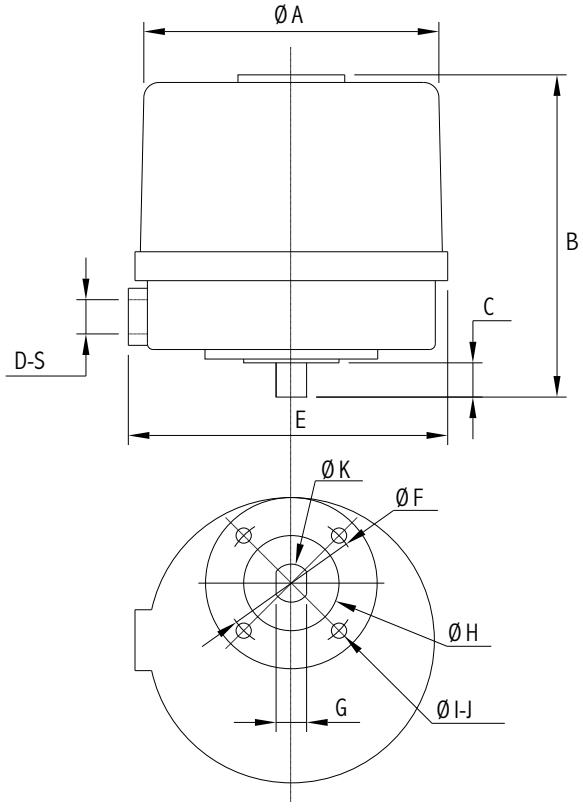
H	I	J Max	N	S	Flange Type
(m8)	.16 (4)	0.67 (17)	45°	½ NPT (m20)	F07



JBM2

	A	B	C	D	E	F	G
inch (mm)	6.06 (154)	6.61 (168)	0.71 (18)	0.04 (1)	6.3 (160)	2.75 (70)	.63 (16)

H	I	J	K	S	Flange Type
1.97 (50)	.16 (4)	(m8)	.78 (20)	½ NPT (m20)	F07



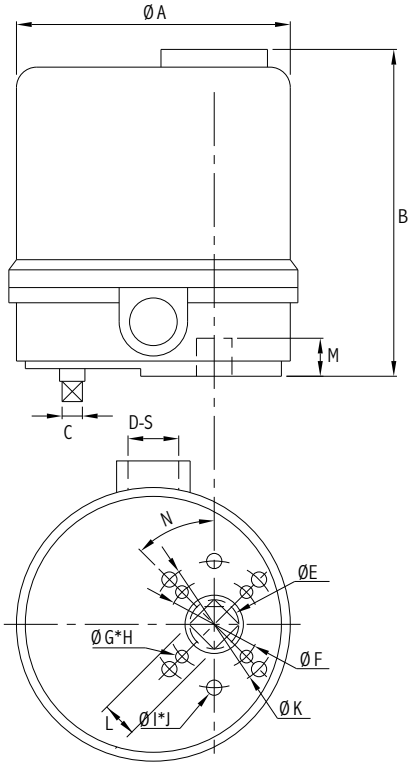
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. Current installation dimension drawings are available on request.

MAJOR DIMENSIONS

JOM1 & JOMB

	A	B	C	D	E	F	G	H
inch (mm)	4.49 (114)	6.10 (155)	0.31 (8)	½ NPT 1	0.75 (19)	1.4 (36)	?? (m5)	4

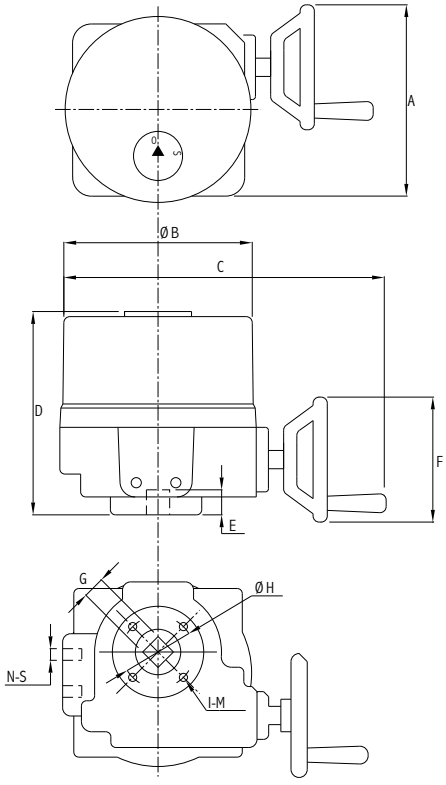
	I	J	K	L	M	N	S	Flange Type
inch (mm)	6	0.23 (6)	1.97 (50)	0.55 (14)	0.59 (15)	45°	½ NPT (m20)	F03/ F05



JOM2 & JOM3

	A	B	C	D	E	F	G Max
inch (mm)	7.87 (200)	7.87 (200)	13 (330)	8.11 (206)	1.18 (30)	5 (125)	.86 (22)

H	I	M	N	S	Flange Type
2.75 (70)	.16 (4)	(m8)	.08 (2)	½ NPT (m20)	F07



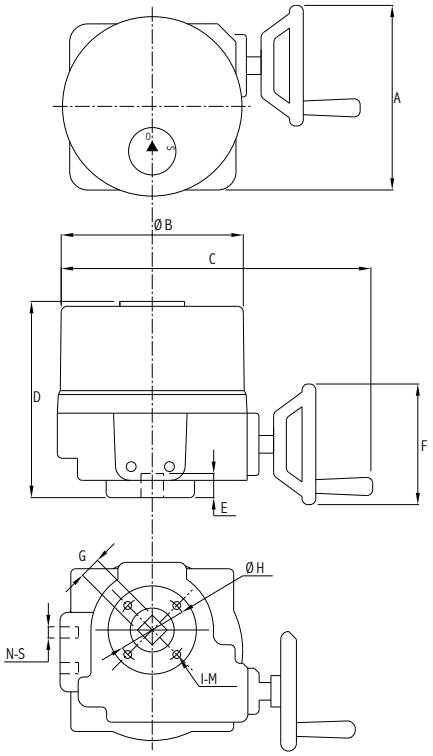
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. Current installation dimension drawings are available on request.

MAJOR DIMENSIONS

JOM4, JOM5 & JOM6

	A	B	C	D	E	F	G Max
inch (mm)	11.8 (300)	9.4 (234)	15 (380)	10.86 (276)	1.57 (40)	7.67 (195)	1.45 (35)

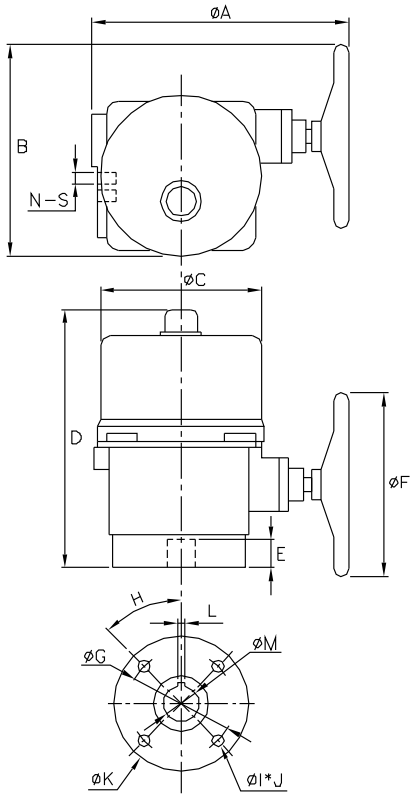
H	I	J	N	S	Flange Type
4 (102)	.16 (4)	(m10)	.08 (2)	½ NPT (m20)	F10



JOM7 & JOM8

	A	B	C	D	E	F	G	H
inch (mm)	17.7 (450)	13.4 (340)	9.2 (234)	16.5 (420)	2.4 (60)	11.6 (295)	5.5 (140)	45°

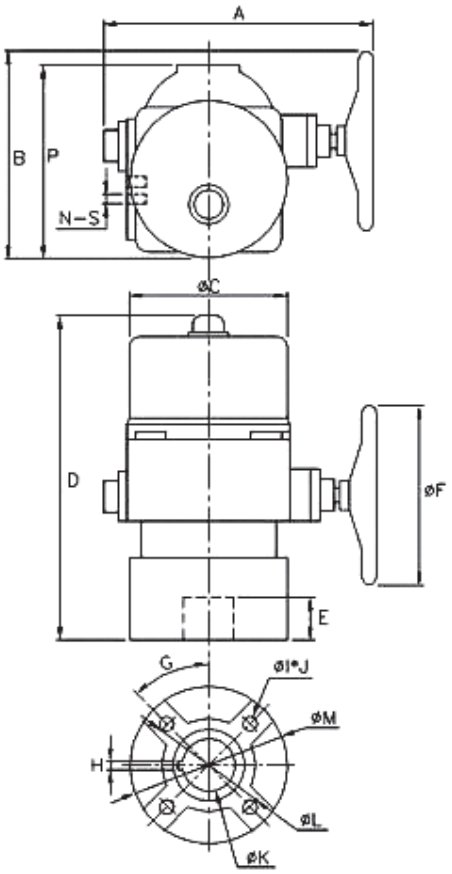
I	J	K	L	M Max.	N	S	Flange Type
(m16)	0.2 (4)	7.1 (180)	0.4 (10)	1.4 (35)	0.1 (2)	½ NPT (m20)	F14



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. Current installation dimension drawings are available on request.

MAJOR DIMENSIONS

JOM9, JOM10, JOM11 & JOM12



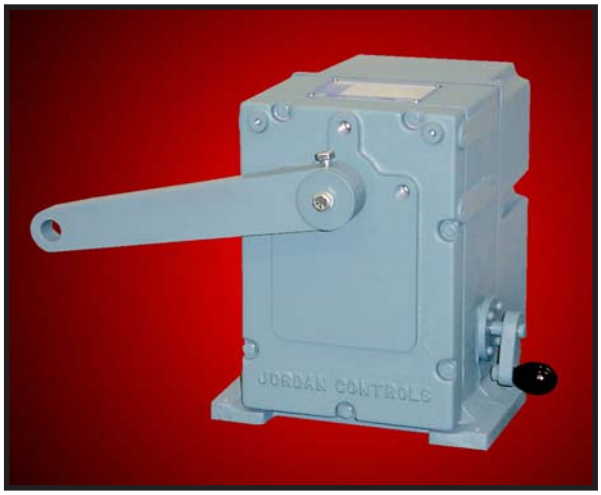
	A	B	C	D	E	F	G	H	I	J	K	L	M	P	N	S	Flange Type
inch (mm)	18.5 (470)	13.8 (350)	10.2 (260)	23.2 (590)	3.9 (100)	15.5 (395)	45°	0.47 (12)	½ NPT (m20)	0.16 (4)	2.9 (75)	6.5 (165)	8.7 (221)	14.2 (360)	.08 (2)	½ NPT (m20)	F16

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. Current installation dimension drawings are available on request.

GENERAL DESCRIPTION

The SM-1700 Series are 90° to 360° rotary actuators internally geared to produce up to 2400 in. lbs. (271 Nm) of torque and offer continuous modulation. This actuator series uses high efficiency spur gears that are designed to meet the exacting requirements for closed-loop, modulating positioning control. The SM-1700 Series is ideally suited for dampers, vanes, valves and other process control applications requiring exact positioning control.

All SM-1700 Series actuators are also available with an internal amplifier (excluding SM-1740 & SM-1760). These amplifiers are all full-featured ac switching devices designed to work seamlessly with the actuator for closed loop control.



FEATURES

- Spur gear train for longer maintenance-free operation
- Permanently lubricated for any position mounting
- Internal or remote-mounted amplifier
- Self-locking drive system to hold last position and prevent backdriving, up to torque rating
- AC input power (single and three phase) or DC input power versions
- Wide ambient temperature range
- Foot, face, or side mounting
- Full two year warranty

BASE MODEL INCLUDES:

- Motor
- Manual override, safely isolated from the motor
- Splined output shaft (20 tooth)
- Cast aluminum NEMA 4 (IP65) & dust ignition-proof enclosure
- Anti-condensation heater and thermostat
- Automatic torque limiting
- Internal spur gear train
- 4-20 mA transmitter for customer use (amplifier models only)
- End of travel switches, not for customer use

POPULAR OPTIONS *(See Pages 197-206)*

- Position switch outputs
- Keyed output shaft
- Drive arm, adapter/clevis kit, and driven arm linkage components
- Auxilliary position limit switches
- Custom mounting and interface hardware
- Local Auto/Manual toggle or key lock rotary switches
- Open-Close-Stop pushbuttons
- Various enclosure coatings
- Local position indicator
- Remote mounted battery back-up, RP-4000

AMPLIFIER FEATURES

- Digital amplifiers
- Isolated, two-wire, 4-20mA feedback
- Pushbutton, customer-adjustable zero and span
- Adjustable deadband
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place
  - Run to preset
- Field selectable dynamic brake
- AC triac switching of AC motors and IGBT for DC motors for increased life
- Easy pushbutton setup

BASIC SPECIFICATIONS FOR SM-1700 SERIES ACTUATORS

Rotation: 0° to 360°, set at factory
(Less than 90°, consult factory)

Drive Train: Spur gearing

Input Power: See specification chart on next page

Torque: Up to 2400 in.lbs. (271 Nm)

Speed: 12 to 60 seconds for 90° rotation
48 to 240 seconds for 360° rotation
For faster speeds, consult factory

Temperature Limits: -40°F to 185°F (-40°C to 85°C)
For greater temperature ranges, consult factory.

Enclosure: Cast aluminum

Environmental Ratings: NEMA Type 4 (IP65), dust ignition proof, Class II, Division 1, Groups E, F & G

Backdrive: Self-locking up to rated torque

Field Wiring: Barrier terminal blocks

Torque Limiting: 10 Amp, SPDT switch, one for each direction (CW & CCW). Switch contacts will open when torque exceeds rated torque for actuator.

Approximate Weight: 40 lbs. (18.1 kg)

Approvals: See page 207

Position Switch Rating: 120/240 Vac, 20 amps

ACTUATOR/AMPLIFIER SPECIFICATIONS

	SM-1715-D, SM-1720-D, SM-1730-D, SM-1740-D, SM-1750-D & SM-1790-D	SM-1760-D
Position Accuracy	1.0 % of full travel	0.25 % of full travel
Repeatability	0.5 % of full travel	0.25 % of full travel
Hysteresis	0.5 % of full travel	0.25 % of full travel
Linearity	+/-1 % of full travel	+/-0.5 % of full travel
Deadband	0.75 % of full travel	0.5 % of full travel
Resolution	0.75 % of full travel	0.5 % of full travel

REMOTE AMPLIFIERS

Actuator	Remote Mounted Amplifier	Catalog Page
SM-1715-N	AD-8900	177
SM-1720-N	AD-8240	173
SM-1730-N	AD-8240	173
SM-1740-N	AD-7540	170
SM-1750-N	AD-8240	173
SM-1760-N	AD-7830/AD-9120	180/184
SM-1790-N	AD-8240	173

SM-1700 SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Maximum Available Torques in. lbs. (Nm)	Speeds For 90° (sec.)	Command Input	Input Impedance in ohms	Feedback	Modulation Rate*	Dynamic Braking	Loss of Power	Loss of Command Signal
		Run	Stall									
SM-1715-N	208/3/50/60	0.5	1.5	1200 (136)	12, 24 or 48	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
	240/3/50/60	0.4	1.3									
	380/3/50	0.3	0.9									
	480/3/50/60	0.2	0.65									
SM-1720-N	120/1/50/60	1.3	1.9	2400 (271)	31 or 60	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
SM-1720-D	120/1/50/60	1.3	1.9	2400 (271)	31 or 60	4-20 mA	200	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
						0-5 Vdc or 0-10 Vdc	100,000					
SM-1730-N	120/1/50/60	0.9	1.2	1200 (136)	12, 24 or 48	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
SM-1730-D	120/1/50/60	0.9	1.2	1200 (136)	12, 24 or 48	4-20 mA	200	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
						0-5 Vdc or 0-10 Vdc	100,000					
SM-1740-N	24 Vdc	1.7	1.9	1000 (113)	16, 31 or 40	N/A	N/A	N/A	4000 starts/hour	N/A	Lock in Place	N/A
SM-1750-N	240/1/50/60	0.45	0.5	1200 (136)	12, 24 or 48	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
SM-1750-D	240/1/50/60	0.45	0.5	1200 (136)	12, 24 or 48	4-20 mA	200	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
						0-5 Vdc or 0-10 Vdc	100,000					
SM-1760-N	90 Vdc	1.7	1.9	1000 (113)	16, 31 or 40	N/A	N/A	N/A	4000 starts/hour	N/A	Lock in Place	N/A
SM-1790-N	240/1/50/60	0.45	0.5	2400 (271)	31 or 60	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
SM-1790-D	240/1/50/60	0.45	0.5	2400 (271)	31 or 60	4-20 mA	200	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
						0-5 Vdc or 0-10 Vdc	100,000					

** Minimum position change of 1%*

SM-1700 SERIES SELECTION CHART

						Selection
1	Basic Model	**SM-1715-N**	3 phase, 208/240/380/480 Vac, (If amplifier desired, see AD-8900			
		SM-1720-N	1 phase, 120 Vac, without built-in amplifier			
		SM-1720-D	SM-1720 with built-in amplifier			
		SM-1730-N	1 phase, 120 Vac, without built-in amplifier			
		SM-1730-D	SM-1730 with built-in amplifier			
		SM-1740-N	24 Vdc, (If amplifier desired, see model AD-7540			
		SM-1750-N	1 phase, 240 Vac, without built-in amplifier			
		SM-1750-D	SM-1750 with built-in amplifier			
		SM-1760-N	90 Vdc, (If amplifier desired, see AD-7830 or AD-9120			
		SM-1790-N	1 phase, 240 Vac, without built-in amplifier			
		SM-1790-D	SM-1790 with built-in amplifier			
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (SM-1720-N, SM-1720-D, SM-1730-N or SM-1730-D)				
		2: 240 Vac, 1 Phase, 50/60 Hz (SM-1750-N, SM-1750-D, SM-1790-N or SM-1790-D)				
		3: 380 Vac, 3 Phase, 50 Hz (SM-1715-N)				
		4: 480 Vac, 3 Phase, 50/60 Hz (SM-1715-N)				
		5: 240 Vac, 3 Phase, 50/60 Hz (SM-1715-N)				
		6: 24 Vdc (SM-1740-N)				
		7: 90 Vdc (SM-1760-N)				
		10: 208 Vac, 3 Phase (SM-1715-N)				
		19: 220 Vac, 1 Phase (SM-1750-D or SM-1790-D)				
3	Speed, sec. (a) / Torque, in. lbs. (Nm)	SM-1720, SM-1790	SM-1715, SM-1730, SM-1750	SM-1740, SM-1760		
4		**60/1800** (203)	**12/1200** (136)	**16/1000** (113)		
		31/1800 (203)	**24/1200** (136)	**31/1000** (113)		
		60/2400 (271)	**48/1200** (136)	**40/1000** (113)		
5	Rotation	Select rotation of 90°, 120° or 360°. (Less than 90°, consult factory)				

(a) Speeds are for 90° of rotation.

SM-1700 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches	
A002	INC/OFF/DEC Toggle Switches	
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002)	
A009	ON/OFF Toggle Power Switch	
Drive Arm/Adapter Clevis/Driven Arm		
D001	Fixed drive arm for splined shaft (old SA1)	
D002	Adjust. drive arm for splined shaft (old SA2)	
D003	Adapter clevis kit used with D001 & D005 (old AK1)	
D004	Adapter clevis kit used with D002 & D006 (old AK2)	
D005	Driven arm used with D001 and D003 (old DA1)	
D006	Driven arm used with D002 and D004 (old DA2)	
D019	Hagan Bailey Pneumatic Replacement (Must provide model number and drive arm dimensions)	
D024	Wedgelock Coupling (valve dims. required)	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not for use with internal amplifiers)	
F003	Potentiometer, 5000 ohms (Not for use with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with internal amplifiers)	
F005	Transmitter, 4-20 mA	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (Not for use with internal amplifiers)	
F999	Special Potentiometer (consult factory)	

Code	Description	Selection
Identification/Certifications		
J001	CE Marking, AC input only (only available for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Position Indicator		
P001	Local Position Indicator (90° rotation only)	
Shaft		
R004	Keyed Shaft, 1" diameter, 1/4" square keyway	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac (Up to 120°)	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac (Up to 120°)	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac (Up to 120°)	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac (Up to 120°)	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S015	Two nonadjustable torque limit switches, SPST, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
Special Mechanical Options		
Y008	Metric mounting holes, 12mm	

SM-1700 SPECIAL OPTIONS

(Consult factory for special requirements)

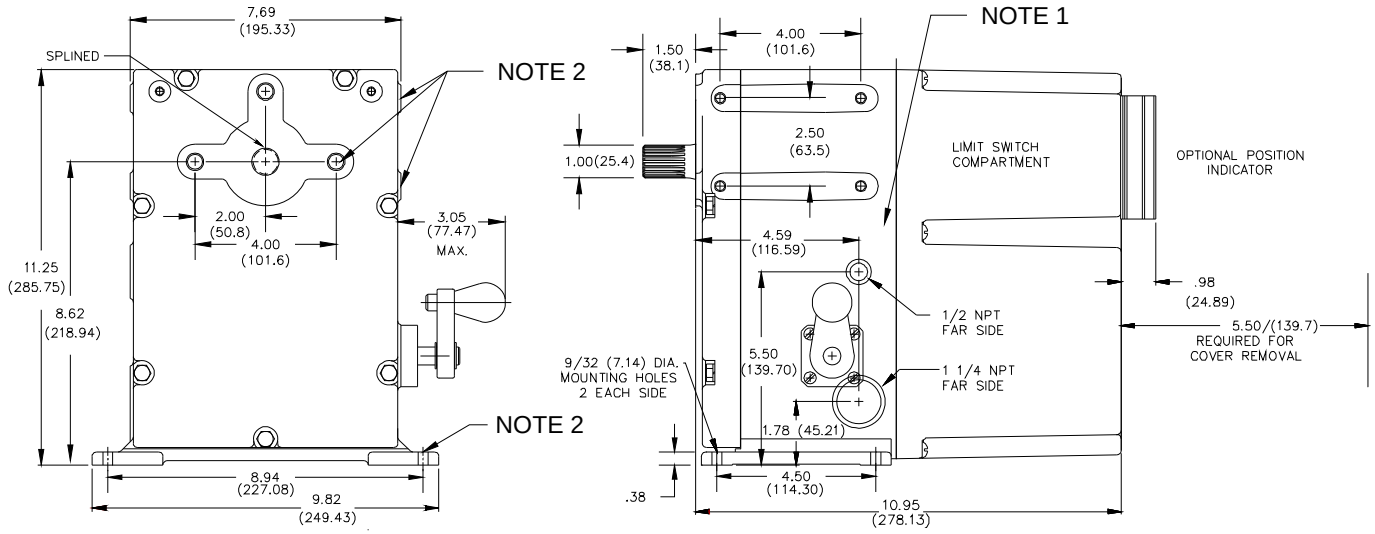
Code	Description	Selection
Toggles, Lights		
A004	Open-Close-Stop Push Buttons - NEMA 4	
A005	Two Pilot Lights - NEMA 4	
A006	Two Position Selector Switches - NEMA 4 (Auto/Manual)	
A007	Three Position Selector Switches - NEMA 4 (INC/OFF/DEC)	
A010	Local Auto/Manual Switches & (INC/OFF/DEC) Switches (A006 & A007)	
A011	Key locks for rotary Auto/Manual switches - NEMA 4	
A012	ON/OFF Rotary Power Switch - NEMA 4	
Connectors		
C001	Plug-In Cannon Type Connector on Enclosure for Power and Control	
C002	Mating Plug-In Cannon Type Connector on Enclosure for Power and Control	

Code	Description	Selection
Feedback		
F008	Contactless 4-20 mA	
F009	Contactless 0-10Vdc	
F010	Anti Backlash Feedback Gearing (single turn)	
Auxiliary Switches		
S009	Two Adjustable Position Switches, SPDT, 10 Amp (360°)	
S010	Two Adjustable Position Switches, DPDT, 10 Amp (360°)	
S012	Two nonadjustable end of travel position switches, SPST, 10 Amp (360°)	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp (360°)	
Special Electrical Options		
X001	24 Vdc power supply, 320 mA, 100-240 Vac input	

For a full description of options, go to the Complete Listing of Options starting on page 197.

SM-1700 MAJOR DIMENSIONS

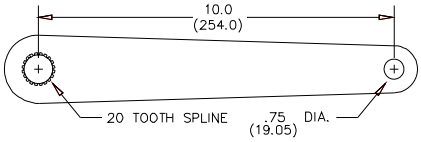
INCHES
(MILLIMETERS)



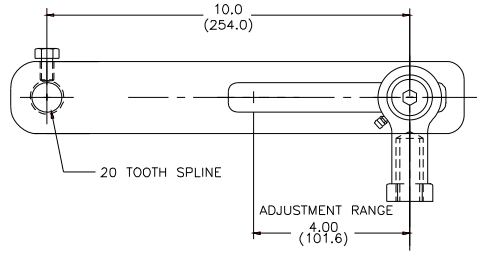
SPLINE DATA:

SPLINE NO.	CHORD	LARGE DIA	ROOT DIA	NO. TEETH
0875-20-2	.204 - .205	.925 - .930	.820 - .825	20

Optional Fixed Drive Arm
D001



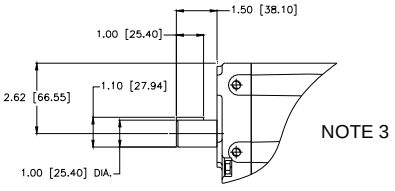
Optional Adjustable Drive Arm
D002



NOTES:

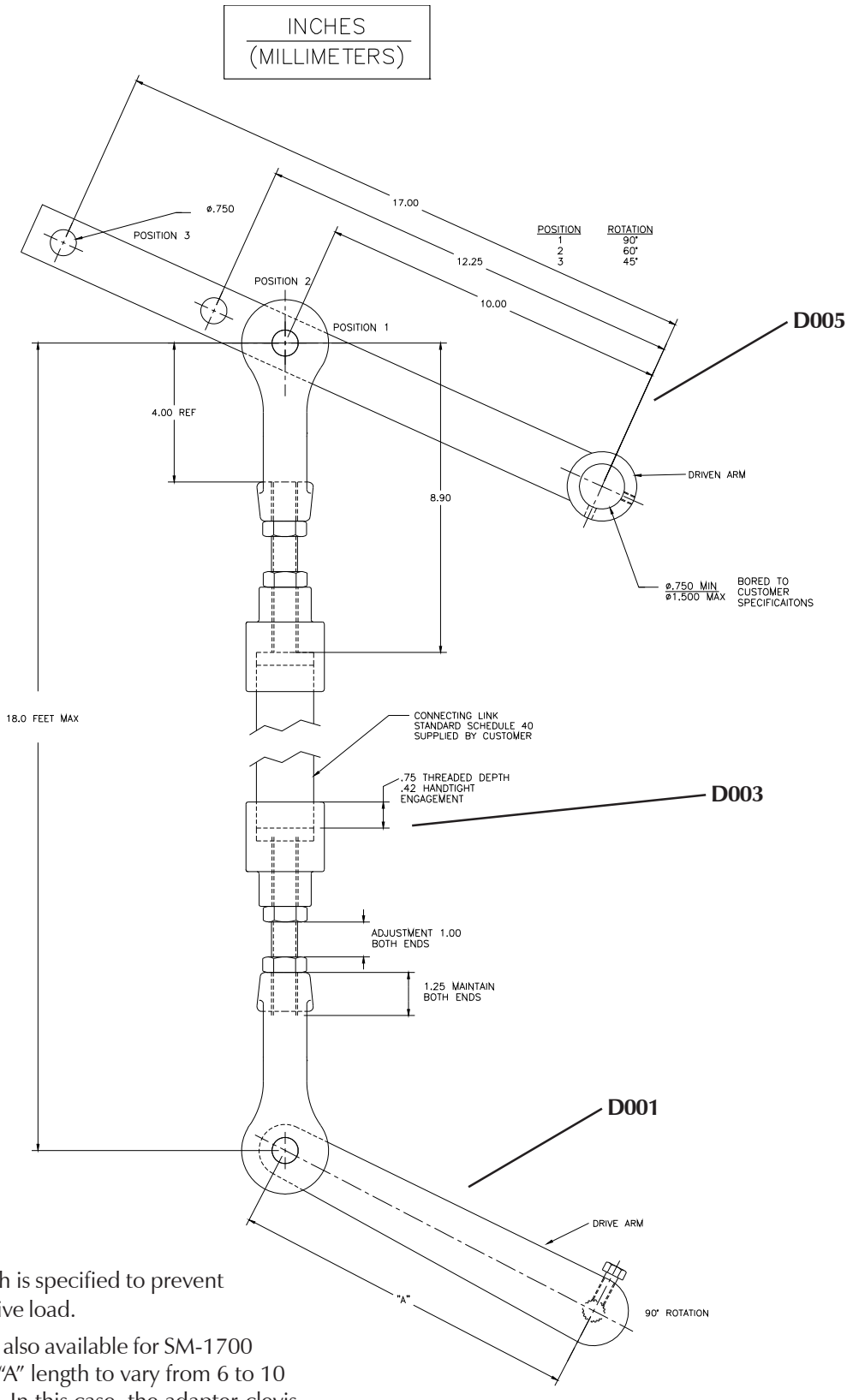
1. The optional local toggle switches are located on these surfaces.
2. The SM-1700 series actuators can be foot, face or side mounted.
3. A keyed output shaft (optional) is available in a one inch (25.4 mm) diameter with 1/4 inch (6.35 mm) square keyway.
4. See page 209 and 211 for use as linear output to increase or decrease torque.

Optional keyed output shaft
R005



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

SM-1700 SERIES DRIVE ARM AND LINKAGE OPTIONS



Notes:

1. Maximum total link length is specified to prevent buckling under compressive load.
2. Adjustable drive arms are also available for SM-1700 Series actuators to allow "A" length to vary from 6 to 10 inches (152 to 254 mm). In this case, the adapter-clevis has rod ball ends with lubricating fittings.
3. Special drive arm lengths are available to meet specific application requirements.

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

GENERAL DESCRIPTION

The SM-5000 Series are quarter turn, rotary actuators internally geared to produce up to 12,500 ft. lbs. (16,950 Nm) of torque, and offer continuous modulation up to 4,000 starts per hour. This series of actuators uses a scotch-yoke output gearing design that provides up to 66% more torque than rated at the ends of travel (0° & 90° positions). The SM-5000 Series is ideally suited for dampers, vanes, valves and other process control applications requiring high torque and exact positioning control.

All SM-5000 Series actuators are also available with an internal amplifier (excluding SM-5140). These amplifiers are all full-featured ac or dc switching devices designed to work seamlessly with the actuator for closed loop control.



SM-5160

FEATURES

- "Scotch-yoke" output torque increases at ends of travel by 66% above rated torque for positive seating and unseating of valves, dampers, etc.
- Permanently lubricated for any position mounting
- Internal amplifier (except SM-5140)
- Self-locking drive system to hold in last position and prevent backdriving up to torque rating
- AC input power (single and three phase) or DC input power versions
- Foot or face mounting
- Full two year warranty

BASE MODEL INCLUDES:

- Motor
- Manual override
- Splined output shaft
- Cast aluminum, painted NEMA 4 (IP65) enclosure
- Anti-condensation heater with thermostat
- Automatic torque limiting
- Internal Scotch-yoke gear train
- 1000 ohm potentiometer feedback (amplifier models only)
- End of travel switches, not for customer use

POPULAR OPTIONS *(See Pages 197-206)*

- Position switch outputs
- Keyed output shaft
- Drive arm, adapter/clevis kit, and driven arm linkage components
- Auxilliary position limit switches
- Custom mounting and interface hardware
- Local Auto/Manual toggle or key lock rotary switches
- Open-Close-Stop pushbuttons
- Pilot lights
- Extended temperature ranges
- Contactless feedback
- Local position indicator
- Battery back-up, RP-4000

SELECT ACTUATOR BY TORQUE

Up to 300 ft. lbs. (407 Nm)	**SM-5100**
300 (407) to 1,000 ft. lbs. (1356 Nm)	**SM-5200**
1,000 (1356) to 3,200 ft. lbs. (4339 Nm)	**SM-5300**
3,200 (4339) to 12,500 ft. lbs. (16,947 Nm)	**SM-5400**

BASIC SPECIFICATIONS FOR SM-5000 SERIES ACTUATORS

Rotation: Up to 90°, set at factory

Drive Train: Scotch-yoke

Temperature Limits: -40° to 150° F (-40° to 65° C)
(For higher temperatures consult factory)

Environmental Ratings:
SM-5100 - NEMA 4 (IP65) Explosion Proof, Class I, Division 1, Groups C& D. Class II, Division 1, Groups E, F & G
SM-5200, SM-5300, SM-5400 - NEMA 4 (IP65)

Enclosure: Cast aluminum with polyurethane coating

Back Drive: Self-locking up to twice the torque rating

Field Wiring: Barrier terminal blocks

Torque Limiting: 20 Amp, SPDT switch, one for each direction (CW & CCW). Switch will change states when torque exceeds rated torque for actuator. (10 Amp switches for SM-5100).

Approximate Weight: SM-5100: 80 lbs. (36 kg)
SM-5200: 210 lbs. (84 kg)
SM-5300: 350 lbs. (147 kg)
SM-5400: 565 lbs. (238 kg)

Approvals: See page 207

Position Switch Rating: 20 Amp, SPDT switch, one for each direction (CW & CCW).

ACTUATOR/AMPLIFIER SPECIFICATIONS

	SM-5115-D, SM-5120-D, SM-5160-D, SM-5190-D, SM-5210-D, SM-5220-D, SM-5260-D, SM-5310-D, SM-5320-D, SM-5360-D, SM-5430-D & SM-5480-D	**SM-5160-D, SM-5260-D, SM-5360-D, & SM-5480-D**
Position Accuracy	1.0 % of full travel	0.25 % of full travel
Repeatability	0.5 % of full travel	0.25 % of full travel
Hysteresis	0.5 % of full travel	0.25 % of full travel
Linearity	+/-1 % of full travel	+/-0.5 % of full travel
Deadband	0.75 % of full travel	0.5 % of full travel
Resolution	0.75 % of full travel	0.5 % of full travel

REMOTE AMPLIFIER CHART

Actuator	**Remote Mounted Amplifier**	**Catalog Page**
SM-5115-N	AD-8900	177
SM-5120-N	AD-8240	173
SM-5140-N	AD-7540	170
SM-5160-N	AD-7830/AD-9120	180/184
SM-5190-N	AD-8240	173
SM-5210-N	AD-8900	177
SM-5220-N	AD-8240	173
SM-5260-N	AD-7830/AD-9120	180/184
SM-5310-N	AD-8900	177
SM-5360-N	AD-7830/AD-9120	180/184
SM-5430-N	AD-8900	177
SM-5480-N	AD-7830/AD-9120	180/184

JORDAN CONTROLS UNIQUE AD-9120 DIGITAL OR AD-7830 ANALOG AMPLIFIER CONTROL

For SM-5160, SM-5260, SM-5360 and SM-5480

GENERAL SPECIFICATIONS COMMON TO AD-7830 & AD-9120 AMPLIFIERS:

1. Drive stage control uses Intelligent Power Module (IPM) for pulse width modulation (PWM) power switching to the motor for longer motor life and cooler operation.
2. Proportional Control - Compares the control signal against the feedback signal and accelerates quickly for a large control change and accelerates slowly for a small control change. Slow acceleration is called "Creep" Control - Very small actuator movements to maintain tight process control.
3. Amplifier provides optimum motor efficiency and lower energy consumption due to features like creep Control
4. Closed loop servo-control between control signal and feedback signal
5. Isolated, two wire 4-20mA loop powered position feedback
6. Power input 120/240 Vac (+/- 10%), 1 phase, 50/60 hz
7. Command Inputs:
   - 4-20mA into 200 ohm shunt
   - 0-10 Vdc into 100K ohm impedance
   - 0-5 Vdc into 100K ohm impedance
8. Bi-Directional (CW or CCW on increase in command signal) Actuator Positioning
9. Over-current, under-voltage, short circuit and over temperature protection onboard
10. Customer adjustable deadband, gain, zero and span
11. Fast amplifier response time (1ms for analog, 8ms for digital) to change in control signal
12. Field selectable loss of signal control: Lock-in-place or go to pre-set position
13. Field adjustable speed control
14. Electronic circuitry temperature range of -40F to 150F
15. Plugable terminal block connectors supplied for quick field wiring terminations
16. Accessory option for -10 to +10 Vdc control signal.

UNIQUE FEATURES OF AD-7830 AMPLIFIER:

- Potentiometer/Dip switch Set-up
- Balance speed control for different load dynamics like overhung loads or back driving type units
- Increase/decrease pushbuttons
- DIP or dry contact auto/manual control
- LED diagnostics of:
  - Loss of signal LOS/fault
  - Increase command
  - Decrease command
  - Null command

UNIQUE FEATURES OF AD-9120 AMPLIFIER:

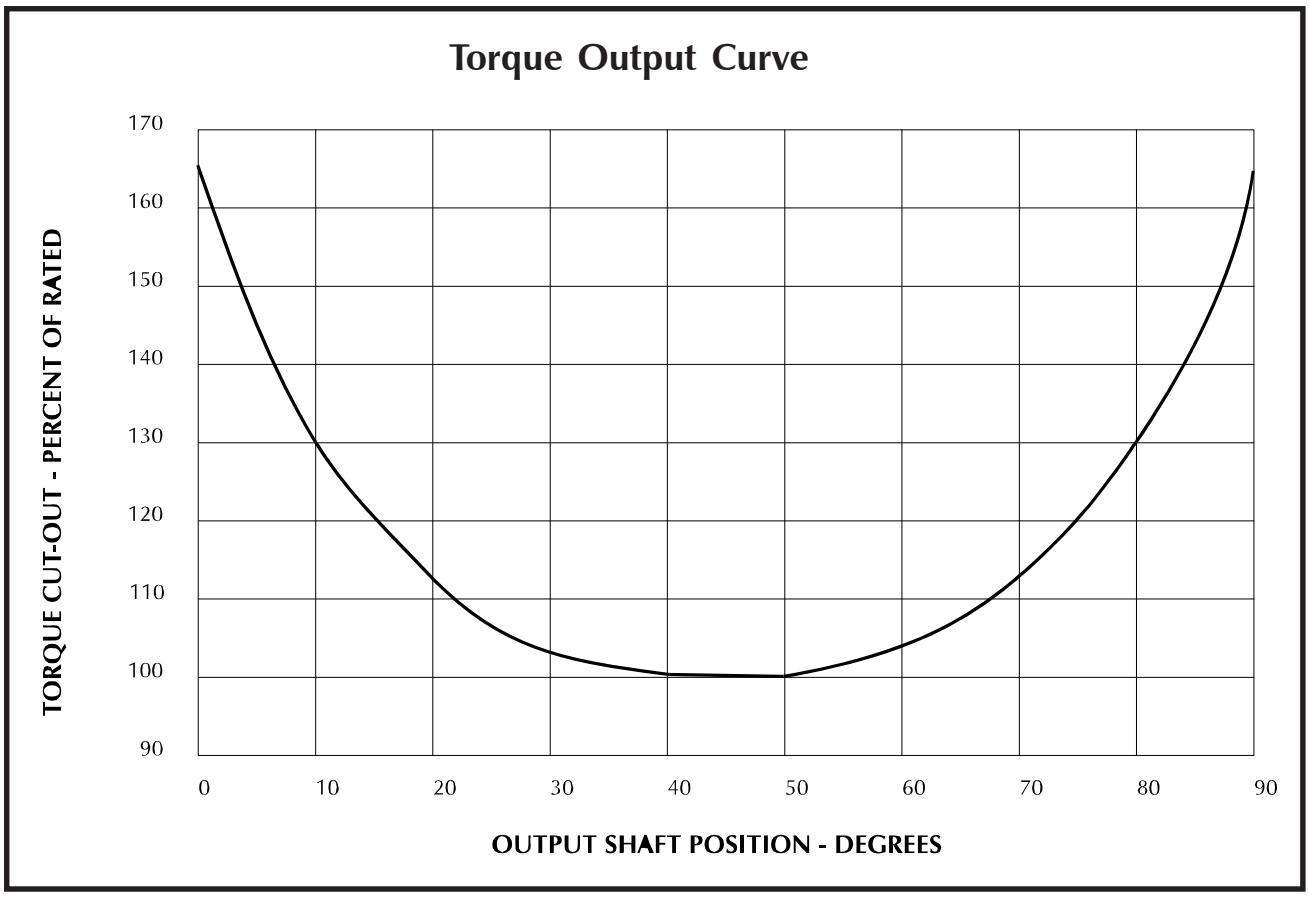
- Microprocessor controlled
- Quick push button set-up
- Prompting vacuum fluorescent display
  - Diagnostics
  - Limit, Torque or Thrust switch fault
  - Intelligent Power Module (IPM) indication for:
    - Over-current
    - Over-temperature
  - Actuator stall condition
  - Loss of command signal
  - Loss of feedback signal
  - Actuator starts
  - Temperature of amplifier
  - Motor dc voltage and current
  - Microprocessor fault
- Onboard "increase" and "decrease" switch
- LED diagnostics for power and fault
  - Loss of 4-20mA signal
  - Loss of position feedback signal
  - MIN/MAX limit switch reached
  - IPM Fault
  - Motor stalled condition

SCOTCH-YOKE DRIVE TORQUE CHARACTERISTICS

The SM-5000 Series actuators utilize a scotch-yoke drive for transforming linear motion from a traveling nut to the rotary motion on the output shaft. The characteristics of the scotch-yoke drive are such that the torque output varies with shaft position as shown below.

ADVANTAGES:

1. Allows use of smaller torque rated actuators for non-linear loads.
2. Aids in linearization of linkage driven loads.
3. Allows actuators to be selected based on position related torque rather than peak plus safety factor.
4. Assures high seating and unseating torque without oversizing of the actuator.



TORQUE OUTPUT CHART

Output Shaft Position (Degrees)	SM-5100 150 ft. lbs. (203Nm)	SM-5100 300 ft. lbs. (407Nm)	SM-5200 600 ft. lbs. (813Nm)	SM-5200 1000 ft. lbs. (1356Nm)	SM-5300 1600 ft. lbs. (2169Nm)	SM-5300 3200 ft. lbs. (4339Nm)
0°	249 (337)	498 (675)	996 (1350)	1660 (2249)	2656 (3599)	5312 (7198)
10°	198 (268)	396 (537)	792 (1074)	1320 (1789)	2112 (2862)	4224 (5724)
20°	170 (230)	340 (461)	678 (919)	1130 (1531)	1808 (2450)	3616 (4900)
30°	155 (210)	310 (420)	618 (838)	1030 (1396)	1648 (2233)	3296 (4466)
40°	150 (203)	300 (407)	600 (813)	1000 (1356)	1600 (2169)	3200 (4339)
45°	150 (203)	300 (407)	600 (813)	1000 (1356)	1600 (2169)	3200 (4339)
50°	150 (203)	300 (407)	600 (813)	1000 (1356)	1600 (2169)	3200 (4339)
60°	155 (210)	310 (420)	618 (838)	1030 (1396)	1648 (2233)	3296 (4466)
70°	170 (230)	340 (461)	678 (919)	1130 (1531)	1808 (2450)	3616 (4900)
80°	198 (268)	396 (537)	792 (1074)	1320 (1789)	2112 (2862)	4224 (5724)
90°	249 (337)	498 (675)	996 (1350)	1660 (2249)	2656 (3599)	5312 (7198)

Output Shaft Position (Degrees)	SM-5400 5000 ft. lbs. (6779Nm)	SM-5400 7500 ft. lbs. (10169Nm)	SM-5400 10000 ft. lbs. (13558Nm)	SM-5400 12500 ft. lbs. (16948Nm)
0°	8300 (11247)	12450 (16870)	16600 (22493)	20750 (28133)
10°	6600 (8943)	9900 (13415)	13200 (17886)	16500 (22371)
20°	5650 (7656)	8475 (11484)	11300 (15312)	14125 (19151)
30°	5150 (6978)	7725 (10467)	10300 (13957)	12875 (17456)
40°	5000 (6779)	7500 (10169)	10000 (13558)	12500 (16948)
45°	5000 (6779)	7500 (10169)	10000 (13558)	12500 (16948)
50°	5000 (6779)	7500 (10169)	10000 (13558)	12500 (16948)
60°	5150 (6978)	7725 (10467)	10300 (13957)	12875 (17456)
70°	5650 (7656)	8475 (11484)	11300 (15312)	14125 (19151)
80°	6600 (8943)	9900 (13415)	13200 (17886)	16500 (22371)
90°	8300 (11247)	12450 (16870)	16600 (22493)	20750 (28133)

SM-5100 SERIES SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Maximum Available Torques ft. lbs. (Nm)	Speeds (sec.)	Command Input	Input Impedance in ohms	Feedback	Modulation Rate *	Dynamic Braking	Loss of Power	Loss of Command Signal
		Run	Stall									
SM-5115-N	208/3/50/60	0.5	1.6	150 (203)	16, 29 or 48	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
	240/3/50/60	0.4	1.3									
	380/3/50	0.3	0.9	300 (407)	29 or 48							
	480/3/50/60	0.2	0.65									
SM-5115-A	208/3/50/60	0.5	1.6	150 (203)	16, 29 or 48	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	240/3/50/60	0.4	1.3									
	380/3/50	0.3	0.9	300 (407)	29 or 48							
	480/3/50/60	0.2	0.65									
SM-5120-N	120/1/50/60	2.5	2.9	150 (203)	16, 29 or 48	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
				300 (407)	29 or 48							
SM-5120-D	120/1/50/60	2.5	2.9	150 (203)	16, 29 or 48	4-20 mA	200	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
				300 (407)	29 or 48	0-5 Vdc or 0-10 Vdc	100,000					
SM-5140-N	24 Vdc	4.5	10	150 (203)	20 or 33	N/A	N/A	N/A	4000 starts/hour	N/A	Lock in Place	N/A
				300 (407)	33							
SM-5160-N	90 Vdc	2.5	6	150 (203)	13, 24 or 39	N/A	N/A	N/A	4000 starts/hour	N/A	Lock in Place	N/A
				300 (407)	24 or 39							
SM-5160-A	120/1/50/60	2.5	6	150 (203)	13, 24 or 39	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	4000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	208/1/50/60											
	240/1/50/60			300 (407)	24 or 39							
	208/3/50/60											
	240/3/50/60											
SM-5160-D	120/1/50/60	2.5	6	150 (203)	13, 24 or 39	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	4000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	208/1/50/60											
	240/1/50/60			300 (407)	24 or 39							
	208/3/50/60											
	240/3/50/60											
SM-5190-N	240/1/50/60	1.2	1.5	150 (203)	16, 29 or 48	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
				300 (407)	29 or 48							
SM-5190-D	240/1/50/60	1.2	1.5	150 (203)	16, 29 or 48	4-20 mA	200	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
				300 (407)	29 or 48	0-5 Vdc or 0-10 Vdc	100,000					

** Minimum position change of 1%*

SM-5100 SERIES SELECTION CHART

						Selection
1	Basic Model	**SM-5115-N**	3 phase, 208/240/380/480 Vac, modulating duty, without built-in amplifier			
		SM-5115-A	SM-5115 with built-in analog amplifier			
		SM-5120-N	1 phase, 120 Vac, modulating duty, without built-in amplifier			
		SM-5120-D	SM-5120 with built-in digital amplifier			
		SM-5140-N	24 Vdc, modulating duty, (If amplifier desired, see AD-7540			
		SM-5160-N	90 Vdc, modulating duty, without built-in amplifier			
		SM-5160-A	SM-5160 with built-in analog amplifier, 120/208/240 Vac, 1 Ph; 208/240 Vac, 3 Ph			
		SM-5160-D	SM-5160 with built-in digital amplifier, 120/208/240 Vac, 1 Ph; 208/240 Vac, 3 Ph			
		SM-5190-N	1 phase, 240 Vac, modulating duty, without built-in amplifier			
		SM-5190-D	SM-5190 above with built-in digital amplifier			
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (SM-5120-N, SM-5120-D, SM-5160-A or SM-5160-D)				
		2: 240 Vac, 1 Phase, 50/60 Hz (SM-5190-D, SM-5160-A or SM-5160-D)				
		3: 380 Vac, 3 Phase, 50 Hz (SM-5115-N or SM-5115-A)				
		4: 480 Vac, 3 Phase, 50/60 Hz (SM-5115-N or SM-5115-A)				
		5: 240 Vac, 3 Phase, 50/60 Hz (SM-5115-N, SM-5115-A, SM-5160-A or SM-5160-D)				
		6: 24 Vdc (SM-5140-N)				
		7: 90 Vdc (SM-5160-N)				
		9: 208 Vac, 1 Phase, 50 Hz (SM-5160-A or SM-5160-D)				
		10: 208 Vac, 3 Phase, 50 Hz (SM-5115-N, SM-5115-A, SM-5160-A or SM-5160-D)				
		19: 220 Vac, 1 Phase (SM-5190-D)				
3	Speed, sec. (a) / Torque, ft. lbs. (Nm)	SM-5115, 5120, 5190	SM-5140	SM-5160		
		16/150 (203)	**20/150 (203)**	**13/150 (203)**		
		29/150 (203)	**33/150 (203)**	**24/150 (203)**		
4		**29/300 (407)**	**33/300 (407)**	**24/300 (407)**		
		48/150 (203)		**39/150 (203)**		
		48/300 (407)		**39/300 (407)**		
5	Rotation	**Specify** required output shaft rotation in degrees. Must be between 33° and 90°.				

(a) Speeds are for 90° of rotation.

SM-5100 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Man. Toggle Switches - NEMA 4	
A002	INC/OFF/DEC Toggle Switches - NEMA 4	
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002) - NEMA 4	
A004	Open-Close-Stop Push Buttons - NEMA 4	
A005	Two Pilot Lights - NEMA 4	
A006	Two Position Selector Switches - NEMA 4 (Auto/Manual)	
A007	Three Position Selector Switches - NEMA 4 (INC/OFF/DEC)	
A009	ON/OFF Toggle Power Switch - NEMA 4	
A010	Local Auto/Manual Switches & (INC/OFF/DEC) Switches - NEMA 4 (A006 & A007)	
A011	Key locks for rotary Auto/Manual switches - NEMA 4	
A012	ON/OFF Rotary Power Switch - NEMA 4	
Drive Arm/Adapter Clevis/Driven Arm		
D001	Fixed (10") drive arm for splined shaft (old SA1)	
D002	Adjustable drive arm for splined shaft (old SA2)	
D003	Adapter clevis kit used with D001 & D005 (old AK1)	
D004	Adapter clevis kit used with D002 & D006 (old AK2)	
D005	Driven arm used with D001 and D003 (old DA1)	
D006	Driven arm used with D002 and D004 (old DA2)	
D020	Hagan Bailey Pneumatic Replacement (Must provide model number and drive arm dimensions)	
D025	Wedgelock Coupling (valve dims. required)	

Code	Description	Selection
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not for use with models with internal amps)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (Not for use with models with internal amplifiers)	
F010	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	
Identification/Certifications		
J001	CE Marking, AC input only (only available for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Position Indicator		
P001	Local Position Indicator	
Shaft		
R004	Keyed Shaft, 1" diameter, 1/4" square keyway	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, not avail. with E003, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, not avail. with E003, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S015	Two nonadjustable torque limit switches, SPST, 10 Amp, 120/240 Vac	
S016	Two nonadjustable torque limit switches SPDT, 10 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp	
Painting/Coating		
W002	Two Part Epoxy	
Special Mechanical Options		
Y009	Metric mounting holes, 12mm	

SM-5100 SPECIAL OPTIONS

(Consult factory for these special requirements)

Code	Description	Selection
Connectors		
C001	Plug-In Cannon Type Connector on Enclosure for Power and Control	
C002	Mating Plug-In Cannon Type Connector on Enclosure for Power and Control	
Enclosure		
E003	Explosion Proof - Class 1, Div. 1, Groups C & D, indoor/outdoor (NEMA 7)	
Feedback		
F008	Contactless 4-20 mA	
F009	Contactless 0-10Vdc	

Code	Description	Selection
3-Phase Controls		
G001	3-Phase Contactor with Overloads	
G002	Control Transformer (120 Vac control voltage)	
G003	Control Transformer (240 Vac control voltage)	
Motor		
M001	3 Phase, 575 Vac	
Special Electrical Options		
X001	24 Vdc power supply, 320 mA, 100-240 Vac input	
X004	Voltage to current converter	
X005	Loop powered 4 to 20 isolator, M-system HSN-IAA	
X006	± 15 Vdc power supply	
X010	Customer command and feedback, 0-10 Vdc or 0-5 Vdc instead of 4-20 mA	
X019	AC Power Sensing with Output	

For a full description of options, go to the Complete Listing of Options starting on page 197.

SM-5200 SERIES SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Maximum Available Torques ft. lbs. (Nm)	Speeds (sec.)	Command Input	Input Impedance in Ohms	Feedback	Modulation Rate *	Dynamic Braking	Loss of Power	Loss of Command Signal
		Run	Stall									
SM-5210-N	208/3/50/60	2.1	11.9	600 (814)	30 or 50	N/A	N/A	N/A	2000 starts/hr	N/A	Lock in Place	N/A
	240/3/50/60	1.8	10.3									
	380/3/50	2.4	4.7	1000 (1356)					20%			
	480/3/50/60	0.9	5.2									
SM-5210-A	208/3/50/60	2.1	11.9	600 (814)	30 or 50	4-20 mA 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	2000 starts/hr	Yes	Lock in Place	Lock in Place or Run to Preset
	240/3/50/60	1.8	10.3									
	380/3/50	2.4	4.7	1000 (1356)					20%			
	480/3/50/60	0.9	5.2									
SM-5220-N	120/1/50/60	7	13.2	600 (814)	30 or 50	N/A	N/A	N/A	2000 starts/hr	N/A	Lock in Place	N/A
	240/1/50/60	3.5	6.6	1000 (1356)					20%			
SM-5220-D	120/1/50/60	7	13.2	600 (814)	30 or 50	4-20 mA	200	1000 ohm pot or 4-20 mA	2000 starts/hr	Yes	Lock in Place	Lock in Place or Run to Preset
	240/1/50/60	3.5	6.6	1000 (1356)		0-5 Vdc or 0-10 Vdc	100,000		20%			
SM-5260-N	90 Vdc	4.7	N/A	600 (814)	28 or 48	N/A	N/A	N/A	4000 starts/hr	N/A	Lock in Place	N/A
				1000 (1356)					20%			
SM-5260-A	120/1/50/60	4.7	N/A	600 (814)	28 or 48	4-20 mA or 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	4000 starts/hr	Yes	Lock in Place	Lock in Place or Run to Preset
	208/1/50/60											
	240/1/50/60											
	208/3/50/60			1000 (1356)					20%			
	240/3/50/60											
SM-5260-D	120/1/50/60	4.7	N/A	600 (814)	28 or 48	4-20 mA or 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	4000 starts/hr	Yes	Lock in Place	Lock in Place or Run to Preset
	208/1/50/60											
	240/1/50/60											
	208/3/50/60			1000 (1356)					20%			
	240/3/50/60											

** Minimum position change of 1%*

SM-5200 SERIES SELECTION CHART

					Selection
1	Basic Model	**SM-5210-N**	3 phase, 208/240/380/480 Vac, modulating duty, without built-in amplifier		
		SM-5210-A	SM-5210 with built-in analog amplifier		
		SM-5220-N	1 phase, 120/240 Vac, modulating duty, without built-in amplifier		
		SM-5220-D	SM-5220 with built-in digital amplifier		
		SM-5260-N	90 Vdc (perm. magnet field), modulating duty, without built-in amplifier		
		SM-5260-A	SM-5260 with built-in analog amplifier, 120/208/240 Vac, 1 Ph; 208/240 Vac, 1 Ph		
		SM-5260-D	SM-5260 with built-in digital amplifier, 120/208/240 Vac, 1 Ph; 208/240 Vac, 1 Ph		
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (SM-5220-N, SM-5220-D, SM-5260-A or SM-5260-D)			
		2: 240 Vac, 1 Phase, 50/60 Hz (SM-5220-N, SM-5220-D, SM-5260-A or SM-5260-D)			
		3: 380 Vac, 3 Phase, 50 Hz (SM-5210-N or SM-5210-A)			
		4: 480 Vac, 3 Phase, 50/60 Hz (SM-5210-N or SM-5210-A)			
		5: 240 Vac, 3 Phase, 50/60 Hz (SM-5210-N, SM-5210-A, SM-5260-A or SM-5260-D)			
		7: 90 Vdc (SM-5260-N)			
		9: 208 Vac, 1 Phase, 50 Hz (SM-5260-A or SM-5260-D)			
		10: 208 Vac, 3 Phase, 50 Hz (SM-5210-N, SM-5210-A, SM-5260-A or SM-5260-D)			
3 / 4	Speed, sec. (a) / Torque, ft. lbs. (Nm)	SM-5210, 5220	SM-5260		
		30/600 (814)	**28/600** (814)		
		50/600 (814)	**48/600 (814)**		
		30/1000 (1356)	**28/1000** (1356)		
		50/1000 (1356)	**48/1000** (1356)		
5	Rotation	**Specify** required output shaft rotation in degrees. Must be between 29° and 90°.			

(a) Speeds are for 90° of rotation.

SM-5200 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Man. Toggle Switches - NEMA 4	
A002	INC/OFF/DEC Toggle Switches - NEMA 4	
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002) - NEMA 4	
A004	Open-Close-Stop Push Buttons - NEMA 4	
A005	Two Pilot Lights - NEMA 4	
A006	Two Position Selector Switches - NEMA 4 (Auto/Manual)	
A007	Three Position Selector Switches - NEMA 4 (INC/OFF/DEC)	
A009	ON/OFF Toggle Power Switch - NEMA 4	
A010	Local Auto/Manual Switches & (INC/OFF/DEC) Switches - NEMA 4 (A006 & A007)	
A011	Key locks for rotary Auto/Manual switches - NEMA 4	
A012	ON/OFF Rotary Power Switch - NEMA 4	
Brake		
B001	Electrically Operated Brake 120 Vac	
B002	Electrically Operated Brake 240 Vac	
Drive Arm/Adapter Clevis/Driven Arm		
D007	Fixed (10") drive arm for splined shaft (old SA1)	
D008	Adjustable drive arm for splined shaft (old SA2)	
D009	Adapter clevis kit used with D007 and D011 (old AK1)	
D010	Adapter clevis kit used with D008 and D012 (old AK2)	
D011	Driven arm used with D007 and D009 (old DA1)	
D012	Driven arm used with D008 and D010 (old DA2)	
D021	Hagan Bailey Pneumatic Replacement (Must provide model number and drive arm dimensions)	
D026	Wedgelock Coupling (valve dim. required)	

Code	Description	Selection
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not for use with models with internal amps)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (Not for use with models with internal amplifiers)	
F010	Anti Backlash Feedback Spring	
F999	Special Potentiometer (consult factory)	
3-Phase Controls		
G001	3-Phase Contactor with Overloads	
G002	Control Transformer (120 Vac control voltage)	
G003	Control Transformer (240 Vac control voltage)	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Position Indicator		
P001	Local Position Indicator	
Shaft		
R005	Keyed Shaft, 1.75" dia., 3/8" square keyway	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S016	Two nonadjustable torque limit switches SPDT, 10 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position SPST, 20 Amp, 120/240 Vac	
S020	Two nonadjustable torque limit switches SPST, 20 Amp, 120/240 Vac	
Painting/Coating		
W002	Two Part Epoxy	
Special Mechanical Options		
Y010	Metric mounting holes, 20mm	

SM-5200 SPECIAL OPTIONS

(Consult factory for these special requirements)

Code	Description	Selection
Connectors		
C001	Plug-In Cannon Type Connector on Enclosure for Power and Control	
C002	Mating Plug-In Cannon Type Connector on Enclosure for Power and Control	
Feedback		
F008	Contactless 4-20 mA	
F009	Contactless 0-10Vdc	
Motor		
M001	3 Phase, 575 Vac	

Code	Description	Selection
Special Electrical Options		
X001	24 Vdc power supply, 320 mA, 100-240 Vac input	
X004	Voltage to current converter	
X005	Loop powered 4 to 20 isolator, M-system HSN-IAA	
X006	± 15 Vdc power supply	
X010	Customer command and feedback, 0-10 Vdc or 0-5 Vdc instead of 4-20 mA	
X019	AC Power Sensing with Output	

For a full description of options, go to the Complete Listing of Options starting on page 197.

SM-5300 SERIES SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Maximum Available Torques ft. lbs. (Nm)	Speeds (sec.)	Command Input	Input Impedance in Ohms	Feedback	Modulation Rate *	Dynamic Braking	Loss of Power	Loss of Command Signal
		Run	Stall									
SM-5310-N	208/3/50/60	2.9	14.5	1600 (2170)	23 or 47	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
	240/3/50/60	2.5	12.5									
	380/3/50	2.2	13.5	3200 (4339)	47							
	480/3/50/60	1.2	6.2									
SM-5310-A	208/3/50/60	2.9	14.5	1600 (2170)	23 or 47	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	240/3/50/60	2.5	12.5									
	380/3/50	2.2	13.5	3200 (4339)	47							
	480/3/50/60	1.2	6.2									
SM-5320-N	120/1/50/60	10	12	1600 (2170)	23 or 47	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
	240/1/50/60	5	6	3200 (4339)	47							
SM-5360-N	90 Vdc	9.5	N/A	1600 (2170)	23 or 47	N/A	N/A	N/A	4000 starts/hour	N/A	Lock in Place	N/A
				3200 (4339)	47							
SM-5360-A	120/1/50/60	9.5	N/A	1600 (2170)	23 or 47	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	4000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	208/1/50/60											
	240/1/50/60			3200 (4339)	47							
	208/3/50/60											
	240/3/50/60											
SM-5360-D	120/1/50/60	9.5	N/A	1600 (2170)	23 or 47	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	4000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	208/1/50/60											
	240/1/50/60			3200 (4339)	47							
	208/3/50/60											
	240/3/50/60											

** Minimum position change of 1%*

SM-5300 SERIES SELECTION CHART

				Selection
1	Basic Model	**SM-5310-N**	3 phase, 208/240/380/480 Vac, modulating duty, without built-in amplifier	
		SM-5310-A	SM-5310 with built-in analog amplifier	
		SM-5320-N	1 phase, 120/240 Vac, modulating duty, without built-in amplifier	
		SM-5360-N	90 Vdc (perm. magnet field), modulating duty, without built-in amplifier	
		SM-5360-A	SM-5360 with built-in analog amplifier, 120/208/240 Vac, 1 Ph; 208/240 Vac, 1 Ph	
		SM-5360-D	SM-5360 with built-in digital amplifier, 120/208/240 Vac, 1 Ph; 208/240 Vac, 1 Ph	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (SM-5320-N, SM-5360-A or SM-5360-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (SM-5320-N, SM-5360-A or SM-5360-D)		
		3: 380 Vac, 3 Phase, 50 Hz (SM-5310-N or SM-5310-A)		
		4: 480 Vac, 3 Phase, 50/60 Hz (SM-5310-N or SM-5310-A)		
		5: 240 Vac, 3 Phase, 50/60 Hz (SM-5310-N, SM-5310-A, SM-5360-A or SM-5360-D)		
		7: 90 Vdc (SM-5360-N)		
		9: 208 Vac, 1 Phase, 50 Hz (SM-5360-A or SM-5360-D)		
		10: 208 Vac, 3 Phase, 50 Hz (SM-5310-N, SM-5310-A, SM-5360-A or SM-5360-D)		
3	Speed, sec. (a) / Torque, ft. lbs. (Nm)	All Models		
		23/1600 (2170)		
4		**47/1600 (2170)**		
		47/3200 (4339)		
5	Rotation	**Specify** required output shaft rotation in degrees. Must be between 70° and 90°.		

(a) Speeds are for 90° of rotation.

SM-5300 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Man. Toggle Switches - NEMA 4	
A002	INC/OFF/DEC Toggle Switches - NEMA 4	
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002) - NEMA 4	
A004	Open-Close-Stop Push Buttons - NEMA 4	
A005	Two Pilot Lights - NEMA 4	
A006	Two Position Selector Switches - NEMA 4 (Auto/Manual)	
A007	Three Position Selector Switches - NEMA 4 (INC/OFF/DEC)	
A009	ON/OFF Toggle Power Switch - NEMA 4	
A010	Local Auto/Manual Switches & (INC/OFF/DEC) Switches - NEMA 4 (A006 & A007)	
A011	Key locks for rotary Auto/Manual switches - NEMA 4	
A012	ON/OFF Rotary Power Switch - NEMA 4	
Drive Arm/Adapter Clevis/Driven Arm		
D013	Fixed (12") drive arm for splined shaft (old SA)	
D014	Adapter clevis kit used with D013 and D015 (old AK)	
D015	Driven arm used with D013 and D014 (old DA)	
D022	Hagan Bailey Pneumatic Replacement (Must provide model number and drive arm dimensions)	
D027	Wedgelock Coupling (valve dim. required)	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not for use with models with internal amps)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (Not for use with models with internal amplifiers)	
F010	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	

Code	Description	Selection
3-Phase Controls		
G001	3-Phase Contactor with Overloads	
G002	Control Transformer (120 Vac control voltage)	
G003	Control Transformer (240 Vac control voltage)	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Position Indicator		
P001	Local Position Indicator	
Shaft		
R006	Keyed Shaft, 2.75" dia., 5/8" square keyway	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S016	Two nonadjustable torque limit switches SPDT, 10 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp, 120/240 Vac	
S020	Two nonadjustable torque limit switches SPST, 20 Amp, 120/240 Vac	
Painting/Coating		
W002	Two Part Epoxy	
Special Mechanical Options		
Y011	Metric mounting holes, 20mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

SM-5300 SPECIAL OPTIONS

(Consult factory for these special requirements)

Code	Description	Selection
Connectors		
C001	Plug-In Cannon Type Connector on Enclosure for Power and Control	
C002	Mating Plug-In Cannon Type Connector on Enclosure for Power and Control	
Feedback		
F008	Contactless 4-20 mA	
F009	Contactless 0-10Vdc	
Motor		
M001	3 Phase, 575 Vac	

Code	Description	Selection
Special Electrical Options		
X001	24 Vdc power supply, 320 mA, 100-240 Vac input	
X004	Voltage to current converter	
X005	Loop powered 4 to 20 isolator, M-system HSN-IAA	
X006	± 15 Vdc power supply	
X010	Command and feedback, 0-10 Vdc or 0-5 Vdc instead of 4-20 mA	
X019	AC Power Sensing with Output	

SM-5400 SERIES SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Maximum Available Torques ft. lbs. (Nm)	Speeds (sec.)	Command Input	Input Impedance in Ohms	Feedback	Modulation Rate *	Dynamic Braking	Loss of Power	Loss of Command Signal
		Run	Stall									
SM-5430-N	208/3/50/60	4.8	37	5000 (6780) or 7500 (10,170	42, 56 or 73	N/A	N/A	N/A	2000 starts/hour	N/A	Lock in Place	N/A
	240/3/50/60	4.2	32	10,000 (13,560)	56 or 73							
	380/3/50/60	2.5	16.9	7500 (10,170) or 12,500 (16,950)	73							
	480/3/50/60	2.1	16									
SM-5430-A	208/3/50/60	4.8	37	5000 (6780) or 7500 (10,170	42, 56 or 73	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	2000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	240/3/50/60	4.2	32	10,000 (13,560)	56 or 73							
	380/3/50/60	2.5	16.9	7500 (10,170) or 12,500 (16,950)	73							
	480/3/50/60	2.1	16									
SM-5480-N	180 Vdc	9.2	N/A	5000 (6780) or 7500 (10,170	40, 54 or 70	N/A	N/A	N/A	4000 starts/hour	N/A	Lock in Place	N/A
				10,000 (13,560)	54 or 70							
				7500 (10,170) or 12,500 (16,950)	70							
SM-5480-A	208/1/50/60	9.2	N/A	5000 (6780) or 7500 (10,170)	40, 54 or 70	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	4000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	240/1/50/60			10,000 (13,560)	54 or 70							
	208/3/50/60			7500 (10,170) or 12,500 (16,950)	70							
	240/3/50/60											
SM-5480-D	208/1/50/60	9.2	N/A	5000 (6780) or 7500 (10,170)	40, 54 or 70	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	1000 ohm pot or 4-20 mA	4000 starts/hour	Yes	Lock in Place	Lock in Place or Run to Preset
	240/1/50/60			10,000 (13,560)	54 or 70							
	208/3/50/60			7500 (10,170) or 12,500 (16,950)	70							
	240/3/50/60											

** Minimum position change of 1%*

SM-5400 SERIES SELECTION CHART

				Selection
1	Basic Model	**SM-5430-N**	3 phase, 208/240/380/480 Vac, modulating duty, without built-in amplifier	
		SM-5430-A	SM-5430 with built-in analog amplifier	
		SM-5480-N	180 Vdc (perm. magnet field), modulating duty, without built-in amplifier	
		SM-5480-A	SM-5480 with built-in analog amplifier, 208/240 Vac, 1 Ph; 208/240 Vac, 1 Ph	
		SM-5480-D	SM-5480 with built-in digital amplifier, 208/240 Vac, 1 Ph; 208/240 Vac, 1 Ph	
2	Input Voltage	**2**: 240 Vac, 1 Phase, 50/60 Hz (SM-5480-A or SM-5480-D)		
		3: 380 Vac, 3 Phase, 50 Hz (SM-5430-N or SM-5430-A)		
		4: 480 Vac, 3 Phase, 50/60 Hz (SM-5430-N or SM-5430-A)		
		5: 240 Vac, 3 Phase, 50/60 Hz (SM-5430-N, SM-5430-A, SM-5480-A or SM-5480-D)		
		8: 180 Vdc (SM-5480-N)		
		9: 208 Vac, 1 Phase, 50 Hz (SM-5480-A or SM-5480-D)		
		10: 208 Vac, 3 Phase, 50 Hz (SM-5430-N, SM-5430-A, SM-5480-A or SM-5480-D)		
3	Speed, sec. (a) / Torque, ft. lbs. (Nm)	SM-5430	SM-5480	
		42/5,000 (6779)	**40/5,000** (6779)	
		42/7,500 (10,168)	**40/7,500** (10,168)	
		56/5,000 (6779)	**54/5,000** (6779)	
		56/7,500 (10,168)	**54/7,500** (10,168)	
4		**56/10,000** (13,558)	**54/10,000** (13,558)	
		73/5,000 (6779)	**70/5,000** (6779)	
		73/7,500 (10,168)	**70/7,500** (10,168)	
		73/12,500 (16,947)	**70/12,500** (16,947)	
5	Rotation	**Specify** required output shaft rotation in degrees. Must be between 70° and 90°.		

(a) Speeds are for 90° of rotation.

SM-5400 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Man. Toggle Switches - NEMA 4	
A002	INC/OFF/DEC Toggle Switches - NEMA 4	
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002) - NEMA 4	
A004	Open-Close-Stop Push Buttons - NEMA 4	
A005	Two Pilot Lights - NEMA 4	
A006	Two Position Selector Switches - NEMA 4 (Auto/Manual)	
A007	Three Position Selector Switches - NEMA 4 (INC/OFF/DEC)	
A009	ON/OFF Toggle Power Switch - NEMA 4	
A010	Local Auto/Manual Switches & (INC/OFF/DEC) Switches - NEMA 4 (A006 & A007)	
A011	Key locks for rotary Auto/Manual switches - NEMA 4	
A012	ON/OFF Rotary Power Switch - NEMA 4	
Drive Arm/Adapter Clevis/Driven Arm		
D016	Fixed (15") drive arm for splined shaft (old SA)	
D017	Adapter clevis kit used with D016 and D018 (old AK)	
D018	Drive arm used with D016 and D017 (old DA)	
D023	Hagan Bailey Pneumatic Replacement (Must provide model number and drive arm dimensions)	
D028	Wedgelock Coupling	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not for use with internal amplifiers)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (Not for use with models with internal amplifiers)	
F010	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	

Code	Description	Selection
3-Phase Controls		
G001	3-Phase Contactor with Overloads	
G003	Control Transformer (240 Vac control voltage)	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Position Indicator		
P001	Local Position Indicator	
Shaft		
R007	Keyed Shaft, 4.25" dia., 1" square keyway	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S016	Two nonadjustable torque limit switches SPDT, 10 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp, 120/240 Vac	
S020	Two nonadjustable torque limit switches SPST, 20 Amp, 120/240 Vac	
Painting/Coating		
W002	Two Part Epoxy	
Special Mechanical Options		
Y012	Metric mounting holes, 26mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

SM-5400 SPECIAL OPTIONS

(Consult factory for these special requirements)

Code	Description	Selection
Connectors		
C001	Plug-In Cannon Type Connector on Enclosure for Power and Control	
C002	Mating Plug-In Cannon Type Connector on Enclosure for Power and Control	
Feedback		
F008	Contactless 4-20 mA	
F009	Contactless 0-10Vdc	
Motor		
M001	3 Phase, 575 Vac	

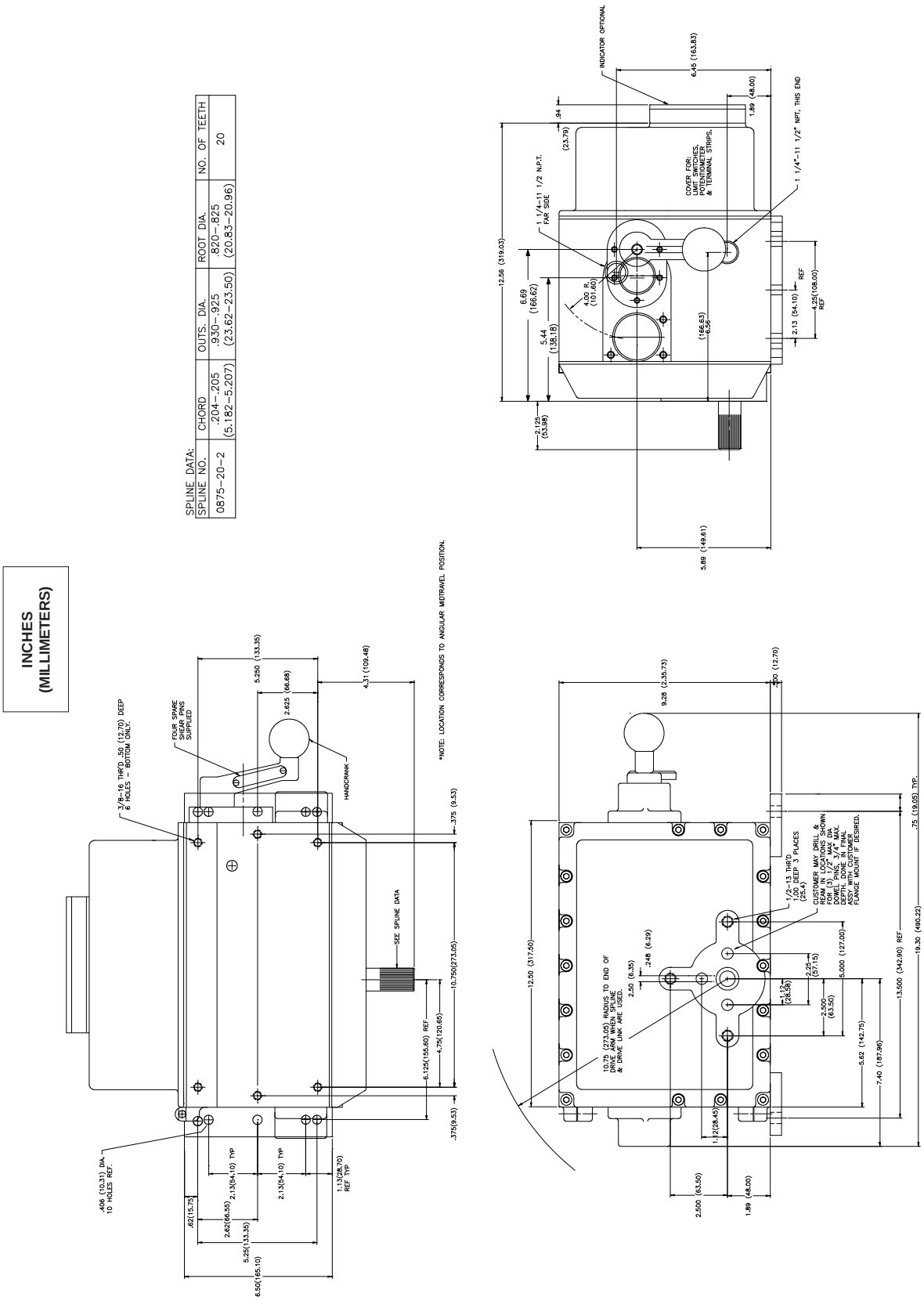
Code	Description	Selection
Special Electrical Options		
X001	24 Vdc power supply, 320 mA, 100-240 Vac input	
X004	Voltage to current converter	
X005	Loop powered 4 to 20 isolator, M-system HSN-IAA	
X006	± 15 Vdc power supply	
X010	Command and feedback, 0-10 Vdc or 0-5 Vdc instead of 4-20 mA	
X019	AC Power Sensing with Output	

SM-5100 MAJOR DIMENSIONS

INCHES
(MILLIMETERS)

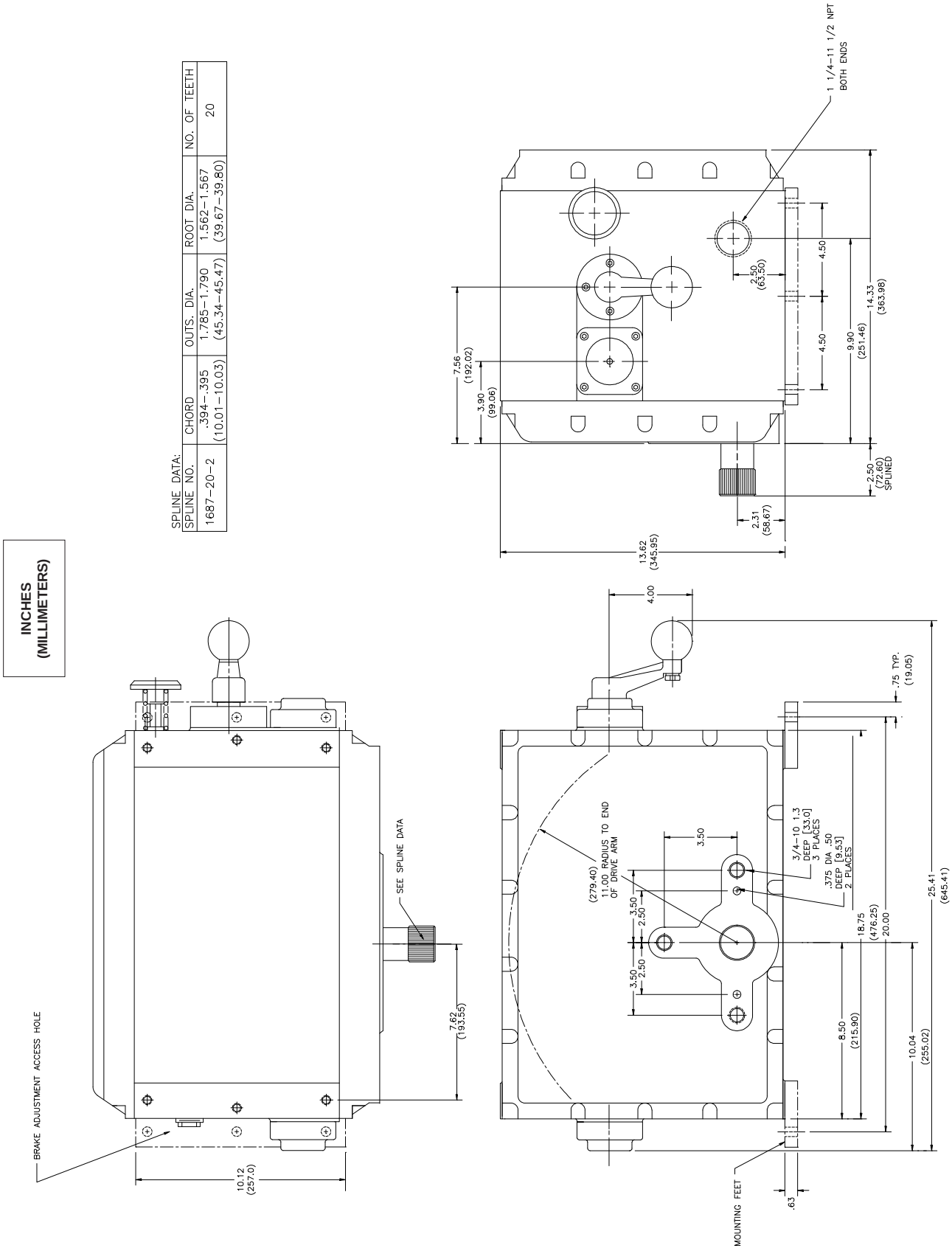
SPLINE DATA:

SPLINE NO.	CHORD	OUTS. DIA.	ROOT DIA.	NO. OF TEETH
0875–20–2	.204–.205 (5.182–5.207)	.930–.925 (23.62–23.50)	.820–.825 (20.83–20.96)	20



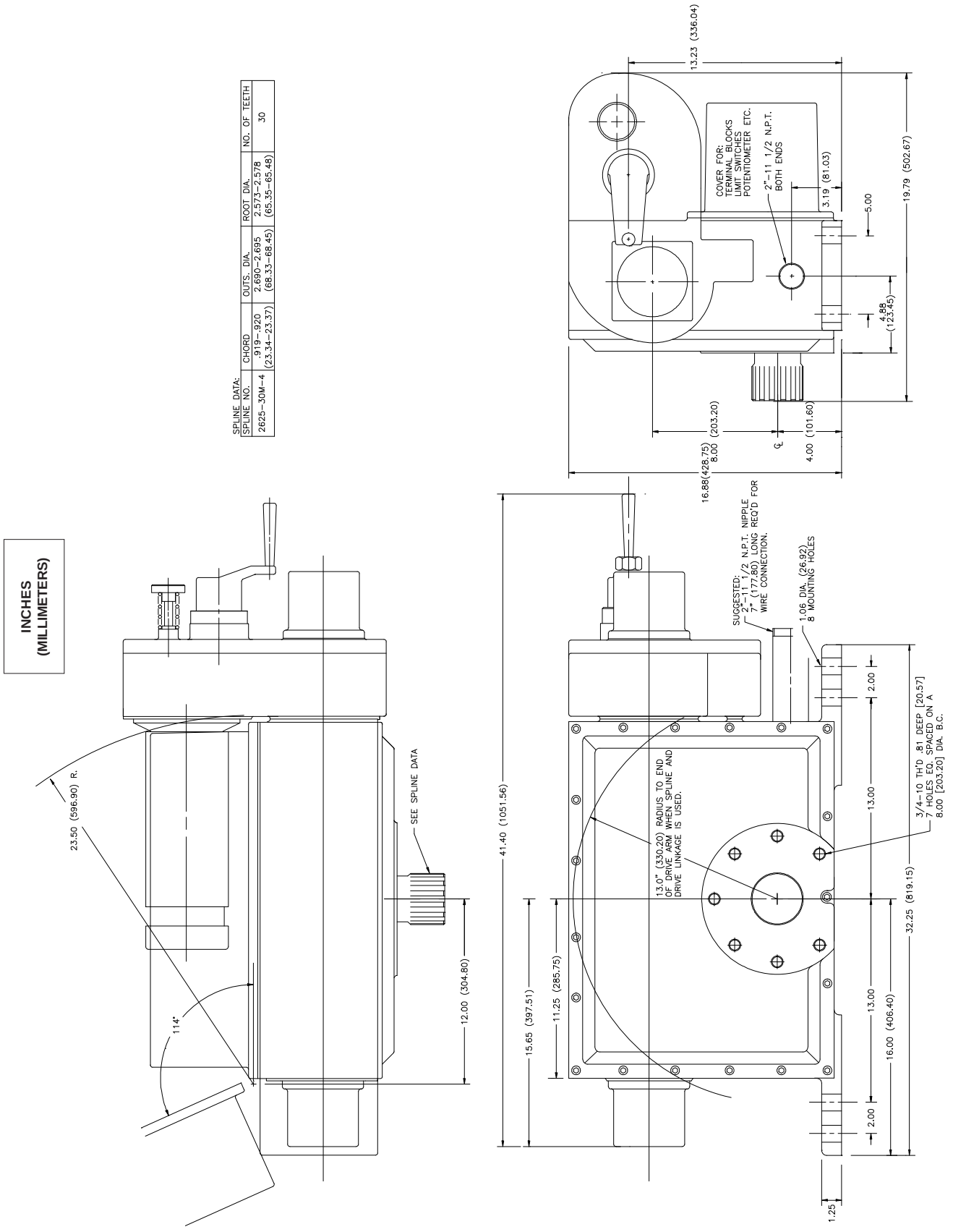
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

SM-5200 MAJOR DIMENSIONS



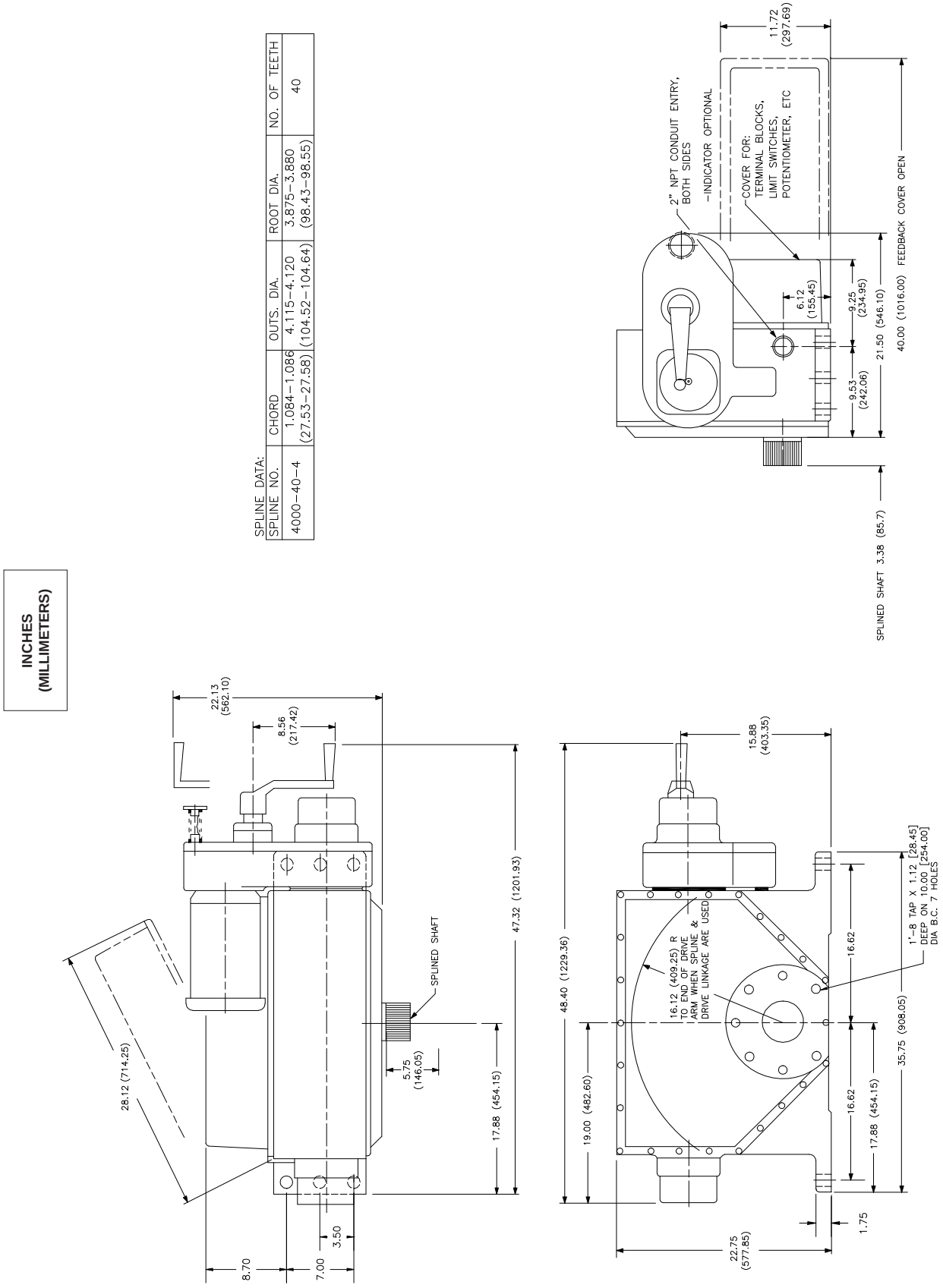
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

SM-5300 MAJOR DIMENSIONS



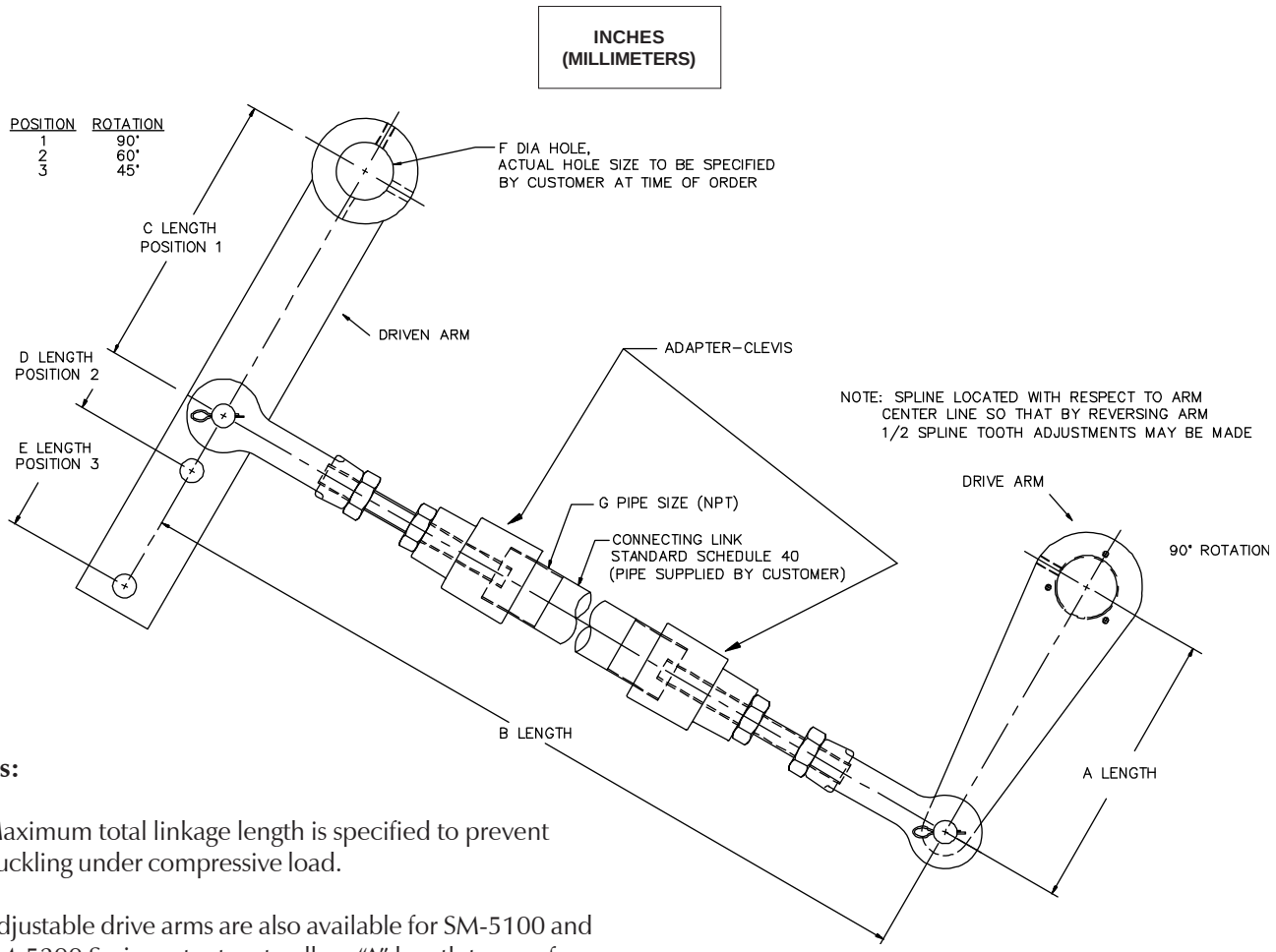
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

SM-5400 MAJOR DIMENSIONS



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

SM-5000 SERIES DRIVE ARM AND LINKAGE OPTIONS



Notes:

1. Maximum total linkage length is specified to prevent buckling under compressive load.

2. Adjustable drive arms are also available for SM-5100 and SM-5200 Series actuators to allow "A" length to vary from 6 to 10 inches (152 to 254 mm). In this case, the adapter-clevis has rod ball ends with lubricating fittings.

3. Special drive arm lengths are available to meet application requirements.

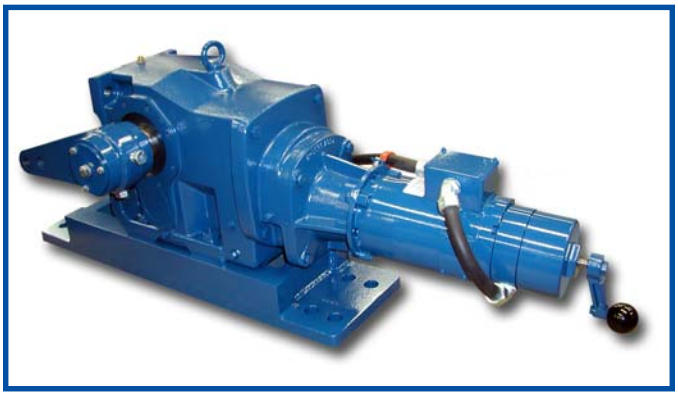
MODEL	"A" LENGTH in. (mm) (See Note 3)	"B" MAXIMUM LENGTH ft. (m)	"C" LENGTH in. (mm)	"D" LENGTH in. (mm)	"E" LENGTH in. (mm)	"F" DIA. HOLE min.-max. in. (mm)	"G" PIPE SIZE
SM-5100	10 (254)	18 (5.5)	10 (254)	12.25 (311)	17 (432)	0.75 - 1.5 (19 - 38.1)	1 - 1/4 (NPT)
SM-5200	10 (254)	30 (9.1)	10 (254)	12.25 (311)	17 (432)	1 - 2 (25.4 - 50.8)	2 - 1/2 (NPT)
SM-5300	12 (305)	18 (5.5)	12 (305)	14.70 (373)	21.50 (546)	1.5 - 3.5 (38.1 - 88.9)	2 - 1/2 (NPT)
SM-5400	15 (381)	18 (5.5)	15 (381)	18.38 (467)	25.50 (648)	2.5-5 (63.5 - 127)	3 - 1/2 (NPT)

Model	Drive Arm Option	Adapter Kit Option	Driven Arm Option
SM-5100	D001	D003	D005
SM-5200	D007	D009	D011
SM-5300	D013	D014	D015
SM-5400	D016	D017	D018

GENERAL DESCRIPTION

The SM-6000 Series are rotary actuators internally geared to produce up to 26,000 ft. lbs. (35,251 Nm) of torque at unrestricted, continuous modulating duty. The SM-6000 Series is ideally suited for dampers, vanes, valves and other process control applications requiring high speed, high torque and exact positioning control.

The SM-6000 Series actuator features an internally mounted digital amplifier which is easily programmed via the keypad and vacuum fluorescent display mounted on the front of the unit.

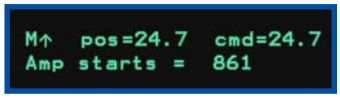


FEATURES

- Automatic torque limiting - protects actuator and driven equipment from damage due to overloads
- Watertight enclosure for outdoor use
- AC input power (single and three phase)
- Anti-backdrive
- Absolute encoder, contactless feedback
- Operating temperature: -40° to 185°F (-40° to 85°C) *Option for 225°F (107°C)*
- Anti-condensation heater
- Key lock Auto/Manual switch located on front panel
- Easy to program keypad located on front panel
  - To setup unit
  - To change parameters



- Easy to read vacuum fluorescent display located on front panel
  - Readout to assist programming
  - Readout of position during operation
  - Brightly lit, is easy to read



- Handcrank for manual operation when power is not available.
- Synthetic oil filled, built for extreme duty use
- Adjustable speed
- Opposite-side shaft extension
- Full two year warranty

BASE MODEL INCLUDES:

- DC Motor
- Manual override
- Electric holding brake
- Absolute encoder, contactless feedback
- Auto/Manual front panel key lock
- Increase and decrease front panel selector switches
- Keyed output shaft
- Anti-condensation heater
- Painted NEMA 4 (IP65) enclosure
- Absolute encoder feedback (amplifier models only)
- End of travels switches, not for customer use

POPULAR OPTIONS *(See Pages 197-206)*

- Drive arm, adapter/clevis kit, and driven arm linkage components
- Additional position limit switches
- Pilot lights, switches, relays
- Extended temperature ranges to 225° F (107° C)
- Motor-up mounting for tight space applications
- Mechanical position stop plate
- Battery back-up, RP-4000

ELECTRICAL SPECIFICATIONS

- **Input Voltage**: 120/1/50-60 ±10%, 240/1/50-60 ±10%, 208/1/50-60 ±10%, 240/3/50-60 ±10% or 208/3/50-60 ±10% amplifier input power.
- **Operating Current**: 10A nominal at 120/1/60 input.
- **Servo Amplifier**: Digital amplifier, internally or remotely mounted with display and keypad mounted on the actuator cover.
- **Electrical Enclosure**: Watertight for outdoor applications. Hinged cover. The enclosure has a front panel mounted keypad, illuminated display, and key lock auto/manual switch.
- **Feedback**: Contactless feedback is standard, potentiometer also available.
- **Command Signal Input**: 4-20 mA, 0-5 Vdc, 0-10 Vdc through customer supplied shielded cable.
- **Terminal Wiring**: Screw type terminals accepting up to #14 AWG solid or stranded copper wire.
- **Position Transmitter**: 4-20 mA isolated, loop powered position transmitter. Requires external power supply. Transmitter ZERO and SPAN are automatically set to actuator ZERO and SPAN, no customer adjustment is required. Operates when power is available to the amplifier.
- **Duty Cycle**: Unrestricted modulating duty
- **Torque Limiting**: Determined through motor current. Torque is limited to approximately 125-150% of actuator rating.
- **Reverse Acting**: ZERO and SPAN points can be set anywhere in actuator operating range.
- **Incremental Control**: Allows INC/DEC switch operation. Optional front cover mounted NEMA style pushbuttons for OPEN and CLOSE operation; keypad operation is standard.
- **Loss of Command Signal**: Selectable Lock-in-place, or move to field setable preset position.

MECHANICAL SPECIFICATIONS

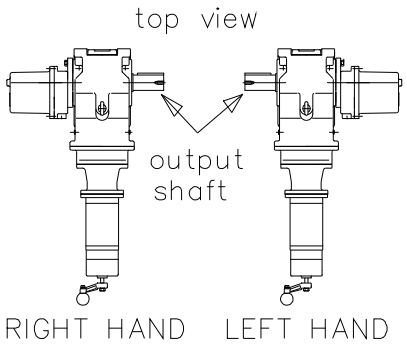
- **Output Torque & Speed**: Refer to selection chart.
- **Output Shaft**: Round with keyway, key supplied.
- **Maximum Travel**: 313 degrees. Greater travel is available as an option; consult factory.
- **Positioning Accuracy**: 0.10% of maximum travel.
- **Repeatability**: 0.10% of maximum travel.
- **Storage Temperature**: -40° F to 185° F (-40° to 85° C)
- **Ambient Operating Temperature**: -40° F to 185° F (-40° to 85° C) with integral electronics. Remote mounted electronics allow the actuator to operate in ambient temperatures to 225° F (107° C).
- **Lubrication**: Synthetic extreme duty industrial oil.
- **Mounting Position**: Base foot mounted only. An optional motor-up mounting is available, consult factory.
- **Conduit Openings**: Two 1/2 inch conduit openings are provided in the enclosure for customer use.
- **PC Board**: Modular PC board set is easily field replaced for service or upgrades.
- **Handwheel**: A handwheel is provided at rear of motor for manual operation.
- **Loss of Power**: Self locking, up to 150% of load.

SM-6000 SERIES SELECTION CHART

				Selection
1	Basic Model	SM-6060-D	Modulating duty, with built-in amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 60 Hz		
		2: 240 Vac, 1 Phase, 50/60 Hz		
		3: 380 Vac, 3 Phase, 50/60 Hz		
		4: 480 Vac, 3 Phase, 50/60 Hz		
		5: 240 Vac, 3 Phase, 50/60 Hz		
		9: 208 Vac, 1 Phase, 50/60 Hz		
		10: 208 Vac, 3 Phase, 50/60 Hz		
		19: 220 Vac, 1 Phase, 50 Hz		
3	Speed, sec. (a) / Torque, ft. lbs. (Nm)	**10/370 (502)**		
		10/550 (746)		
		10/800 (1085)		
		10/1400 (1898)		
		10/2500 (3389)		
4		**10/4400 (5965)**	*Not available for 120 Vac input voltage*	
		10/6200 (8406)	*Not available for 120 Vac input voltage*	
		12/8000 (10846)	*Not available for 120 Vac input voltage*	
		18/11000 (14914)	*Not available for 120 Vac input voltage*	
		26/16500 (22371)	*Not available for 120 Vac input voltage*	
		42/26000 (35251)	*Not available for 120 Vac input voltage*	
5	Rotation	**Specify** required output shaft rotation in degrees. Must be between 25° and 300°.		
6	Output Shaft	Specify left hand (**L**), right hand (**R**) or vertical (**V**) orientation.		

(a) Speeds are for 90° of rotation.

Output Shaft Extension



See page 209 and 211 for use as linear output to increase or decrease torque.

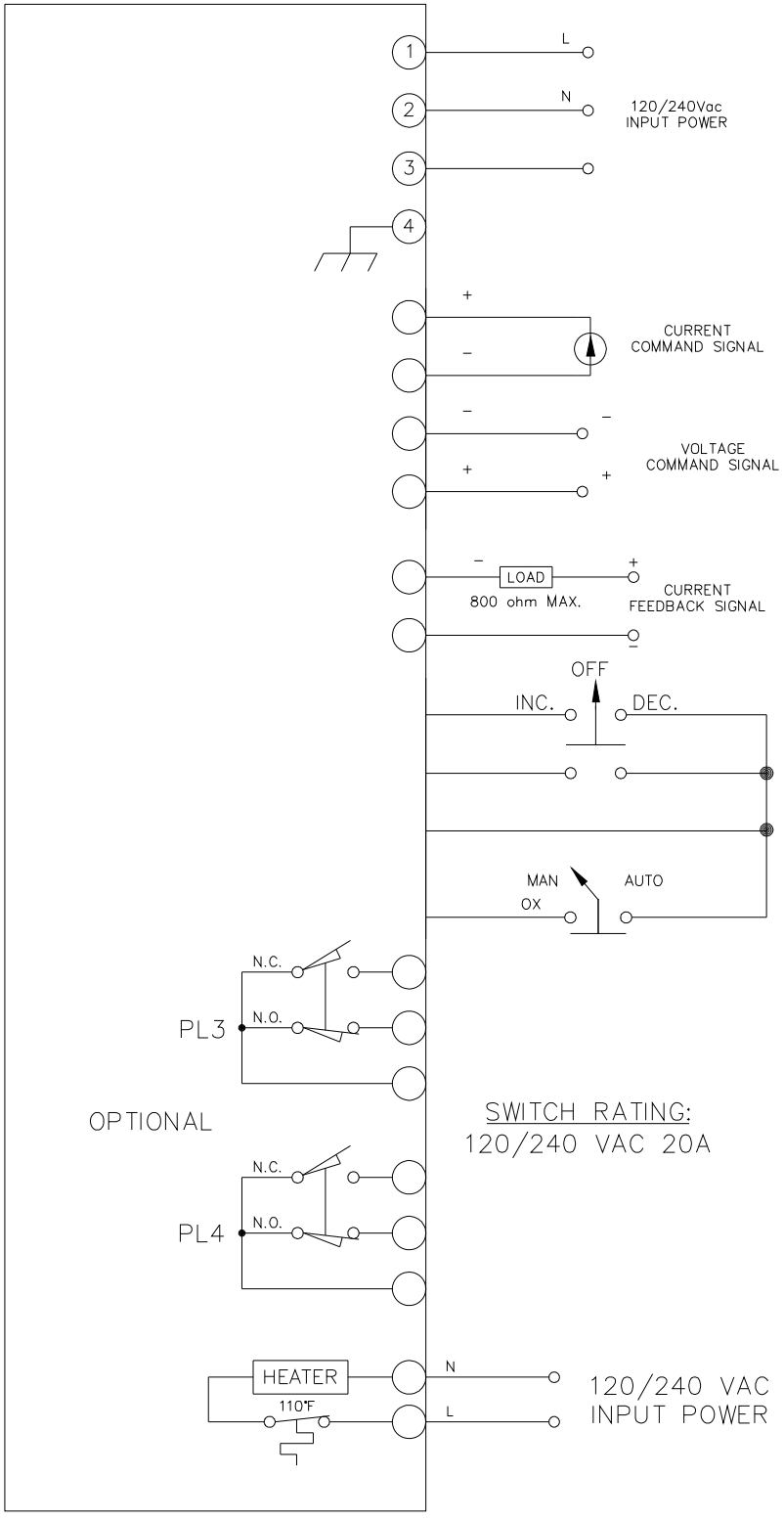
SM-6000 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A005	Two Pilot Lights - NEMA 4 (open/close)	
A012	ON/OFF Rotary Power Switch - NEMA 4	
A016	Fault Indicator Light	
A017	Power Indicator Light	
A018	Auto/Manual Indicator Light	
Connectors		
C001	Plug-In Cannon Type Connector on Enclosure for Power and Control	
C002	Mating Plug-In Cannon Type Connector on Enclosure for Power and Control	
Drive Arm/Adapter Clevis/Driven Arm		
D059	Keyed drive arm for 370 ft. lb. unit	
D060	Keyed drive arm for 800 ft. lb. unit	
D061	Keyed drive arm for 1400 ft. lb. unit	
D062	Keyed drive arm for 2500 ft. lb. unit	
D063	Keyed drive arm for 4400 ft. lb. unit	
D064	Keyed drive arm for 6200 ft. lb. unit	
D065	Keyed drive arm for 8000 & 11000 ft. lb. unit	
D066	Keyed drive arm for 16500 ft. lb. unit	
D067	Keyed drive arm for 26000 ft. lb. unit	
D068	Adapter assembly kit for 370 ft. lb. unit	
D069	Adapter assembly kit for 800 & 1400 ft. lb. unit	
D070	Adapter assembly kit for 2500 & 4400 ft. lb. unit	
D071	Adapter assembly kit for 6200, 8000, 11000 & 16500 ft. lb. unit	
D072	Adapter assembly kit for 26000 ft. lb. unit	
D073	Driven arm for 370 ft. lb. unit	
D074	Driven arm for 800 & 1400 ft. lb. unit	
D075	Driven arm for 2500 & 4400 ft. lb. unit	
D076	Driven arm for 6200, 8000, 11000 & 16500 ft. lb. unit	
D077	Driven arm for 26000 ft. lb. unit	
D078	Pedestal replacement & drive arm to replace SM-5100	
D079	Pedestal replacement & drive arm to replace SM-5200	
D080	Pedestal replacement & drive arm to replace SM-5300	
D081	Pedestal replacement & drive arm to replace SM-5400	
D082	Pedestal replacement & drive arm to replace Hagan Bailey	
D083	Adjustable length drive arm for 370 ft. lb. unit	
D084	Adjustable length drive arm for 800 ft. lb. unit	
D085	Adapter clevis kit for 370 ft. lb. unit (Used with D083)	
D086	Adapter clevis kit for 800 ft. lb. unit (Used with D084)	
D124	Adjustable length drive arm for 550 ft. lb. unit	
D125	Adapter clevis kit for 550 ft. lb. unit (Used with D124)	
D126	Driven arm for 550 ft. lb. unit	

Code	Description	Selection
Position Feedback		
F005	Transmitter, 4-20 mA output	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, not avail. with E003, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, not avail. with E003, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp	
Special Electrical Options		
X025	Relay contact to detect overtorque (stall)	
X026	Remote Amplifier for SM-6000 (185° F)	
X027	Remote Amplifier for SM-6000 (225° F), requires potentiometer feedback	
X033	Loss of line power relay (Consult factory)	
X034	Loss of signal relay (Consult factory)	
X055	Step down transformer for 480 to 240Vac. Close coupled, 4400 ft. lbs. and higher	
X057	Step down transformer for 480 to 240Vac. Remote-mounted, up to 2500 ft. lbs.	
X070	Torque overload indicator (Consult factory)	
Special Mechanical Options		
Y023	Extended rotation beyond 300°	
Y024	Mechanical stop for 370 ft. lb. unit	
Y025	Mechanical stop for 800 ft. lb. unit	
Y026	Mechanical stop for 1400 ft. lb. unit	
Y027	Mechanical stop for 2500 ft. lb. unit	
Y028	Mechanical stop for 4400 ft. lb. unit	
Y029	Mechanical stop for 6200 & 8000 ft. lb. unit	
Y030	Mechanical stop for 11000 ft. lb. unit	
Y031	Mechanical stop for 16500 ft. lb. unit	
Y032	Mechanical stop for 26000 ft. lb. unit	
Y999	Explosion-proof enclosure	

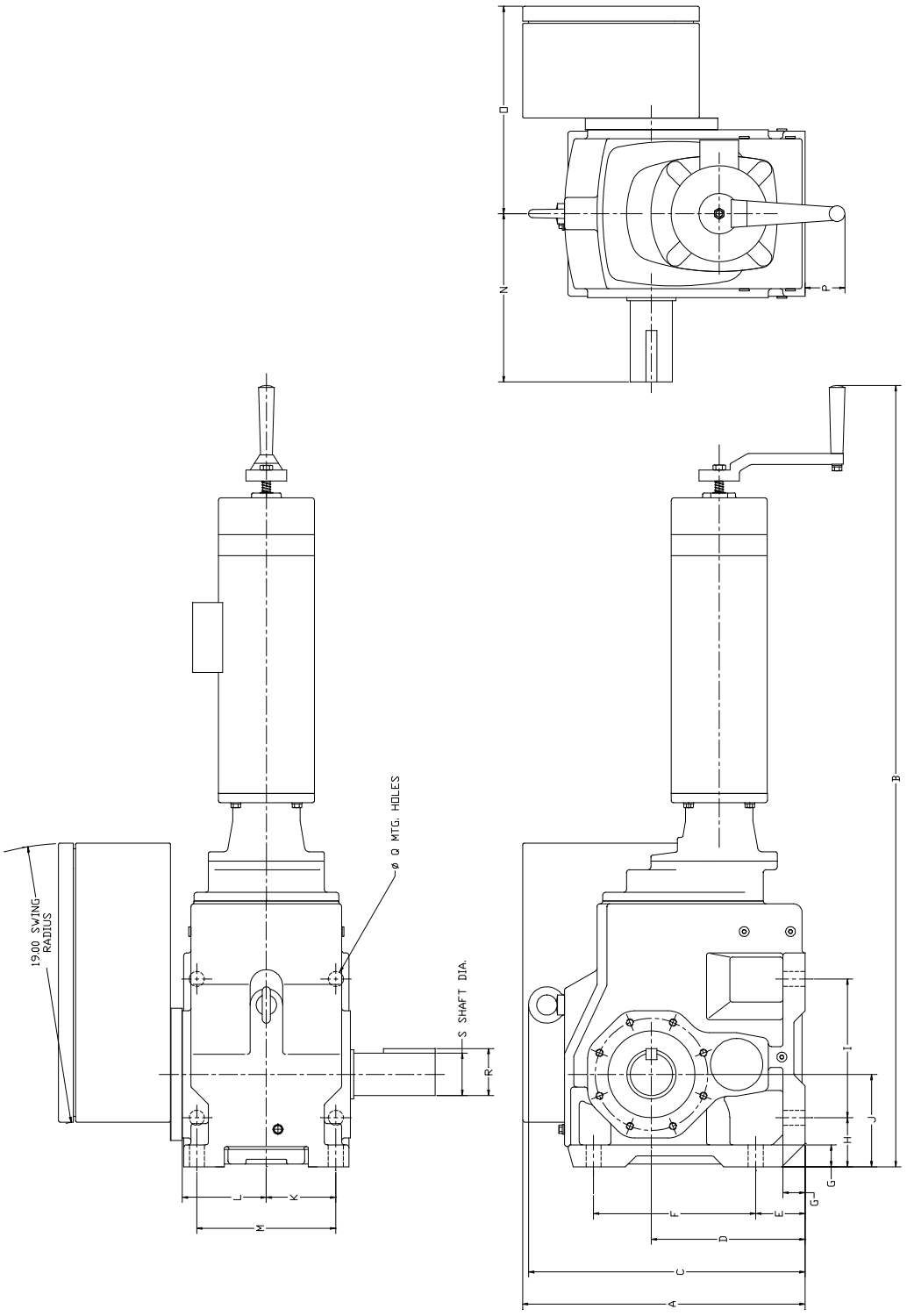
For a full description of options, go to the Complete Listing of Options starting on page 197.

TYPICAL WIRING DIAGRAM



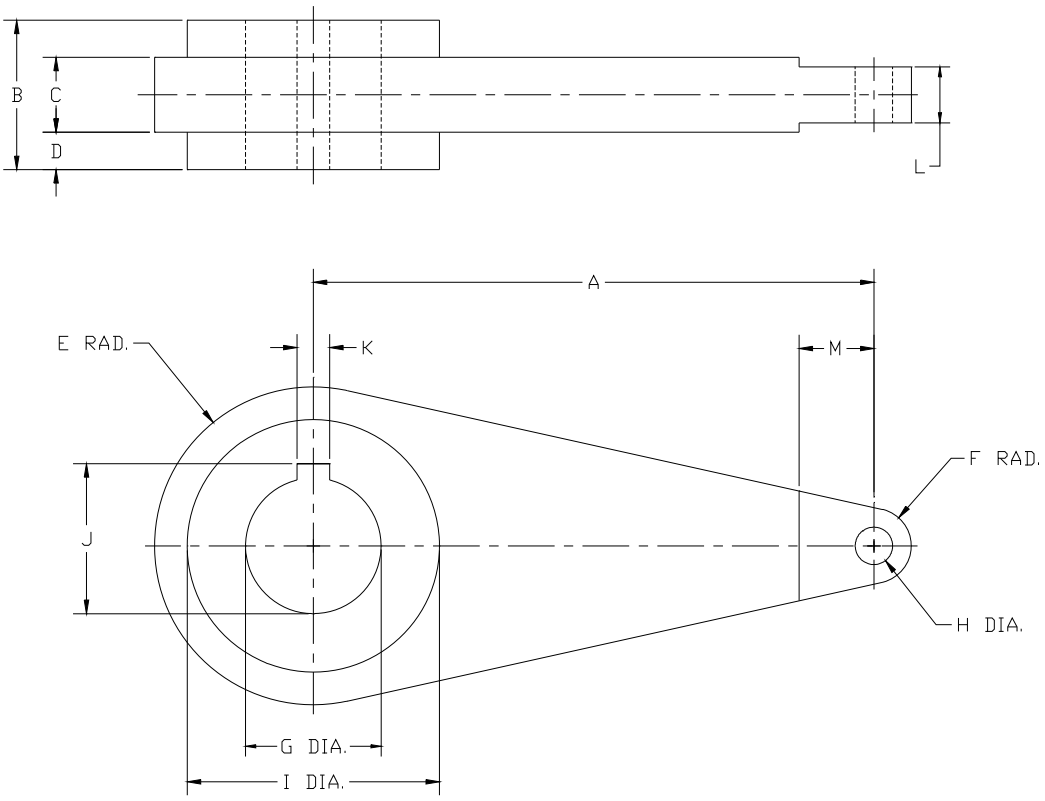
These dimensions and specifications are subject to change without notice.
Current drawings and specifications are available upon request.

SM-6000 MAJOR DIMENSIONS



Speed/ Torque	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	Key
10/370	13.16	34.84	9.02	4.41	1.46	5.12	.63	1.42	5.12	2.80	2.36	2.80	4.72	5.32	11.21	1.14	.43	1.36	1.250/1.249	.25 x .25 x 1.63 LG.
10/800	14.26	36.85	10.61	5.51	1.97	5.12	.95	2.76	4.72	3.94	2.76	3.39	5.51	6.34	11.80	.19	.55	1.51	1.375/1.374	.31 x .31 x 2.13 LG.
10/1400	15.82	38.09	12.99	7.09	2.56	7.48	.99	2.94	5.91	4.41	3.25	3.94	6.50	7.72	12.35	-	.71	1.92	1.750/1.749	.38 x .38 x 2.50 LG.
10/2500	17.10	45.74	15.35	8.35	3.03	8.66	1.26	3.03	7.09	5.20	3.54	4.53	7.09	9.45	12.94	-	.88	2.65	2.375/2.374	.63 x .63 x 4.00 LG.
10/4400	19.18	53.21	18.78	10.43	3.335	11.02	1.50	3.35	9.45	6.30	4.73	5.71	9.45	11.46	14.12	2.70	1.02	3.20	2.875/2.874	.75 x .75 x 4.50 LG.
10/6200	21.80	63.37	21.97	12.40	4.13	12.20	1.65	4.13	11.02	7.87	5.32	6.69	10.63	13.66	15.10	-	1.30	4.01	3.625/3.624	.88 x .88 x 5.50 LG.
12/8000	23.50	70.88	27.91	14.76	4.33	15.75	1.77	4.33	13.78	8.86	6.46	7.87	12.99	16.46	16.28	-	1.54	4.82	4.375/4.374	1.00 x 1.00 x 7.25 LG.
18/11000	23.50	70.88	27.91	14.76	4.33	15.75	1.77	4.33	13.78	8.86	6.46	7.87	12.99	16.46	16.28	-	1.54	4.82	4.375/4.374	1.00 x 1.00 x 7.25 LG.
26/16500	26.47	73.44	32.46	17.72	4.72	18.90	1.97	4.33	14.96	9.84	8.27	9.37	16.54	17.99	17.78	-	1.54	5.30	4.750/4.749	1.25 x 1.25 x 7.25 LG.
42/26000	28.44	80.41	36.60	19.69	4.53	21.25	1.97	4.53	21.25	12.40	9.45	11.61	18.90	21.85	20.02	-	1.30	6.05	5.500/5.499	1.25 x 1.25 x 8.38 LG.

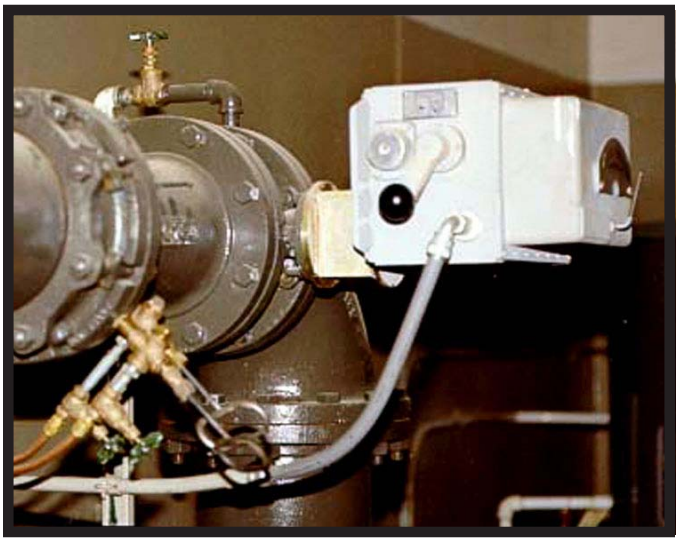
SM-6000 DRIVE ARM DIMENSIONS



Speed/Torque	A	B	C	D	E	F	G	H	I	J	K	L	M
10/370	10.00	1.00	-	-	1.08	.75	1.25	.75	-	1.39	.25	1.00	-
10/800	10.00	1.50	-	-	1.50	1.00	1.38	1.00	-	1.54	.31	1.50	-
10/1400	10.00	1.50	-	-	1.50	1.00	1.75	1.00	-	1.95	.38	1.50	-
10/2500	12.00	2.50	1.50	.50	2.63	1.00	2.38	1.00	4.50	2.68	.63	1.50	-
10/4400	12.00	2.50	1.50	.50	2.63	1.00	2.88	1.00	4.50	3.23	.75	1.50	-
10/6200	15.00	4.00	2.00	1.00	4.25	1.00	3.63	1.00	6.75	4.04	.88	1.50	2.00
12/8000 & 18/11,000	15.00	4.00	2.00	1.00	4.25	1.00	4.38	1.00	6.75	4.85	1.00	1.50	2.00
26/16,500	15.00	4.00	2.00	1.00	4.25	1.00	4.75	1.00	6.75	5.33	1.25	1.50	2.00
42/26,000	18.00	5.00	2.00	1.50	4.63	1.50	5.50	1.25	7.75	6.08	1.25	2.00	-

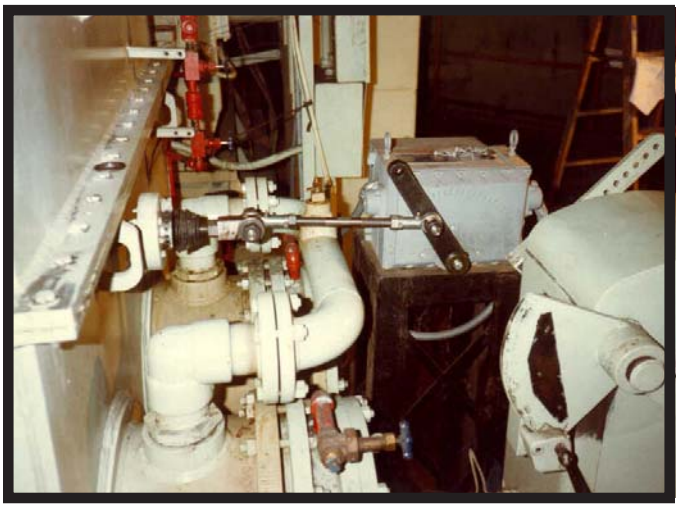
These dimensions and specifications are subject to change without notice.
Current drawings and specifications are available upon request.

ROTARY ACTUATOR APPLICATIONS



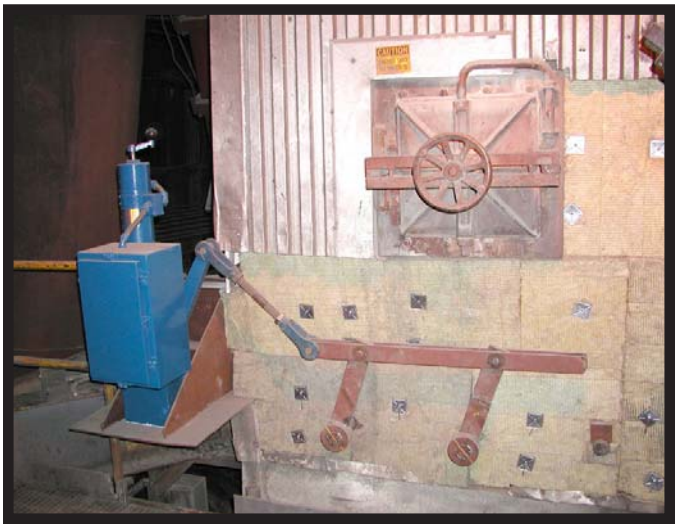
SM-5120 on a Butterfly Valve

- Controlling a butterfly valve at a water purification plant
- Mineral buildup can be a problem at the ends of travel, and the Scotch Yoke torque design solved this problem
- Environment is very moist and requires a NEMA 4 enclosure rating
- Unit calibrated at the factory for 90 Degrees rotation



SM-5120 on a Hydraulic Drive Scoop Tube (Pedestal Mount)

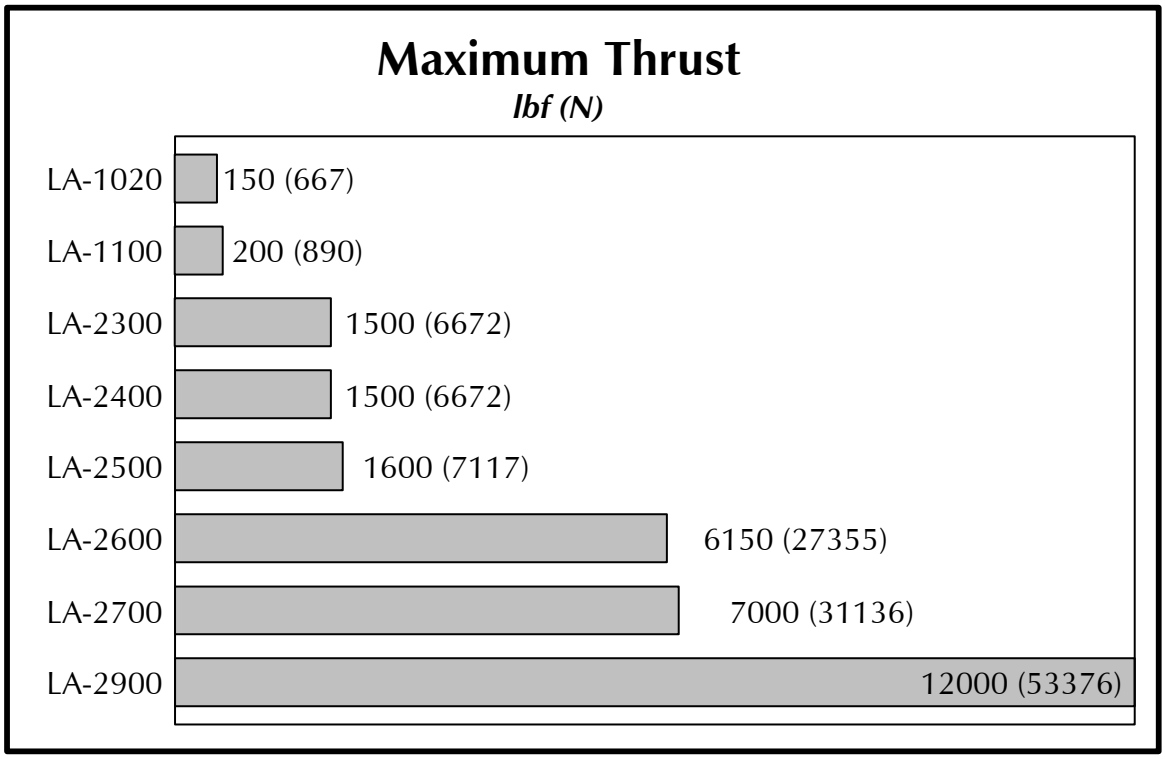
- Hydraulic variable speed drive with scoop tube position controlled by the Jordan actuator.
- The drive controls the speed of the boiler feed water pump.
- Moderate duty cycle application requires a characterization of the feedback.
- A fabricated mounting pedestal was provided to match the mounting of the old Bailey actuator.



SM-6000 on a Gas Recirculation Damper

- High temperature operation reduces costly repairs and unscheduled maintenance.
- Digital amplifier with easy-to-read display for easy access to status and diagnostic information.
- Pushbutton setup for fast and easy commissioning.
- Vertical Mounting.

THRUST RANGES



MODEL SERIES SELECTIONS

Model Series	Performance					Power Input Voltage**			Catalog Page #
	Max.Thrust lbf (N)	Stroke - inches (mm)		Speed - in./sec. (mm/sec.)		1 Phase 50/60 Hz	DC	3 Phase 50/60 Hz	
		Minimum	Maximum	Minimum	Maximum				
LA-1020	150 (667)	0.75 (19)	18 (457)	0.2 (5.1)	0.67 (17)	115/230	12-36	—	60
LA-1100	200 (890)	0.75 (19)	18 (457)	0.033 (0.84)	5 (127)	120/240	24	—	64
LA-2300	1500 (6672)	6 (152)	36 (914)	0.2 (5.1)	0.8 (20.3)	120/240	—	240/480	70
LA-2400	1500 (6672)	6 (152)	24 (610)	0.1 (2.54)	1 (25.4)	120/240	24	240/480	70
LA-2500	1600 (7117)	6 (152)	24 (610)	0.9 (22.9)	0.9 (22.9)	120/240	—	240/480	70
LA-2600	6150 (27355)	6 (152)	30 (762)	0.5 (12.7)	0.8 (20.3)	120/240	—	240/480	70
LA-2700	7000 (31136)	6 (152)	30 (762)	0.6 (15.2)	1.6 (40.6)	120/240	—	240/480	70
LA-2900	12000 (53376)	6 (152)	30 (762)	0.25 (6.4)	0.25 (6.4)	240*	180	240/480	70

** Requires Jordan Controls amplifier for AC input*
*** +/- 10%*

LA-1000 SERIES LINEAR ACTUATORS

GENERAL DESCRIPTION

The LA-1000 Series are full-featured, linear actuators designed to meet the exacting requirements for closed-loop modulating positioning control. Designed for low thrust linear applications, these actuators are capable of accepting analog current and voltage command signals. The brushless stepper motor design provides smooth, highly accurate positioning, with positive position-lock when not in motion. The LA-1000 Series is ideally suited for small dampers, vanes and slide gates requiring exact position control and continuous modulation.

All LA-1000 Series actuators come with a standard digital internal amplifier. These amplifiers are all full-featured ac switching devices designed to work seamlessly with the actuator for closed loop control.



FEATURES

- Permanently lubricated for any position mounting
- Amplifier supplies current to hold last position and prevent backdriving, up to thrust rating under power
- AC or DC input power versions
- 4 to 20 mA position, loop-powered, feedback signal
- Field selectable adjustments for:
  - speed
  - deadband
  - zero and span
  - command signal type
  - standard or reverse acting
  - manual-auto operation
  - output shaft position on loss of signal
- Wide ambient temperature range
- Full one year warranty

BASE MODEL INCLUDES:

- Motor (stepper)
- Manual override
- Dual stroke arms with bellows
- Drive shaft
- Split range command input
- Amplifier
- Electronic thrust limiting
- Cast aluminum NEMA 4 (IP65) & dust ignition-proof enclosure
- Four adjustable position switches (40 mA at 40 Vdc)
- Internal spur gear train
- 4-20 mA transmitter for customer use

POPULAR OPTIONS *(See Pages 197-206)*

- Signal Conversion Module: Convert 40 mA, 40 Vdc low level to 5 Amp, 120/240 Vac current rated position limit switches
- Low current at null (24 Vdc only)
- Switching input powered ac or dc positioning for positioning actuator using ac or dc remote voltage control
- Process Variable Controller to control one process variable - 115/230 Vac only
- Custom mounting and interface hardware
- Local Auto/Manual and INC/OFF/DEC toggle switches (Close-coupled, NEMA 4)
- Various enclosure coatings
- Battery backup to position actuator on loss of ac power
- ATEX Approval

SPECIFICATIONS

Thrust: Up to 150 lbf. (10 Nm)

Speed: See selection chart on next page.

Input Voltage: See selection chart on next page.

Linear Travel: 0.75 inch (19 mm) to 18 inches (457 mm), 3 inch increments. *For greater strokes, consult factory.*

Temperature Limits: -40°F to 150°F (-40°C to 65°C) *For greater temperature ranges, consult factory.*

Environmental Ratings: Explosion-proof for Class I, Division 1, Groups C and D; dust-ignition-proof for Class II, Division 1, Groups E, F, G. Also rated NEMA 4 (IP65) indoor/outdoor.

Command Input:
Full Range: 4-20 mA into 200 ohms
Split Range: 4-12 and 12-20 mA into 200 ohms
Voltage: 0-5 Vdc or 0-10 Vdc into 100,000 ohms
Switch: Dry contact closures

Field Wiring: To barrier terminal blocks

Thrust Limiting: Current sensing PWM

Positioning Accuracy: 1.5% for full range

Feedback: 4-20 mA, customer supplied (loop power)

Loss of Signal: Lock-in-place or run to preset.

Loss of Power: Lock-in-place at last position.

Direction: Extend or retract

Modulation Rate: Unrestricted modulating duty

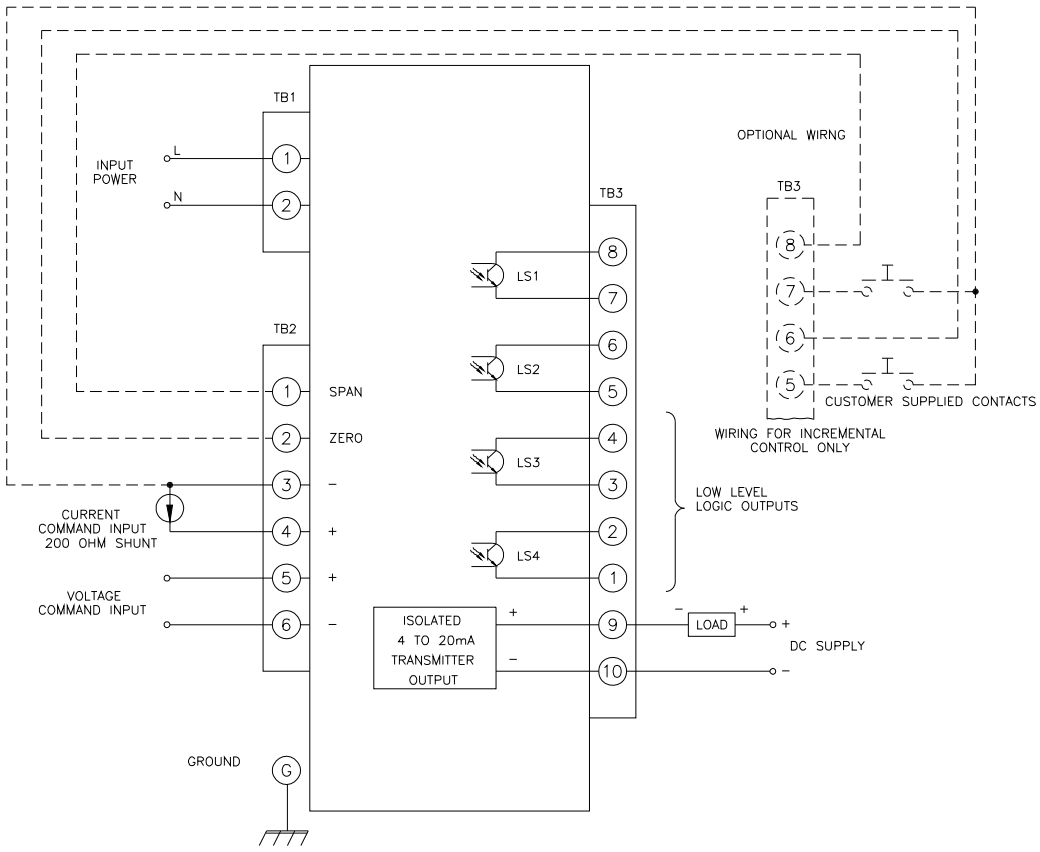
Backdrive: Self-locking up to thrust rating

Approximate Weight: LA-1020: 14 lbs. (6.4 kg)

Approvals: See page 207

Position Switch Rating: 40mA at 40 Vdc

TYPICAL WIRING DIAGRAM



These dimensions and specifications are subject to change without notice. Current drawings and specifications are available upon request.

LA-1000 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-1020-D**	1 phase, 120/240 Vac, with built-in digital amplifier	
2	Input Voltage	**1**: 115 Vac, 1 Phase, 50/60 Hz		
		2: 230 Vac, 1 Phase, 50/60 Hz		
		6: 24-36 Vdc		
		12: 12-36 Vdc, maximum output thrust is limited - consult factory		
		13: 24 Vac		
3	Speed in./min. (mm/min.) / Thrust, lbf. (N)	**35/25** (111)		
		30/50 (222)		
4		**20/100** (445)		
		10/150 (667)		
5	Linear Travel	**Specify Stroke** - 0.75 inch (19 mm) to 18 inches (457 mm)(Rack bellows is standard) (3 inch increments)		

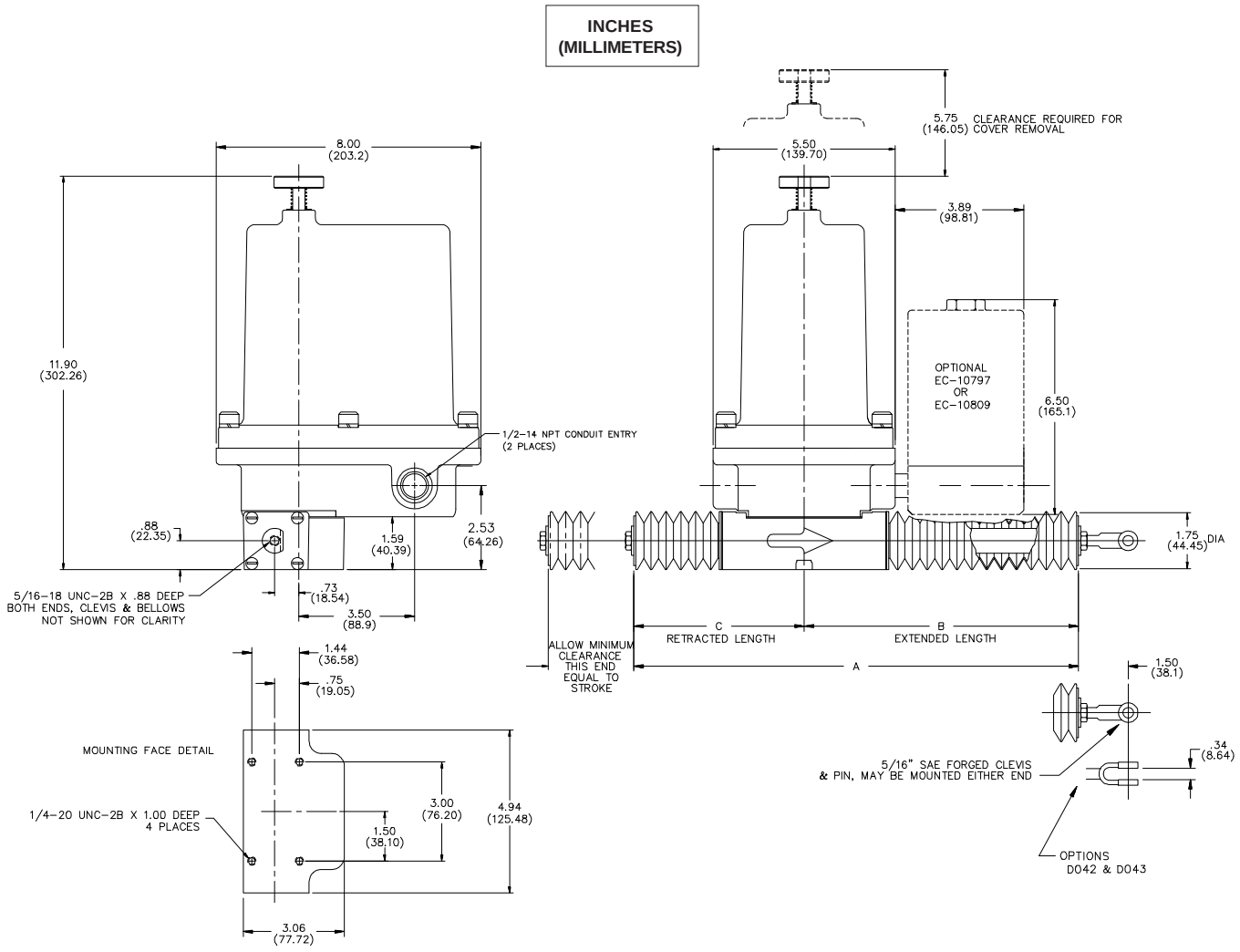
LA-1000 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure) NEMA 4	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure) NEMA 4	
Drive Arm/Adapter Clevis/Driven Arm		
D042	Clevis, one end	
D043	Clevis, both ends	
Identification/Certifications		
J001	CE Marking	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
J006	ATEX Approval (Input voltages 1, 2 & 6) Consult factory for option compatibility.	

For a full description of options, go to the Complete Listing of Options starting on page 197.

Code	Description	Selection
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	
Special Electrical Options		
X012	Signal conversion module, installed in a close-coupled, NEMA 4 enclosure (not availabe with X015)	
X013	2 wire incremental dc input	
X014	3 wire incremental 120/240 Vac input	
X015	Battery Backup, only for 120/240 Vac, installed in a close-coupled, NEMA 4 enclosure (not availabe with X012)	
X017	Process variable	
X018	Low Current (24 Vdc only)	
Special Mechanical Options		
Y014	Metric mounting holes, 6mm	

LA-1000 MAJOR DIMENSIONS



STROKE	A	B	C
0.75-3 (19-76)	12.38 (314)	7.69 (195)	4.69 (119)
3-6 (76-152)	17.38 (441)	11.69 (297)	5.69 (145)
6-9 (153-229)	22.38 (568)	15.69 (399)	6.69 (170)
9-12 (229-305)	27.38 (695)	19.69 (500)	7.69 (195)
12-15 (305-381)	32.38 (882)	23.69 (602)	8.69 (221)
15-18 (381-457)	42.38 (1076)	31.69 (805)	10.69 (272)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

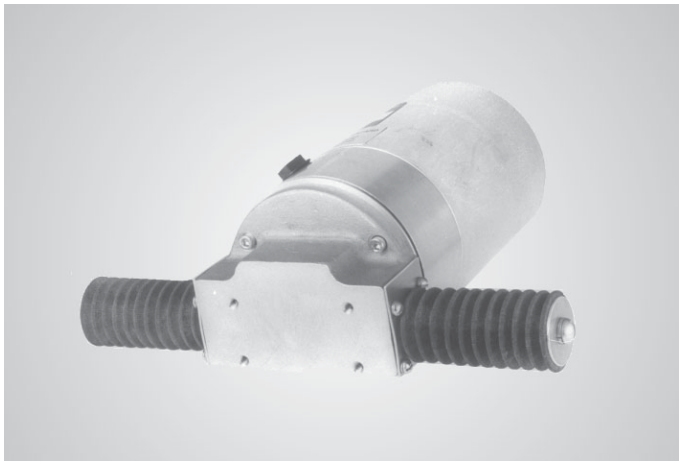
LA-1100 SERIES LINEAR ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The LA-1100 Series are linear, bi-directional actuators, designed to meet the exacting requirements for closed-loop modulating positioning control. They can be applied to a variety of sliding stem valves to provide remote valve positioning for control of gases, steam or other liquids in various process control applications.

All LA-1100 Series actuators are also available with an internal amplifier (AC voltage only). These amplifiers are all full-featured ac switching devices designed to work seamlessly with the actuator for closed loop control.



FEATURES

- Permanently lubricated for any position mounting
- A wide variety of speed/thrust combinations
- AC or DC input power versions
- Wide ambient temperature range
- Full one year warranty

AMPLIFIER FEATURES

- Isolated, two-wire, 4-20mA feedback
- Customer-adjustable deadband
- Customer-adjustable zero and span
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place - Run to preset
- Field selectable dynamic brake
- AC triac switching of motor for increased life

BASE MODEL INCLUDES:

- Motor
- End of travel switches, not for customer use
- Aluminum NEMA 4 (IP65), explosion-proof enclosure with screw-on cover
- Rack bellows on both ends
- Internal spur and rack and pinion gear train
- 1000 ohm potentiometer feedback (amplifier models only)

POPULAR OPTIONS *(See Pages 197-206)*

- Potentiometer or 4 to 20 mA position feedback
- Anti-condensation heater and thermostat
- Custom mounting and interface hardware
- Handcrank (LA-1500 only)

SPECIFICATIONS

	LA-1100	LA-1500
Thrust Range lbf. (N)	3.5 to 200 (16 to 890)	74 to 500 (329 to 2224)
Stroke in. (mm)	0.75 to 18 (19 to 457)	0.75 to 21 (19 to 533)
Speed in./min. (mm/min.)	2 to 61 (51 to 1549)	1 to 56 (25 to 1422)
Approx. Weight lbs. (kg)	15 to 20 (7 to 9)	20 to 27 (9 to 12)

Temperature Limits: -40°F to 150°F (-40°C to 65°C)
For greater temperature ranges, consult factory.

Field Wiring: To barrier terminal blocks

Approvals: See page 207

Position Switch Rating: 120/240 Vac, 10 Amp

Additional Specifications on next page

LA-1100/LA-1500 SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Command Input	Input Impedance in ohms	Feedback	Modulation Rate *
		Run	Stall				
LA-1110-N	120/1/50/60	0.25	0.26	N/A	N/A	N/A	2000 starts/hr
LA-1110-D	120/1/50/60	0.25	0.26	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA loop powered	2000 starts/hr
LA-1140-N	24 Vdc	1.2	4.8	N/A	N/A	N/A	4000 starts/hr
LA-1170-N	240/1/50/60	0.14	0.15	N/A	N/A	N/A	2000 starts/hr
LA-1170-D	240/1/50/60	0.14	0.15	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA loop powered	2000 starts/hr

**** Rated for continuous modulating service at 1% minimum position change.***

N/A = Not Applicable

REMOTE AMPLIFIER CHART

Actuator	Remote Mounted Amplifier	Catalog Page
LA-1110-N	AD-8240	173
LA-1140-N	AD-7530	170
LA-1170-N	AD-8240	173

LA-1100 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-1110-N**	1 phase, 120 Vac, 50/60 Hz, without built-in digital amplifier	
		LA-1110-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier	
		LA-1140-N	24 Vdc (If amplifier desired, see remote model AD-7530	
		LA-1170-N	1 phase, 240 Vac, 50/60 Hz, without built-in digital amplifier	
		LA-1170-D	1 phase, 240 Vac, 50/60 Hz, with built-in digital amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-1110-N, LA-1110-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-1170-N, LA-1170-D)		
		6: 24 Vdc (LA-1140-N, LA-1180-N)		
3 / 4	Speed in./min. (mm/min.) / Thrust lbf. (Nm)	LA-1110 LA-1170	LA-1140	
		2/200 (51/890)	**2.25/200** (57/890)	
		4/200 (102/890)	**4.5/200** (114/890)	
		8/112 (203/489)	**9/200** (229/890)	
		15/56 (381/249)	**18/200** (457/890)	
		31/28 (787/125)	**36/140** (914/623)	
		61/14 (1549/62)	**75/70** (1905/311)	
5	Linear Travel	**Specify** Travel - 0.75 inch (19 mm) to 18 inches (457 mm) stroke available. (3 inch increments)		

LA-1100 STANDARD OPTIONS

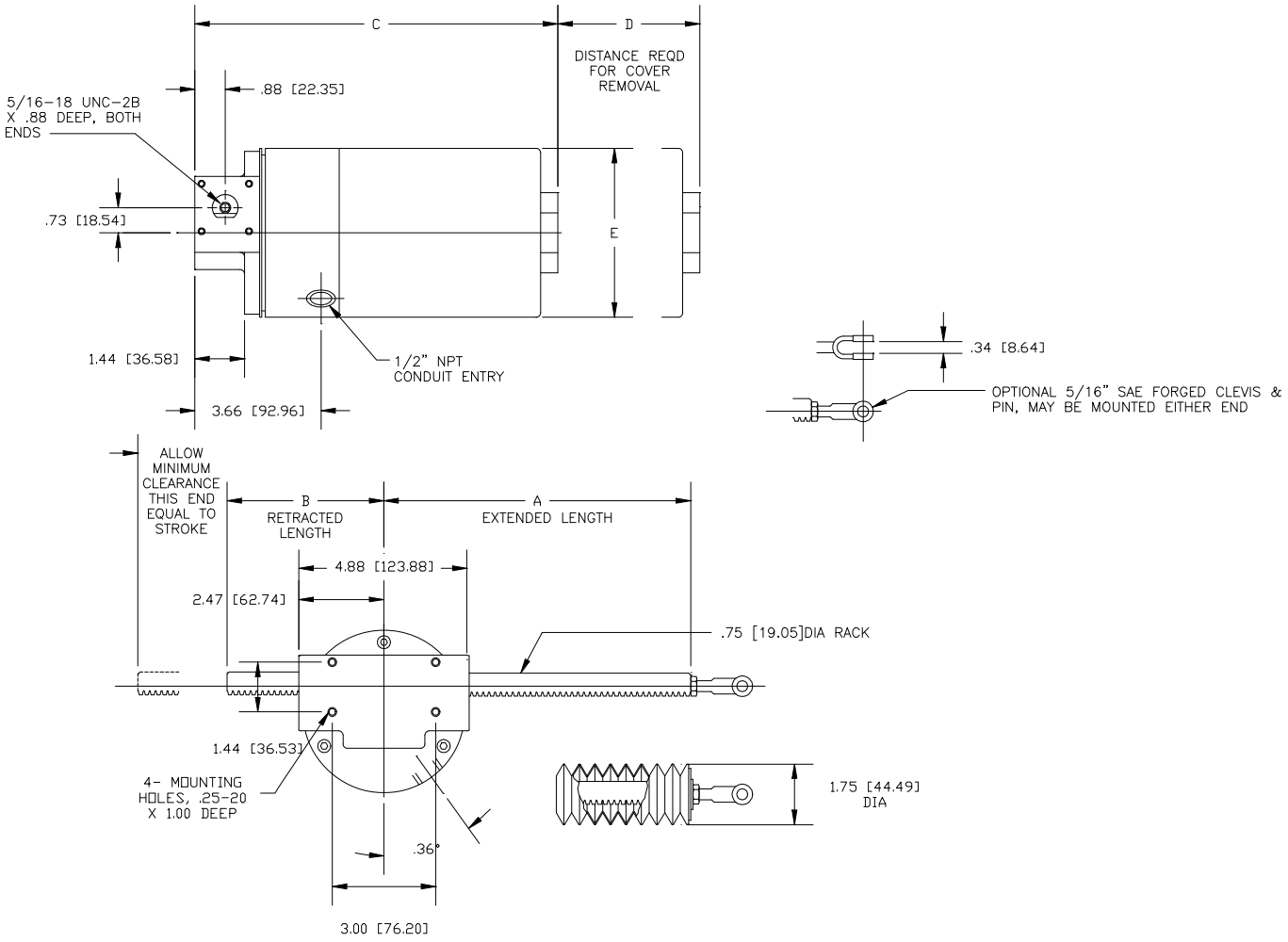
Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure)	
Drive Arm/Adapter Clevis/Driven Arm		
D042	Clevis, one end	
D043	Clevis, both ends	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not for use with units with internal amps)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA	
F006	One Potentiometer, 1000 ohms plus Transmitter, 4-20 mA (not for use with units with internal amps)	
F010	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	

Code	Description	Selection
Identification/Certifications		
J001	CE Marking, AC input only (available for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S005	Aluminum Cam (*Up to 5.85 turns)*	
S006	Aluminum Feedback Frame	
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	
Special Mechanical Options		
Y015	Metric mounting holes, 6mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-1100 MAJOR DIMENSIONS

INCHES (MILLIMETERS)



Stroke in. (mm)	With Rack Bellows		Without Rack Bellows	
	A	B	A	B
0.75 -3	7.69	4.69	5.62	2.62
(19-76)	(195)	(119)	(143)	(67)
3-6	11.69	5.69	8.62	2.62
(76-152)	(297)	(145)	(219)	(67)
6-9	15.69	6.69	11.62	2.62
(142-229)	(399)	(170)	(295)	(67)
9-12	19.69	7.69	14.62	2.62
(229-305)	(500)	(195)	(371)	(67)
12-15	23.69	8.69	17.62	2.62
(305-381)	(602)	(221)	(448)	(67)
15-18	27.69	9.69	20.62	2.62
(381-457)	(703)	(246)	(524)	(67)

C		D		E
W/out Amp	With Amp	W/out Amp	With Amp	
10.53	12.06	6.28	7.81	4.88
(267)	(306)	(162)	(198)	(124)

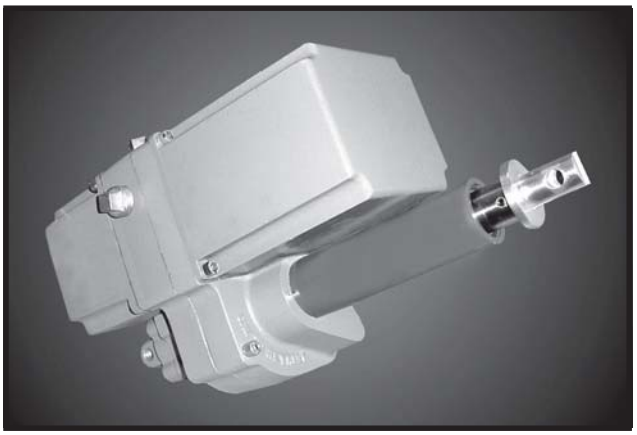
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LA-2000 SERIES LINEAR ACTUATORS

GENERAL DESCRIPTION

The LA-2000 Series are self-contained, bi-directional, linear actuators internally geared to produce up to 12,000 lbs. (53,376 N) of thrust, and offer continuous modulating duty up to 2,000 starts per hour. A variety of speeds, thrusts and motors are available which make the LA-2000 Series ideally suited for reliable positioning of dampers, combustion controls, gates, diverters, variable speed drives, or other linear positioning applications.

Most LA-2000 Series actuators are also available with an internal amplifier. These amplifiers are all full-featured ac or dc switching devices designed to work seamlessly with the actuator for closed loop control.



LA-2320

FEATURES

- Permanently lubricated for any position mounting *
- Internal or remote mounted amplifier
- Self-locking drive system to hold in last position and prevent backdriving up to thrust rating
- AC input power (single and three phase) or DC input power versions
- Wide ambient temperature range
- Two year warranty on LA-2300, LA-2400, LA-2500 and LA-2900; one year warranty on LA-2600 and LA-2700 models

BASE MODEL INCLUDES:

- Motor
- Manual override (except LA-2300)
- Automatic thrust limiting (except LA-2300)
- End of travel switches, not for customer use
- Aluminum NEMA 4 (IP65) enclosure
- Clevis mounting
- Anti-condensation heater and thermostat (except LA-2300)
- Internal spur gear train and acme power thread
- 4-20 mA transmitter for customer use (amplifier models only)

** Check with factory for vertical mounting*

AMPLIFIER FEATURES

- Analog or digital available
- Isolated, two-wire, 4-20mA output
- Push button set-up for digital amplifier
- Customer-adjustable zero and span
- Adjustable deadband
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place
  - Run to preset
- Digital amplifier proportional control
- Dynamic brake

POPULAR OPTIONS *(See Pages 197-206)*

- Auxilliary position limit switches
- Trunnion mount (Not avail. on LA-2900 models) **
- Flange mount for LA-2300, LA-2400, LA-2600 & LA-2700 models (Not avail. in 6 inch stroke for LA-2600 & LA-2700) **
- Anti-condensation heater and thermostat (LA-2300 only)
- Integral reversing starter with overload protection (LA-2600 & 2700, 3 phase models)
- Potentiometer, contactless or 4 to 20 mA position feedback
- Battery back-up, RP-4000
- Waterproof tube and handcrank, used when tube is in up position and subject to moisture intrusion **

*** Check with factory for special considerations*

ACTUATOR/AMPLIFIER SPECIFICATIONS

	Models with Internal (D) or External Analog Amplifier
Position Accuracy	1.0 % of full travel
Repeatability	0.5 % of full travel
Hysteresis	0.5 % of full travel
Linearity	+/-1 % of full travel
Deadband	0.75 % of full travel
Resolution	0.75 % of full travel

REMOTE AMPLIFIER CHART

Actuator	Remote Mounted Amplifier	Catalog Page
LA-2320-N	AD-8240	173
LA-2390-N	AD-8240	173
LA-2410-N	AD-8240	173
LA-2415-N	AD-8900	177
LA-2420-N	AD-8240	173
LA-2440-N	AD-7530	170
LA-2460-N	AD-7830/AD-9120	180/184
LA-2490-N	AD-8240	173
LA-2510-N	AD-8900	177
LA-2520-N	AD-8240	173
LA-2610-N	AD-8900	177
LA-2620-N	AD-8240	173
LA-2710-N	AD-8900	177
LA-2720-N	AD-8240	173
LA-2910-N	AD-8900	177
LA-2980-N	AD-9120	184

BASIC SPECIFICATIONS FOR LA-2000 SERIES ACTUATORS

Stroke: 6 to 36 in. (152-762 mm) **

Temperature Limits: -40° to 150°F (-40° to 65°C) **

Position Limit Switches: 10 Amp SPDT switch (LA-2300, LA-2400), 20 Amp SPDT switch (LA-2500, LA-2600, LA-2700 & LA-2900), one for each end of travel

Environmental Ratings:

Model Series	Watertight	Explosion-proof Gas	Dust-Ignition Proof
LA-2300	S	NA	S
LA-2400	S	NA	S
LA-2500	S	S	S
LA-2600	S	O	O
LA-2700	S	O	O
LA-2900	S	NA	NA

KEYS: S: Standard O: Optional NA: Not Available
ENCLOSURE DEFINTIONS:
Watertight = NEMA Type 4 (IP65) Indoor/outdoor
Explosionproof Gas = NEMA Type 7 Explosion proof, Class I, Division 1, Groups C & D, indoor/outdoor
Dust-Ignition Proof = NEMA Type 9 Explosion proof, Class II, Division 1, Groups E, F & G, indoor/outdoor

Back Drive: Self-locking up to thrust rating

Field Wiring: To barrier terminal blocks

Thrust Limiting: 20 Amp, SPDT switch, one each extend and retract. Switch will change states when thrust exceeds rated thrust for actuator. (10 Amp for LA-2400) (Not available for LA-2300)

Field Wiring: To barrier terminal blocks

Approvals: See page 207

Position Switch Rating:
LA-2300 & LA-2400: 120/240 Vac, 10 amp
LA-2500, LA-2600, LA-2700 & LA-2900: 120/240 Vac, 20 amp

Approximate Weight: LA-2300: 40 lbs. (18 kg)
LA-2400: 40 lbs. (18 kg)
LA-2500: 105 lbs. (48 kg)
LA-2600: 185 lbs. (84 kg)
LA-2700: 185 lbs. (84 kg)
LA-2900: 450 lbs. (204 kg)

*** Check with factory for special considerations*

LA-2300 & LA-2400 SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Maximum Thrust lbf. (N)		Speed in./sec. mm/sec.	Command Input	Input Impedance in ohms	Feed back	Modulation Rate *	Dynamic Braking	Loss of Power	Loss of Command Signal
		Run	Stall	Running	Breakaway								
LA-2320-N	120/1/50/60	2.5	2.9	750 (3336)	1500 (6672)	2.0 (50.8)	N/A	N/A	N/A	600 starts/hour	N/A	L.I.P.	N/A
LA-2320-D	120/1/50/60	2.5	2.9	750 (3336)	1500 (6672)	2.0 (50.8)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	600 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2390-N	240/1/50/60	1.1	1.6	750 (3336)	1500 (6672)	2.0 (50.8)	N/A	N/A	N/A	600 starts/hour	N/A	L.I.P.	N/A
LA-2390-D	240/1/50/60	1.1	1.6	750 (3336)	1500 (6672)	2.0 (50.8)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	600 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2315-N	240/480/3/50/60	0.4/0.2	1.3/0.65	1500 (6672)	N/A	0.7 (18)	N/A	N/A	N/A	2000 starts/hour	N/A	L.I.P.	N/A
	380/3/50	0.3	0.9	1500 (6672)	N/A	0.6 (15)							
LA-2410-N	120/1/50/60	0.9	1.2	800 (3558)	N/A	0.7 (18)	N/A	N/A	N/A	2000 starts/hour	N/A	L.I.P.	N/A
LA-2410-D	120/1/50/60	0.9	1.2	800 (3558)	N/A	0.7 (18)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	2000 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2415-N	240/480/3/50/60	0.4/0.2	1.3/0.65	1500 (6672)	N/A	0.7 (18)	N/A	N/A	N/A	2000 starts/hour	N/A	L.I.P.	N/A
	380/3/50	0.3	0.9	1500 (6672)	N/A	0.6 (15)							
LA-2420-N	120/1/50/60	2.5	2.9	1500 (6672)	N/A	0.7 (18)	N/A	N/A	N/A	600 starts/hour	N/A	L.I.P.	N/A
LA-2420-D	120/1/50/60	2.5	2.9	1500 (6672)	N/A	0.7 (18)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	600 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2440-N	24 Vdc	1.7	7.5	550 (2446)	N/A	1 (25)	N/A	N/A	N/A	2000 starts/hour	N/A	L.I.P.	N/A
LA-2450-N	240/1/50/60	0.45	0.5	800 (3558)	N/A	0.7 (18)	N/A	N/A	N/A	2000 starts/hour	N/A	L.I.P.	N/A
LA-2450-D	240/1/50/60	0.45	0.5	800 (3558)	N/A	0.7 (18)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	2000 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2460-N	90 Vdc	1.7	1.9	550 (2446)	N/A	1 (25)	N/A	N/A	N/A	2000 starts/hour	N/A	L.I.P.	N/A
LA-2490-N	240/1/50/60	1.1	1.6	1500 (6672)	N/A	0.7 (18)	N/A	N/A	N/A	600 starts/hour	N/A	L.I.P.	N/A
LA-2490-D	240/1/50/60	1.1	1.6	1500 (6672)	N/A	0.7 (18)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	600 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P

**** Minimum position change of 1% (5% for LA-2300 models)***
L.I.P. = Lock-in-Place
R.T.P. = Run to Position
N/A = Not Applicable

LA-2500 & LA-2600 SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Maximum Thrust lbf. (N)		Speed in./sec. mm/sec.	Command Input	Input Impedance in ohms	Feed back	Modulation Rate *	Dynamic Braking	Loss of Power	Loss of Command Signal
		Run	Stall	Running	Breakaway								
LA-2510-N	240/480/3/50/60	3/1.5	7.5/3.75	1600 (7117)	N/A	0.9 (23)	N/A	N/A	N/A	1200 starts/hour	N/A	L.I.P.	N/A
	380/3/50	2.4	4.7			0.8 (19)							
LA-2520-D	120/240/1/50/60	6/3	15/7.5	1600 (7117)	N/A	0.9 (23)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	1200 starts/hour		L.I.P.	L.I.P. or R.T.P
LA-2520-E	120/240/1/50/60	6/3	15/7.5	1600 (7117)	N/A	0.9 (23)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	1200 starts/hour		L.I.P.	L.I.P. or R.T.P
LA-2520-N	120/240/1/50/60	6/3	15/7.5	1600 (7117)	N/A	0.9 (23)	N/A	N/A	N/A	1200 starts/hour	N/A	L.I.P.	N/A
LA-2610-N	208/240/480/3/50/60	1.5/3	7.5/3.75	2500 (11120)	N/A	0.6 (15)	N/A	N/A	N/A	1200 starts/hour	N/A	L.I.P.	N/A
	380/3/50	2.4	4.7			0.5 (13)							
	208/240/480/3/50/60	3.2/1.6	24/12	3600 (16013)	N/A	0.6 (15)							
	380/3/50	2.2	13.5			0.5 (13)							
	208/240/480/3/50/60	4.2/2.1	29/14.5	4700 (20906)	N/A	0.8 (20)							
	380/3/50	2.5	16.9			0.7 (17)							
	208/240/480/3/50/60	6.2/3.1	40/20	6150 (27355)	N/A	0.5 (13)							
	380/3/50	3.45	25			0.4 (11)							
LA-2610-A	240/3/50/60	1.5/3	7.5/3.75	2500 (11120)	N/A	0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	2.4	4.7			0.5 (13)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	1.5/3	7.5/3.75			0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	1.5/3	7.5/3.75			0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2610-A	240/3/50/60	3.2/1.6	24/12	3600 (16013)	N/A	0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	2.2	13.5			0.5 (13)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	3.2/1.6	24/12			0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	3.2/1.6	24/12			0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2610-A	240/3/50/60	4.2/2.1	29/14.5	4700 (20906)	N/A	0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	2.5	16.9			0.7 (17)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	4.2/2.1	29/14.5			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	4.2/2.1	29/14.5			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2610-A	240/3/50/60	6.2/3.1	40/20	6150 (27355)	N/A	0.5 (13)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	3.45	25			0.4 (11)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	6.2/3.1	40/20			0.5 (13)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	6.2/3.1	40/20			0.5 (13)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	1200 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2620-N	120/240/1/50/60	6/3	15/7.5	2500 (11120)	N/A	0.6 (15)	N/A	N/A	N/A	1200 starts/hour	N/A	L.I.P.	N/A
	120/240/1/50/60	6.4/3.2	33/16.5	3600 (16013)	N/A	0.6 (15)							
LA-2620-D	120/240/1/50/60	6/3	15/7.5	2500 (11120)	N/A	0.5 (13)	4-20 mA, 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA, loop-powered	1200 starts/hour		L.I.P.	L.I.P. or R.T.P

**** Minimum position change of 1%***

L.I.P. = Lock-in-Place

R.T.P. = Run to Position

N/A = Not Applicable

LA-2700 & LA-2900 SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Maximum Thrust lbf. (N)		Speed in./sec. mm/sec.	Command Input	Input Impedance in ohms	Feed back	Modulation Rate *	Dynamic Braking	Loss of Power	Loss of Command Signal
		Run	Stall	Running	Breakaway								
LA-2710-N	208/240/480/3/50/60	3/1.5	7.5/3.75	2500 (11120)	3600 (16013)	0.6 (15)	N/A	N/A	N/A	300 starts/hour	N/A	L.I.P.	N/A
	380/3/50	2.4	4,7			0.5 (13)							
	208/240/480/3/50/60	2.4/1.2	17/8.5	3600 (16013)	4700 (20906)	0.8 (20)							
	380/3/50	2	9.7			0.7 (17)							
	208/240/480/3/50/60	3.2/1.6	24/12	1875 (8340)	4675 (20794)	0.6 (15)							
	380/3/50	2.2	13.5			0.5 (13)							
	208/240/480/3/50/60	3.2/1.6	24/12	4700 (20906)	6150 (27355)	0.8 (20)							
	380/3/50	2.2	13.5			0.7 (17)							
	208/240/480/3/50/60	4.2/2.1	29/14.5	2350 (10453)	5900 (26243)	0.8 (20)							
	380/3/50	2.5	16.9			0.7 (17)							
	208/240/480/3/50/60	6.2/3.1	40/20	2800 (12454)	7000 (31136)	0.8 (20)							
	380/3/50	3.45	25			0.7 (17)							
LA-2710-A	240/3/50/60	3/1.5	7.5/3.75	2500 (11120)	3600 (16013)	0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	2.4	4,7			0.5 (13)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	3/1.5	7.5/3.75			0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	3/1.5	7.5/3.75			0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2710-A	240/3/50/60	2.4/1.2	17/8.5	3600 (16013)	4700 (20906)	0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	2	9.7			0.7 (17)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	2.4/1.2	17/8.5			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	2.4/1.2	17/8.5			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2710-A	240/3/50/60	3.2/1.6	24/12	1875 (8340)	4675 (20794)	0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	2.2	13.5			0.5 (13)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	3.2/1.6	24/12			0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	3.2/1.6	24/12			0.6 (15)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2710-A	240/3/50/60	3.2/1.6	24/12	4700 (20906)	6150 (27355)	0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	2.2	13.5			0.7 (17)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	3.2/1.6	24/12			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	3.2/1.6	24/12			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2710-A	240/3/50/60	4.2/2.1	29/14.5	2350 (10453)	5900 (26243)	0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	2.5	16.9			0.7 (17)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	4.2/2.1	29/14.5			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	4.2/2.1	29/14.5			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2710-A	240/3/50/60	6.2/3.1	40/20	2800 (12454)	7000 (31136)	0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	380/3/50	3.45	25			0.7 (17)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	480/3/50/60	6.2/3.1	40/20			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
	208/3/50/60	6.2/3.1	40/20			0.8 (20)	4-20 mA, 0-5 Vdc or 0-10 Vdc	475	4-20 mA, loop-powered	300 starts/hour	Yes	L.I.P.	L.I.P. or R.T.P
LA-2720-N	120/240/1/50/60	5.4/2.7	30/15	2500 (11120)	3600 (16013)	0.6 (15)	N/A	N/A	N/A	300 starts/hour	N/A	L.I.P.	N/A
	120/240/1/50/60	6/3	37.5/18.75	2500 (11120)	3600 (16013)	1.2 (30)							
	120/240/1/50/60	6.4/3.2	33/16.5	3600 (16013)	4700 (20906)	0.8 (20)							
LA-2910-N	240/480/3/50/60	4.21/2.1	29/14.5	12,000 (53376)	N/A	0.25 (6)	N/A	N/A	N/A	2000 starts/hour	N/A	L.I.P.	N/A
	380/3/50	2.5	16.9			0.21 (5)							
LA-2980-N	180 Vdc	9.2	NA	12,000 (53376)	N/A	0.25 (6)	N/A	N/A	N/A	2000 starts/hour	N/A	L.I.P.	N/A

** 10% position changes per hour*
L.I.P. = Lock-in-Place
R.T.P. = Run to Position
N/A = Not Applicable

LA-2300 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-2320-N**	1 phase, 120 Vac, without built-in digital amplifier	
		LA-2320-D	LA-2320 above with built-in digital amplifier	
		LA-2390-N	1 phase, 240 Vac, without built-in digital amplifier	
		LA-2390-D	LA-2390 above with built-in digital amplifier	
		LA-2315-N	3 phase, 208/240/380/480 Vac, without built-in analog amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-2320-N, LA-2320-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-2390-N, LA-2390-D)		
		3: 380 Vac, 3 Phase, 50 Hz (LA-2315-N)		
		4: 480 Vac, 3 Phase, 50/60 Hz (LA-2315-N)		
		5: 240 Vac, 3 Phase, 50/60 Hz (LA-2315-N)		
		10: 208 Vac, 3 Phase, 50 Hz (LA-2315-N)		
3	Speed in./s. (mm/s.) / Breakaway Thrust lbf. (N)	LA-2320-N, LA-2320-D, LA-2390-N & LA-2390-D		
		6"-24"	6"-18"	6"-30"	6"-36"
		0.2/1000/500 (5.1/4448/2224)	**0.4/1500/750** (10.2/6672/3336)	**0.8/750/340** (20.3/3336/1512)	**2.0/260/130** (50.8/1156/578)
4	Breakaway Thrust lbf. (N) / Thrust lbf. (N)	LA-2315-N			
		6"-18"	6"-36"	6"-36"	6"-36"
		0.2/1500/750 (5.1/6672/3336)	**0.4/600/300** (10.2/2668/1334)	**0.8/380/190** (20.3/1690/845)	**2.0/150/75** (50.8/667/334)
5	Linear Travel	(Travel less than the selected maximum is established by limit switch settings for models without amp.)			
		6: 6 inches (152 mm) (specify desired max. travel)			
		12: 12 inches (305 mm) (specify desired max. travel)			
		18: 18 inches (457 mm) (specify desired max. travel)			
		24: 24 inches (610 mm) (specify desired max. travel)			
		30: 30 inches (762 mm) (specify desired max. travel)			
		36: 36 inches (914 mm) (specify desired max. travel)			

LA-2300 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches	
A009	ON/OFF Toggle Power Switch	
Drive Arm/Adapter Clevis/Driven Arm		
D030	Trunnion Mount, not available with 6 inch stroke	
D034	Flange Mount, not avail. with 6 inch stroke	
D047	Special front clevis (provide dimensions)	
D048	Special rear clevis (provide dimensions)	
D055	Bellows for Linear Actuators	
D056	Vertical Moisture Shield	
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not available for units with internal amplifiers)	
F003	Potentiometer, 5000 ohms (not available for units with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (not available for units with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (not available for units with internal amplifiers)	
F999	Special Potentiometer (consult factory)	

Code	Description	Selection
Heater		
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certification		
J001	CE Marking, AC input only, (only available for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	

For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-2400 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-2410-N**	1 phase, 120 Vac, modulating duty, without built-in digital amplifier	
		LA-2410-D	LA-2410 above with built-in digital amplifier	
		LA-2415-N	3 phase, 208/240/380/480 Vac, modulating duty, without built-in analog amplifier	
		LA-2420-N	1 phase, 120 Vac, intermittent duty, without built-in digital amplifier	
		LA-2420-D	LA-2420 above with built-in digital amplifier	
		LA-2440-N	24 Vdc (perm. magnet field), modulating duty, without built-in analog amplifier	
		LA-2450-N	1 phase, 240 Vac, modulating duty, without built-in digital amplifier	
		LA-2450-D	LA-2450 above with built-in digital amplifier	
		LA-2460-N	90 Vdc (perm. magnet field), modulating duty, without built-in amplifier	
		LA-2490-N	1 phase, 240 Vac, intermittent duty, without built-in digital amplifier	
		LA-2490-D	LA-2490 above with built-in digital amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-2410-N, LA-2410-D, LA-2420-N and LA-2420-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-2450-N, LA-2450-D, LA-2490-N and LA-2490-D		
		3: 380 Vac, 3 Phase, 50 Hz (LA-2415-N)		
		4: 480 Vac, 3 Phase, 50/60 Hz (LA-2415-N)		
		5: 240 Vac, 3 Phase, 50/60 Hz (LA-2415-N)		
		6: 24 Vdc voltage designator (LA-2440-N)		
		7: 90 Vdc voltage designator (LA-2460-N)		
		10: 208 Vac, 3 Phase, 50 Hz (LA-2415-N)		
		19: 220 Vac, 1 Phase (LA-2450-D and LA-2490-D)		

		LA-2410-N, LA-2410-D LA-2415-N LA-2450-N, LA-2450-D		LA-2420-N, LA-2420-D LA-2490-N, LA-2490-D				LA-2440-N LA-2460-N	
3	Speed in./s. (mm/s.)	6"-18"	24"	6"	6"-12"	12"-18"	24"	6"-24"	
4	Thrust lbf. (N)	**0.1/800** (2.5/3558)	**0.1/550** (2.5/2446)	**0.1/1500** (2.5/6672)	**0.1/1200** (2.5/5338)	**0.1/1000** (2.5/4448)	**0.1/800** (2.5/3558)	**0.2/550** (5.1/2446)	
		0.2/800 (5.1/3558)	**0.2/550** (5.1/2446)	**0.2/1500** (5.1/6672)	**0.2/1200** (5.1/5338)	**0.2/1000** (5.1/4448)	**0.2/800** (5.1/3558)	**0.25/550** (6.4/2446)	
		0.4/200 (10.2/890)	**0.4/200** (10.2/890)		**0.4/550** (10.2/2446)	**0.4/550** (10.2/2446)	**0.4/550** (10.2/2446)	**0.6/150** (15.2/667)	
		0.7/200 (17.8/890)	**0.7/200** (17.8/890)		**0.7/450** (17.8/2002)	**0.7/450** (17.8/2002)	**0.7/450** (17.8/2002)	**1/150** (25.4/667)	

5	Linear Travel *	(Travel less than the selected maximum is established by limit switch settings for models without amp.)	
		6: 6 inches (152 mm), specify desired max. travel, (1500 lbf. maximum)	
		12: 12 inches (305 mm), specify desired max. travel, (800 lbf. maximum)	
		18: 18 inches (457 mm), specify desired max. travel, (800 lbf. maximum)	
		24: 24 inches (610 mm), specify desired max. travel, (550 lbf. maximum)	

** Contact factory for greater stroke lengths*

LA-2400 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches	
A009	ON/OFF Toggle Power Switch	
Drive Arm/Adapter Clevis/Driven Arm		
D031	Trunnion Mount, not available with 6 inch stroke	
D044	Flange Mount, not avail. with 6 inch stroke	
D045	Female front clevis	
D046	Female rear clevis	
D047	Special front clevis (provide dimensions)	
D048	Special rear clevis (provide dimensions)	
D055	Bellows for Linear Actuators	
D056	Vertical Moisture Shield	
D058	Rotating clevis	
PositionFeedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not available for units with internal amplifiers)	
F003	Potentiometer, 5000 ohms (not available for units with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (not available for units with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (not available for units with internal amplifiers)	
F999	Special Potentiometer (consult factory)	

Code	Description	Selection
Identification/Certification		
J001	CE Marking, AC input only (only available for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	

For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-2500 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-2510-N**	3 phase, 208/240/380/480 Vac, without built-in analog amplifier	
		LA-2520-N	1 phase, 120/240 Vac, without built-in digital amplifier	
		LA-2520-D	LA-2520 above with built-in digital amplifier	
		LA-2520-E	LA-2520 above with built-in "QuikSet" amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-2520-N, LA-2520-D or LA-2520-E)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-2520-N, LA-2520-D or LA-2520-E)		
		3: 380 Vac, 3 Phase, 50 Hz (LA-2510-N)		
		4: 480 Vac, 3 Phase, 50/60 Hz (LA-2510-N)		
		5: 240 Vac, 3 Phase, 50/60 Hz (LA-2510-N)		
		10: 208 Vac, 3 Phase, 50 Hz (LA-2510-N)		
3	Speed in./s. / Thrust lbf. (N)			
		0.9/800 (23/3558)		
		0.9/1200 (23/5338)		
4		**0.9/1600** (23/7117)		
5	Linear Travel *	**6**: 6 inches (152.4 mm)		
		12A: 12 inches (304.8 mm) (clevis to clevis dimension of 24.88 inches (631.9 mm), ITT equivalent)		
		12B: 12 inches (304.8 mm) (clevis to clevis dimension of 25.38 inches (644.6 mm), LA-2601 equivalent)		
		12C: 12 inches (304.8 mm) (standard clevis to clevis dimension of 26.50 inches (673 mm) - LA-2800 equivalent)		
		12D: 12 inches (304.8 mm) (clevis to clevis dimension of 26.59 inches (675.3mm), B&W MC-107xx equivalent)		
		18: 18 inches (457.2 mm)		
		24: 24 inches (609.6 mm)		

** Contact factory for greater stroke lengths*

LA-2500 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002)	
A009	ON/OFF Toggle Power Switch	
Brake		
B004	Drag Brake (800 lbs. maximum)	
Drive Arm/Adapter Clevis/Driven Arm		
D032	Trunnion Mount (not avail. for 6" stroke)	
D049	Special front clevis (provide dimensions)	
D050	Special rear clevis (provide dimensions)	
D055	Bellows for Linear Actuators	
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not avail. for units with internal amplifiers)	
F003	Potentiometer, 5000 ohms (not available for units with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (not available for units with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (not available for units with internal amplifiers)	
F999	Special Potentiometer (consult factory)	
Identification/Certification		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	

Code	Description	Selection
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac (Not available with F002 or F006)	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac (Not available with F002 or F006)	
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S017	Two nonadjustable thrust limit switches SPDT, 10 Amp, 120/240 Vac	
S021	Two nonadjustable thrust limit switches SPST, 20 Amp, 120/240 Vac	
Conduit Entries		
V001	1/2 inch Conduit Entry Additional (NEMA 4)	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	

For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-2600 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-2610-N**	3 phase, 208/240/380/480 Vac, intermittent duty, without built-in analog amplifier	
		LA-2610-A	LA-2610 above with built-in analog amplifier, intermittent duty	
		LA-2620-N	1 phase, 120/240 Vac, intermittent duty, without built-in digital amplifier	
		LA-2620-D	LA-2620 above with built-in digital amplifier, intermittent duty	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-2620-N, LA-2620-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-2620-N, LA-2620-D)		
		3: 380 Vac, 3 Phase, 50 Hz (LA-2610-N, LA-2610-A)		
		4: 480 Vac, 3 Phase, 50/60 Hz (LA-2610-N, LA-2610-A)		
		5: 240 Vac, 3 Phase, 50/60 Hz (LA-2610-N, LA-2610-A)		
		10: 208 Vac, 3 Phase, 50/60 Hz (LA-2610-N, LA-2610-A)		
3	Speed in./s. (mm/s.) / Thrust lbf. (N)	Available as 1 or 3 phase		
		0.6/2500 (15/11120) (With AD-8130, 0.5/2500)		
		0.6/3600 (15/16013) (Not available with LA-2620-A)		
4		Available as 3 phase only		
		0.8/4700 (20/20906)		
		0.5/6150 (20/27355)		
5	Linear Travel *	Up to 3600 lbf.	Greater than 3600 lbf.	
		6: 6 inches (152 mm)	**6**: 6 inches (152 mm)	
		12: 12 inches (305 mm)	**12**: 12 inches (305 mm)	
		18: 18 inches (457 mm)	**18**: 18 inches (457 mm)	
		24: 24 inches (610 mm)	**24**: 24 inches (610 mm)	
		30: 30 inches (762 mm)	**30**: 30 inches (762 mm)	

** Contact factory for greater stroke lengths*

LA-2600 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002)	
A009	ON/OFF Toggle Power Switch	
Drive Arm/Adapter Clevis/Driven Arm		
D033	Trunnion Mount, up to 3600 lbf. (Not available for 6" stroke	
D035	Flange Mount, up to 3600 lbf. (Not available for 6" stroke	
D051	Special front clevis (provide dimensions)	
D052	Special rear clevis (provide dimensions)	
Enclosure		
E003	Explosion-Proof - Class 1, Div. 1, Groups C & D, Class II, Div. 1, Groups E, F & G (Only available for 2500 and 3600 lbf. Models)	
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not avail. for units with internal amplifiers)	
F003	Potentiometer, 5000 ohms (not available for units with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (not available for units with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (not available for units with internal amplifiers)	
F999	Special Potentiometer (consult factory)	

Code	Description	Selection
3-Phase Controls		
G001	3-Phase Contactor with Overloads	
G002	Control Transformer	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp, 120/240 Vac	
S021	Two nonadjustable thrust limit switches SPST, 20 Amp, 120/240 Vac	
S022	Two nonadjustable thrust limit switches SPDT, 20 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	

For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-2700 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-2710-N**	3 phase, 208/240/380/480 Vac, intermittant duty, without built-in analog amplifier	
		LA-2710-A	LA-2710 above with built-in digital amplifier, intermittant duty	
		LA-2720-N	1 phase, 120/240 Vac, intermittant duty, without built-in amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-2720-N)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-2720-N)		
		3: 380 Vac, 3 Phase, 50 Hz (LA-2710-N, LA-2710-A)		
		4: 480 Vac, 3 Phase, 50/60 Hz (LA-2710-N, LA-2710-A)		
		5: 240 Vac, 3 Phase, 50/60 Hz (LA-2710-N, LA-2710-A)		
		10: 208 Vac, 3 Phase, 50/60 Hz (LA-2710-N, LA-2710-A)		
3	Speed (a) in./s. (mm/s.)	Available as 1 or 3 phase		
		0.6/2500/3600 (15/11120/16013)		
		0.8/3600/4700 (15/16013/20906)		
4	Thrust, lbf. (N) Running / Thrust, lbf. (N) Breakaway	Available as 3 phase only		
		0.6/1875/4675 (15/8340/20794)		
		0.8/2350/5900 (20/10452/26243)		
		0.8/4700/6150 (20/20906/27355)		
		0.8/2800/7000 (20/12454/31136)		
5	Linear Travel *	Up to 3600 lbf.	Greater than 3600 lbf.	
		6: 6 inches (152 mm)	**6**: 6 inches (152 mm)	
		12: 12 inches (305 mm)	**12**: 12 inches (305 mm)	
		18: 18 inches (457 mm)	**18**: 18 inches (457 mm)	
		24: 24 inches (610 mm)	**24**: 24 inches (610 mm)	
		30: 30 inches (762 mm)	**30**: 30 inches (762 mm)	

** Contact factory for greater stroke lengths*

LA-2700 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002)	
A009	ON/OFF Toggle Power Switch	
Drive Arm/Adapter Clevis/Driven Arm		
D033	Trunnion Mount, up to 3600 lbf. (Not available for 6" stroke	
D035	Flange Mount, up to 3600 lbf. (Not available for 6" stroke	
D051	Special front clevis (provide dimensions)	
D052	Special rear clevis (provide dimensions)	
Enclosure		
E003	Explosion-Proof - Class 1, Div. 1, Groups C & D, Class II, Div. 1, Groups E, F & G	
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not avail. for units with internal amplifiers)	
F003	Potentiometer, 5000 ohms (not available for units with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (not available for units with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (not available for units with internal amplifiers)	
F999	Special Potentiometer (consult factory)	

Code	Description	Selection
3-Phase Controls		
G001	3-Phase Contactor with Overloads	
G002	Control Transformer	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp, 120/240 Vac	
S021	Two nonadjustable thrust limit switches SPST, 20 Amp, 120/240 Vac	
S022	Two nonadjustable thrust limit switches SPDT, 20 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	

For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-2900 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-2910-N**	3 phase, 240/380/480 Vac, modulating duty, without built-in analog amplifier	
		LA-2980-N	180 Vdc (permanent magnet field), modulating duty, without built-in analog amplifier	
2	Input Voltage	**3**: 380 Vac, 3 Phase, 50 Hz (LA-2910-N)		
		4: 480 Vac, 3 Phase, 50/60 Hz (LA-2910-N)		
		5: 240 Vac, 3 Phase, 50/60 Hz (LA-2910-N)		
		8: 180 Vdc (LA-2980-N)		
3	Speed in./s. (mm/s.) / Thrust lbf. (N)	**0.25/2000** (635/8896)		
		0.25/4000 (635/17792)		
		0.25/6000 (635/26688)		
4		**0.25/8000** (635/35584)		
		0.25/10,000 (635/44480)		
		0.25/12,000 (635/53376)		
5	Linear Travel *	**6**: 6 inches (152.4 mm)		
		12: 12 inches (304.8 mm)		
		18: 18 inches (457.2 mm)		
		24: 24 inches (609.6 mm)		
		30: 30 inches (762 mm)		

** Contact factory for greater stroke lengths*

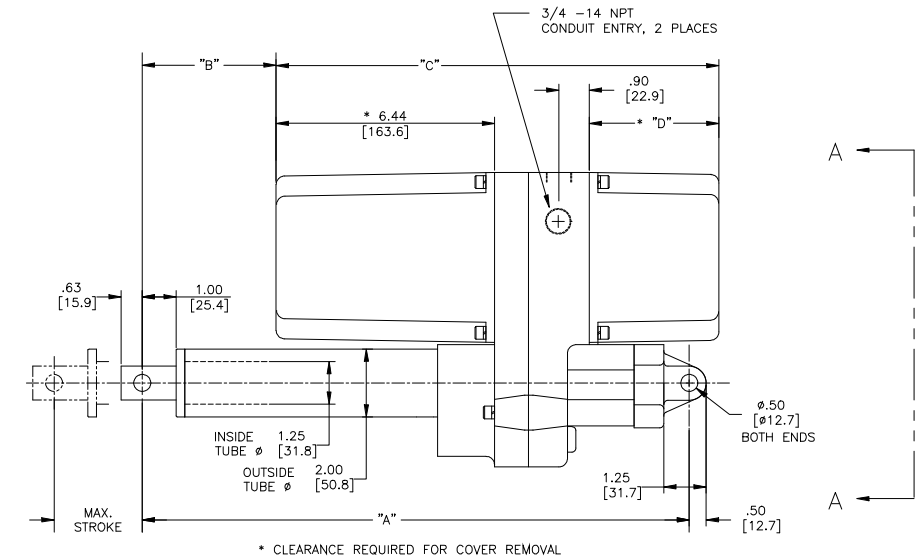
LA-2900 STANDARD OPTIONS

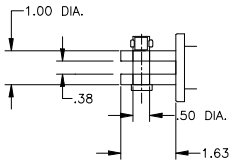
Code	Description	Selection
Drive Arm/Adapter Clevis/Driven Arm		
D053	Special front clevis (provide dimensions)	
D054	Special rear clevis (provide dimensions)	
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each	
F003	Potentiometer, 5000 ohms	
F004	Potentiometer, 10,000 ohms	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter	
F999	Special Potentiometer (consult factory)	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp, 120/240 Vac	
S021	Two nonadjustable thrust limit switches SPST, 20 Amp, 120/240 Vac	
S022	Two nonadjustable thrust limit switches SPDT, 20 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	

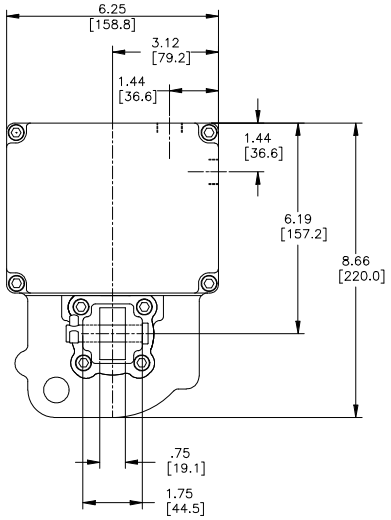
For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-2300 MAJOR DIMENSIONS

INCHES (MILLIMETERS)



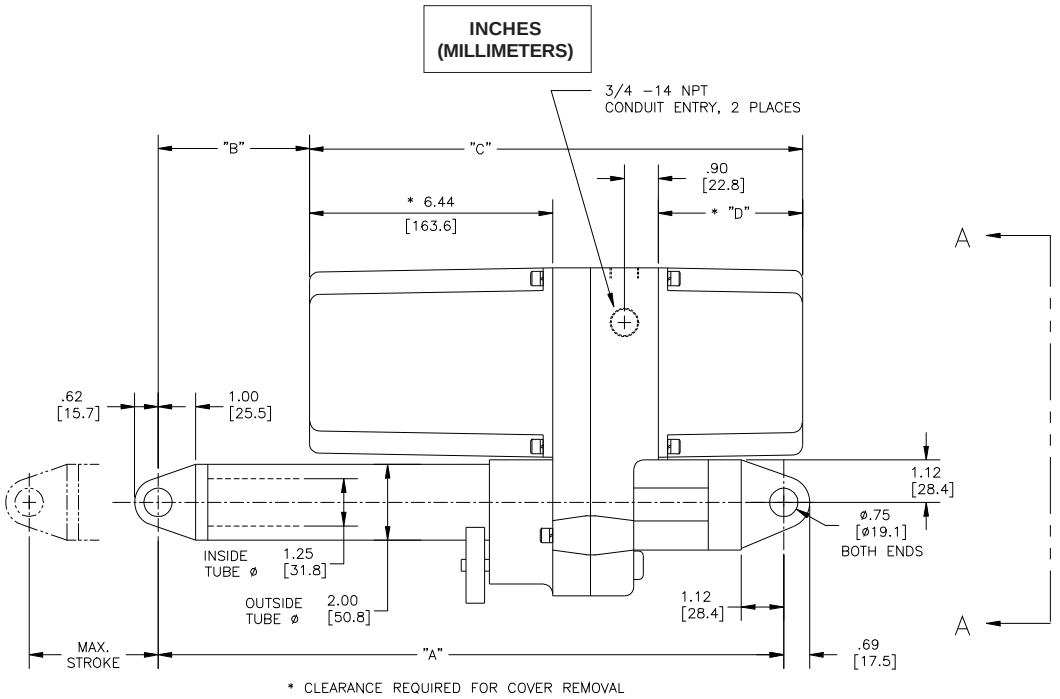


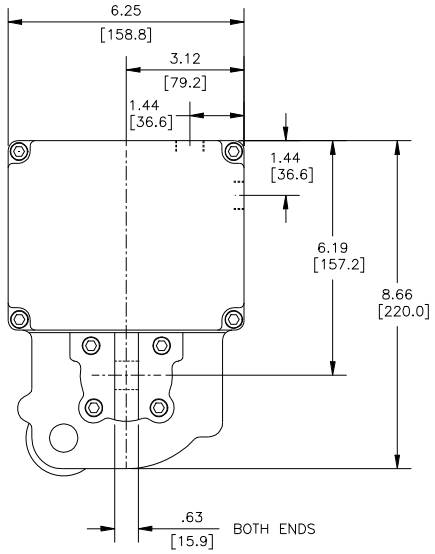


Stroke in. (mm)	A	B	C	D	
				Without Amp	With amp
6 (152)	15.43 (392)	2.83 (72)	6 max. (152)	3.81 (97)	6.44 (164)
12 (305)	21.43 (544)	8.83 (224)	12 max. (305)	3.81 (97)	6.44 (164)
18 (457)	27.43 (697)	14.83 (377)	18 max. (457)	3.81 (97)	6.44 (164)
24 (609)	33.43 (849)	20.83 (529)	24 max. (609)	3.81 (97)	6.44 (164)
30 (762)	39.43 (1002)	26.83 (681)	30 max. (762)	3.81 (97)	6.44 (164)
36 (914)	45.43 (1154)	32.83 (834)	36 max. (914)	3.81 (97)	6.44 (164)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LA-2400 MAJOR DIMENSIONS



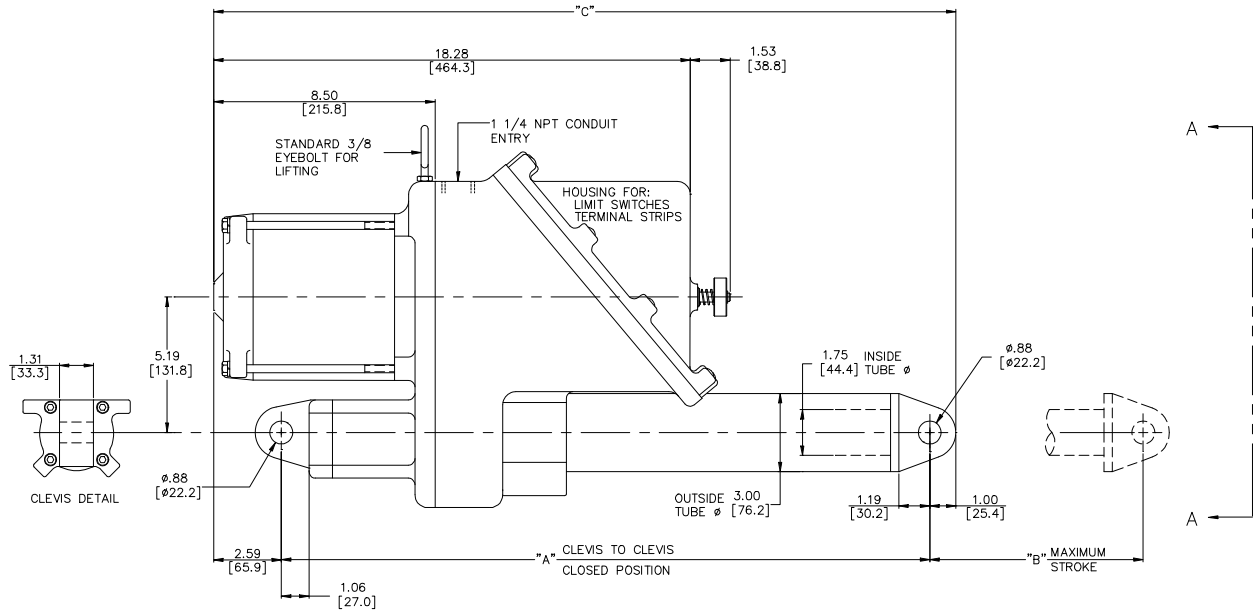


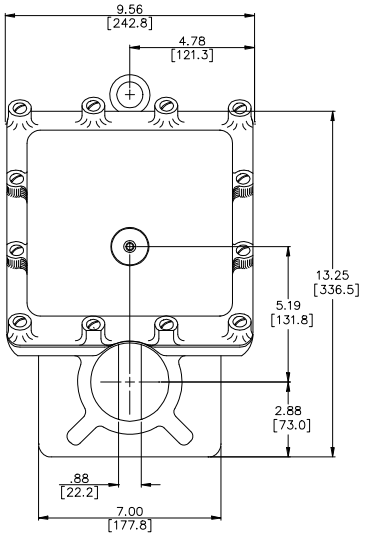
Stroke in. (mm)	A	B	C		D	
			Without Amp	With amp	Without Amp	With amp
2 (51) to 6 (152)	16.61 (422)	4.01 (102)	13.1 (333)	15.73 (400)	3.81 (97)	6.44 (164)
6.01 (153) to 12 (305)	22.61 (574)	10.01 (254)	13.1 (333)	15.73 (400)	3.81 (97)	6.44 (164)
12.01 (305) to 18 (457)	28.61 (727)	16.01 (407)	13.1 (333)	15.73 (400)	3.81 (97)	6.44 (164)
18.01 (457) to 24 (610)	34.61 (879)	22.01 (559)	13.1 (333)	15.73 (400)	3.81 (97)	6.44 (164)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LA-2500 MAJOR DIMENSIONS

INCHES (MILLIMETERS)





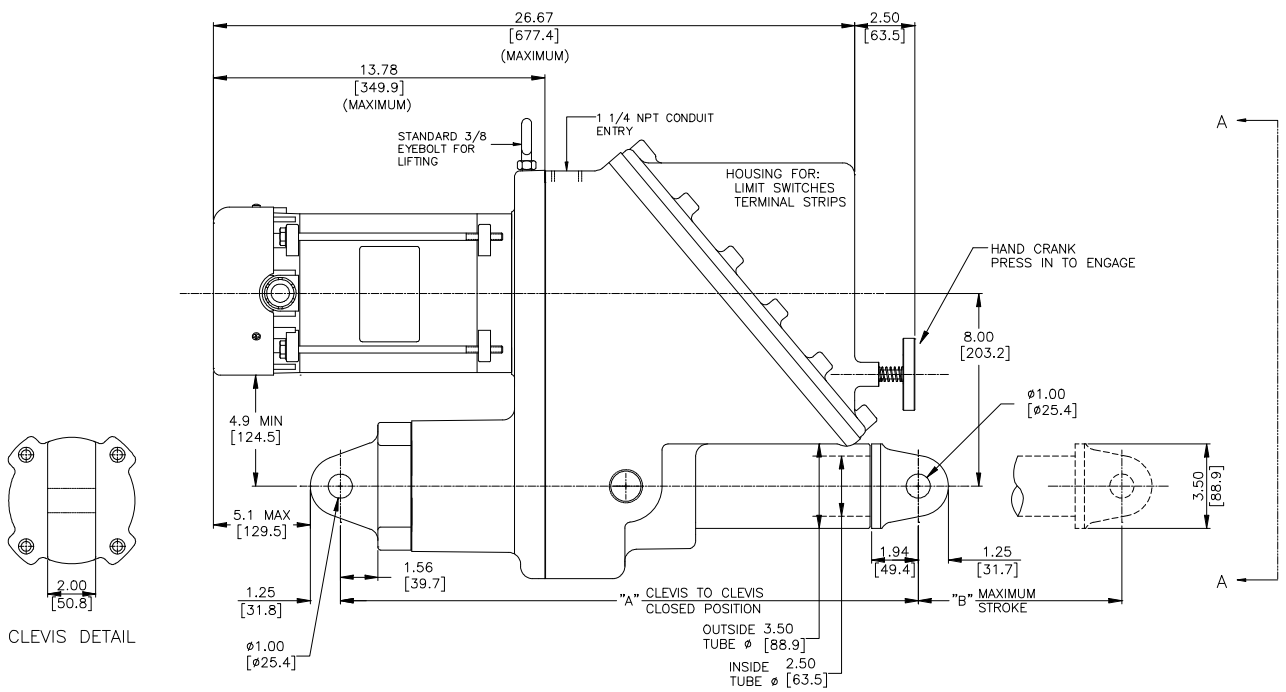
VIEW A–A

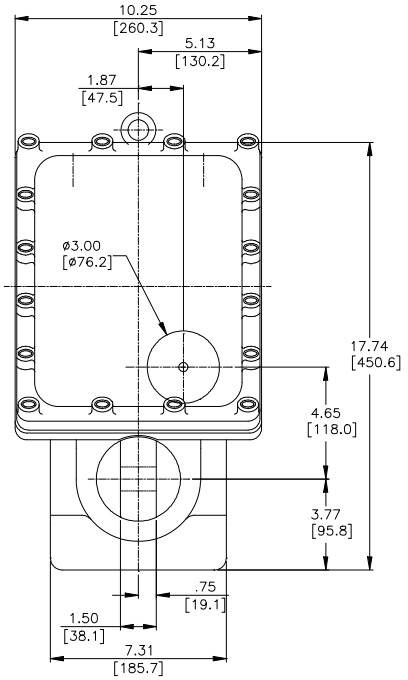
Stroke in. (mm)	A	B Linear Travel	C
6 (152)	20.5 (521)	6	24.09 (612)
12 (305)	24.88 (632)	12A	28.47 (723)
	25.38 (645)	12B	28.97 (736)
	26.55 (673)	12C	30.09 (764)
	26.59 (675)	12D	30.18 (767)
18 (457)	32.5 (826)	18	36.09 (917)
24 (610)	39.5 (1003)	24	43.09 (1094)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LA-2600/LA-2700 MAJOR DIMENSIONS

INCHES (MILLIMETERS)



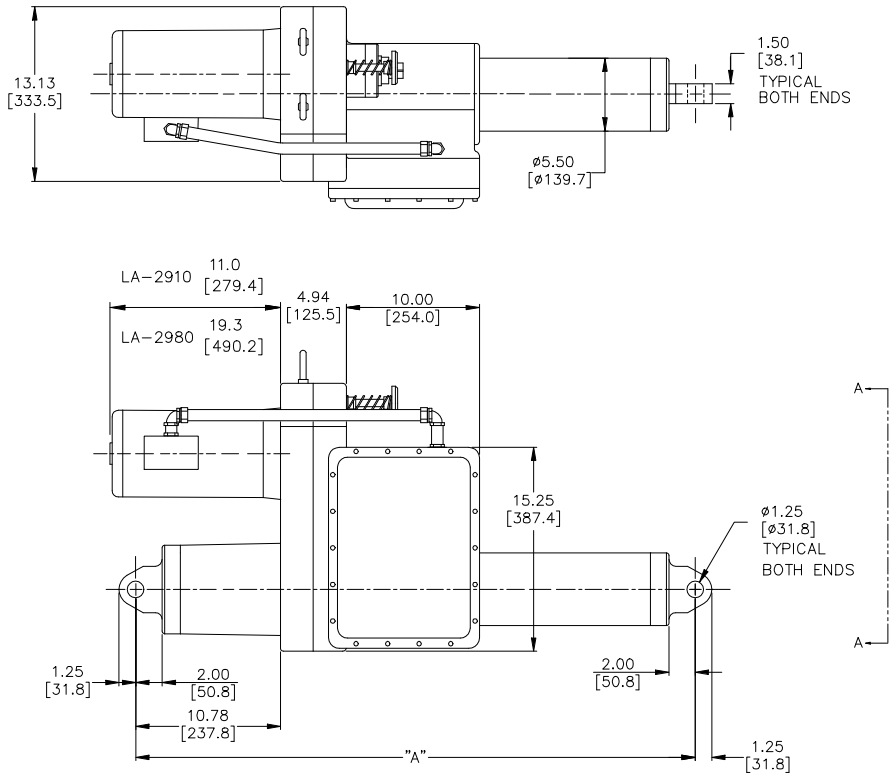


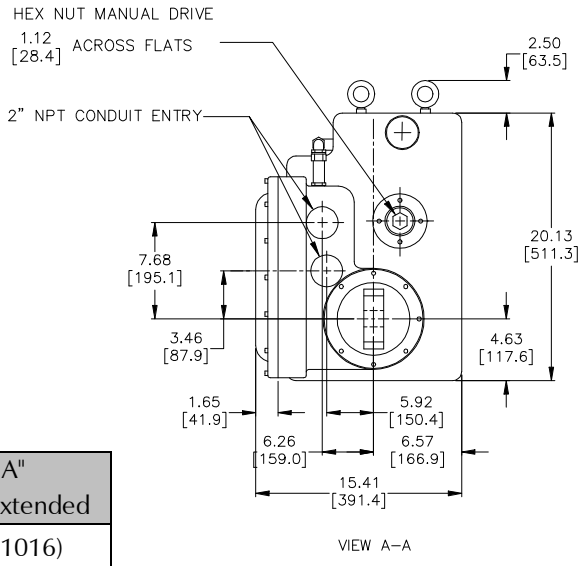
MAXIMUM THRUST -lbf (N)			Dimension (Stroke Length)				
LA-2600	LA-2700	"B"	6 in. (152 mm)	12 in. (305 mm)	18 in. (457 mm)	24 in. (610 mm)	30 in. (762 mm)
2500 (11120)	-	"A"	22.41 (569)	28.41 (722)	35.35 (898)	42.29 (1074)	49.23 (1250)
3600 (16013)	3600/4675 (16013/20794)		25.02 (636)	31.02 (788)	37.96 (964)	44.90 (1140)	51.84 (1317)
4700 (20906)	4700/5900 (20906/26243)		24.10 (612)	30.10 (765)	37.04 (941)	43.98 (1117)	50.92 (1293)
6150 (27355)	6150/7000 (27355/31136)		25.95 (659)	31.95 (812)	38.89 (988)	45.83 (1164)	52.77 (1340)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LA-2900 MAJOR DIMENSIONS

INCHES (MILLIMETERS)





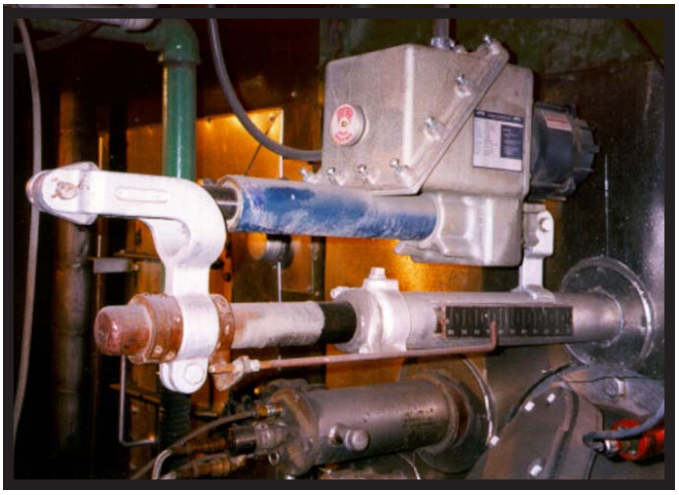
Stroke in. (mm)	"A" Fully Retracted	"A" Fully Extended
6 (152)	34 (864)	40 (1016)
12 (305)	40 (1016)	52 (1321)
18 (457)	48 (1219)	66 (1676)
24 (610)	54 (1372)	78 (1981)
30 (762)	60 (1524)	90 (2286)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.



LA-2440 on a Butterfly Damper

- Butterfly damper modulating air flow to the baghouse.
- Four dampers work together to provide balance and draft control in continuous operation.
- AD-7540 and 24Vdc motor in LA-2440 provide fine tuning with speed control.
- Replaced a troublesome pneumatic cylinder and control system.



LA-2520 on a Burner Air Shroud Damper

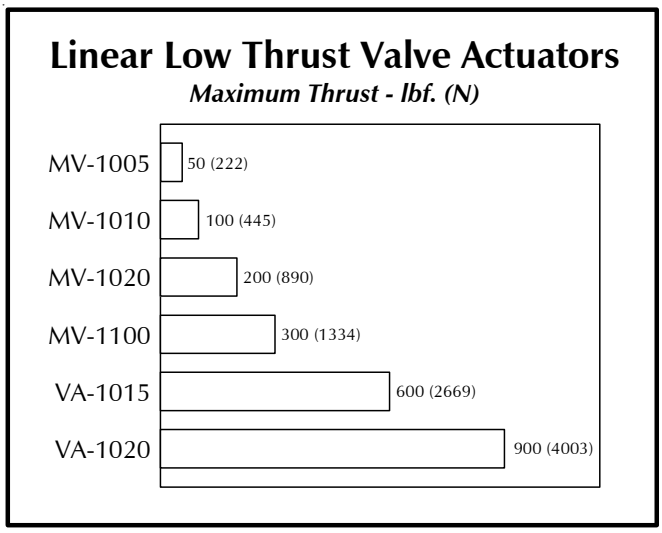
- A design standard for electric operated burner air shroud, diffuser disc, or over fire air port.
- Positioning control, low duty cycle, with 4-20mA control.
- Manual handwheel.

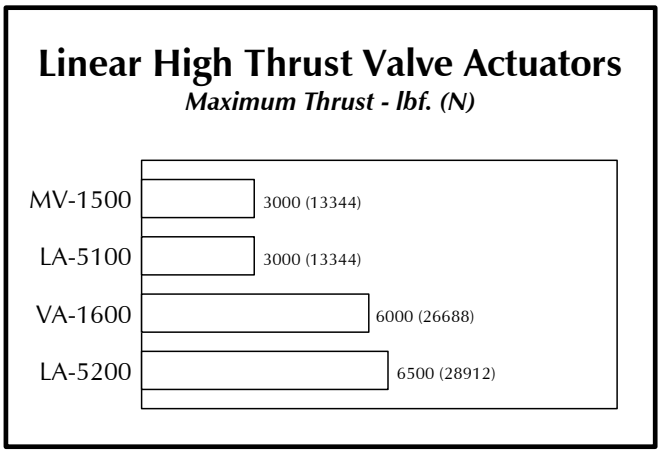


LA-2910 on FD Inlet Vane Damper

- Controls the Inlet Vane Dampers for the FD Fan
- Fan directs air through pollution control equipment at a steel mill's electric furnace
- Replaced the hydraulic cylinder that leaked and was unreliable in cold weather
- Application environment is outdoors and requires a NEMA 4 enclosure rating
- The unit is capable of modulating duty

THRUST RANGES





MODEL SERIES SELECTION CHART

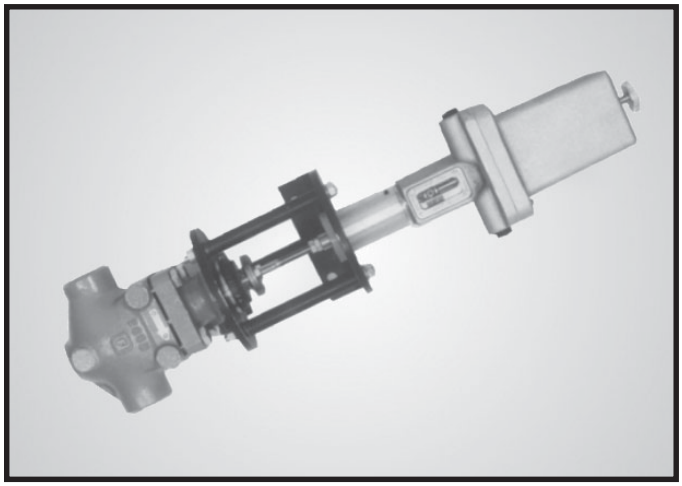
Model Series	Performance					Power Input Voltage		Catalog Page #
	Max.Thrust lbf (N)	Stroke - inches (mm)		Speed - sec./in. (sec./mm)		1 Phase 50/60 Hz	DC	
		Minimum	Maximum	Minimum	Maximum			
MV-1005	50 (222)	0.09 (2.3)	1.38 (35)	11.1 (.44)	11.1 (.44)	115/230	N/A	88
MV-1010	100 (445)	0.09 (2.3)	1.38 (35)	25 (0.99)	4 (0.16)	115/230	24-36	88
MV-1020	200 (890)	0.09 (2.3)	1.38 (35)	8 (00.3)	4 (0.16)	115/230	12-36	88
MV-1100	300 (1334)	0.25 (6.4)	0.75 (19)	72 (2.83)	7 (0.28)	115/230	24	97
VA-1010	600 (2669)	0.34 (8.6)	1.5 (38.1)	357 (14)	29 (11.4)	115/230	24-36	93
VA-1020	900 (4003)	0.34 (8.6)	1.5 (38.1)	71 (2.8)	29 (11.4)	115/230	12-36	93
MV-1500	3000 (13344)	0.25 (6.4)	4 (101.6)	40 (1.57)	11 (0.43)	120/240	24	97
LA-5100	3000 (13344)	0.75 (19)	2.25 (57.2)	22 (0.87)	6 (0.12)	120/240	24	106
VA-1600	6000 (26688)	0.25 (6.4)	2 (51)	64 (2.52)	32 (1.26)	120/240	24	103
LA-5200	6500 (28912)	1 (25.4)	4 (101.6)	13 (0.51)	5 (0.12)	120/240	90	106

MV-1000 SERIES LINEAR VALVE ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The MV-1000 series are full-featured actuators designed to be used with low thrust control valves, capable of accepting standard analog current and voltage control signals. The brushless stepper motor design provides smooth, highly accurate positioning, with positive position-lock when not in motion. The MV-1000 Series is ideally suited for valve, regulator and metering pumps requiring exact position control and unrestricted continuous modulation.



FEATURES

- Permanently lubricated for any position mounting
- Amplifier supplies current to hold last position and prevent backdriving, up to thrust rating under power
- AC or DC input power versions
- 4 to 20 mA position, loop-powered, feedback signal
- Field selectable adjustments for:
  - speed
  - deadband
  - zero and span
  - command signal type
  - standard or reverse acting
  - manual-auto operation
  - output shaft position on loss of signal
- Wide ambient temperature range
- Full one year warranty

BASE MODEL INCLUDES:

- Motor (stepper)
- Manual override (except MV-1005)
- Amplifier
- Electronic thrust limiting
- Cast aluminum NEMA 4 (IP65) & dust ignition-proof enclosure
- Four adjustable position switches (40 mA at 40 Vdc)
- Valve position indicator
- Internal steel gear train
- 4-20 mA transmitter for customer use

POPULAR OPTIONS *(See Pages 197-206)*

- Cast yoke base, maximum valve yoke boss diameter - 2.125 in. (53.98 mm)
- Signal Conversion Module: Convert 40 mA, 40 Vdc low level to 5 Amp, 120/240 Vac current rated position limit switches
- Low current at null (24 Vdc only)
- Switching input powered ac or dc positioning for positioning actuator using ac or dc remote voltage control
- Process Variable Controller to control one process variable - 120/240 Vac only
- Custom mounting and interface hardware
- Local Auto/Manual and INC/OFF/DEC toggle switches (Close-coupled, NEMA 4)
- Exd IIB (Hydrogen) enclosure
- Various enclosure coatings
- Battery backup to position actuator on loss of ac power
- Factory mounting to valve
- ATEX Approval
  (ATEX units have M20 conduit openings)

SPECIFICATIONS

Thrust: Up to 200 lbf. (890 N)

Speed: See selection chart on next page

Input Power: See selection chart on next page

Linear Travel: 0.09 inch (2.3 mm) to 1.375 inches (35 mm), adjustable.

Temperature Limits: -40°F to 150°F (-40°C to 65°C)
For greater temperature ranges, consult factory.

Environmental Ratings: Explosion-proof for Class I, Division 1, Groups C and D; dust-ignition-proof for Class II, Division 1, Groups E, F, G. Also rated NEMA 4 (IP65) indoor/outdoor. (Optional Exd IIB enclosure)

Command Input:
Full Range: 4-20 mA into 200 ohms
Split Range: 4-12 or 12-20 mA into 200 ohms
Voltage: 0-5 Vdc and 0-10 Vdc into 200,000 ohms
Switch: Dry contact closures

Field Wiring: To barrier terminal blocks

Thrust Limiting: Current sensing PWM

Positioning Accuracy: 1.5% for full range

Feedback: 4-20 mA, customer supplied (loop power)

Loss of Signal: Lock-in-place or run to preset.

Loss of Power: Lock-in-place at last position.

Direction: Extend or retract

Modulation Rate: Unrestricted modulating duty

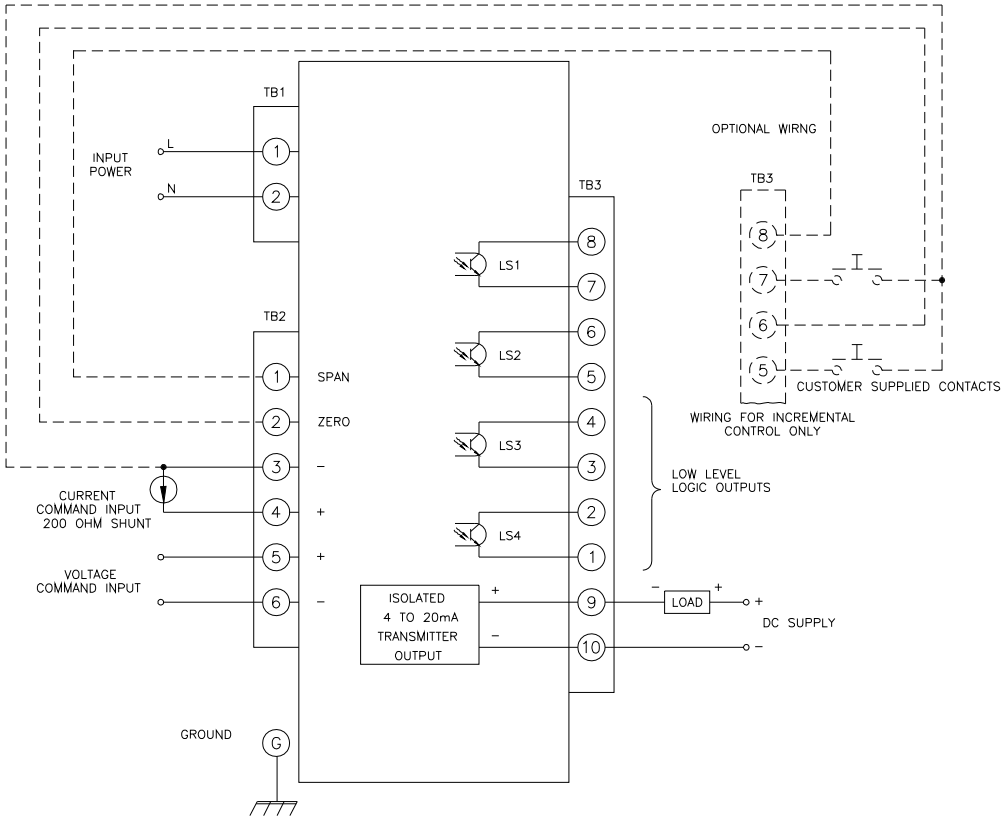
Backdrive: Self-locking up to thrust rating

Approximate Weight *(without valve)*: MV-1005: 12 lbs. (5.4 kg)
MV-1010: 12 lbs. (5.4 kg)
MV-1020: 16 lbs. (7.3 kg)

Approvals: See page 207

Position Switch Rating: 40 mA at 40 Vdc

TYPICAL WIRING DIAGRAM



These dimensions and specifications are subject to change without notice. Current drawings and specifications are available upon request.

MV-1000 SERIES SELECTION CHART

					Selection
1	Basic Model	**MV-1010-D**	100 pounds thrust output (445N), with built-in amplifier (threaded base)		
		MV-1020-D	200 pounds thrust output (890N), with built-in amplifier (threaded base)		
		MV-1005-0102	Threaded base, no mounting hardware, 50 lbs thrust output (222N), 4 to 20 mA input		
		MV-1005-0103	Threaded base, no mounting hardware, 50 lbs thrust output (222N) 4 to 20 mA input and feedback		
2	Input Voltage	**1**: 115 Vac, 1 Phase, 50/60 Hz (All Models)			
		2: 230 Vac, 1 Phase, 50/60 Hz (All Models)			
		6: 24-36 Vdc, (Not available for MV-1005)			
		12: 12-36 Vdc (MV-1020-D), maximum output trust is limited - consult factory			
		13: 24 Vac, (Not available for MV-1005)			
3	Speed in./sec. / Thrust, lbf. (N)	**MV-1010**	**M1020**	**MV-1005**	
		0.25/25 (111)	**0.25/100 (445)**	**0.09/50 (222)**	
4		**0.09/50 (222)**	**0.13/200 (890)**		
		0.07/75 (334)			
		0.05/100 (445)			
5	Linear Travel	**Specify Stroke** - 0.09 inch (2.3 mm) to 1.375 inches (35 mm)			
6*	Mounting Yoke Type	**T**: Universal mounting kit. Maximum value yoke boss diameter - MV-1005 & MV-1010=1.25 in. (31.75 mm) MV-1020=2.25 in. (57.15 mm) or mechanical tubing bracket (unlimited yoke boss diameter)			
		Y: Cast yoke base. Maximum valve yoke boss diameter - 2.125 in. (53.98 mm) Not available for MV-1005-0102 & -0103			

**See valve mounting options L001 through L003*

MV-1000 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure) NEMA 4	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure) NEMA 4	
Enclosure		
E005	Exd IIB enclosure for use in Hydrogen atmosphere (AC input only)	
Identification/Certifications		
J001	CE Marking (AC input only)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
J006	ATEX Approval (Input voltages 1, 2 & 6) M20 Conduit Openings. Consult factory for option compatibility.	

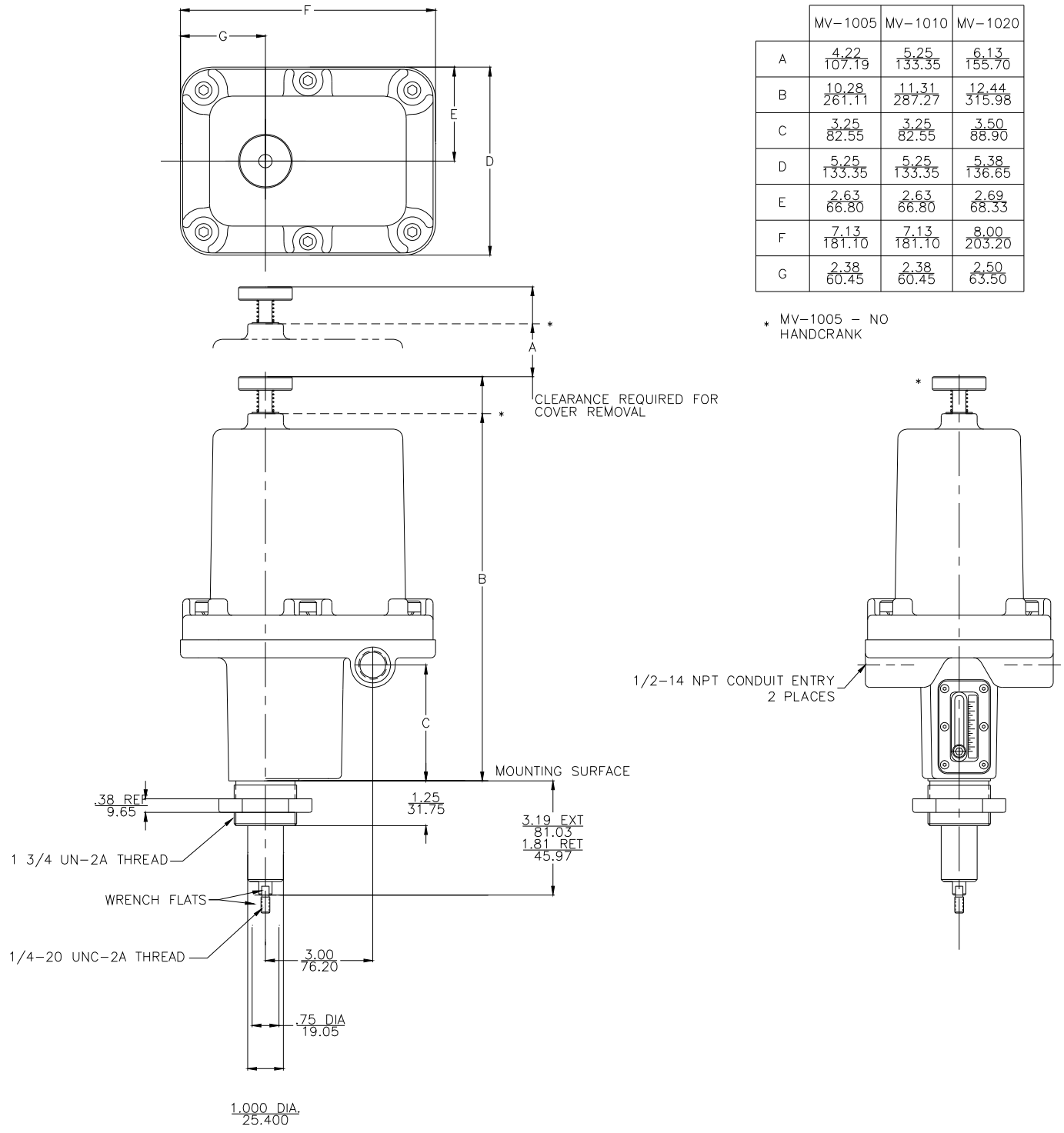
For a full description of options, go to the Complete Listing of Options starting on page 197.

Code	Description	Selection
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	
Special Electrical Options		
X012	Signal conversion module, installed and wired to actuator	
X013	2 wire incremental dc input	
X014	3 wire incremental 115/230 Vac input	
X015	Battery Backup, only for 115/230 Vac, not available with X012	
X017	Process variable (MV-1020 only)	
X018	Low Current (24 Vdc only)	

MV-1000 MAJOR DIMENSIONS

Threaded Base

INCHES
MILLIMETERS



	MV-1005	MV-1010	MV-1020
A	4.22 / 107.19	5.25 / 133.35	6.13 / 155.70
B	10.28 / 261.11	11.31 / 287.27	12.44 / 315.98
C	3.25 / 82.55	3.25 / 82.55	3.50 / 88.90
D	5.25 / 133.35	5.25 / 133.35	5.38 / 136.65
E	2.63 / 66.80	2.63 / 66.80	2.69 / 68.33
F	7.13 / 181.10	7.13 / 181.10	8.00 / 203.20
G	2.38 / 60.45	2.38 / 60.45	2.50 / 63.50

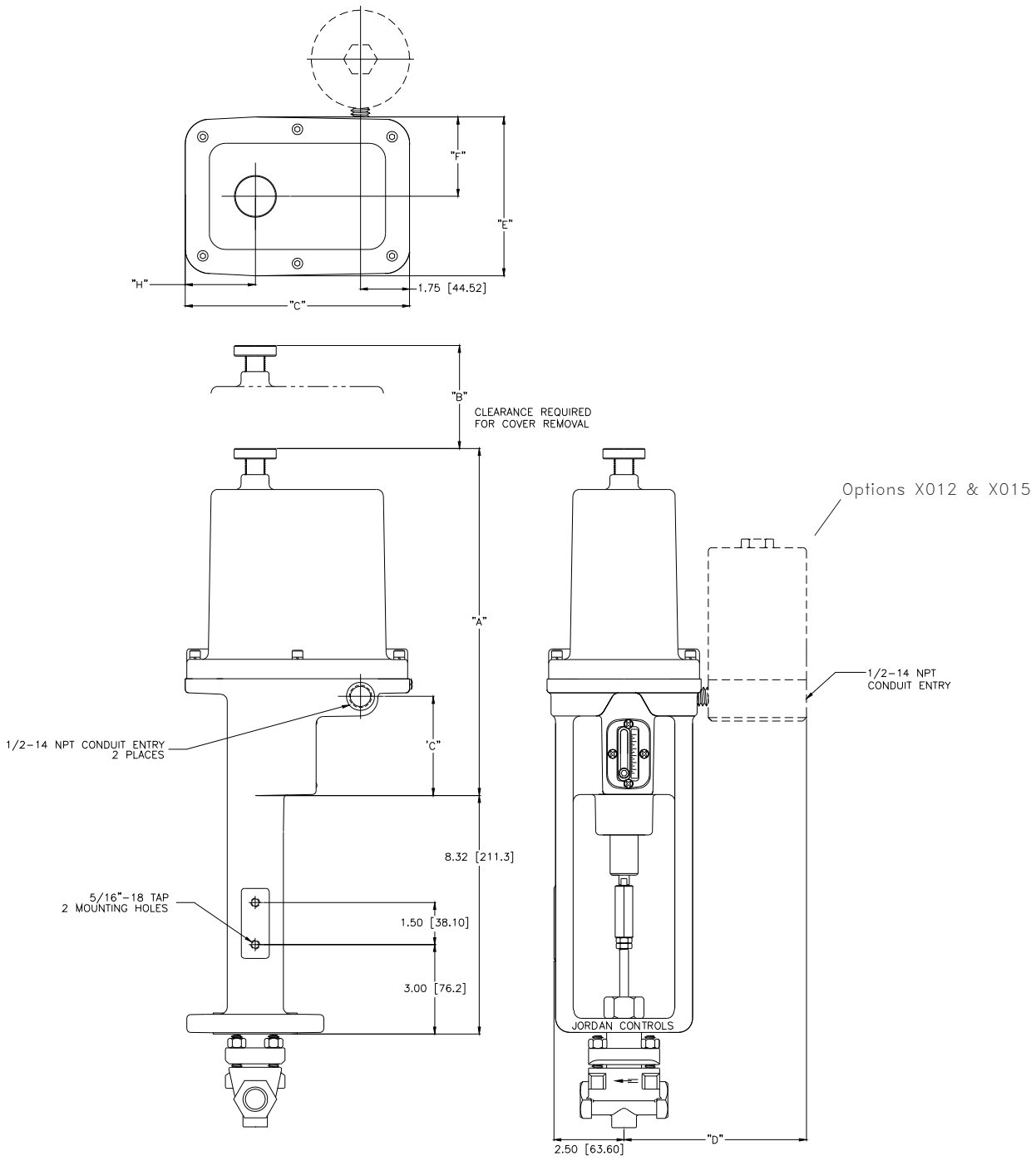
* MV-1005 – NO HANDCRANK

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

MV-1000 MAJOR DIMENSIONS

Cast Yoke Base

INCHES
MILLIMETERS



	A	B	C	D	E	F	G	H
MV-1010	11.43 (290.3)	5.25 (133.4)	3.40 (86.4)	6.52 (165.6)	5.39 (136.9)	2.69 (68.3)	7.12 (180.8)	2.37 (60.2)
MV-1020	12.37 (314.2)	6.13 (155.7)	3.53 (89.7)	6.64 (168.7)	5.66 (143.8)	2.83 (71.9)	8.00 (203.2)	2.50 (63.5)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

VA-1000 SERIES LINEAR VALVE ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The VA-1000 series are full-featured actuators designed to be used with low thrust control valves, capable of accepting standard analog current and voltage control signals. The brushless stepper motor design provides smooth, highly accurate positioning, with positive position-lock when not in motion. The VA-1000 Series is ideally suited for valve, regulator and metering pumps requiring exact position control and unrestricted continuous modulation.



FEATURES

- Permanently lubricated for any position mounting
- Amplifier supplies current to hold last position and prevent backdriving, up to thrust rating under power
- AC or DC input power versions
- 4 to 20 mA position, loop-powered, feedback signal
- Field selectable adjustments for:
  - speed
  - deadband
  - zero and span
  - command signal type
  - standard or reverse acting
  - manual-auto operation
  - output shaft position on loss of signal
- Wide ambient temperature range
- Full one year warranty

BASE MODEL INCLUDES:

- Motor (stepper)
- Manual override
- Amplifier
- Electronic thrust limiting
- Cast aluminum NEMA 4 (IP65) & dust ignition-proof enclosure
- Four adjustable position switches (40 mA at 40 Vdc)
- Valve position indicator
- Internal steel gear train
- 4-20 mA transmitter for customer use

POPULAR OPTIONS *(See Pages 197-206)*

- Signal Conversion Module: Convert 40 mA, 40 Vdc low level to 5 Amp, 120/240 Vac current rated position limit switches
- Low current at null (24 Vdc only)
- Switching input powered ac or dc positioning for positioning actuator using ac or dc remote voltage control
- Process Variable Controller to control one process variable
- Custom mounting and interface hardware
- Local Auto/Manual and INC/OFF/DEC toggle switches (Close-coupled, NEMA 4)
- Various enclosure coatings
- Battery backup to position actuator on loss of ac power
- Factory mounting to valve
- ATEX Approval

SPECIFICATIONS

Thrust: Up to 900 lbf. (4003 N)

Speed: See selection chart on next page

Input Power: See selection chart on next page

Linear Travel: 0.34 inch (8.63 mm) to 1.5 inches (38.1 mm), adjustable.

Temperature Limits: -40°F to 150°F (-40°C to 65°C)
For greater temperature ranges, consult factory.

Environmental Ratings: Explosion-proof for Class I, Division 1, Groups C and D; dust-ignition-proof for Class II, Division 1, Groups E, F, G. Also rated NEMA 4 (IP65) indoor/outdoor. (Optional Group B)

Command Input:
Full Range: 4-20 mA into 200 ohms
Split Range: 4-12 or 12-20 mA into 200 ohms
Voltage: 0-5 Vdc or 0-10 Vdc into 200,000 ohms
Switch: Dry contact closures

Field Wiring: To barrier terminal blocks

Thrust Limiting: Current sensing PWM

Positioning Accuracy: 1.5% for full range

Feedback: 4-20 mA, customer supplied (loop power)

Loss of Signal: Lock-in-place or run to preset.

Loss of Power: Lock-in-place at last position.

Direction: Extend or retract

Modulation Rate: Unrestricted modulating duty

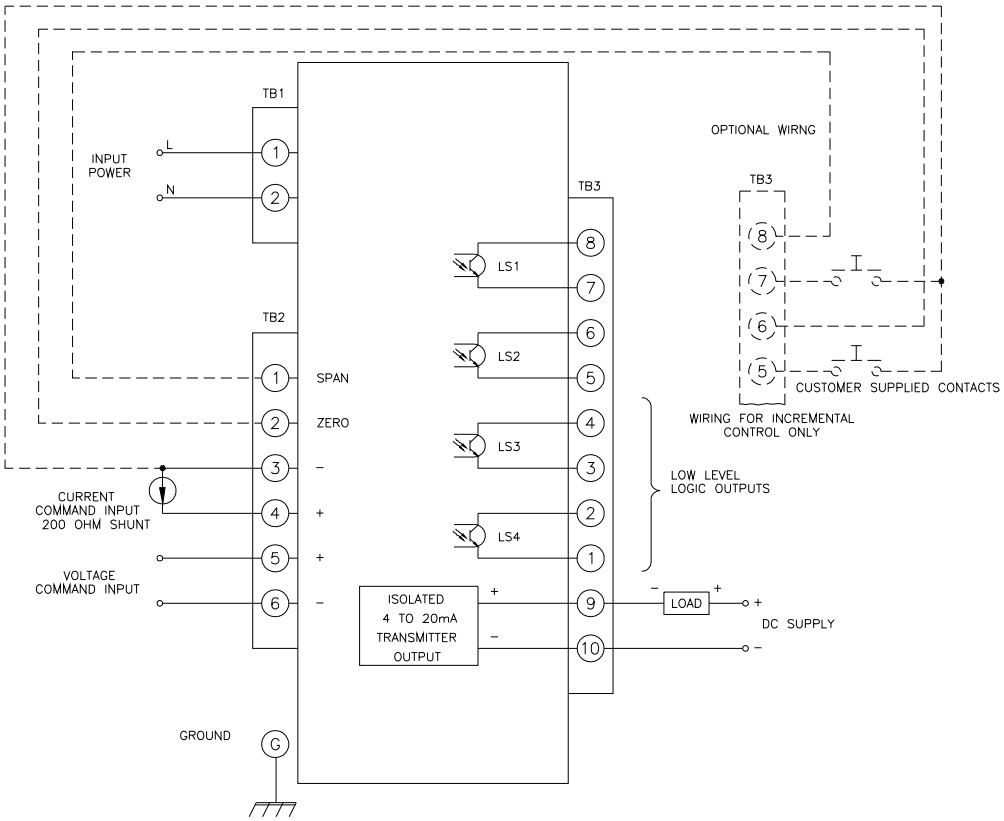
Backdrive: Self-locking up to thrust rating

Approximate Weight: VA-1015: 20 lbs. (9 kg)
(without valve) VA-1020: 24 lbs. (10.9 kg)

Approvals: See page 207

Position Switch Rating: 40 mA at 40 Vdc

TYPICAL WIRING DIAGRAM



These dimensions and specifications are subject to change without notice. Current drawings and specifications are available upon request.

VA-1000 SERIES SELECTION CHART

				Selection
1	Basic Model	**VA-1015-D**	Up to 600 pounds thrust output (2,669 N), with built-in amplifier	
		VA-1020-D	Up to 900 pounds thrust output (4,003 N), with built-in amplifier	
2	Input Voltage	**1**: 115 Vac, 1 Phase, 50/60 Hz (All Models)		
		2: 230 Vac, 1 Phase, 50/60 Hz (All Models)		
		6: 24-36 Vdc, (All Models)		
		12: 12-36 Vdc (VA-1020-D), maximum output thrust is limited - consult factory		
		13: 24 Vac (All Models)		
3	Speed in./sec. (mm/sec.) / Thrust, lbf. (N)	**VA-1015**	**VA-1020**	
		0.034/220 (979)	**0.034/220 (979)**	
		0.020/350 (1557)	**0.028//300 (1334)**	
4		**0.014/440 (1957)**	**0.025/420 (1868)**	
		0.0028/600 (2669)	**0.020/660 (2936)**	
			0.014/900 (4003)	
5	Linear Travel	**Specify Stroke** - 0.34 inch (8.63 mm) to 1.5 inches (38.1 mm)		

For valve mounting options, see L001 through L003 in the Options section.

VA-1000 STANDARD OPTIONS

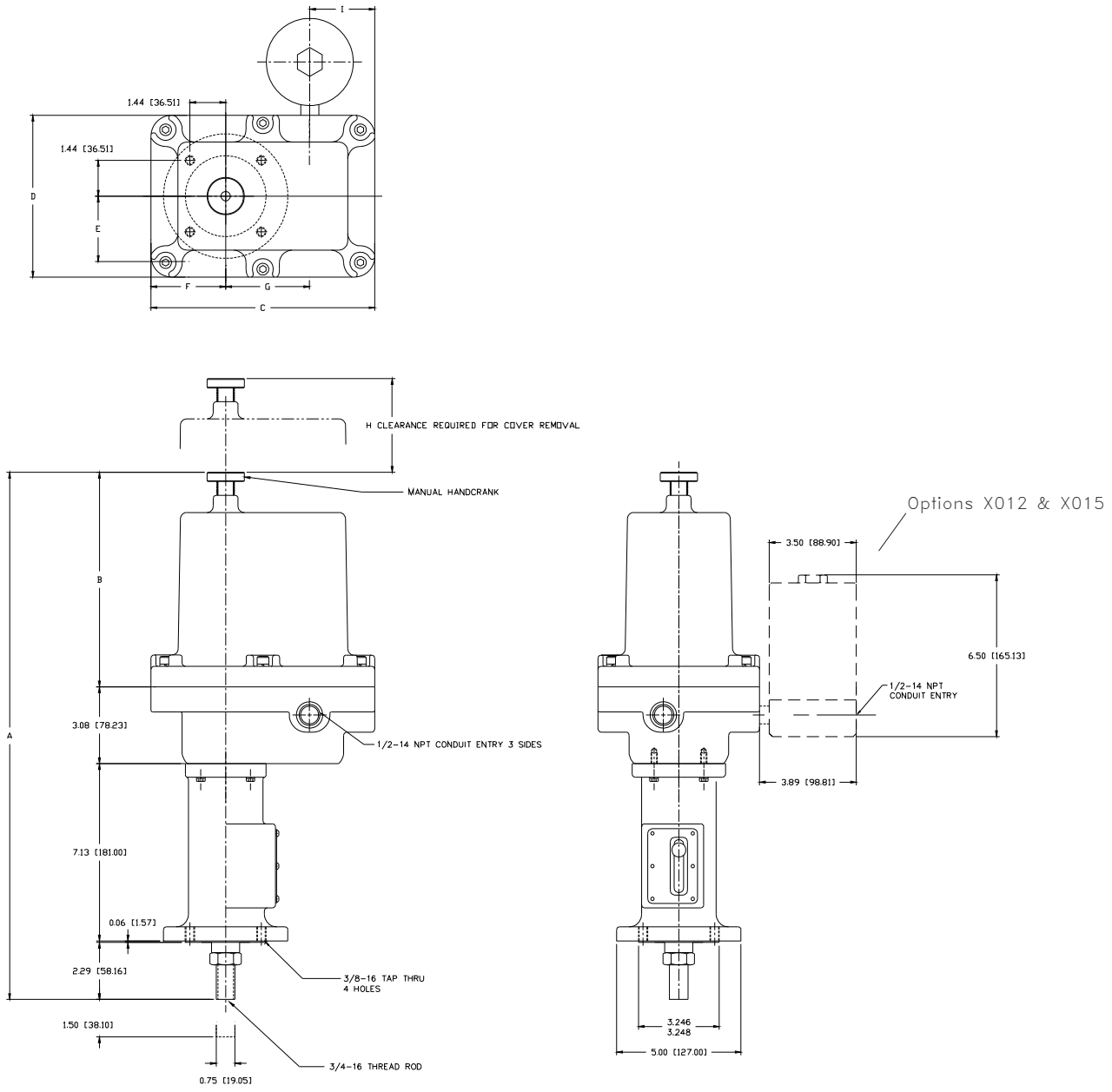
Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure)	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure)	
Identification/Certifications		
J001	CE Marking (AC input only)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
J006	ATEX Approval (Input voltages 1, 2 & 6) Consult factory for option compatibility.	
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	

Code	Description	Selection
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	
Special Electrical Options		
X012	Signal conversion module, installed and wired to actuator	
X013	2 wire incremental dc input	
X014	3 wire incremental 120/240 Vac input	
X015	Battery Backup, only for 120/240 Vac, not available with X012	
X017	Process variable (VA-1020 only)	
X018	Low Current (24 Vdc only)	

For a full description of options, go to the Complete Listing of Options starting on page 197.

VA-1000 MAJOR DIMENSIONS

INCHES
MILLIMETERS



	A	B	C	D	E	F	G	H	I
VA-1015	18.9 (480.06)	7.35 (186.69)	7.13 (181.1)	5.25 (133.35)	2.63 (66.8)	2.38 (60.45)	3.00 (76.20)	5.25 (133.35)	1.75 (44.45)
VA-1020	21.3 (541.02)	8.25 (209.55)	8.00 (203.2)	5.5 (139.7)	2.75 (69.85)	2.5 (63.5)	3.5 (88.9)	5.75 (146.05)	2.00 (50.8)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

MV-1100/MV-1500 SERIES LINEAR VALVE ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The MV-1100 and MV-1500 Series are linear, bi-directional actuators, designed to meet the exacting requirements for closed-loop modulating positioning control. They can be applied to a variety of sliding stem valves to provide remote valve positioning for control of gases, steam or other liquids in various process control applications.

All MV-1100 and MV-1500 Series actuators are also available with an internal amplifier (AC voltage only). These amplifiers are all full-featured ac switching devices designed to work seamlessly with the actuator for closed loop control.



FEATURES

- Permanently lubricated for any position mounting
- A wide variety of speed/thrust combinations
- AC or DC input power versions
- Cushioned valve seating
- Wide ambient temperature range
- Full one year warranty

BASE MODEL INCLUDES:

- Motor
- End of travel switches, not for customer use
- Explosion proof enclosure
- Internal spur gear train
- 4-20 mA transmitter for customer use (amplifier models only)

AMPLIFIER FEATURES

- Isolated, two-wire, 4-20mA feedback
- Customer-adjustable deadband
- Customer-adjustable zero and span
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place
  - Run to preset
- Field selectable dynamic brake
- AC triac switching of motor for increased life

POPULAR OPTIONS *(See Pages 197-206)*

- Potentiometer or 4 to 20 mA position feedback
- Custom mounting and interface hardware
- Anti-condensation heater and thermostat
- Motor brake (MV-1500 only)
- Handcrank (MV-1500 only)
- Battery backup to position actuator on loss of ac power

SPECIFICATIONS

	MV-1100	MV-1500
Thrust Range lbf. (N)	50 to 300 (222 to 1334)	300 to 3000 (1334 to 13344)
Stroke in. (mm)	0.25 to 0.75 (6.35 to 19.05)	0.25 to 4.0 (6.35 to 101.6)
Speed sec./in. (sec./mm)	7 to 72 (0.27 to 2.83)	11 to 40 (0.43 to 1.57)
Approx. Weight lbs. (kg)	16 (7.2)	40 (18.1)

Temperature Limits: -40°F to 150°F (-40°C to 65°C)
For greater temperature ranges, consult factory.

Field Wiring: To barrier terminal blocks

Approvals: See page 207

Position Switch Rating: 120/240 Vac, 10 Amp

The MV-1100 and MV-1500 Series do not have overthrust protection. Overthrusting the actuator will cause permanent damage.

MV-1100/MV-1500 SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Command Input	Input Impedance in ohms	Feedback	Modulation Rate *
		Run	Stall				
MV-1110-N	120/1/50/60	0.25	0.26	N/A	N/A	N/A	2000 starts/hr
MV-1110-D	120/1/50/60	0.25	0.26	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA loop-powered	2000 starts/hr
MV-1140-N	24 Vdc	1.2	4.8	N/A	N/A	N/A	4000 starts/hr
MV-1170-N	240/1/50/60	0.14	0.15	N/A	N/A	N/A	2000 starts/hr
MV-1170-D	240/1/50/60	0.14	0.15	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA loop-powered	2000 starts/hr
MV-1510-N	120/1/50/60	0.44	0.65	N/A	N/A	N/A	2000 starts/hr
MV-1510-D	120/1/50/60	0.44	0.65	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA loop-powered	2000 starts/hr
MV-1530-N	120/1/50/60	0.9	1.2	N/A	N/A	N/A	2000 starts/hr
MV-1530-D	120/1/50/60	0.9	1.2	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA loop-powered	2000 starts/hr
MV-1540-N	24 Vdc	1.7	1.9	N/A	N/A	N/A	4000 starts/hr
MV-1550-N	240/1/50/60	0.45	0.5	N/A	N/A	N/A	2000 starts/hr
MV-1550-D	240/1/50/60	0.45	0.5	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA loop-powered	2000 starts/hr
MV-1570-N	240/1/50/60	0.27	0.4	N/A	N/A	N/A	2000 starts/hr
MV-1570-D	240/1/50/60	0.27	0.4	4-20 mA 0-5 Vdc or 0-10 Vdc	200 100,000	4-20 mA loop-powered	2000 starts/hr

**** Minimum position change of 1%***

N/A = Not Applicable

REMOTE AMPLIFIER CHART

Actuator	Remote Mounted Amplifier	Catalog Page
MV-1110-N	AD-8240	173
MV-1140-N	AD-7530	170
MV-1170-N	AD-8240	173
MV-1510-N	AD-8240	173
MV-1530-N	AD-8240	173
MV-1540-N	AD-7530	170
MV-1550-N	AD-8240	173
MV-1570-N	AD-8240	173

MV-1100 SERIES SELECTION CHART

				Selection
1	Basic Model	**MV-1110-N**	1 phase, 120 Vac, 50/60 Hz, without built-in digital amplifier	
		MV-1110-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier	
		MV-1140-N	24 Vdc (If amplifier desired, see remote model AD-7530	
		MV-1170-N	1 phase, 240 Vac, 50/60 Hz, without built-in digital amplifier	
		MV-1170-D	1 phase, 240 Vac, 50/60 Hz, with built-in digital amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (MV-1110-N, MV-1110-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (MV-1170-N, MV-1170-D)		
		6: 24 Vdc (MV-1140-N)		
3	Speed sec./in. / Thrust lbf. (N)	MV-1110 MV-1170	MV-1140	
		72/**300** (1334)	**14**/**300** (1334)	
4		**36**/**200** (889)	**7**/**200** (889)	
		18/**100** (445)		
		9/**50** (222)		
5	Linear Travel	**Specify** Travel - 0.25 inch (6.35 mm) to 0.75 inches (19.05 mm) stroke available.		

For valve mounting options, see L001 through L003 in the Options section.

MV-1100 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure)	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure)	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not available with amplifier)	
F003	Potentiometer, 5000 ohms (not available with amplifier)	
F004	Potentiometer, 10,000 ohms (not available with amplifier)	
F005	Transmitter, 4-20 mA output	
F006	One Pot, 1000 ohms plus 4-20 mA Transmitter, not available with amplifier	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certifications		
J001	CE Marking (AC only (only for units with amplifiers	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	

Code	Description	Selection
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	
Auxiliary Switches		
S007	Aluminum Feedback Frame (Multi Turn)	
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	

For a full description of options, go to the Complete Listing of Options starting on page 197.

MV-1500 SERIES SELECTION CHART

					Selection
1	Basic Model	**MV-1510-N**	1 phase, 120 Vac, 50-60 Hz, without built-in digital amplifier		
		MV-1510-D	1 phase, 120 Vac, 50-60 Hz, with built-in digital amplifier		
		MV-1530-N	1 phase, 120 Vac, 50-60 Hz, without built-in digital amplifier		
		MV-1530-D	1 phase, 120 Vac, 50-60 Hz, with built-in digital amplifier		
		MV-1540-N	24 Vdc, (If amplifier desired, see model AD-7540		
		MV-1550-N	1 phase, 240 Vac, 50-60 Hz, without built-in digital amplifier		
		MV-1550-D	1 phase, 240 Vac, 50-60 Hz, with built-in digital amplifier		
		MV-1570-N	1 phase, 240 Vac, 50-60 Hz, without built-in digital amplifier		
		MV-1570-D	1 phase, 240 Vac, 50-60 Hz, with built-in digital amplifier		
2	Input	**1**: 120 Vac, 1 Phase, 50/60 Hz (MV-1510-N, MV-1510-D, MV-1530-N, MV-1530-D)			
		2: 240 Vac, 1 Phase, 50/60 Hz (MV-1550-N, MV-1550-D, MV-1570-N, MV-1570-D)			
		6: 24 Vdc, (MV-1540-N)			
3	Speed sec./in.	MV-1510 MV-1570	MV-1530 MV-1550	MV-1540	
		40/1200 (5338)	34/3000 (13345)	**23/1600** (7117)	
4	Thrust lbf. (N)	**23/600** (2669)	**32/2000** (8896)	**16/1200** (5338)	
		14/400 (1779)	**32/1600** (7117)	**11/800** (3558)	
		11/300 (1334)	**23/1200** (5338)	**8/600** (2669)	
			14/1100 (4893)		
			11/800 (3558)		
5	Linear Travel	**Specify** Travel - 0.25 inch (6.35 mm) to 4.0 inches (101.6 mm) stroke available.			
6*	Mounting Yoke Type	**S**: Standard length: Strokes to 1.5 inches (38 mm), soft seating extend position only. Maximum thrust 1200 lbf (5338 N). Maximum yoke boss 2.8 inches (71 mm).			
		L: Extended length: Strokes to 2 inches (51 mm). Bi-directional soft seating. Maximum thrust 2000 lbf (8896 N). Maximum yoke boss 2.8 inches (71 mm)			
		EL: Extended length: Strokes to 4 inches (102 mm). Bi-directional soft seating. Maximum thrust 3000 lbf (13345 N). Maximum yoke boss > 3.6 inches (90 mm).			

**See valve mounting options L001 through L003*

MV-1500 STANDARD OPTIONS

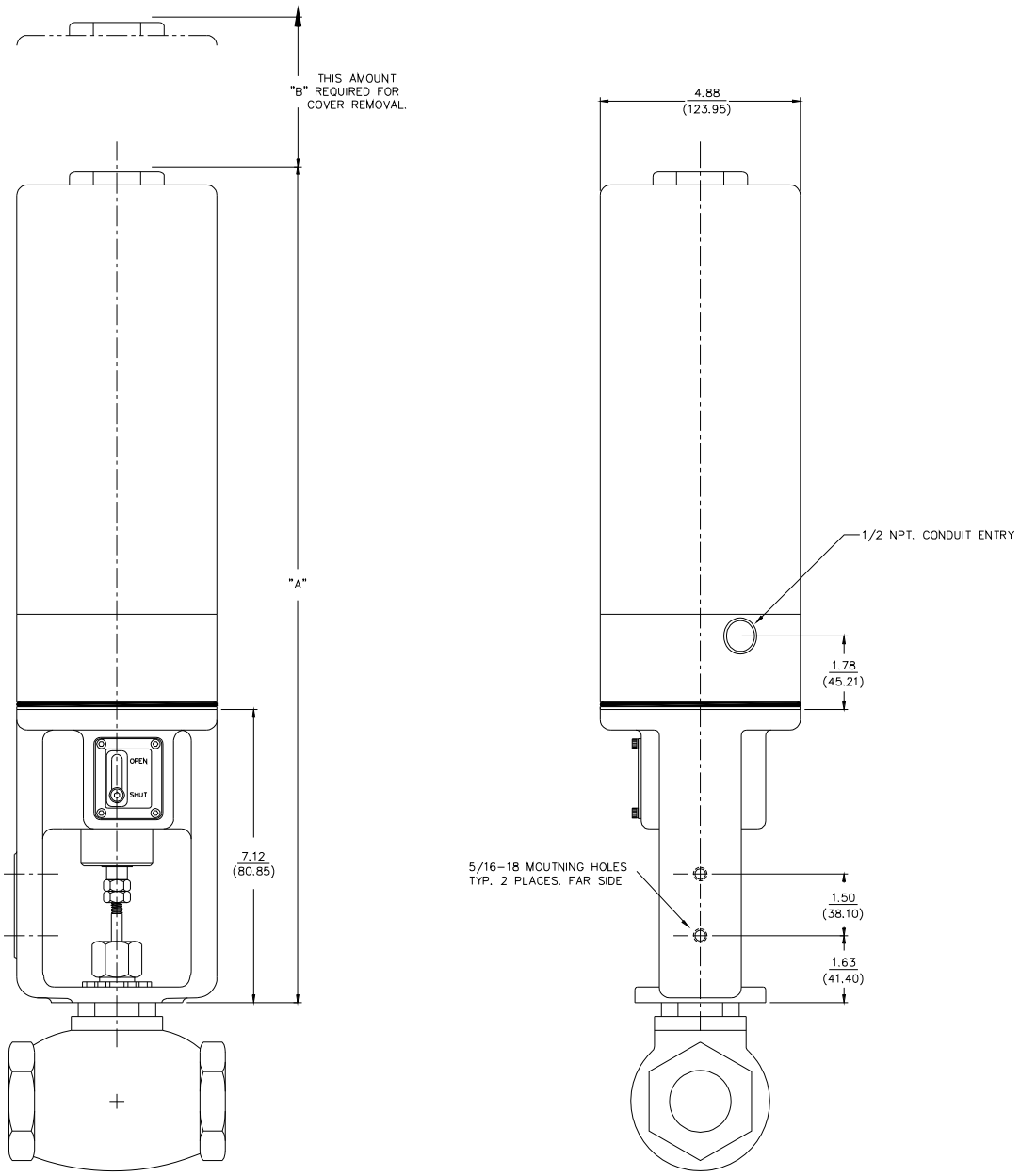
Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure)	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure)	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not available with amplifier)	
F003	Potentiometer, 5000 ohms (not available with amplifier)	
F004	Potentiometer, 10,000 ohms (not available with amplifier)	
F005	Transmitter, 4-20 mA output	
F006	One Pot, 1000 ohms plus 4-20 mA Transmitter, not available with amplifier	
F010	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certifications		
J001	CE Marking, AC only (only for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	

Code	Description	Selection
Handcrank		
K001	Manual Handcrank (Top-mount)	
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	
Auxiliary Switches		
S005	Aluminum Cam	
S006	Aluminum Feedback Frame	
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	

For a full description of options, go to the Complete Listing of Options starting on page 197.

MV-1100 MAJOR DIMENSIONS

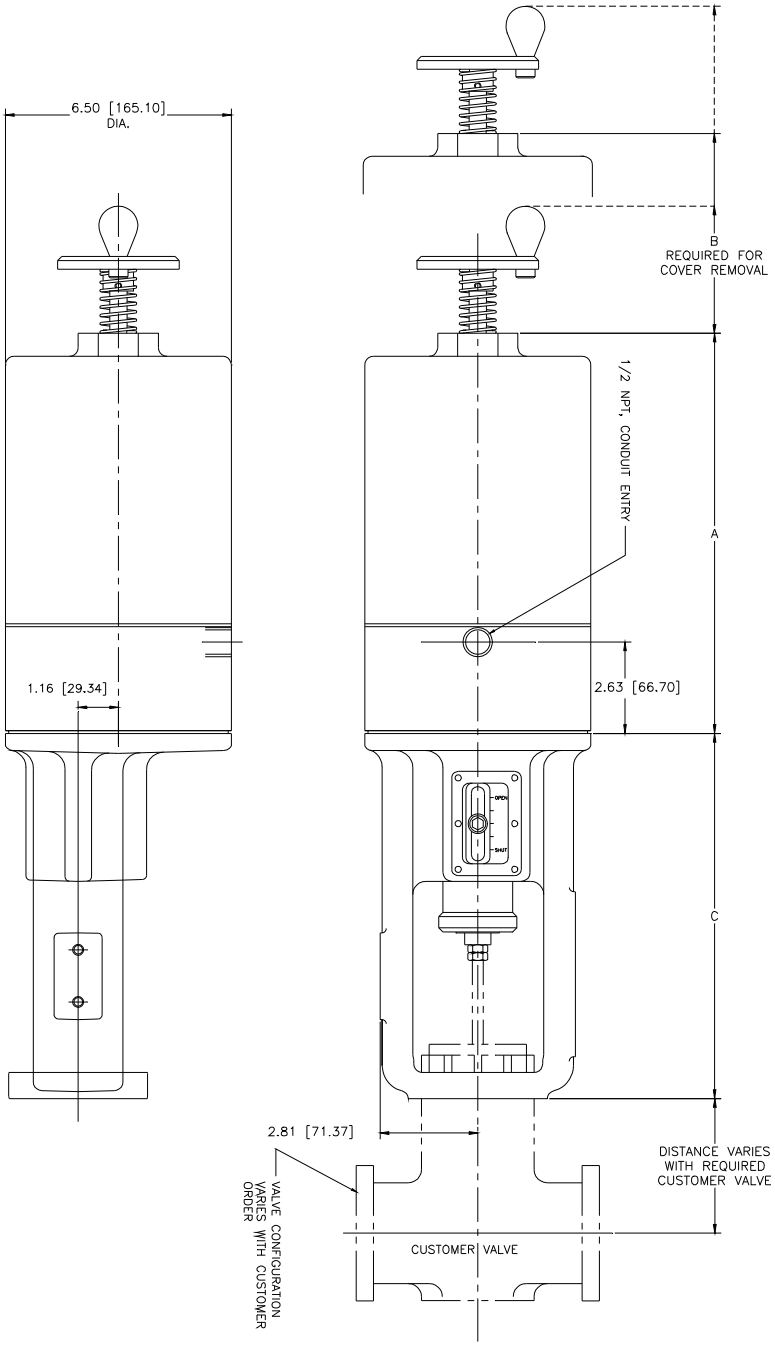
INCHES
MILLIMETERS



	A	B
MV-1110-N MV-1140-N MV-1170-N	16.20 (411.48)	7.81 (198.37)
MV-1110-D MV-1170-D	17.25 (438.15)	7.81 (198.37)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

MV-1500 MAJOR DIMENSIONS



	A	B	C		
			Standard	*Long*	*Extra Long*
MV-1510-N MV-1530-N MV-1540-N MV-1550-N MV-1570-N	11.50 (292.10)	8.36 (212.34)	10.50 (266.70)	15.55 (394.97)	Consult Factory
MV-1510-D MV-1530-D MV-1550-D MV-1570-D	13.18 (334.77)	10.04 (255.02)	10.50 (266.70)	15.55 (394.97)	Consult Factory

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

VA-1600 SERIES LINEAR VALVE ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The VA-1600 Series actuators are linear, bi-directional actuators, designed to meet the exacting requirements for closed-loop modulating positioning control. They can be applied to a variety of sliding stem valves to provide remote valve positioning for control of gases, steam or other liquids in various process control applications.

All VA-1600 Series actuators are also available with an internal amplifier (AC voltage only). These amplifiers are all full-featured ac switching devices designed to work seamlessly with the actuator for closed loop control.



FEATURES

- Permanently lubricated for any position mounting
- A wide variety of speed/thrust combinations
- AC or DC input power versions
- Wide ambient temperature range
- Cushioned valve seating
- Full one year warranty

AMPLIFIER FEATURES

- Isolated, two-wire, 4-20mA feedback
- Customer-adjustable deadband
- Customer-adjustable zero and span
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place
  - Run to preset
- Field selectable dynamic brake
- AC triac switching of motor for increased life

BASE MODEL INCLUDES:

- Motor
- End of travel switches, not for customer use
- Aluminum NEMA 4 (IP65), explosion proof enclosure
- Internal spur gear train
- 4-20 mA transmitter for customer use (amplifier models only)

POPULAR OPTIONS *(See Pages 197-206)*

- Potentiometer or 4-20 mA position feedback
- Custom mounting and interface hardware
- Local Auto/Manual toggle switches (Close-coupled, NEMA 4)
- Anti-condensation heater and thermostat
- Manual handwheel

SPECIFICATIONS

Thrust: Up to 6,000 lbf. (26,688 N)

Speed: See selection chart on next page.

Linear Travel: 0.25 inch (6 mm) to 2.0 inches (51 mm), adjustable.

Temperature Limits: -40°F to 150°F (-40°C to 65°C) *For greater temperature ranges, consult factory.*

Environmental Ratings: Explosion-proof for Class I, Division 1, Groups C and D; dust-ignition-proof for Class II, Division 1, Groups E, F, G. Also rated NEMA 4 (IP65) indoor/outdoor.

Feedback: 4-20 mA, customer supplied (loop power)

Field Wiring: To barrier terminal blocks

Positioning Accuracy: 1.5% for full range

Loss of Signal: Lock-in-place or drive to preset signal position.

Loss of Power: Lock-in-place at last position.

Modulation Rate: Modulating duty

Backdrive: Self-locking up to thrust rating

Approximate Weight: 72 lbs. (33 kg), *without valve & mounting hardware*

VA-1600 SERIES SELECTION CHART

					Selection
1	Basic Model	**VA-1630-N**	1 phase, 120 Vac, 50/60 Hz, without built-in amplifier		
		VA-1630-D	1 phase, 120 Vac, 50/60 Hz, with built-in digital amplifier		
		VA-1650-N	1 phase, 240 Vac, 50/60 Hz, without built-in amplifier		
		VA-1650-D	1 phase, 240 Vac, 50/60 Hz, with built-in digital amplifier		
		VA-1680-N	24 Vdc, modulating duty, without built-in amplifier		
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (VA-1630-N, VA-1630-D)			
		2: 240 Vac, 1 Phase, 50/60 Hz (VA-1650-N, VA-1650-D)			
		6: 24 Vdc (VA-1680-N)			
3	Speed sec./in. / Thrust lbf. (N)	VA-1630 VA-1650		VA-1680	
		32/3000 (13344)		**32/3000** (13344)	
4		**32/4000** (17792)		**32/4000** (17792)	
		64/5000 (22240)		**32/5000** (22240)	
		64/6000 (26688)		**32/6000** (26688)	
5	Linear Travel	**Specify** Travel - 0.25 inch (6.35 mm) to 2.0 inches (51 mm) stroke available.			

**See valve mounting options L001 through L003*

VA-1600 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure)	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure)	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not available for units with internal amp)	
F003	Potentiometer, 5000 ohms (not available for units with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (not available for units with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (not available for units with internal amplifiers)	
F010	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certifications		
J001	CE Marking, AC input only (only for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Handcrank		
K001	Manual Handcrank	

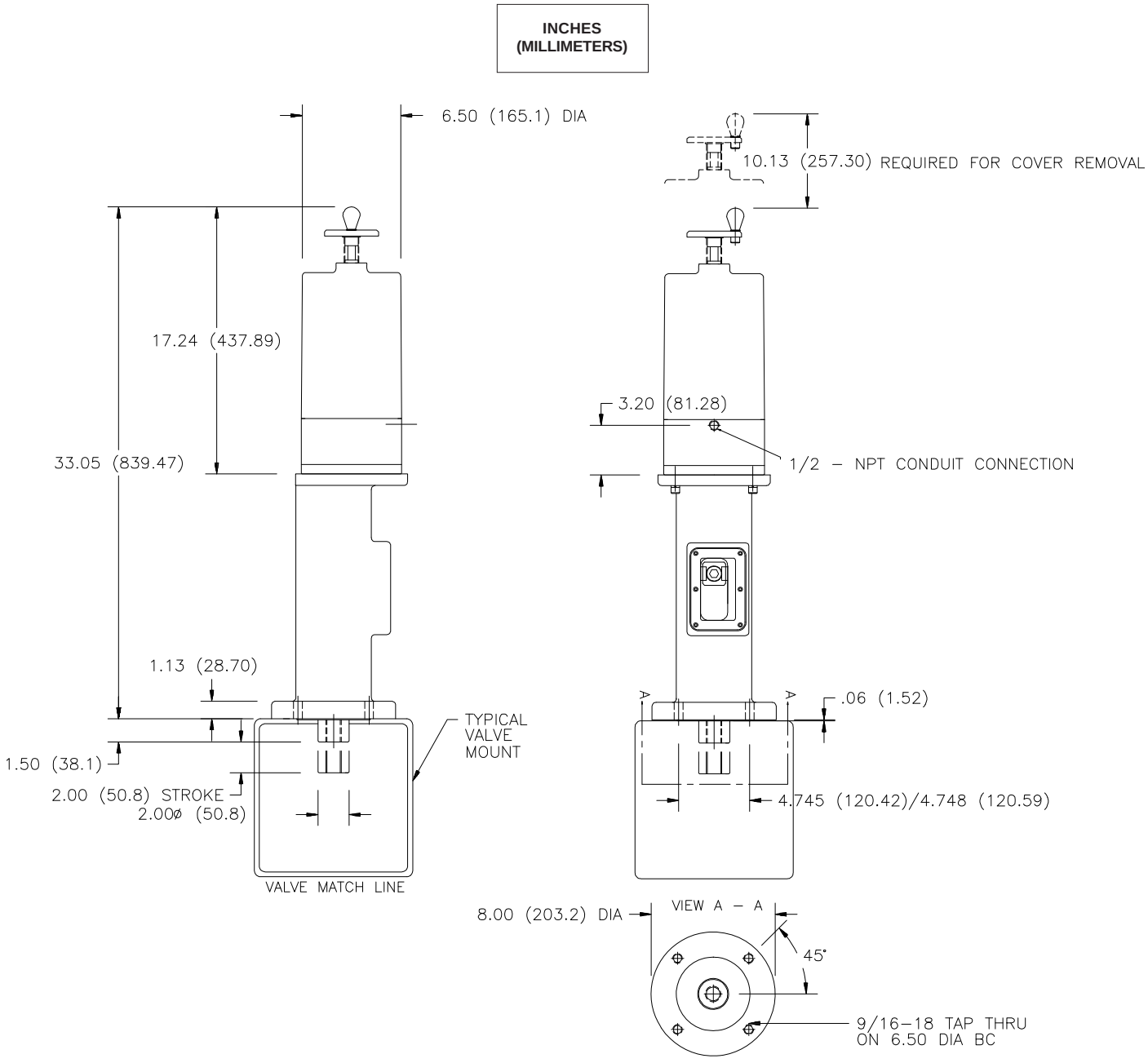
Code	Description	Selection
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	
Auxiliary Switches		
S005	Aluminum Cam	
S006	Aluminum Feedback Frame	
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	

For a full description of options, go to the Complete Listing of Options starting on page 197.

REMOTE AMPLIFIER CHART

Actuator	Remote Mounted Amplifier	Catalog Page
VA-1630-N	AD-8240	173
VA-1650-N	AD-8240	173
VA-1680-N	AD-7540	170

VA-1600 MAJOR DIMENSIONS



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LA-5000 SERIES LINEAR VALVE ACTUATORS

GENERAL DESCRIPTION

The LA-5000 Series are bi-directional, linear actuators internally geared to produce up to 6,500 lbs. (28,913 N) of thrust, and offer continuous modulating duty up to 4,000 starts per hour. This series of actuators combine the scotch-yoke output gearing design of the SM-5000 Series with a three-point toggle mechanism to convert rotary motion to linear motion. The LA-5000 Series is ideally suited for any sliding stem valve or other applications requiring a high-thrust, short stroke linear actuator.

All LA-5000 Series actuators are also available with an internal amplifier (excluding LA-5140). They are all full-featured ac or dc switching devices designed to work with the actuator for closed loop control.

FEATURES

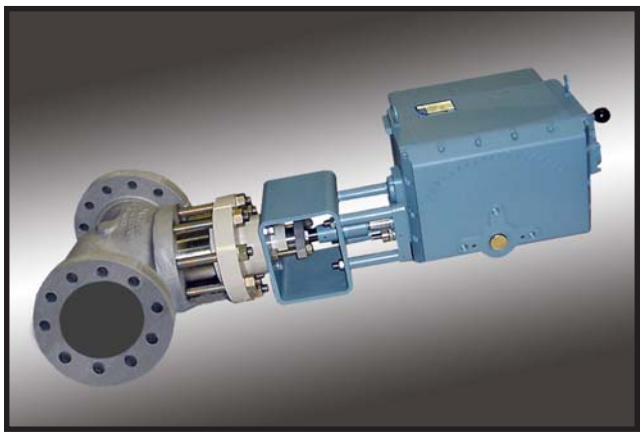
- "Scotch-yoke" final output increases thrust at ends of travel for positive seating and unseating of valves.
- Permanently lubricated for any position mounting
- Internal or remote mounted amplifier
- Self-locking drive system to hold in last position and prevent backdriving up to thrust rating
- Manual handcrank, safely isolated from motor, with overload indication and shear pin protection
- AC input power (single and three phase) or DC input power versions
- Wide ambient temperature range
- All mounting and interface hardware supplied
- Full two year warranty

BASE MODEL INCLUDES:

- Motor
- Manual override
- Cast aluminum, painted NEMA 4 (IP65) enclosure
- Anti-condensation heater with thermostat
- Automatic thrust limiting
- Internal Scotch-yoke gear train
- 4-20 mA transmitter for customer use (amplifier models only)
- End of travel switches, not for customer use

SELECT BY THRUST & STROKE

Actuator Series	Minimum Stroke inch (mm)	Maximum Stroke inch (mm)	Maximum Thrust lbf. (N)
LA-5100	0.75 (19)	2.25 (57)	3000 (13344)
LA-5200	1.00 (25)	4.00 (102)	6500 (28912)



LA-5120

POPULAR OPTIONS *(See Pages 197-206)*

- Split command input
- Position switch outputs
- Auxilliary position limit switches
- Electric brake
- Local Auto/Manual toggle or key lock rotary switches
- Open-Close-Stop pushbuttons
- Pilot lights
- Extended temperature ranges
- Local position indicator
- Thrust or position seating of valve
- Battery back-up, RP-4000

AMPLIFIER FEATURES

- Digital or analog available
- Isolated, two-wire, 4-20mA output
- Push button set-up for digital amplifier
- Customer-adjustable zero and span
- Adjustable deadband
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place - Run to preset
- Digital amplifier proportional control
- Soft seat at ends of travel
- AC triac switching of AC motors and IGBT for DC motors for increased life

Due to the complexities of the LA-5000 Series, please consult factory for exact model specification and pricing.

BASIC SPECIFICATIONS FOR LA-5000 SERIES ACTUATORS

Stroke: 0.75 in. (19mm) to 5.25 in. (133mm)

Drive Train: Scotch-yoke

Temperature Limits: -40° to 150° F (-40° to 65° C)
(For higher temperatures consult factory)

Environmental Ratings:
LA-5100 - NEMA 4 (IP65) Explosion Proof, Class I, Division 1, Groups C& D. Class II, Division 1, Groups E, F & G
LA-5200 - NEMA 4 (IP65)

Enclosure: Cast aluminum with polyurethane coating

Back Drive: Self-locking up to twice the thrust rating

Field Wiring: To barrier terminal blocks

Thrust Limiting: 10 Amp, SPDT switch, one for each direction (Extend & Retract). Switch will change states when thrust exceeds rated thrust for actuator. (20 Amp switches for LA-5200).

Approximate Weight: LA-5100: 125 lbs. (57 kg)
LA-5200: 245 lbs. (111 kg)

Approvals: See page 207

Position Switch Rating: 120/240 Vac, 20 Amp

ACTUATOR/AMPLIFIER SPECIFICATIONS

	LA-5115-A, LA-5120-D, LA-5160-A, LA-5190-D, LA-5210-A, LA-5220-D & LA-5260-A	LA-5160-D & LA-5260-D
Position Accuracy	1.0 % of full travel	0.25 % of full travel
Repeatability	0.5 % of full travel	0.25 % of full travel
Hysteresis	0.5 % of full travel	0.25 % of full travel
Linearity	+/-1 % of full travel	+/-0.5 % of full travel
Deadband	0.75 % of full travel	0.5 % of full travel
Resolution	0.75 % of full travel	0.5 % of full travel

See page 31 for additional features of Jordan Controls analog and digital amplifiers.

REMOTE AMPLIFIER CHART

Actuator	Remote Mounted Amplifier	Catalog Page
LA-5115-N	AD-8900	177
LA-5120-N	AD-8240	173
LA-5140-N	AD-7540	170
LA-5160-N	AD-7830/AD-9120	180/184
LA-5190-N	AD-8240	173
LA-5210-N	AD-8900	177
LA-5220-N	AD-8240	173
LA-5260-N	AD-7830/AD-9120	180/184

LA-5100 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-5115-N**	3 phase, 208/240/380/480 Vac, modulating duty, without built-in amplifier	
		LA-5115-A	LA-5115 with built-in analog amplifier	
		LA-5120-N	1 phase, 120 Vac, modulating duty, without built-in amplifier	
		LA-5120-D	LA-5120 with built-in digital amplifier	
		LA-5140-N	24 Vdc, modulating duty, (If amplifier desired, see AD-7540	
		LA-5160-N	90 Vdc, modulating duty, without built-in amplifier	
		LA-5160-A	LA-5160 with built-in analog amplifier, 120/208/240 Vac, 1 Ph; 208/240 Vac, 3 Ph	
		LA-5160-D	LA-5160 with built-in digital amplifier, 120/208/240 Vac, 1 Ph; 208/240 Vac, 3 Ph	
		LA-5190-N	1 phase, 240 Vac, modulating duty, without built-in amplifier	
		LA-5190-D	LA-5190 above with built-in digitalamplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-5120-N, LA-5120-D, LA-5160-A or LA-5160-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-5190-D, LA-5160-A or LA-5160-D)		
		3: 380 Vac, 3 Phase, 50 Hz (LA-5115-N or LA-5115-A)		
		4: 480 Vac, 3 Phase, 50/60 Hz (LA-5115-N or LA-5115-A)		
		5: 240 Vac, 3 Phase, 50/60 Hz (LA-5115-N, LA-5115-A, LA-5160-A or LA-5160-D)		
		6: 24 Vdc (LA-5140-N)		
		7: 90 Vdc (LA-5160-N)		
		9: 208 Vac, 1 Phase, 50 Hz (LA-5160-A or LA-5160-D)		
		10: 208 Vac, 3 Phase, 50 Hz (LA-5115-N, LA-5115-A, LA-5160-A or LA-5160-D)		
3 / 4	Speed / Thrust	**Consult** Factory for Model Designation and Pricing		
5	Linear Travel	**Consult** Factory for Model Designation and Pricing		

**See valve mounting options L001 through L003*

LA-5100 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches	
A002	INC/OFF/DEC Toggle Switches	
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002)	
A004	Open-Close-Stop Push Buttons - NEMA 4	
A005	Two Pilot Lights - NEMA 4	
A006	Two Position Auto/Manual Selector Switches - NEMA 4	
A007	Three Position INC/OFF/DEC Selector Switches - NEMA 4	
A009	ON/OFF Toggle Power Switch	
A010	Local Auto/Manual Switches & (INC/OFF/DEC) Switches - NEMA 4 (A006 & A007)	
A011	Key locks for Auto/Manual rotary switches - NEMA 4	
A012	ON/OFF Rotary Power Switch - NEMA 4	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not for use with models with internal amplifiers)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (Not for use with models with internal amplifiers)	
F010	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	
Identification/Certifications		
J001	CE Marking, AC input only (only available for units with amplifiers)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	

Code	Description	Selection
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	
Position Indicator		
P001	Local Position Indicator	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S015	Two nonadjustable thrust limit switches SPST, 10 Amp, 120/240 Vac	
S016	Two nonadjustable thrust limit switches SPDT, 10 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp	
Painting/Coating		
W002	Two Part Epoxy	

LA-5100 SPECIAL OPTIONS

(Consult factory for special requirements)

Code	Description	Selection
Connectors		
C001	Plug-In Cannon Type Connector on Enclosure for Power and Control	
C002	Mating Plug-In Cannon Type Connector on Enclosure for Power and Control	
Enclosure		
E003	Explosion Proof - Class 1, Div. 1, Groups C & D, indoor/outdoor (NEMA 7)	
Feedback		
F008	Contactless 4-20 mA	
F009	Contactless 0-10Vdc	
3-Phase Controls		
G001	3-Phase Contactor with Overloads	
G002	Control Transformer (120 Vac control voltage)	
G003	Control Transformer (240 Vac control voltage)	

Code	Description	Selection
Motor		
M001	3 Phase, 575 Vac	
Special Electrical Options		
X001	24 Vdc power supply, 320 mA, 100-240 Vac input	
X004	Voltage to current converter	
X005	Loop powered 4 to 20 isolator, M-system HSN-IAA	
X006	± 15 Vdc power supply	
X010	Customer command and feedback, 0-10 Vdc or 0-5 Vdc instead of 4-20 mA	
X019	AC Power Sensing with Output	

For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-5200 SERIES SELECTION CHART

				Selection
1	Basic Model	**LA-5210-N**	3 phase, 208/240/380/480 Vac, modulating duty, without built-in amplifier	
		LA-5210-A	LA-5210 with built-in analog amplifier	
		LA-5220-N	1 phase, 120/240 Vac, modulating duty, without built-in amplifier	
		LA-5220-D	LA-5220 with built-in digital amplifier	
		LA-5260-N	90 Vdc (perm. magnet field), modulating duty, without built-in amplifier	
		LA-5260-A	LA-5260 with built-in analog amplifier, 120/208/240 Vac, 1 Phase	
		LA-5260-D	LA-5260 with built-in digital amplifier, 120/208/240 Vac, 1 Phase	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-5220-N, LA-5220-D, LA-5260-A or LA-5260-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-5220-N, LA-5220-D, LA-5260-A or LA-5260-D)		
		3: 380 Vac, 3 Phase, 50 Hz (LA-5210-N or LA-5210-A)		
		4: 480 Vac, 3 Phase, 50/60 Hz (LA-5210-N or LA-5210-A)		
		5: 240 Vac, 3 Phase, 50/60 Hz (LA-5210-N or LA-5210-A)		
		7: 90 Vdc (LA-5260-N)		
		9: 208 Vac, 1 Phase, 50 Hz (LA-5260-A or LA-5260-D)		
		10: 208 Vac, 3 Phase, 50 Hz (LA-5210-N or LA-5210-A)		
3 / 4	Speed / Thrust	**Consult** Factory for Model Designation and Pricing		
5	Linear Travel	**Consult** Factory for Model Designation and Pricing		

**See valve mounting options L001 through L003*

LA-5200 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches	
A002	INC/OFF/DEC Toggle Switches	
A003	Local Auto/Manual Toggle Switches & INC/OFF/DEC Toggles (A001 & A002)	
A004	Open-Close-Stop Push Buttons - NEMA 4	
A005	Two Pilot Lights - NEMA 4	
A006	Two Position Auto/Manual Selector Switches - NEMA 4	
A007	Three Position INC/OFF/DEC Selector Switches - NEMA 4	
A009	ON/OFF Toggle Power Switch	
A010	Local Auto/Manual Switches & (INC/OFF/DEC) Switches - NEMA 4 (A006 & A007)	
A011	Key locks for Auto/Manual rotary switches - NEMA 4	
A012	ON/OFF Rotary Power Switch - NEMA 4	
Brake		
B001	Electrically Operated Brake 120 Vac	
B002	Electrically Operated Brake 240 Vac	
Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (Not for use with models with internal amplifiers)	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (Not for use with models with internal amplifiers)	
F010	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	
3-Phase Controls		
G001	3-Phase Contactor with Overloads	
G002	Control Transformer (120 Vac control voltage)	
G003	Control Transformer (240 Vac control voltage)	

Code	Description	Selection
Identification/Certifications		
J002	Stainless Steel Tags	
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	
Position Indicator		
P001	Local Position Indicator	
Auxiliary Switches		
S001	Two Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S002	Two Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S003	Four Adjustable Position Switches, SPDT, 20 Amp, 120/240 Vac	
S004	Four Adjustable Position Switches, DPDT, 20 Amp, 120/240 Vac	
S014	Two nonadjustable end of travel position switches, SPDT, 20 Amp, 120/240 Vac	
S016	Two nonadjustable thrust limit switches SPDT, 10 Amp, 120/240 Vac	
S019	Two nonadjustable end of travel position switches, SPST, 20 Amp, 120/240 Vac	
S020	Two nonadjustable thrust limit switches SPST, 20 Amp, 120/240 Vac	
Painting/Coating		
W002	Two Part Epoxy	

LA-5200 SPECIAL OPTIONS

(Consult factory for special requirements)

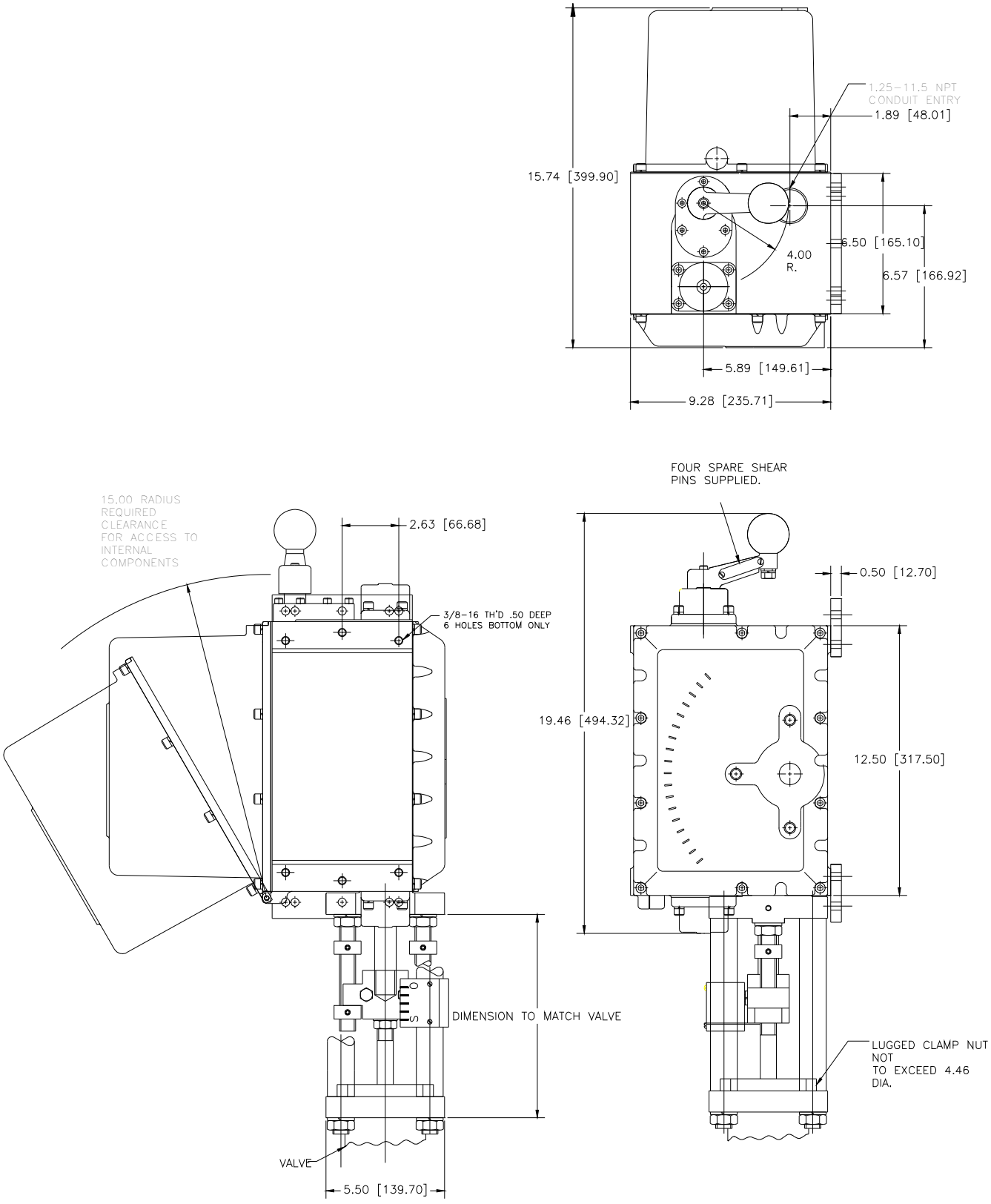
Code	Description	Selection
Connectors		
C001	Plug-In Cannon Type Connector on Enclosure for Power and Control	
C002	Mating Plug-In Cannon Type Connector on Enclosure for Power and Control	
Feedback		
F008	Contactless 4-20 mA	
F009	Contactless 0-10Vdc	
Motor		
M001	3 Phase, 575 Vac	

Code	Description	Selection
Special Electrical Options		
X001	24 Vdc power supply, 320 mA, 100-240 Vac input	
X004	Voltage to current converter	
X005	Loop powered 4 to 20 isolator, M-system HSN-IAA	
X006	± 15 Vdc power supply	
X010	Customer command and feedback, 0-10 Vdc or 0-5 Vdc instead of 4-20 mA	
X019	AC Power Sensing with Output	

For a full description of options, go to the Complete Listing of Options starting on page 197.

LA-5100 MAJOR DIMENSIONS

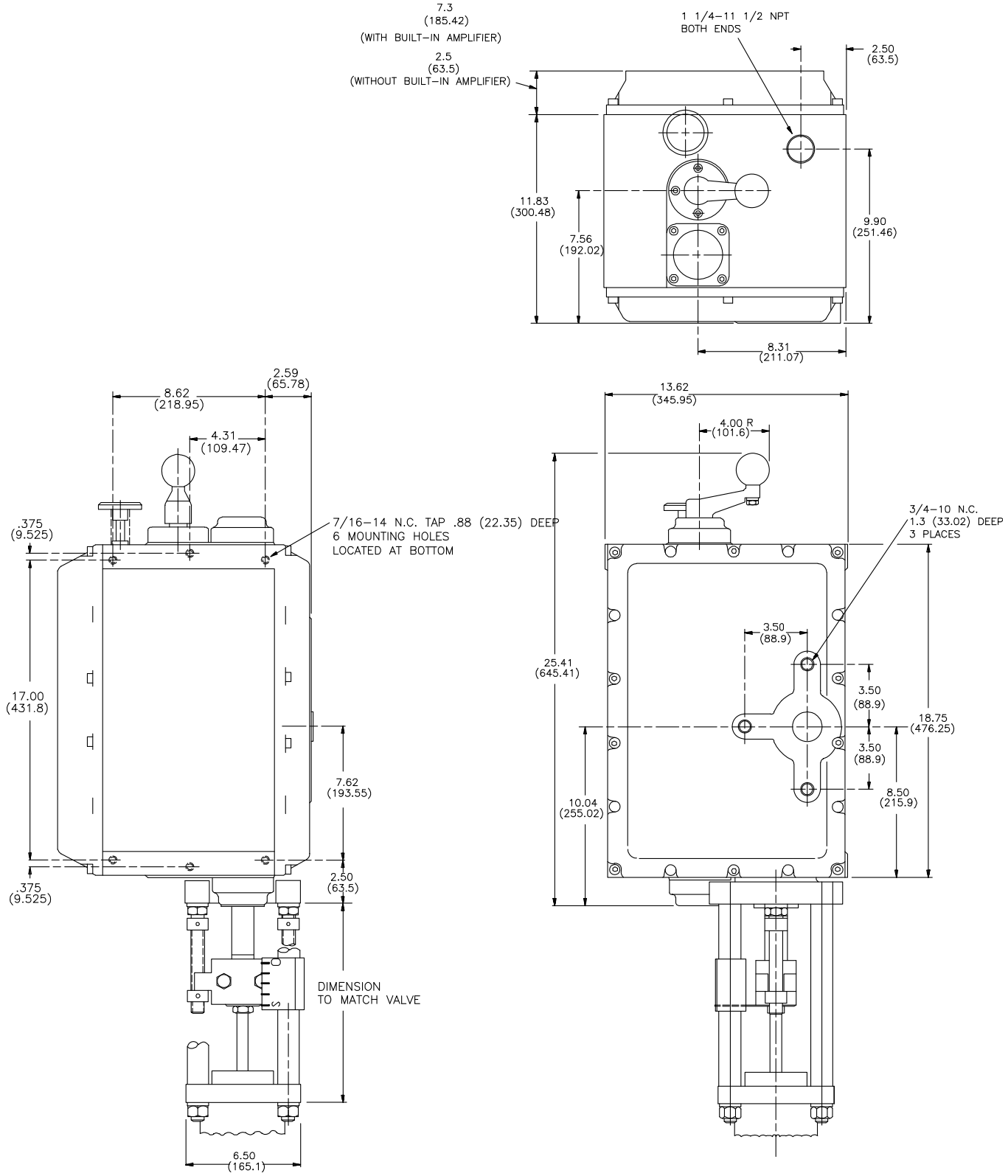
INCHES (MILLIMETERS)



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LA-5200 MAJOR DIMENSIONS

INCHES (MILLIMETERS)



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LINEAR VALVE ACTUATOR APPLICATIONS



LA-5120 used for sulfur mining

- Steam control valve for underground injection to melt sulfur and pump it to the surface.
- Remote location in West Central Texas.
- Air lines for a pneumatic operator were not practical for this remote location.



VA-1020 Special High Thrust Unit - 2,400 lbs

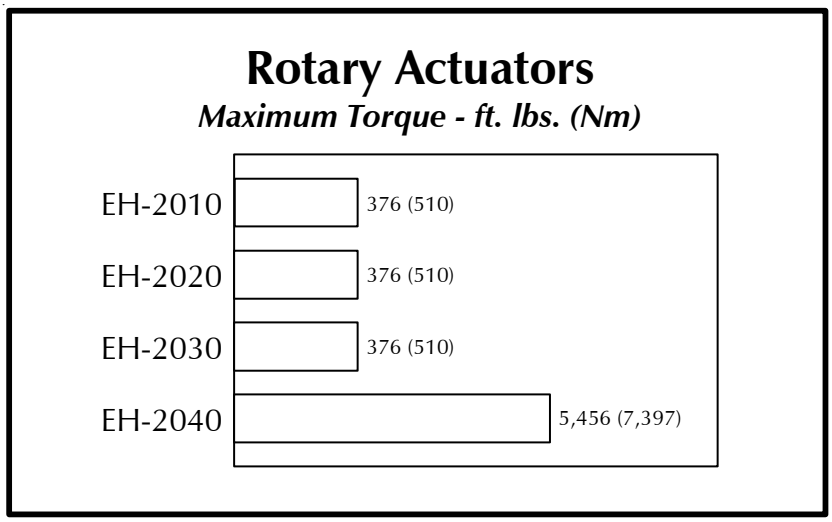
- Light Crude Oil Injection control valve used to extract heavy crude oil.
- Remote location.
- Available power was limited, and customer took advantage of the Sleeper Circuit.
- Customer wanted to use the VA-1020 because it was so similar to the SM-1020 used successfully on Gas Lift Choke Valves.



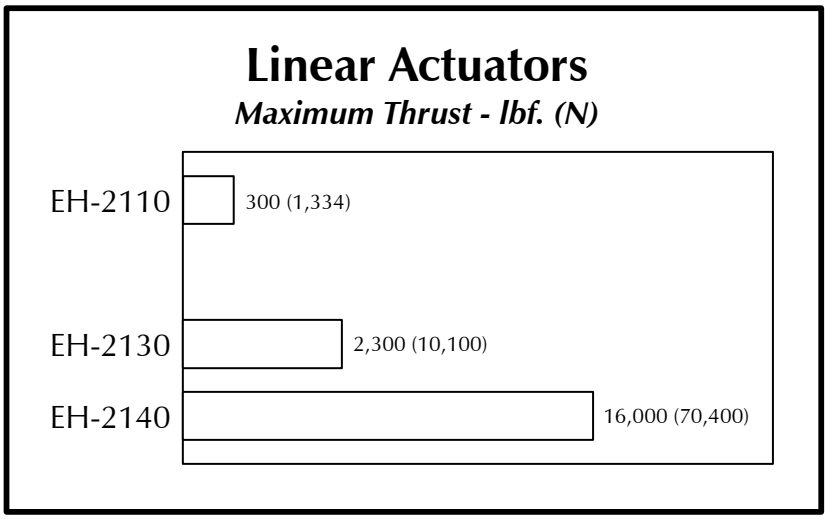
MV-1020 used for chemical feed

- Modulating duty controls the position of needle valves.
- Through the DCS, flow rate is accurately controlled as ALUM is added to the raw water system.
- Application environment is outdoors and requires a NEMA 4 enclosure rating.

TORQUE RANGES



THRUST RANGES



EH-2010 ROTARY ELECTRO-HYDRAULIC ACTUATORS

GENERAL DESCRIPTION

The EH-2010 Series, On-Off, Rotary Electro-Hydraulic actuator produces a high amount of torque in a small package with all electronics built-in. The internal construction assures a leak-proof system.

It is ideally suited for 90° rotation for on-off control of ball valves, butterfly valves, plug valves and dampers. This actuator combines the precision of hydraulic controls with the reliability of spring power for simplicity of operation. The EH-2010 is an economical actuator for On-Off, 90° applications.



FEATURES

- Torque: 212 to 4513 in. lbs. (24 to 510 Nm)
- On-Off, 90 degree rotation
- Mounting above, below or horizontal to valve
- NEMA 4 (IP65) enclosure
- Lock-in-place
- Leak-proof hydraulic system
- Low maintenance and long-life
- 2 adjustable position switches 5A, 230 Vac, SPDT
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Fast speed emergency shutdown
- Spring return on loss of power
- Custom mounting and interface hardware
- Manual override
- Low temperature
- Thermal jacket for very low and very high temperature applications

GENERAL SPECIFICATIONS:

Rotation: 90°, set at factory

Input Power: 120 or 230 Vac, 1Ph., 50/60 Hz

Speed & Torque: See table 1

Duty Cycle: 80%

Enclosure: Aluminum A1 Si 12, DIN1725

Temperature Limits: -13°F to 167°F (-25°C to 75°C)

Environmental Ratings: NEMA 4 (IP65)

Paint Finish: 2-part epoxy, 150 microns thick

Loss of Power: Fails in last position or in the direction of spring return

Adjustable Position Switches:
Two, each rated at 5 Amp at 230 Vac, SPDT

Hydraulic Fluid: 10c ST hydraulic fluid

Hydraulic Pump: One or two oscillating

Mounting: Vertical or horizontal (See instruction manual)

Cable Entries: 1, 1/2" NPT

Valve Mounting Type: ISO-5211

Weight: See Table 3

Consult factory for wiring information.

EH-2010 SERIES SELECTION CHART

				Selection
1	Basic Model	**EH-2010-N**	On-Off actuator, no amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz		
		2: 230 Vac, 1 Phase, 50/60 Hz		
		16: 110 Vac, 1 Phase, 50/60 Hz		
3	Speed/ Torque	**A**		
		B		
		C	***See Table 1 for Speed/Torque Selection***	
		D		
		E		
		F		
		G		
4	Rotation	90°		
5	Close/Open *	C: Spring force clockwise, lock-in-place		
		CCW: Spring force counter clockwise, lock-in-place		

* As viewed from actuator output shaft

EH-2010 STANDARD OPTIONS

Code	Description	Selection
	Electro-Hydraulic	
N004	Failsafe, Spring return on loss of power	
N006	Quick Failsafe (see Table 2 for speed select)	
N015	Mounting direction, if not above valve	
N016	Silicone oil for low temperature operation, -40° F (-40° C)	
N017	Valve mounting kit	
N019	Thermal jacket for low, -40° F (-40° C) and high, 482° F (250°C) (for two hours) temperature applications	
N021	Geared manual override (speed/torque A-C)	
N022	Geared manual override (speed/torque D-E)	
N023	Geared manual override (speed/torque F-G)	
N026	Bio-degradable fluid for food/beverage	

This table should be used to select the torque and speed of the actuator. Actuator has two torques for each direction of the shaft. These torques are start and finish torque with operating torque typically Spring Stroke Direction Finish of Stroke.

Selection Code	Hydraulic Stroke Direction			Spring Stroke Direction		
	Speed in Sec.	Torque in-lb (Nm)		Speed in Sec.	Torque in-lb (Nm)	
		Start	***Finish***		***Start***	***Finish***
A	16	300(34)	150(17)	3	335(38)	210(24)
B	22	442(50)	200(23)	4	530(60)	336(38)
C	37	700(79)	360(41)	6	740(84)	460(52)
D	38	1,500(170)	750(85)	12	1,645(186)	1,035(117)
E	66	2,475(280)	1,240(140)	20	2,920(330)	1,855(210)
F	115	4,390(496)	2,175(246)	30	5,255(594)	3,080(348)
G	165	6645(751)	3,285(371)	45	7,300(825)	4,513(510)

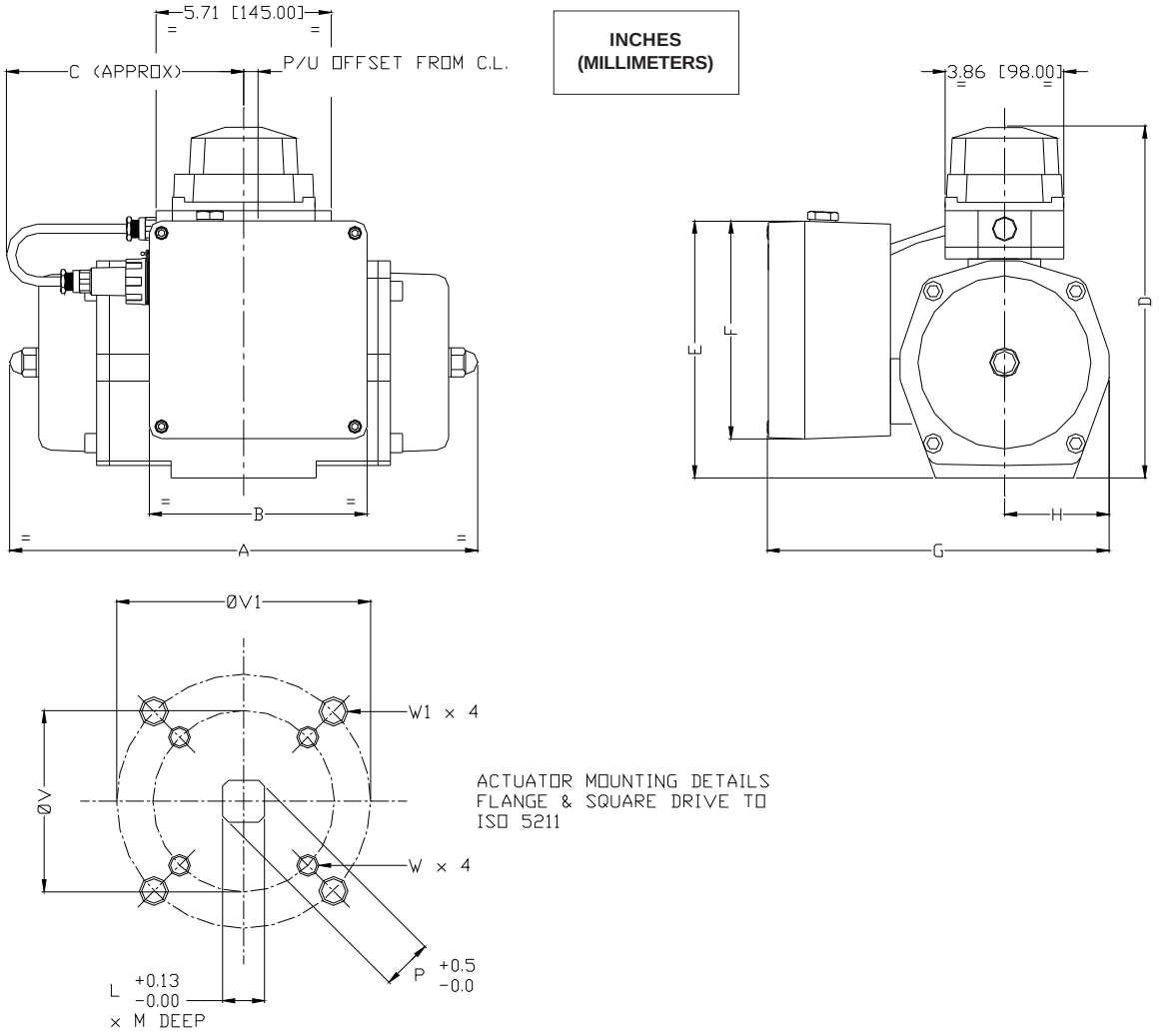
Table 1

This table should be used when selecting option N006.

Speed/Torque Code	Speed in Sec.
A	N/A
B	N/A
C	1.5
D	2.5
E	4
F	5
G	9

Table 2

EH-2010 MAJOR DIMENSIONS



	Dimensions - in. (mm)						
	Selection Code A	Selection Code B	Selection Code C	Selection Code D	Selection Code E	Selection Code F	Selection Code G
A	8.3 (204)	9.8 (249)	10.5 (267)	14.1 (360)	15.2 (387)	18.8 (477)	20.3 (517)
B	7.9 (200)	7.9 (200)	7.9 (200)	7.1 (180)	7.1 (180)	7.1 (180)	7.1 (180)
C	5.5 (140)	5.5 (140)	5.5 (140)	7.1 (180)	7.1 (180)	7.1 (180)	7.1 (180)
D	3.6 (93)	4.1 (105)	4.6 (118)	5.6 (143)	7.1 (181)	8.6 (220)	10.2 (259)
E	4.3 (110)	4.3 (110)	4.3 (110)	4.3 (110)	4.3 (110)	4.3 (110)	4.3 (110)
F	3.5 (90)	3.5 (90)	3.5 (90)	4.3 (110)	4.3 (110)	4.3 (110)	4.3 (110)
G	3.8 (96)	4.2 (107)	4.6 (117)	5.4 (137)	7.1 (182)	8.5 (217)	9.5 (242)
H	2 (53)	2.2 (58)	2.5 (63)	2.9 (73)	3.7 (95)	4.4 (113)	4.9 (126)
L	0.55 (14)	0.55 (14)	0.74 (19)	0.86 (22)	1.06 (27)	1.0 (27)	1.4 (36)
M	1.1 (27)	1.2 (30)	1.1 (27)	1.4 (37)	1.4 (37)	1.4 (38)	2 (51)
P	0.7 (18)	0.7 (18)	0.9 (25)	1.1 (28)	1.4 (36)	1.4 (36)	1.9 (48)
V	1.9 (50)	1.9 (50)	1.9 (50)	2.7 (70)	2.7 (70)	4 (102)	4 (102)
V1	2.7 (70)	2.7 (70)	2.7 (70)	4 (102)	4 (102)	4.9 (125)	5.5 (140)
W	1/4-20 x 0.39	1/4-20 x 0.39	1/4-20 x 0.39	5/16-18 x 0.51	5/16-18 x 0.51	3/8-16 x 0.63	3/8-16 x 0.63
W1	5/16-18 x 0.39	5/16-18 x 0.39	5/16-18 x 0.51	3/8-16 x 0.63	3/8-16 x 0.63	1/2-13 x 0.78	5/8-11 x 0.98
Weight lb. (kg)	16.5 (7.5)	18.7 (8.5)	20.9 (9.5)	37.4 (17)	70.5 (32)	103.6 (47)	127.8 (58)

Table 3

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

EH-2020 ROTARY ELECTRO-HYDRAULIC ACTUATORS

Jordan CONTROLS

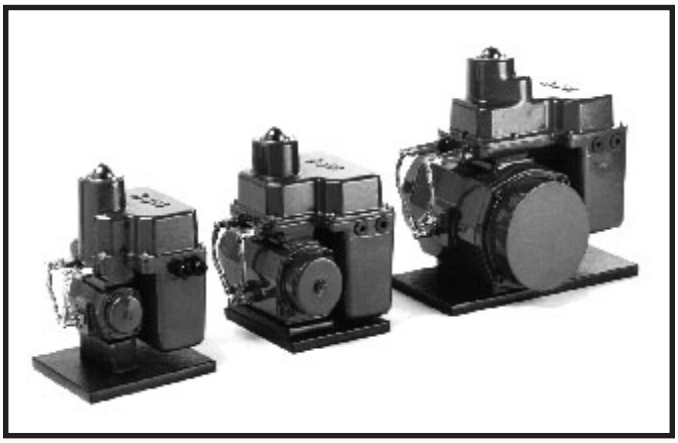
GENERAL DESCRIPTION

The EH-2020 Series Rotary Electro-Hydraulic actuator produces a high amount of torque in a small package with all electronics built-in. The internal construction assures a leak-proof system.

It is ideally suited for 90° rotation for both modulating or on-off control of ball valves, butterfly valves, plug valves and dampers. This actuator combines the precision of hydraulic controls with the reliability of spring power for simplicity of operation.

These actuators are available in the same small package with spring return for fail-safe to close or open on loss of power.

The EH-2020 accepts an analog command signal to accurately position the actuator while providing maximum resolution and speed of response. With resolution better than 0.2% of a degree, the actuator offers superior control when compared to conventional electric or pneumatic actuators.



FEATURES

- Torque: 212 to 4513 in. lbs. (24 to 510 Nm)
- Mounting above, below or horizontal to valve
- NEMA 4 (IP65) enclosure
- Lock-in-place on loss of power
- 90 degree rotation
- Leak-proof hydraulic system
- Internal electronics with easy setup
- Unrestricted continuous duty
- 4-20mA command and feedback
- Low maintenance and long-life
- 2 adjustable position switches, 5A, 230 Vac, SPDT
- Automatic torque limiting
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Fast speed emergency shutdown
- Spring fail safe to open or close on loss of power
- Auto/Manual switches
- Additional position switch, 5A, 230 Vac, SPDT
- Manual override
- Remote mounted electronics *(See AD-2000)*
- Anti-condensation heater

GENERAL SPECIFICATIONS:

Rotation: 90°, set at factory

Input Power: 110, 120 or 230 Vac, 1Ph., 50/60 Hz

Speed & Torque: See Table 1

Temperature Limits:
EH-2020-N: -13°F to 167°F (-25°C to 75°C)
EH-2020-A: -13°F to 131°F (-25°C to 55°C)

Environmental Ratings: NEMA 4 (IP65)

Paint Finish: 2-part epoxy, 150 microns thick

Command Signal: 4-20 mA, 20-4 mA, 0-10 Vdc, 10-0 Vdc or On-Off (switch main power)

Feedback Signal: 4-20 mA or 0-10 Vdc

Deadband Adjustment: 0-2.5% of range

Loss of Power: Fails in last position

Modulation Rate: Unrestricted Modulating Duty

Position Accuracy: Less than 0.2%

Repeatability: Less than 0.25%

Adjustable Position Switches: 2, 5A, 230 Vac, SPDT

Hydraulic Fluid: 10c ST hydraulic fluid

Automatic Torque Limiting: Explosion-proof pressure switch. CENELEC approved.

Hydraulic Pump: Two oscillating pumps

Mounting: Vertical or horizontal (See instruction manual)

Cable Entries: 3, 1/2″ NPT

Valve Mounting Type: ISO-5211

Weight: See Table 3

Consult factory for wiring information.

EH-2020 SERIES SELECTION CHART

				Selection
1	Basic Model	**EH-2020-N**	On-Off actuator, no amplifier (can use remote electronics to modulate)	
		EH-2020-A	Modulating actuator with amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz		
		2: 230 Vac, 1 Phase, 50/60 Hz		
		16: 110 Vac, 1 Phase, 50/60 Hz		
3	Speed/ Torque	**A**		
		B		
		C	***See Table 1 for Speed/Torque Selection***	
		D		
		E		
		F		
		G		
4	Rotation	90°		
5	Close/Open *	C: Spring force clockwise, lock-in-place		
		CCW: Spring force counter clockwise, lock-in-place		

* As viewed from actuator output shaft

EH-2020 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches, NEMA 4	
Heater		
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Electro-Hydraulic		
N004	Failsafe - Spring return on loss of power	
N006	Quick Failsafe (see Table 2 for speed select)	
N007	Manual override (336 in. lbs. or less)	
N008	Manual override (460 to 1035 in. lbs.)	
N009	Manual override (1858 to 4513 in. lbs.)	
N011	0-10 Vdc Command	
N015	Mounting direction, if not above valve	
N016	Silicone oil for low temperature operation, -40° F (-40° C)	
N017	Valve mounting kit	
N025	M20 threaded cable entry	
N026	Bio-degradable fluid for food/beverage	
Auxiliary Switches		
S023	1 adjust. position switch, 5A, 230 Vac, SPDT	

This table should be used to select the torque and speed of the actuator. Actuator has two torques, one for each direction of the shaft. These torques are start and finish torque with required operating torque typically using Spring Stroke Direction/Finish.

Selection Code	Hydraulic Stroke Direction			Spring Stroke Direction		
	Speed in Sec.	Torque in-lb (Nm)		Speed in Sec.	Torque in-lb (Nm)	
		Start	*Finish*		*Start*	*Finish*
A	18	300(34)	150(17)	4	335(38)	210(24)
B	24	440(50)	200(23)	5	530(60)	336(38)
C	18	700(79)	360(41)	6.5	740(84)	460(52)
D	38	1,500(170)	750(85)	15	1,645(186)	1,035(117)
E	66	2,475(280)	1,240(140)	25	2,920(330)	1,855(210)
F	115	4,390(496)	2,175(246)	32	5,255(594)	3,080(348)
G	165	6645(751)	3,285(371)	58	7,300(825)	4,513(510)

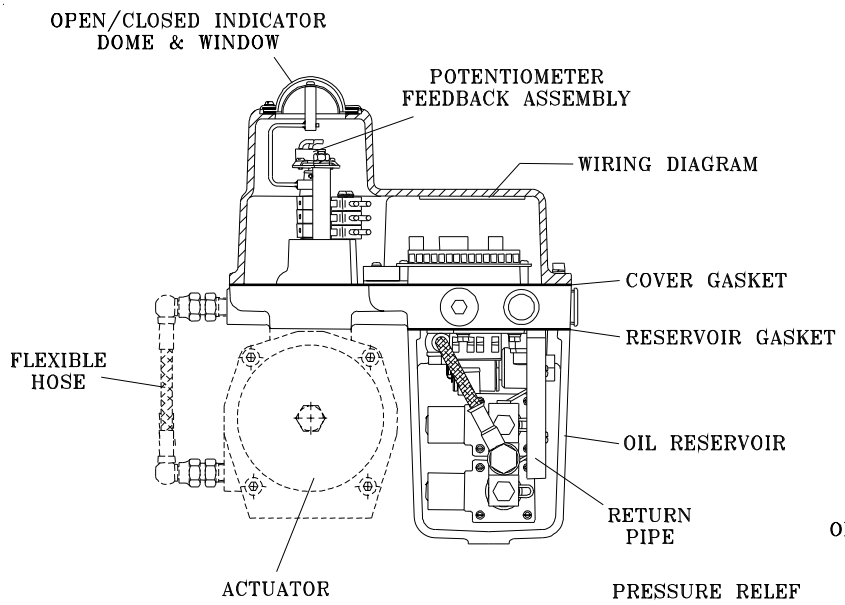
Table 1

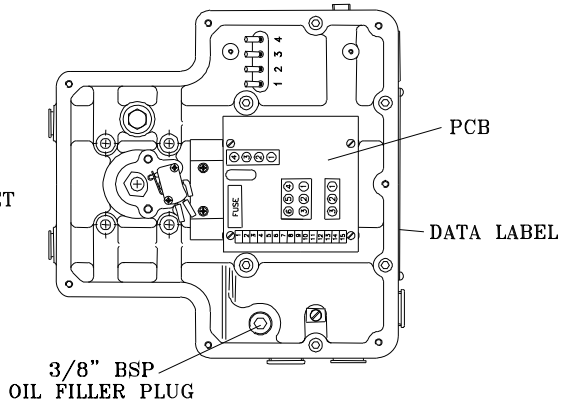
This table should be used when selecting option N006.

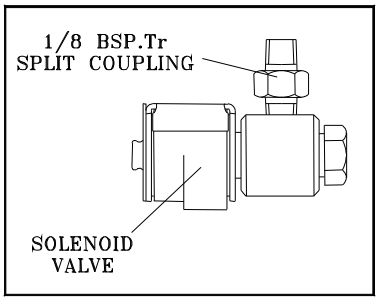
Speed/Torque Code	Speed in Sec.
A	N/A
B	N/A
C	1.5
D	2.5
E	4
F	5
G	9

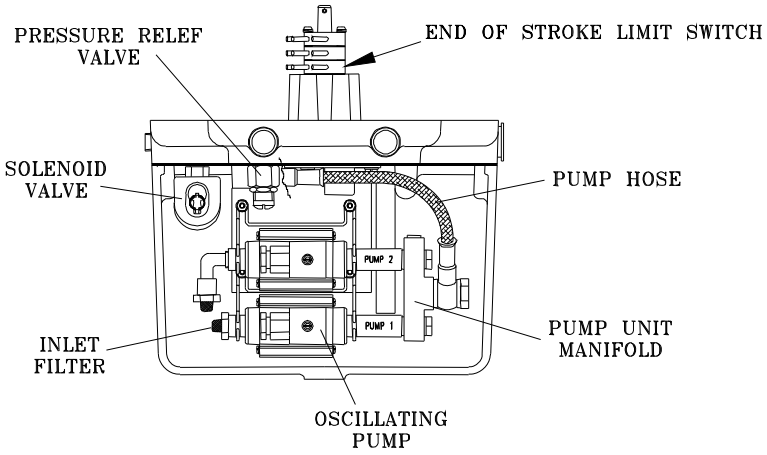
Table 2

EH-2020 PARTS IDENTIFICATION

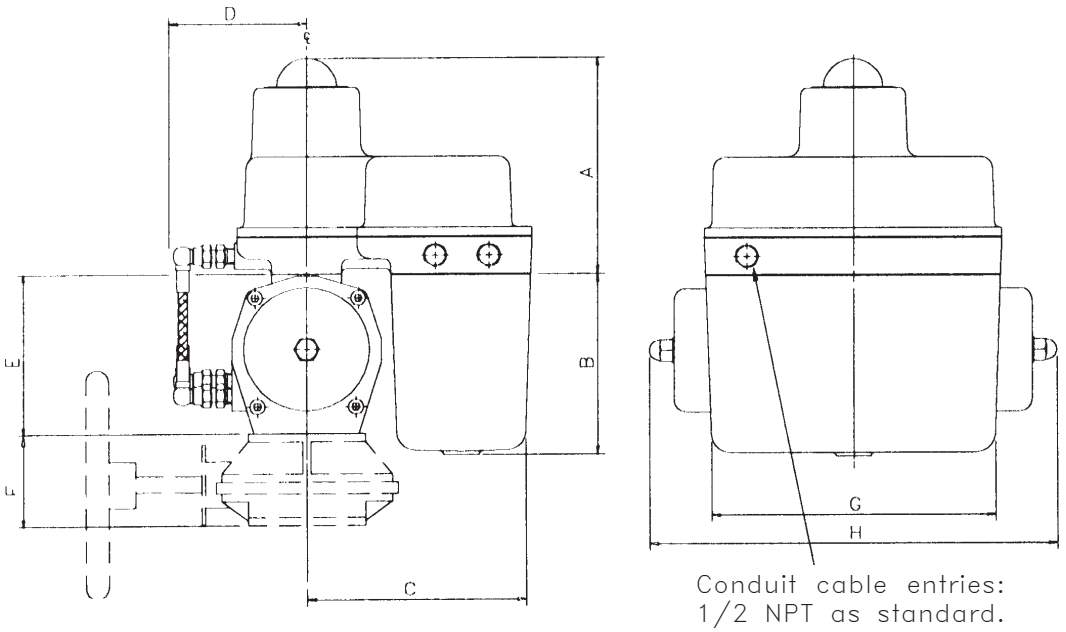








EH-2020 MAJOR DIMENSIONS



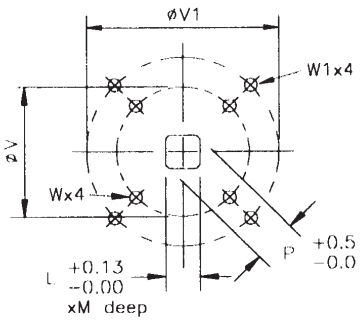


Table 3

	Dimensions - in. (mm)						
	Selection Code A	Selection Code B	Selection Code C	Selection Code D	Selection Code E	Selection Code F	Selection Code G
A	7.2 (185)	7.2 (185)	7.3 (187)	7.3 (187)	8.2 (210)	8.2 (210)	8.2 (210)
B	5.2 (133)	5.2 (133)	6.2 (158)	6.2 (158)	6.2 (158)	6.2 (158)	6.2 (158)
C	6.2 (157)	6.2 (157)	7.6 (193)	7.6 (193)	9.8 (249)	9.8 (249)	9.8 (249)
D	3.7 (95)	3.9 (100)	4.1 (105)	4.5 (115)	5.4 (137)	6.1 (155)	6.6 (168)
E	3.6 (93)	4.1 (105)	4.6 (118)	5.6 (143)	7.1 (181)	8.6 (220)	10.2 (259)
F	2.7 (70)	2.7 (70)	3.5 (89)	3.5 (89)	4.9 (124)	4.9 (124)	4.9 (124)
G	8.6 (220)	8.6 (220)	10.1 (257)	10.1 (257)	10.1 (257)	10.1 (257)	10.1 (257)
H	8 (204)	9.8 (249)	10.5 (267)	14.1 (360)	15.2 (387)	18.7 (477)	20.3 (517)
L	0.55 (14)	0.55 (14)	0.74 (19)	0.86 (22)	1.06 (27)	1.0 (27)	1.4 (36)
M	1.1 (27)	1.2 (30)	1.1 (27)	1.4 (37)	1.4 (37)	1.4 (38)	2 (51)
P	0.7 (18)	0.7 (18)	0.9 (25)	1.1 (28)	1.4 (36)	1.4 (36)	1.4 (36)
V	1.9 (50)	1.9 (50)	1.9 (50)	2.7 (70)	2.7 (70)	4 (102)	4 (102)
V1	2.7 (70)	2.7 (70)	2.7 (70)	4 (102)	4 (102)	4.9 (125)	5.5 (140)
W	1/4-20 x 0.39	1/4-20 x 0.39	1/4-20 x 0.39	5/16-18 x 0.51	5/16-18 x 0.51	3/8-16 x 0.63	3/8-16 x 0.63
W1	5/16-18 x 0.39	5/16-18 x 0.39	5/16-18 x 0.51	3/8-16 x 0.63	3/8-16 x 0.63	1/2-13 x 0.78	1/2-13 x 0.78
Weight lb. (kg)	44 (20)	46 (21)	49 (22)	57 (26)	90 (41)	123 (56)	148 (67)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

EH-2030 ROTARY ELECTRO-HYDRAULIC ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The EH-2030 Series Rotary Electro-Hydraulic actuator produces a high amount of torque in a small package with all electronics built-in. The internal construction assures a leak-proof and explosion-proof system.

It is ideally suited for 90° rotation for both modulating or on-off control of ball valves, butterfly valves, plug valves and dampers. This actuator combines the precision of hydraulic controls with the reliability of spring power for simplicity of operation.

These actuators are available in the same small package with spring return for fail-safe to close or open on loss of power.

The EH-2030 accepts an analog command signal to accurately position the actuator while providing maximum resolution and speed of response. With resolution better than 0.2% of a degree, the actuator offers superior control when compared to conventional electric or pneumatic actuators.



FEATURES

- Torque: 460 to 4513 in. lbs. (52 to 510 Nm)
- Mounting above, below or horizontal to valve
- Explosion-proof enclosure
- Lock-in-place on loss of power
- 90 degree rotation
- Leak-proof hydraulic system
- Very few moving parts
- Internal electronics with easy setup
- Unrestricted continuous duty
- Anti-condensation heater
- 4-20mA command and feedback
- Low maintenance and long-life
- 2 adjustable position switches, 5A, 230 Vac, SPDT
- Automatic torque limiting
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Fast speed emergency shutdown
- Spring fail safe to open or close on loss of power
- Auto/Manual switches
- Manual override
- Remote mounted electronics *(See AD-2000)*

GENERAL SPECIFICATIONS:

Rotation: 90°, set at factory

Input Power: 110, 120 or 230 Vac, 1Ph., 50/60 Hz

Speed & Torque: See Table 1

Temperature Limits:
EH-2030-N: -13°F to 167°F (-25°C to 75°C)
EH-2030-A: -13°F to 131°F (-25°C to 55°C)

Environmental Ratings:
EEx dme IIB T3 Ta -35° C to 60° or
Class I, Zone 1 AEx md IIB T3 Ta -35° C to 60°

Paint Finish: 2-part epoxy, 150 microns thick

Command Signal: 4-20 mA, 20-4 mA, 0-10 Vdc, 10-0 Vdc or On-Off (switch main power)

Feedback Signal: 4-20 mA or 0-10 Vdc

Deadband Adjustment: 0-2.5% of range

Loss of Power: Fails in last position

Modulation Rate: Unrestricted Modulating Duty

Position Accuracy: Less than 0.2%

Repeatability: Less than 0.25%

Adjustable Position Switches: 2, 5A, 230 Vac, SPDT

Automatic Torque Limiting: Explosion-proof pressure switch. CENELEC approved.

Hydraulic Fluid: 10c ST hydraulic fluid

Hydraulic Pump: Two oscillating pumps

Mounting: Vertical or horizontal (See instruction manual)

Cable Entries: 3, 1/2" NPT

Valve Mounting Type: ISO-5211

Weight: See Table 3

Consult factory for wiring information.

EH-2030 SERIES SELECTION CHART

				Selection
1	Basic Model	**EH-2030-N**	On-Off actuator, no amplifier (can use remote electronics to modulate)	
		EH-2030-A	Modulating actuator with amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz		
		2: 230 Vac, 1 Phase, 50/60 Hz		
		16: 110 Vac, 1 Phase, 50/60 Hz		
3	Speed/ Torque	**C**		
		D		
		E	***See Table 1 for Speed/Torque Selection***	
		F		
		G		
4	Rotation	90°		
5	Close/Open *	C: Spring force clockwise, lock-in-place		
		CCW: Spring force counter clockwise, lock-in-place		

* As viewed from actuator output shaft

EH-2030 STANDARD OPTIONS

Code	Description	Selection
	Toggles, Lights	
A017	Local Auto/Manual Toggle Switches, X-Proof	
	Electro-Hydraulic	
N004	Failsafe - Spring return on loss of power	
N006	Quick Failsafe (see Table 2 for speed select)	
N008	Manual override (460 to 1855 in. lbs.)	
N009	Manual override (1855 to 4513 in. lbs.)	
N011	0-10 Vdc Command	
N015	Mounting direction, if not above valve	
N016	Silicone oil for low temperature operation, -40° F (-40° C)	
N017	Valve mounting kit	
N025	M20 threaded cable entry	
N026	Bio-degradable fluid for food/beverage	

This table should be used to select the torque and speed of the actuator. Actuator has two torques, one for each direction of the shaft. These torques are start and finish torque with required operating torque typically using Spring Stroke Direction/Finish.

Selection Code	Hydraulic Stroke Direction			Spring Stroke Direction		
	Speed in Sec.	Torque in-lb (Nm)		Speed in Sec.	Torque in-lb (Nm)	
		Start	*Finish*		*Start*	*Finish*
C	18	700(79)	360(41)	7	740(84)	460(52)
D	38	1,500(170)	750(85)	15	1,645(186)	1,035(117)
E	66	2,475(280)	1,240(140)	25	2,920(330)	1,855(210)
F	120	4,390(496)	2,175(246)	35	4,903(554)	3,080(348)
G	165	6645(751)	3,285(371)	58	7,300(825)	4,513(510)

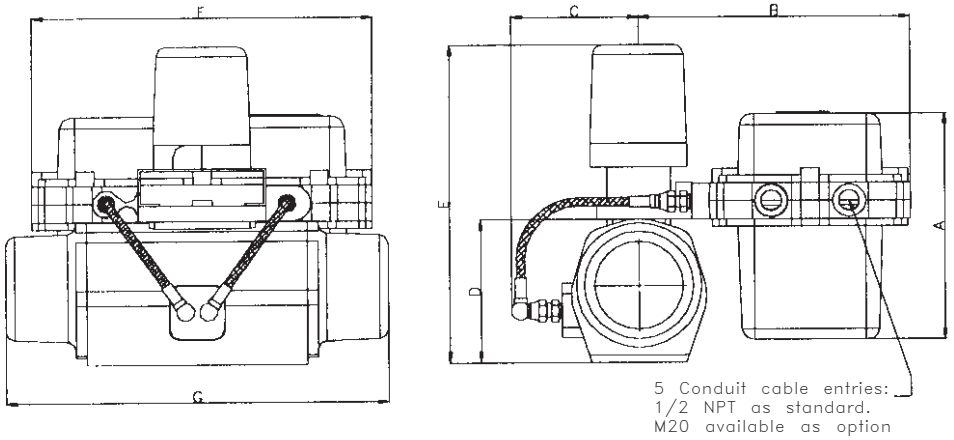
Table 1

This table should be used when selecting option N006.

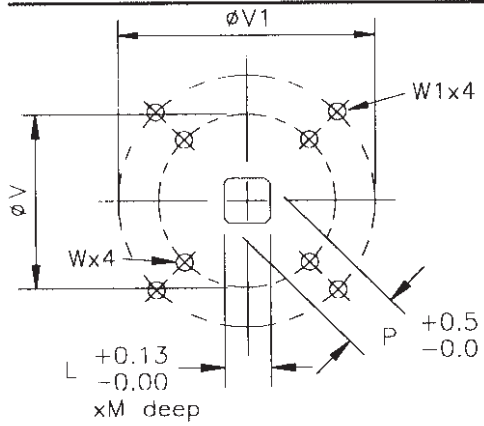
Speed/Torque Code	Speed in Sec.
C	1.5
D	2.5
E	4
F	5
G	9

Table 2

EH-2030 MAJOR DIMENSIONS



ACTUATOR MOUNTING DETAILS



FLANGE & SQUARE DRIVE TO ISO 5211

Table 3

	Dimensions - in. (mm)				
	Selection Code C	Selection Code D	Selection Code E	Selection Code F	Selection Code G
A	8.7 (221)				
B	10.4 (265)	10.8 (275)	11.7 (298)	12.4 (315)	12.9 (327)
C	4.4 (113)	4.8 (123)	5.7 (145)	6.4 (163)	6.9 (176)
D	4.6 (118)	5.6 (143)	7.1 (181)	8.6 (220)	10.2 (259)
E	11 (281)	12 (306)	13.5 (344)	15.1 (383)	16.6 (422)
F	12.6 (322)				
G	10.5 (267)	14.1 (360)	15.2 (387)	18.8 (477)	20.3 (517)
L	0.74 (19)	0.86 (22)	1.06 (27)	1.0 (27)	1.4 (36)
M	1.1 (27)	1.4 (37)	1.4 (37)	1.4 (38)	2 (51)
P	0.9 (25)	1.1 (28)	1.4 (36)	1.4 (36)	1.9 (48)
V	1.9 (50)	2.7 (70)	2.7 (70)	4 (102)	4 (102)
V1	2.7 (70)	4 (102)	4 (102)	4.9 (125)	5.5 (140)
W	1/4-20 x 0.39	5/16-18 x 0.51	5/16-18 x 0.51	3/8-16 x 0.63	3/8-16 x 0.63
W1	5/16-18 x 0.51	3/8-16 x 0.63	3/8-16 x 0.63	1/2-13 x 0.78	1/2-13 x 0.78
Weight lb. (kg)	49 (22)	57 (26)	75 (34)	101 (46)	126 (57)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

EH-2040 ROTARY ELECTRO-HYDRAULIC ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The EH-2040 Series Rotary Electro-Hydraulic actuator produces a high amount of torque in a small package with all electronics built-in. The internal construction makes this a leak-proof system.

It is ideally suited for 90° rotation for both modulating or on-off control of ball valves, butterfly valves, plug valves and dampers. This actuator combines the precision of hydraulic controls with the reliability of spring power for simplicity of operation.

These actuators are available in the same small package with spring return for fail-safe to close or open on loss of power.

The EH-2040 accepts an analog command signal to accurately position the actuator while providing maximum resolution and speed of response. With resolution better than 0.2% of a degree, the actuator offers superior control when compared to conventional electric or pneumatic actuators.

FEATURES

- Torque: 376 to 5,456 ft. lbs. (510 to 7,397 Nm)
- Mounting above, below or horizontal to valve
- NEMA 4 (IP65) enclosure
- Lock-in-place on loss of power
- 90 degree rotation
- Leak-proof hydraulic system
- Internal electronics with easy setup
- Unrestricted continuous duty
- 4-20mA command and feedback
- Low maintenance and long-life
- 2 adjustable position switches, 5A, 230 Vac, SPDT
- Automatic torque limiting
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Fast speed emergency shutdown
- Spring fail safe to open or close on loss of power
- Auto/Manual switches
- Anti-condensation heater
- Manual override
- Explosion proof - FM approved
- Split range command input
- Remote mounted electronics *(See AD-2000)*



GENERAL SPECIFICATIONS:

Rotation: 90°, set at factory
Input Power: 110 or 230 Vac, 1Ph., 50 Hz
120 Vac, 1Ph., 60 Hz.
Speed & Torque: See Table 1
Power Consumption:
Running: 350VA (220 watts) max.
Null: 43VA (20 watts)
Temperature Limits:
EH-2040-N: -13°F to 167°F (-25°C to 75°C)
EH-2040-A: -13°F to 131°F (-25°C to 55°C)
Environmental Ratings: NEMA 4 (IP65); Optional explosion-proof Class I Div. 1 Groups C&D, Class II Div. 1 Groups E, F & G, FM approved.
Paint Finish: 2-part epoxy, 150 microns thick
Enclosure:
Power Unit: Cast aluminum BS1490 Grade Lm27 (ASTM B179 Grade 328.1)
Actuator housing & piston: Carbon Steel ASTM A-283 Gr. D
Command Signal: 4-20 mA, 20-4 mA, 0-10 Vdc, 10-0 Vdc or On-Off (switch main power)
Feedback Signal: 4-20 mA or 0-10 Vdc
Deadband Adjustment: 0-2.5% of range
Loss of Power: Fails in last position
Modulation Rate: Unrestricted Modulating Duty
Position Accuracy: Less than 0.2% of a degree
Repeatability: Less than 0.25%
Adjust. Position Switches: 2, 5 A, 230 Vac, SPDT
Hydraulic Fluid: 10c ST hydraulic fluid
Hydraulic Pump: Two oscillating pumps
Mounting: Vertical (option for horizontal)
Cable Entries: 3, 1/2" NPT
Deviation Alarm: If actuator hunts for null position for a programmable time (30 sec. to 10 minutes), a 250 Vac, 1 A, SPST output is activated.
Weight: 321 to 420 lbs. (146 to 191 kg)

Consult factory for wiring information.

EH-2040 SERIES SELECTION CHART

				Selection
1	Basic Model	**EH-2040-N**	On-Off actuator, no amplifier, (can use remote electronics to modulate)	
		EH-2040-A	Modulating actuator with amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 60 Hz		
		2: 110/230 Vac, 1 Phase, 50 Hz		
3	Speed/ Torque	**A**		
		B		
		C	***See Table 1 for Speed/Torque Selection***	
		D		
		E		
		F		
		G		
4	Rotation	90°		
5	Close/Open *	C: Spring force clockwise, lock-in-place		
		CCW: Spring force counter clockwise, lock-in-place		

* As viewed from actuator output shaft

EH-2040 STANDARD OPTIONS

Code	Description	Selection
	Toggles, Lights	
A001	Local Auto/Manual Toggle Switches, NEMA 4	
A017	Local Auto/Manual Toggle Switches, X-Proof	
	Heater	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
	Electro-Hydraulic	
N004	Failsafe - Spring return on loss of power	
N006	Quick speed ESD failsafe	
N011	0-10 Vdc Command	
N015	Mounting direction, if not above valve	
N016	Silicone oil for low temperature operation, -40° F (-40° C)	
N017	Valve mounting kit	
N024	Explosion-proof, Class 1, Zone 1 Aex md IIBT3Ta	
N026	Bio-degradable fluid for food/beverage	
N027	Quick speed ESD failsafe - explosion proof	
N030	Geared manual overdrive, Code A	
N031	Geared manual overdrive, Code B	
N034	Split range command input	
N036	Geared manual overdrive, Code C	

This table should be used to select the torque and speed of the actuator. Actuator has two torques, one for each direction of the shaft. These torques are start and finish torque with required operating torque typically using Spring Stroke Direction/Finish.

Selection Code	Hydraulic Stroke Direction			Spring Stroke Direction		
	Speed in Sec.	Torque ft-lb (Nm)		Speed in Sec.	Torque ft-lb (Nm)	
		Start	*Finish*		*Start*	*Finish*
A	39	554 (751)	273 (371)	39	608 (825)	376 (510)
B	60	911 (1,235)	472 (640)	50	1,047 (1,420)	661 (897)
C	80	1,566 (2124)	1,060 (1,438)	60	1,364 (1,849)	855 (1,160)
D	82	1,331 (1,805)	609 (826)	82	1,923 (2,607)	1,336 (1,812)
E	152	3,261 (4,422)	2,102 (2,850)	152	2,817 (3,820)	1,756 (2,381)
F	255	5,221 (7,079)	2,711 (3,676)	255	5,519 (7,483)	3,321 (4,504)
G	331	5,765 (7,816)	3,742 (5,073)	331	7,110 (9,640)	5,456 (7,397)

Table 1

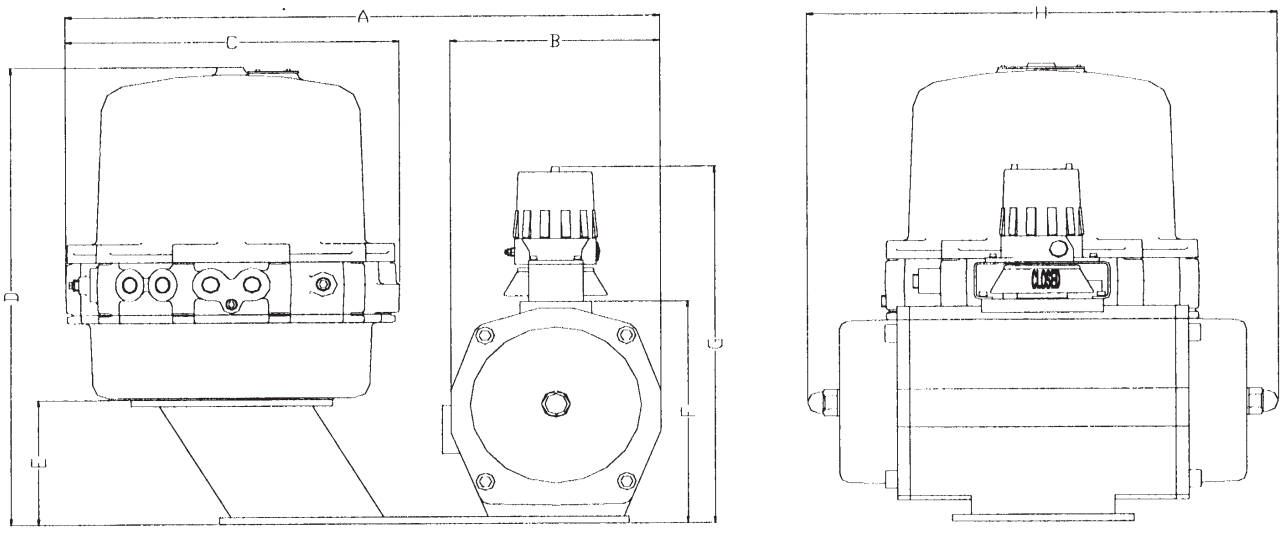
This table should be used when selecting option N006 & N027.

Speed/Torque Code	Speed (in./sec.)
A	N/A
B	8
C	8
D	12
E	20
F	40
G	50

Table 2

EH-2040 MAJOR DIMENSIONS

(for selection codes A, B & C)

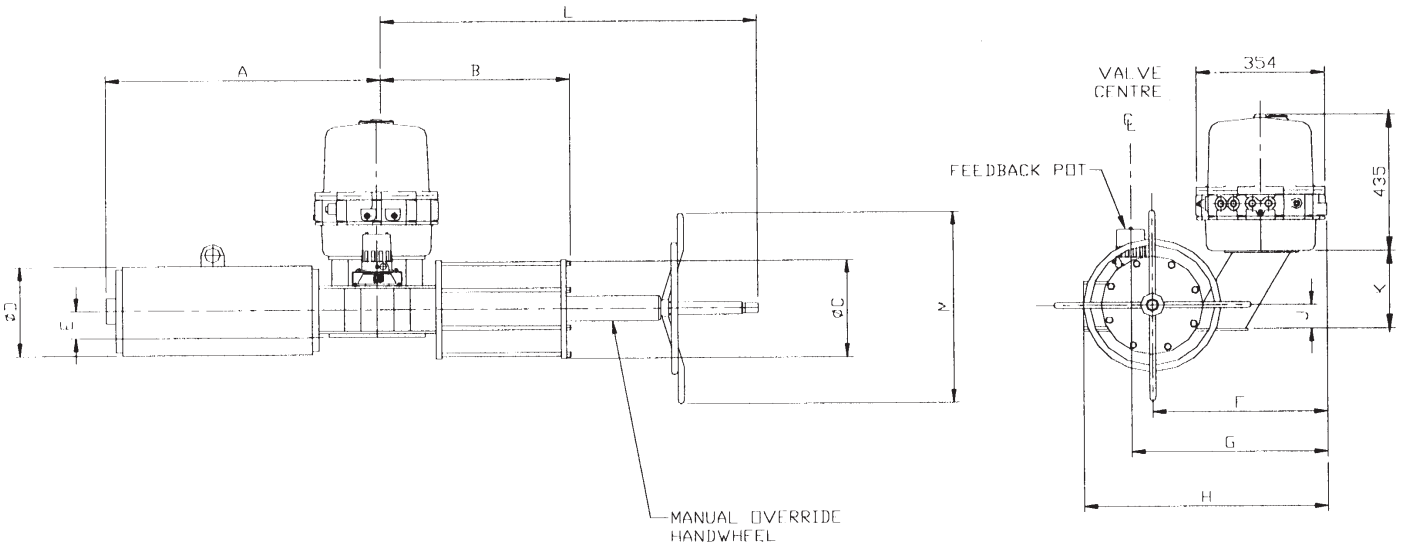


Dimensions - in. (mm)			
	Selection Code A	Selection Code B	Selection Code C
A	25.6 (652)	27 (686)	30.3 (771)
B	9.1 (231)	10.4 (265)	13.7 (350)
C	13.9 (354)	13.9 (354)	13.9 (354)
D	21.6 (550)	21.6 (550)	21.6 (550)
E	5.9 (150)	5.9 (150)	5.9 (150)
F	10.2 (259)	11.7 (297)	14 (356)
G	16.9 (429)	18.3 (467)	20.7 (526)
H	20.3 (517)	25.1 (637)	22.4 (570)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

EH-2040 MAJOR DIMENSIONS

(for selection codes E, F, G & H)



Dimensions - in. (mm)				
	Selection Code A	Selection Code B	Selection Code C	Selection Code D
A	30.7 (780)	34.6 (880)	41.5 (1055)	66.3 (1685)
B	26.7 (680)	28 (710)	27.8 (705)	27.8 (705)
C	12.2 (310)	13.9 (355)	16.3 (415)	16.3 (415)
D	12.8 (325)	12.8 (325)	16.1 (410)	16.1 (410)
E	3.5 (90)	3.5 (90)	3.9 (100)	5.1 (130)
F	218 (554)	22.6 (576)	23.8 (606)	23.8 (606)
G	24.3 (619)	26 (661)	27.8 (706)	29 (736)
H	27.8 (707)	28.9 (734)	31.7 (806)	34.1 (866)
J	3.1 (80)	3.1 (80)	3.7 (93)	4 (102)
K	9.6 (245)	10.5 (268)	10.5 (268)	10.5 (268)
L	44.9 (140)	46 (1170)	59 (1500)	59 (1500)
M	23.6 (600)	23.6 (600)	23.6 (600)	23.6 (600)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

EH-2110 LINEAR ELECTRO-HYDRAULIC ACTUATORS

GENERAL DESCRIPTION

The EH-2110 Series Linear Electro-Hydraulic actuator produces a high amount of thrust in a small package, with all electronics built-in. The design of the internal components assures a leak-proof system.

This actuator is ideally suited for modulating or on-off control of all types of globe valves.

These actuators combine the precision of hydraulic controls with the reliability of spring power for simplicity of operation.

They are available in the same small package with spring return for fail-safe to close or open on loss of power.

The EH-2110 Series accepts an analog command signal to accurately position the actuator while providing maximum resolution and speed of response. With resolution better than 0.25%, these actuators offer superior control when compared to conventional electric or pneumatic actuators.

FEATURES

- Thrusts from 150 to 300 lbf. (667 to 1334 N)
- NEMA 4 enclosure
- Lock-in-place on loss of power
- Stroke distance up to 1.625 inches (42 mm)
- Leak-proof hydraulic system
- Internal electronics with easy setup
- Unrestricted continuous duty
- 4-20mA command and feedback
- Very few moving parts
- Low maintenance and long-life
- Low power consumption
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Spring fail safe for emergency shutdown
- Manual override
- Low temperature
- 1 or 2 adjustable position switches, 5A, 230 Vac, SPDT
- Local Auto/Manual Switches
- Bio-degradable oil for food & beverage applications
- Anti-condensation heater



BASIC SPECIFICATIONS

Stroke Length: Up to 1.625 in. (42 mm)

Input Power: 110, 120 or 230 Vac, 1Ph., 50/60 Hz

Power Consumption:
Operating: 160 VA (81 Watts) Max.
Null: 15 VA (6 Watts) Max.

Speed & Thrust: See Table 1

Temperature Limits:
EH-2110-N: -13°F to 167°F (-25°C to 75°C)
EH-2110-A: -13°F to 131°F (-25°C to 55°C)

Enclosure: Cast aluminum B.S. 1490 Grade LM27 (ASTM. B179 Grade 328.1)

Environmental Ratings: NEMA 4 (IP65)

Paint Finish: 2-part epoxy, 150 microns thick

Command Signal: 4-20 mA or On-Off (switch main power). *Options*: 20-4 mA, 0-10 Vdc, 10-0 Vdc

Feedback Signal: 4-20 mA

Deadband Adjustment: 0-2.5% of range

Modulation Rate: Unrestricted Modulating Duty

Resolution: Less than 0.25%

Repeatability: Less than 0.1%

Failure Mode: Fail in last position

Hydraulic Fluid: 10c ST hydraulic fluid

Hydraulic Pump: Oscillating pumps

Mounting: Vertical or horizontal

Cable Entry: 2, 1/2" NPT

Weight: 30 lbs. (14 kg)

Consult factory for wiring information.

EH-2110 SERIES SELECTION CHART

				Selection
1	Basic Model	**EH-2110-N**	On-Off actuator, no amplifier (can use remote electronics to modulate)	
		EH-2110-A	Modulating actuator with amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz		
		2: 230 Vac, 1 Phase, 50/60 Hz		
		16: 110 Vac, 1 Phase, 50/60 Hz		
3	Speed/Thrust	**A**		
		B	***See Table 1 for Speed/Thrust Selection***	
		C		
4	Linear Travel	**Specify**, 0.5 to 1.625 in. (12.7 to 41.3 mm)		
5	Close/Open *	**E**: Spring to extend, lock-in-place		
		R: Spring to retract, lock-in-place		

* As viewed from actuator output shaft

EH-2110 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches, NEMA 4	
Heater		
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	

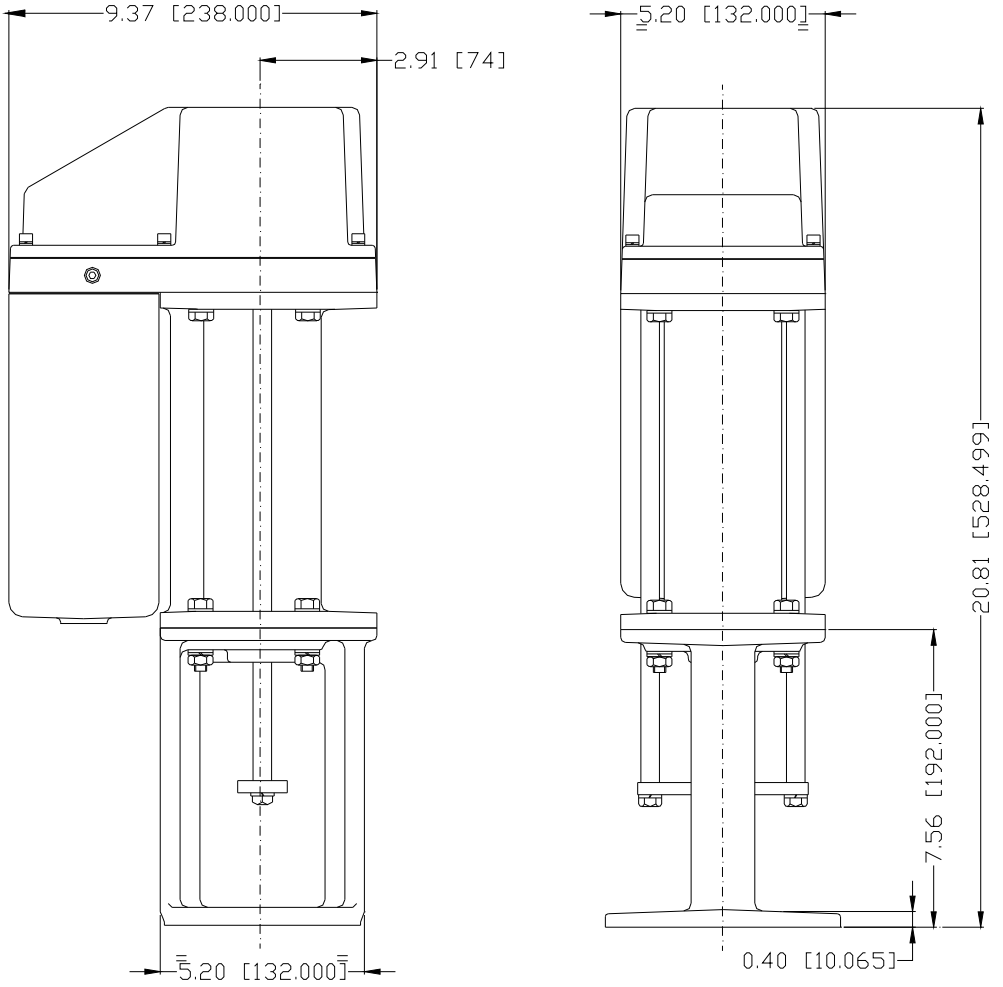
Code	Description	Selection
Electro-Hydraulic		
N011	0-10 Vdc Command	
N013	Failsafe - Spring return on loss of power	
N015	Mounting direction, if not above valve	
N016	Silicone oil for low temperature operation, -40° F (-40° C)	
N017	Valve mounting kit	
N025	M20 threaded cable entry	
N026	Bio-degradable fluid for food/beverage	
N032	Overthrust protection using a pressure switch	
N035	Manual override	
Auxiliary Switches		
S022	2 adjust. position switches, 5A, 230 Vac, SPDT	
S023	1 adjust. position switch, 5A, 230 Vac, SPDT	

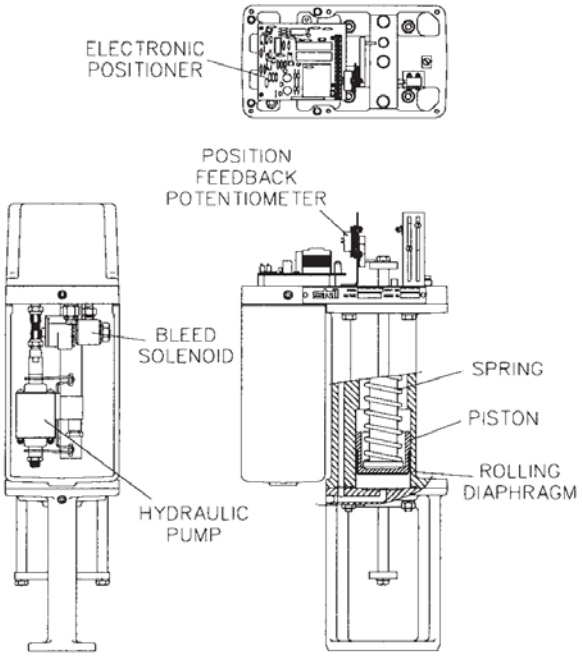
This table should be used to select the thrust and speed of the actuator. Actuator has two thrusts for each direction of the shaft. These thrusts are start and finish thrust with operating thrust typically is Spring Stroke Direction Finish of Stroke.

Selection Code	Hydraulic Stroke Direction			Spring Stroke Direction		
	Speed Sec./inch	Thrust in lbf (N)		Speed Sec./inch	Thrust in lbf (N)	
		Start	*Finish*		*Start*	*Finish*
A	11	440(1957)	384(1708)	1.85	206(916)	150(667)
B	11	290(1289)	150(667)	1.85	440(1957)	300(1334)
C	11	340(1512)	223(992)	1.85	367(1632)	250(1112)

Table 1

EH-2110 MAJOR DIMENSIONS





These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

EH-2130 LINEAR ELECTRO-HYDRAULIC ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The EH-2130 Series Linear Electro-Hydraulic actuator produces a high amount of thrust in a small package, with all electronics built-in. The internal construction makes this a leak-proof system.

This actuator is ideally suited for modulating or on-off control of all types of globe valves.

These actuators combine the precision of hydraulic controls with the reliability of spring power for simplicity of operation.

They are available in the same small package with spring return for fail-safe to close or open on loss of power.

The EH-2130 Series accepts an analog command signal to accurately position the actuator while providing maximum resolution and speed of response. With resolution better than 0.25%, these actuators offer superior control when compared to conventional electric or pneumatic actuators.

FEATURES

- Thrusts from 400 to 2,300 lbf. (1,760 to 10,100 N)
- NEMA 4 enclosure
- Lock-in-place on loss of power
- Stroke distance up to 2.5 inches (64 mm)
- Leak-proof hydraulic system
- Internal electronics with easy setup
- Split range command input
- 2 adjustable position switches, 5A, 230 Vac, SPDT
- 4-20mA command and feedback
- Unrestricted continuous duty
- Low maintenance and long-life
- Low power consumption
- Anti-condensation heater
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Spring fail safe for emergency shutdown
- Explosion-proof
- Manual override
- Low temperature
- Remote mounted electronics *(See AD-2000)*
- Automatic thrust limiting



BASIC SPECIFICATIONS

Stroke Length: Up to 2.5 in. (64mm)
Input Power: 110, 120 or 230 Vac, 1Ph., 50/60 Hz
Power Consumption:
Operating: 305 VA (156 Watts) Max.
Null: 15 VA (6 Watts) Max.
Speed & Thrust: See Table 1
Temperature Limits:
EH-2130-N: -13°F to 167°F (-25°C to 75°C)
EH-2130-A: -13°F to 131°F (-25°C to 55°C)
Enclosure: Cast aluminum (ASTM. B179 Grade 328.1)
Environmental Ratings: NEMA 4 (IP65),
Option: Explosion proof Class 1, Zone 1 AEx md IIB T3 Ta
Paint Finish: 2 part epoxy, 150 microns thick
Command Signal: 4-20 mA or On-Off (switch main power). *Options*: 20-4 mA, 0-10 Vdc, 10-0 Vdc
Feedback Signal: 4-20 mA or 0-10 Vdc
Deadband Adjustment: 0-2.5% of range
Modulation Rate: Unrestricted Modulating Duty
Resolution: Less than 0.25%
Repeatability: Less than 0.25%
Adjustable Position Switches: 1, 5 A, 230 Vac, SPDT
Failure Mode: Fail in last position
Hydraulic Fluid: 10c ST hydraulic fluid
Hydraulic Pump: Oscillating pumps, dual redundant hydraulic pumps
Approvals: CE Mark
Conformance Directives: Low Voltage - 73/23/EEC
EMC - 89/336/EEC, 92/31/EEC & 95/54/EC
Machinery - 98/37/EC; Pressure Equipment - 97/23/EC
Mounting: Vertical or horizontal
Cable Entry: 3, 1/2" NPT
Weight: See table 3

Consult factory for wiring information.

EH-2130 SERIES SELECTION CHART

				Selection
1	Basic Model	**EH-2130-N**	On-Off actuator, no amplifier (can use remote electronics to modulate)	
		EH-2130-A	Modulating actuator with amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz		
		2: 230 Vac, 1 Phase, 50/60 Hz		
		16: 110 Vac, 1 Phase, 50/60 Hz		
3	Speed/Thrust	**A**		
		B	***See Table 1 for Speed/Thrust Selection***	
		C		
		D		
		E		
4	Linear Travel	**Specify**, 0.5 to 2.5 in. (12.7 to 64 mm)		
5	Close/Open *	**E**: Spring to extend, lock-in-place		
		R: Spring to retract, lock-in-place		

* As viewed from actuator output shaft

EH-2130 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches, NEMA 4	
A017	Local Auto/Manual Toggle Switches, X-Proof	
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	

Code	Description	Selection
Electro-Hydraulic		
N011	0-10 Vdc Command	
N013	Failsafe - spring return on loss of power	
N015	Mounting direction, if not above valve	
N016	Silicone oil for low temperature operation, -40° F (-40° C)	
N017	Valve mounting kit	
N018	Manual override hand pump, NEMA 4	
N024	Explosion-proof, Class 1, Zone 1 Aex md IIBT3Ta	
N024	Explosion proof enclosure, -31° F - 140° F (-35° C - 60° C)	
N025	M20 threaded cable entry	
N026	Bio-degradable fluid	
N027	Quick speed ESD failsafe - Explosion proof	
N028	Overthrust protection using a pressure switch - NEMA 4 & Explosion-proof	
N039	Manual override hand pump, Explosion-proof	

This table should be used to select the thrust and speed of the actuator. Actuator has two thrusts for each direction of the shaft. These thrusts are start and finish thrust with operating thrust typically is Spring Stroke Direction Finish of Stroke.

Selection Code	Hydraulic Stroke Direction			Spring Stroke Direction		
	Speed Sec./inch	Thrust in lbf (N)		Speed Sec./inch	Thrust in lbf (N)	
		Start	*Finish*		*Start*	*Finish*
A	7.2	440(1,940)	250(1,100)	3	590(2,600)	400(1,760)
B	12.8	700(3,100)	450(1,980)	6	850(3,700)	600(2,640)
C	19.2	1,100(4,840)	700(3,980)	10	2,250(9,900)	1,000(4,400)
D	28.8	2,000(8,800)	1,000(4,400)	12	2,500(11,000)	1,500(6,600)
E	47	3,200(14,100)	1,700(7,500)	20	3,800(16,700)	2,300(10,100)

Table 1

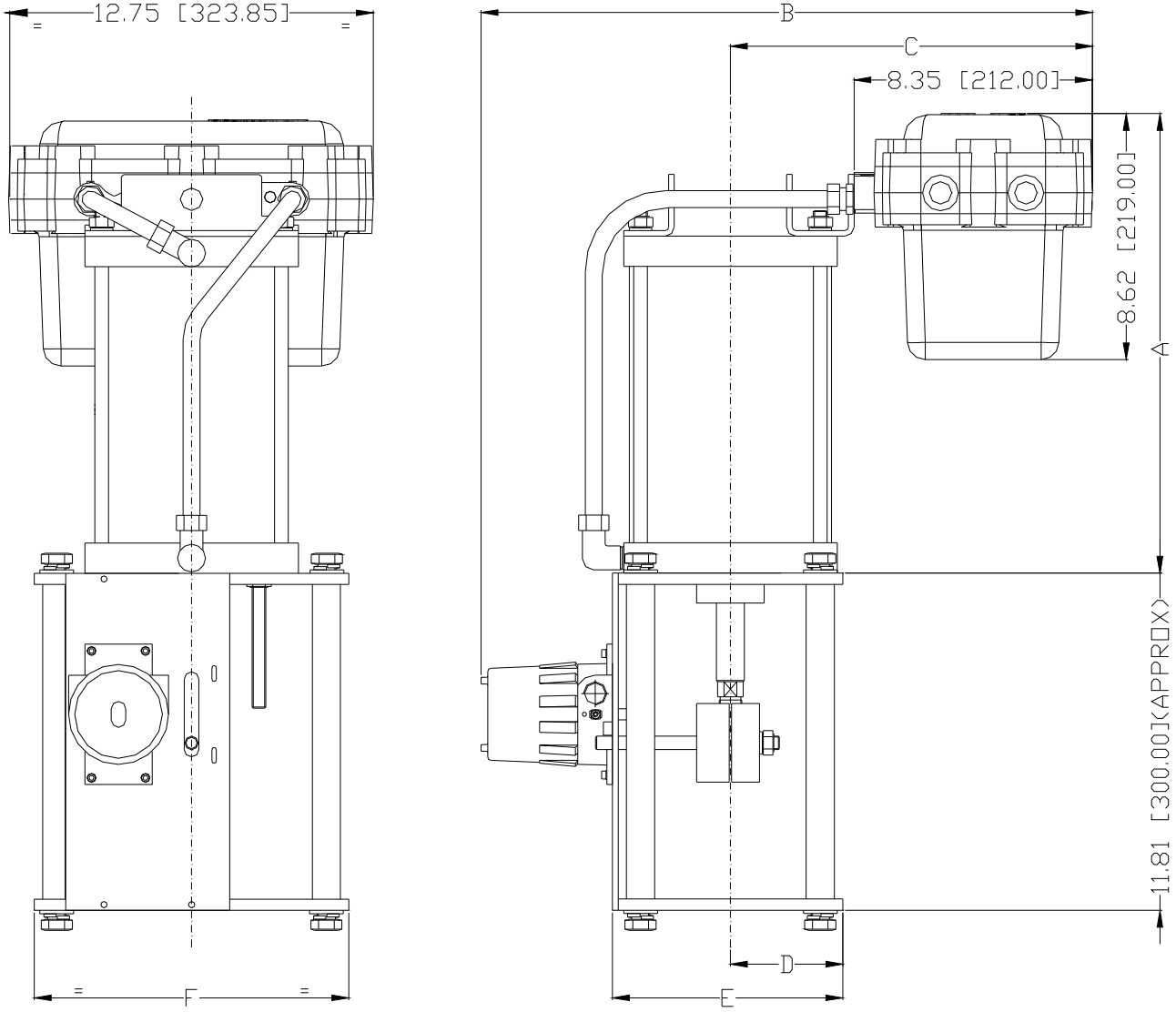
This table should be used when selecting option N027.

Speed/Torque Code	Speed (sec./in.)
A	0.75
B	1.5
C	2.5
D	3
E	5

Table 2

EH-2130 MAJOR DIMENSIONS

INCHES (MILLIMETERS)



	Dimensions - in. (mm)				
	Selection Code A	Selection Code B	Selection Code C	Selection Code D	Selection Code E
A	15.4 (395)	13 (330)	13 (330)	13 (330)	13 (330)
B	19.5 (495)	16.9 (430)	16.9 (430)	16.9 (430)	16.9 (430)
C	11.8 (300)	11.8 (300)	11.8 (300)	11.8 (300)	11.8 (300)
D	6.5 (165)	7.5 (190)	8.2 (210)	8.2 (210)	10 (255)
E	10.6 (270)	11.6 (295)	12.4 (315)	12.4 (315)	14.1 (360)
F	3.5 (90)	5.5 (140)	7.1 (180)	7.1 (180)	10.6 (270)
Weight lb (kg)	121 (55)	165 (75)	187 (85)	220 (100)	287 (130)

Table 3

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

EH-2140 LINEAR ELECTRO-HYDRAULIC ACTUATORS

Jordan CONTROLS

GENERAL DESCRIPTION

The EH-2140 Series Linear Electro-Hydraulic actuator produces a high amount of thrust in a small package, with all electronics built-in. The internal construction makes this a leak-proof system.

This actuator is ideally suited for modulating or on-off control of all types of globe valves.

These actuators combine the precision of hydraulic controls with the reliability of spring power for simplicity of operation.

They are available in the same small package with spring return for fail-safe to close or open on loss of power.

The EH-2140 Series accepts an analog command signal to accurately position the actuator while providing maximum resolution and speed of response. With resolution better than 0.25%, these actuators offer superior control when compared to conventional electric or pneumatic actuators.

FEATURES

- Thrusts from 2,000 to 16,000 lbf. (8,800 to 70,400 N)
- NEMA 4 enclosure
- Lock-in-place on loss of power
- Stroke distance up to 4 inches (102 mm)
- Leak-proof hydraulic system
- Internal electronics with easy setup
- 4-20mA command and feedback
- Unrestricted continuous duty
- Low maintenance and long-life
- Low power consumption
- 1 adjustable position switches, 5A, 230 Vac, SPDT
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Spring fail safe for emergency shutdown
- Explosion-proof
- Manual override
- Low temperature
- Remote mounted electronics *(See AD-2000)*
- Bio-degradable oil for food & beverage applications
- Anti-condensation heater
- Automatic thrust limiting
- Split range command input



BASIC SPECIFICATIONS

Stroke Length: Up to 4 in. (102 mm)
Input Power: 110 or 230 Vac, 1Ph., 50 Hz
120 Vac, 1Ph., 60 Hz
Power Consumption:
Operating: 350 VA (220 Watts) Max.
Null: 43 VA (20 Watts) Max.
Speed & Thrust: See Table 1
Temperature Limits:
EH-2140-N: -13°F to 167°F (-25°C to 75°C)
EH-2140-A: -13°F to 131°F (-25°C to 55°C)
Enclosure: Cast aluminum (ASTM. B179 Grade 328.1)
Environmental Ratings: NEMA 4 (IP65)
Option:
Explosion proof - FM approved, Class I, Div 1, Groups C&D, Class II, Div 1, Groups E, F & G
Paint Finish: 2 part epoxy, 150 microns thick
Command Signal: 4-20 mA or On-Off (switch main power). *Options*: 20-4 mA, 0-10 Vdc, 10-0 Vdc
Feedback Signal: 4-20 mA or 0-10 Vdc
Deadband Adjustment: 0-2.5% of range
Modulation Rate: Unrestricted Modulating Duty
Resolution: Less than 0.25%
Repeatability: Less than 0.25%
Adjustable Position Switches: 1, 5 A, 230 Vac, SPDT
Hydraulic Fluid: 10c ST hydraulic fluid
Hydraulic Pump: Motorized vane pump
Mounting: Vertical or horizontal
Approvals: CE Mark
Cable Entry: 3, 1/2" NPT
Deviation Alarm: If actuator hunts for null position for a programmable time (30 sec. to 10 minutes), a 250 Vac, 1 A, SPST output is activated.
Weight: See table 3

Consult factory for wiring information.

EH-2140 SERIES SELECTION CHART

				Selection
1	Basic Model	**EH-2140-N**	On-Off actuator, no amplifier (can use remote electronics to modulate)	
		EH-2140-A	Modulating actuator with amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz		
		2: 230 Vac, 1 Phase, 50/60 Hz		
		16: 110 Vac, 1 Phase, 50/60 Hz		
3	Speed/Thrust	**A**		
		B		
		C		
		D	***See Table 1 for Speed/Thrust Selection***	
		E		
		F		
		G		
4	Linear Travel	**Specify, 0.5 to 4 in. (12.7 to 102 mm)**		
5	Close/Open *	**E**: Spring to extend, lock-in-place		
		R: Spring to retract, lock-in-place		

* As viewed from actuator output shaft

EH-2140 STANDARD OPTIONS

Code	Description	Selection
Toggles, Lights		
A001	Local Auto/Manual Toggle Switches, NEMA 4	
A017	Local Auto/Manual Toggle Switches, X-Proof	
Heater		
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Valve Mounting		
L001	Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.	
L002	Valve provided by customer for mounting and set-up with actuator by Jordan Controls.	
L003	Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.	

Code	Description	Selection
Electro-Hydraulic		
N006	Quick speed ESD failsafe - NEMA 4	
N011	0-10 Vdc Command	
N013	Failsafe - spring return on loss of power	
N015	Mounting direction, if not above valve	
N016	Silicone oil for low temperature operation, -40° F (-40° C)	
N017	Valve mounting kit	
N018	Manual override hand pump	
N024	Explosion-proof, Class 1, Zone 1 Aex md IIBT3Ta	
N025	M20 threaded cable entry	
N026	Bio-degradable fluid	
N027	Quick speed ESD failsafe - Explosion-proof	
N028	Overthrust protection using a pressure switch - NEMA 4 & Explosion-proof	
N030	4-12, 12-20 mA command input	

This table should be used to select the thrust and speed of the actuator. Actuator has two thrusts for each direction of the shaft. These thrusts are start and finish thrust with operating thrust typically Spring Stroke Direction Finish of Stroke.

Selection Code	Speed Sec./inch	Hydraulic Stroke Direction Thrust in lbf (N)		Spring Stroke Direction Thrust in lbf (N)	
		Start	***Finish***	***Start***	***Finish***
A	9.1	4,500(19,800)	2,200(9,680)	4,300(18,900)	2,000(8,800)
B	14.3	6,800(30,000)	2,800(12,300)	7,500(33,000)	3,500(15,400)
C	23.4	12,000(53,000)	6,000(26,000)	11,000(48,000)	5,000(22,000)
D	23.4	10,000(44,000)	2,000(8,800)	15,000(66,000)	7,000(30,800)
E	28.6	10,800(47,500)	2,000(8,800)	18,000(79,200)	10,000(44,000)
F	37.7	14,500(63,800)	2,500(11,000)	25,000(110,000)	13,000(57,200)
G	47.7	19,000(83,600)	3,000(13,200)	32,000(141,000)	16,000(70,400)

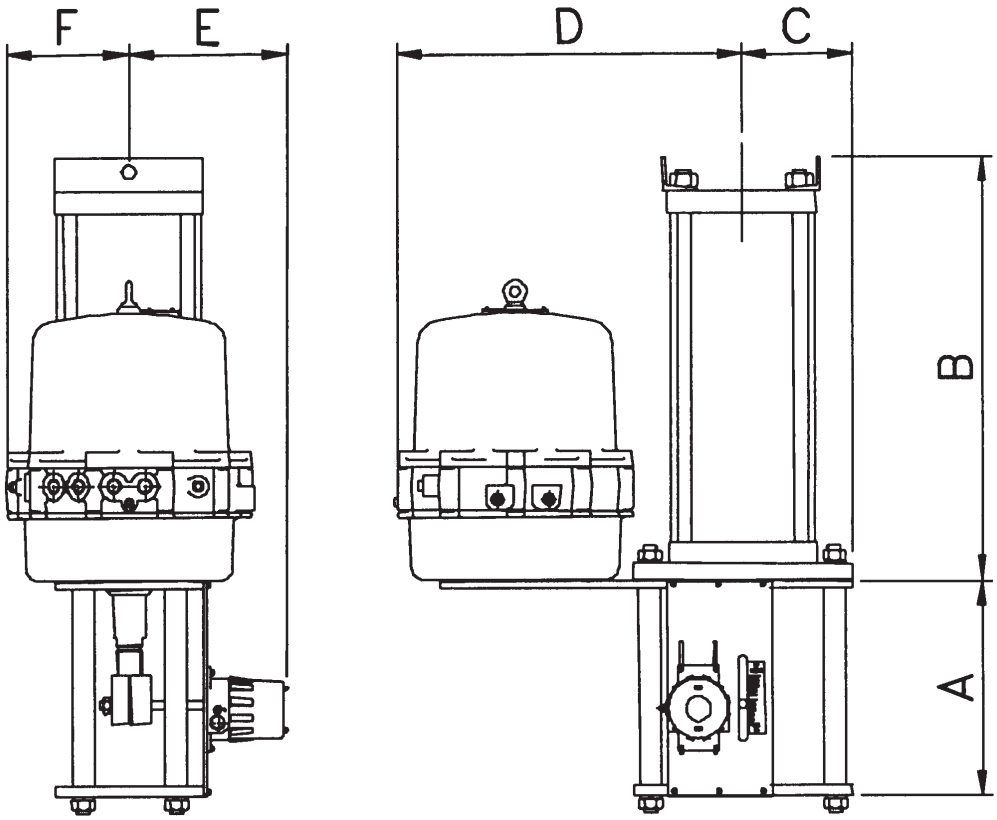
Table 1

This table should be used when selecting option N006 & N027.

Speed/Torque Code	Speed (sec./in.)
A	2.3
B	3.6
C	5.85
D	5.85
E	7.15
F	9.4
G	12

Table 2

EH-2140 MAJOR DIMENSIONS



	Dimensions - in. (mm)						
	Selection Code A	Selection Code B	Selection Code C	Selection Code D	Selection Code E	Selection Code F	Selection Code G
A	15.7 (400)	15.7 (400)	15.7 (400)	15.7 (400)	15.7 (400)	15.7 (400)	15.7 (400)
B	24.4 (620)	27.1 (690)	28.3 (720)	31.5 (800)	39.3 (1000)	39.3 (1000)	43.3 (1100)
C	6.3 (160)	7.5 (190)	9.1 (230)	9.1 (230)	9.1 (230)	9.1 (230)	9.1 (230)
D	20.1 (510)	21.6 (550)	23.6 (600)	23.6 (600)	23.6 (600)	23.6 (600)	23.6 (600)
E	9.1 (230)	9.8 (250)	11.4 (290)	11.4 (290)	11.4 (290)	11.4 (290)	11.4 (290)
F	7.1 (180)	7.1 (180)	7.1 (180)	7.1 (180)	7.1 (180)	7.1 (180)	7.1 (180)
Weight lb. (kg)	397 (180)	507 (230)	551 (250)	573 (260)	639 (290)	705 (320)	771 (350)

Table 3

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

AD-2000 SERIES REMOTE AMPLIFIERS

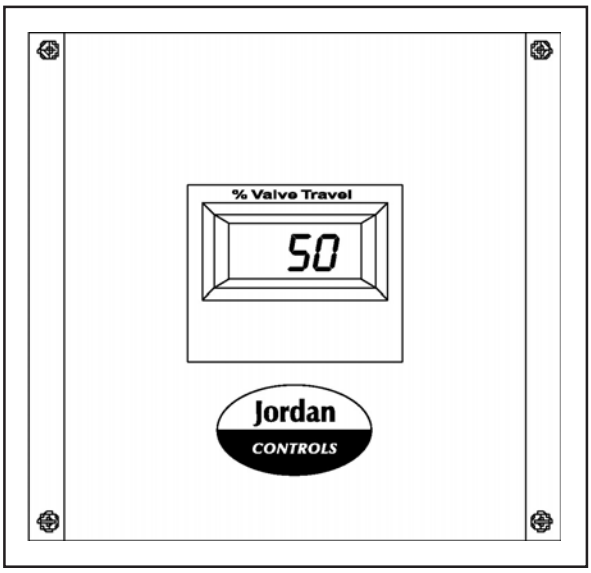
GENERAL DESCRIPTION

This remote mounted amplfier is specifically designed to operate with all EH-2000 Series actuators.

The AD-2000 has a liquid crystal display showing percentage of open for valve or damper position.

This amplifier accepts the command signal from a controller and provides a voltage output to move the actuator. The potentiometer position feedback from the actuator inputs to the AD-2000.

The AD-2000 will feedback to the customer's controller the position of the valve or damper with a 4-20 mA or 0-10 Vdc signal. When using the AD-2000, order the EH-2000 actuator without an amplifier.



FEATURES

- Accepts analog and voltage command signals
- Split range input
- Adjustable ZERO, SPAN and deadband
- 4-20 mA position feedback
- 120 or 240 Vac, field selectable
- Watetight enclosure
- Local calibration for remote actuators located in adverse conditions
- Full one year warranty

SPECIFICATIONS

Power:
Voltage Input: 120/240 Vac, 1Ph., 50/60 Hz.
Power Consumption: 3 VA
Fuse Protection: 5 Amperes

Command Signal Inputs: 4 to 20 mA current command into a 50 ohm impedence, or 0-10 Vdc. Split range input (4-12 mA, 12-20 mA) is avalable.

Position Feedback Signal: 4-20 mA or 0-10 Vdc (The output is not isolated from the signal input)

Position Signal Output: 4-20 mA or 0-10 Vdc

CE Marking: Conforms to requirements of EMC Directive and Low Voltage Directive

Enclosure: Cast Aluminum

Enclosure Rating: NEMA 13 (IP65)

Enclosure Rating With Output Indicator: NEMA 12 (IP55)

Cable Entry: 4, NPT

Operating Temperature: -13° to 131°F (-25° to 55°C)

Adjustable: ZERO, SPAN and deadband setting

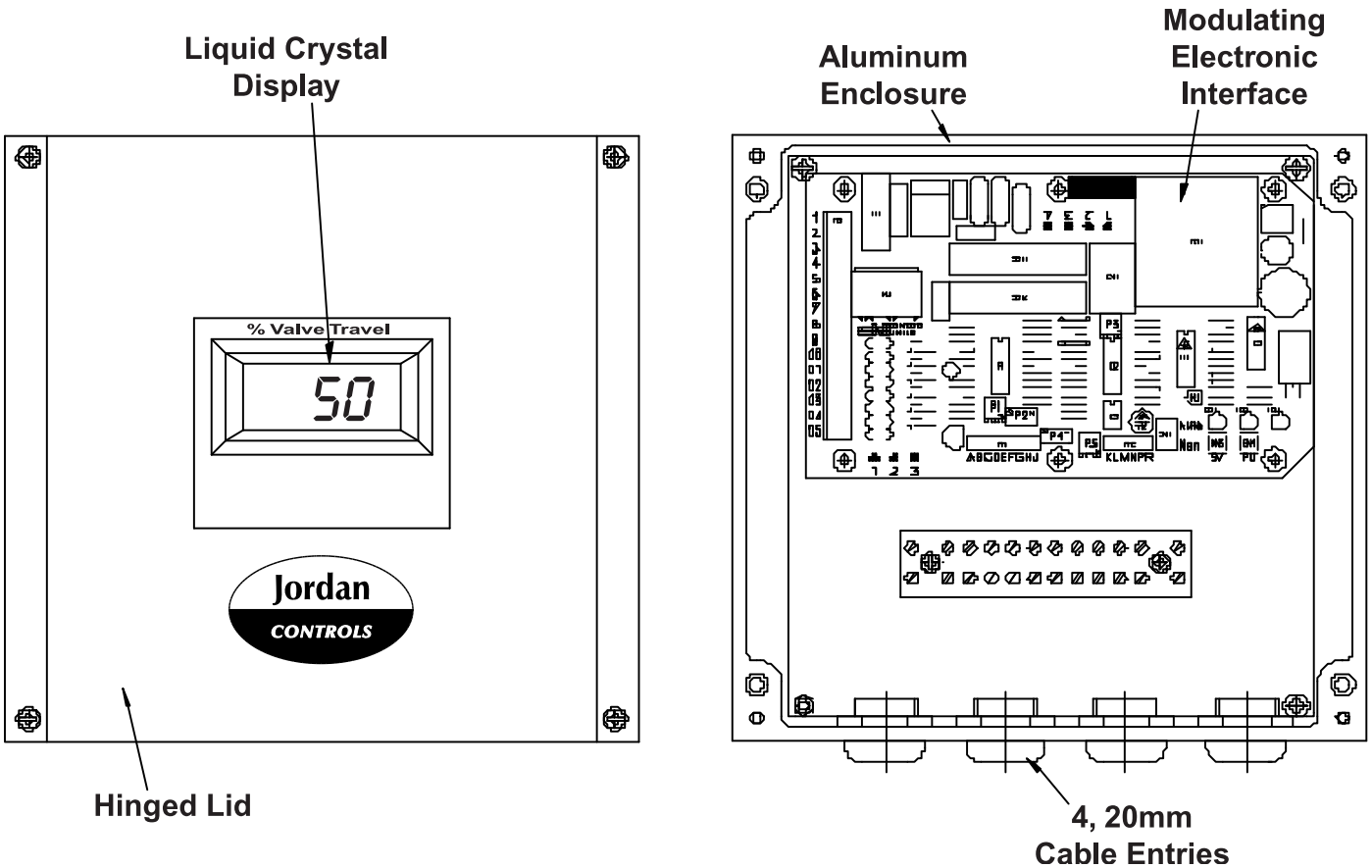
Approximate Weight: 4 lbs. (1.81 kg)

Remote Mounting Distance: 50 feet or less. (Consult factory for longer runs.)

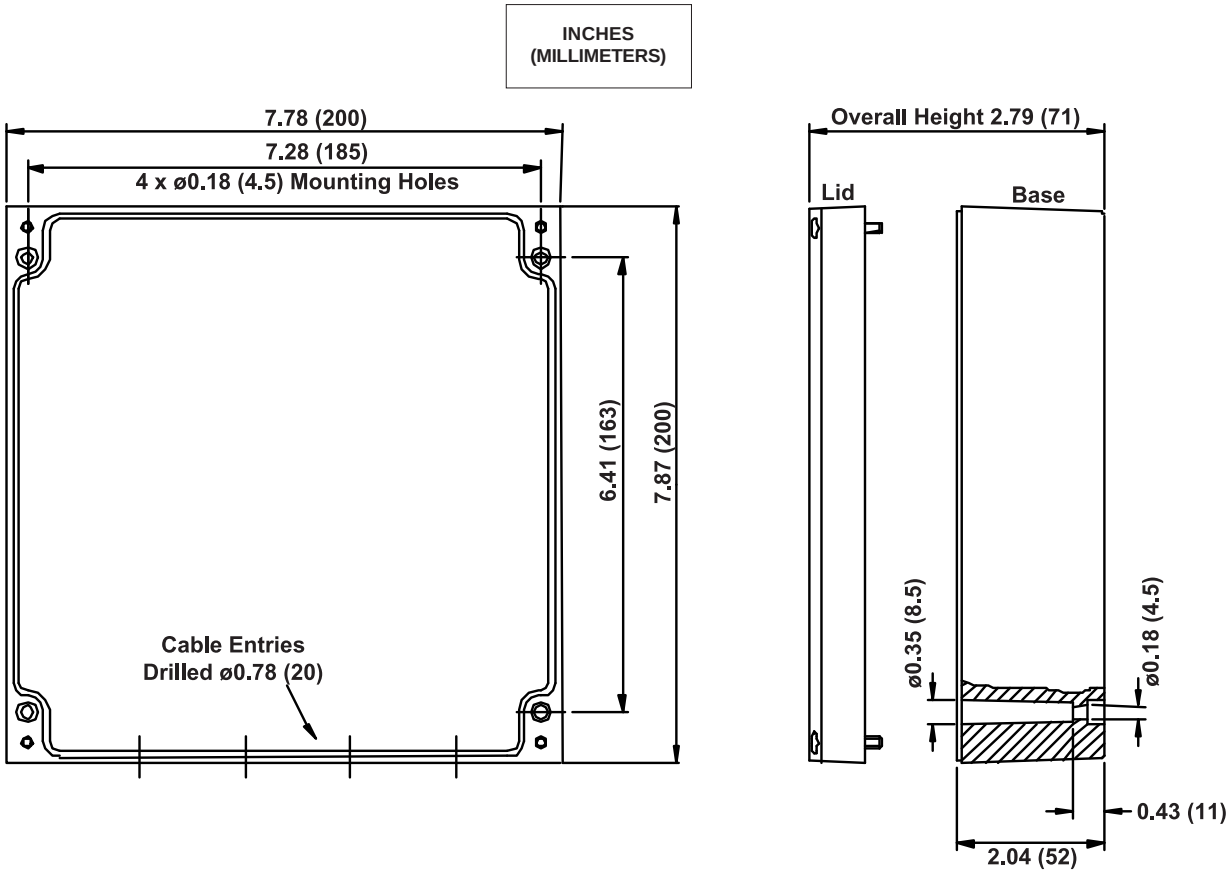
AD-2000 SERIES SELECTION CHART

			Selection
1	Basic Model	**AD-2000-P**	
2	Voltage Input	**1**: 120 Vac, 1Ph., 50/60 Hz.	
		2: 240 Vac, 1Ph., 50/60 Hz.	
3	Actuator Controlled	**EH-2XXX** (Complete model designation by filling in Xs)	
4	Input Signal	**V**: 0-10 Vdc	
		M: 4-20 mA	

COMPONENT LOCATION

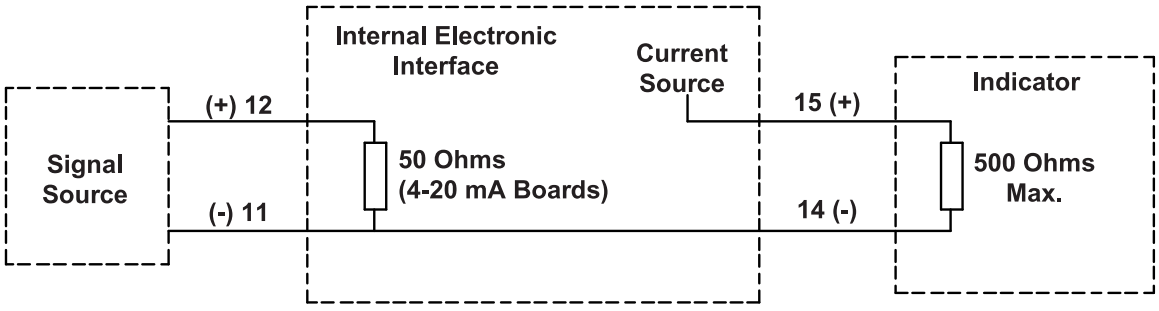


MAJOR DIMENSIONS



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

TYPICAL WIRING DIAGRAM



NOTES:

1. Maximum wire run from amplifier to actuator must not exceed 50 feet.
2. Shielded wiring is required with the shield grounded at source common for all low level circuits. This includes command & feedback signals and position torque limit switches.
3. Wire size must allow for minimal voltage drop in wiring to the actuator motor, but not smaller than 14 AWG.

Due to wide variations in the terminal numbering of actuator products, actual wiring should follow the print supplied with the actuator and amplifier.

SAMPLE ELECTRO-HYDRAULIC APPLICATIONS



EH-2020 on a Globe Valve

- Totally self-contained, leakproof system with manifold.
- High speed, 3 second shift time for 90° rotation.
- High accuracy positioning at varying load conditions.
- Used to control a vacuum effect to develop Mach 1 (and faster) air speeds for weapons testing.



EH-2130 on a Control Valve

- Low-pressure, leak-proof, self-contained system.
- Mounted on a 6 inch control valve for outflow/back wash control on pressurized filter beds.
- Application required accurate control along with spring fail safe on loss of power.



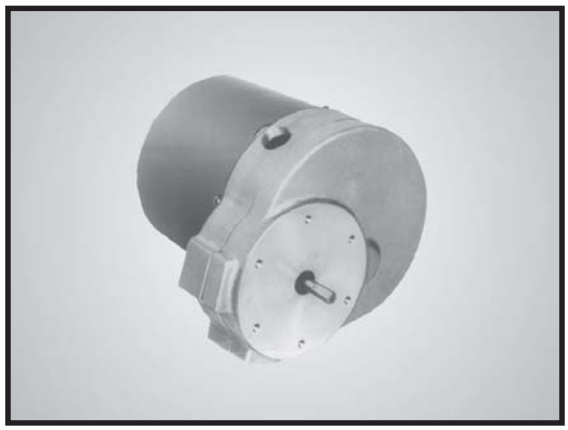
EH-9100 special high torque, high speed, emergency shutdown

- 36 in stroke in 5 - 7 seconds with shade for direct sunlight protection
- 36 inch gate valve in oil pipeline.
- Anti-water Hammer Feature.
- Emergency shutdown in-place testing

GENERAL DESCRIPTION

The TA-1200 throttle actuator is a high response electric actuator which is used with Jordan Controls AD-7530 servo amplifier to control throttle linkage in engine test stands. The actuator is a self-contained, bi-directional, electrically operated unit with output torque of 45 inch-pounds (5.1Nm). It contains a high response servomotor with precision potentiometer feedback and an output shaft rotation of 90 degrees in 0.25 seconds under full load.

The remote mounted AD-7530 servo amplifier accepts a potentiometer or an analog computer input command and produces a proportional dc output to drive the TA-1200.



TA-1200 FEATURES

- NEMA Type 12 enclosure with threaded conduit entry and terminal strip wiring
- Permanently lubricated for any position mounting
- 1000 ohm potentiometer
- Anti-backlash feedback for maximum accuracy
- Heavy-duty, high response permanant magnet servo motor
- Positioning accuracy - better than 0.25% of full stroke
- Cushioned mechanical end stops
- Full one year warranty

TA-1200 SPECIFICATIONS

Torque: Up to 45 in.lbs. (5.1 Nm)

Rotation: 90°

Speed: 0.25 sec. for 90°

Temperature Limits: 32°F to 131°F (0°C to 55°C)
For greater temperature ranges, consult factory.

Environmental Ratings: NEMA Type 12

Input Power: See selection chart on next page

Field Wiring: To barrier terminal blocks

Backdrive: Self-locking with amplifier

Approximate Weight: 20 lbs. (8.1 kg)

AD-7530 FEATURES

- Current limiting overload protection
- Fast response
- Null output

AD-7530 SPECIFICATIONS

Power Input: 120/240 Vac, 1 phase, 50/60 Hz into 175 VA

Signal Inputs:
Command: 0 to 10 Vdc, 4 to 20 mA
Position Feedback: 1000 ohm potentiometer

Approximate Weight: 15 lbs. (6.8 kg)

POPULAR OPTIONS *(See Pages 197-206)*

- Calibrated drive arm
- Anti-condensation heater and thermostat
- Increased rotation to 135°
- 1/2 inch output shaft
- NEMA 4 enclosure for AD-7530

TA-1200 SERIES SELECTION CHART

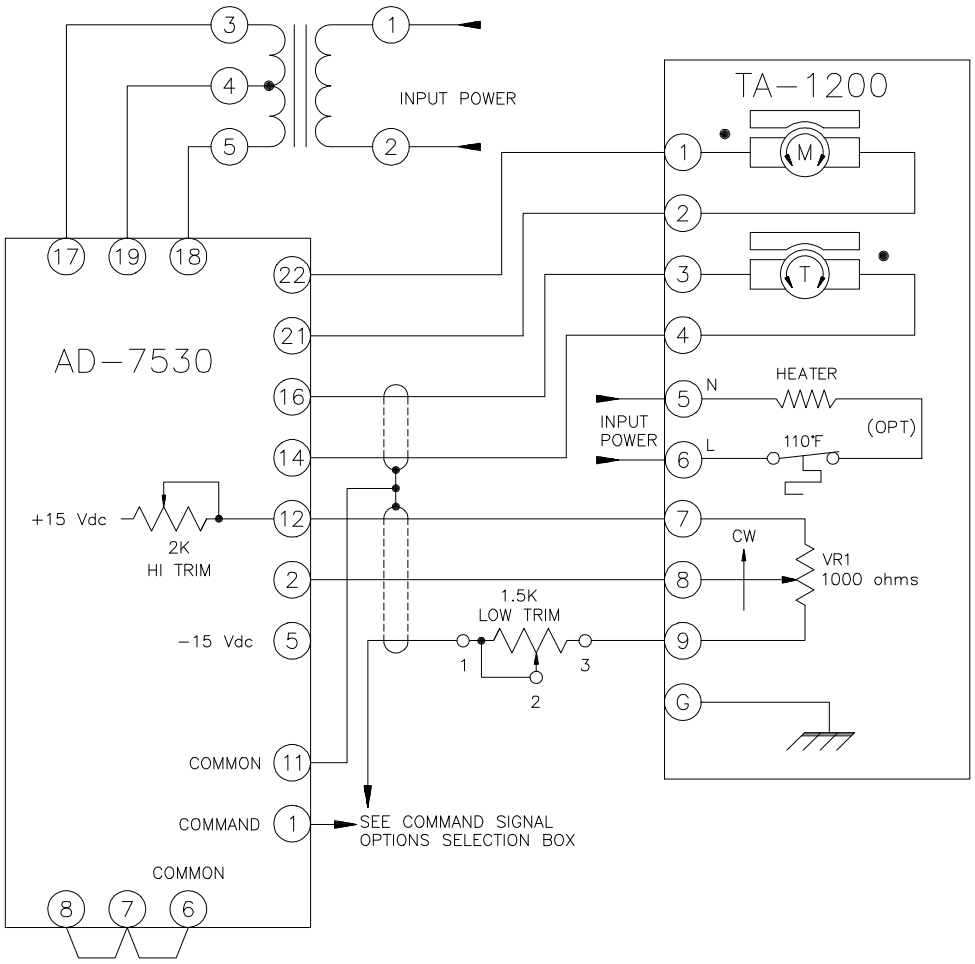
				Selection
1	Basic Model	**TA-1200**	High speed rotary actuator	
		TA-1200-R	High speed rotary actuator with chassis mounted AD-7530	
		TA-1200-E	High speed rotary actuator with separately mounted AD-7530 in a NEMA 4 enclosure	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (TA-1200-R & TA-1200-E)		
		2: 240 Vac, 1 Phase, 50/60 Hz (TA-1200-R & TA-1200-E)		
		6: 24 Vdc (TA-1200)		
3	Input Signal	**P**: 1000 ohm potentiometer (Required for TA-1200-R or TA-1200-E)		
		V: 0 to 10 Vdc (Required for TA-1200-R or TA-1200-E)		
		A: 4 to 20 mA dc (Required for TA-1200-R or TA-1200-E)		
		N: None		

TA-1200 STANDARD OPTIONS

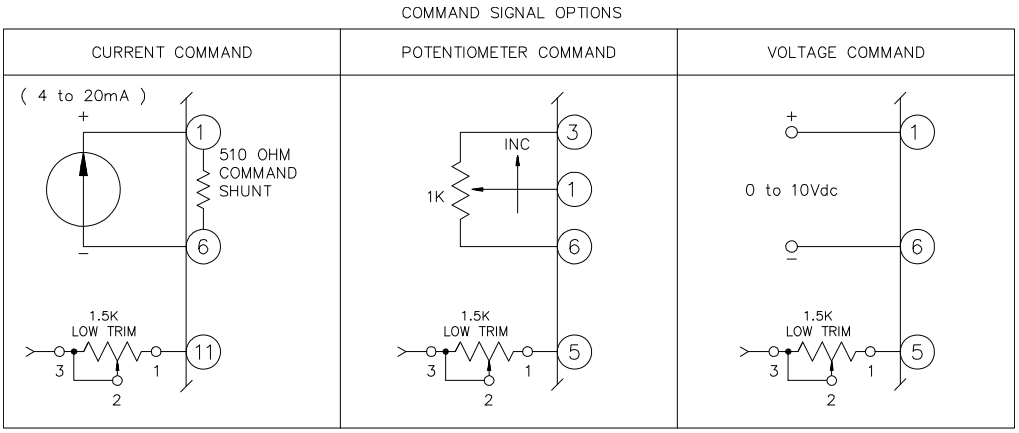
Code	Description	Selection
Drive Arm/Adapter Clevis/Driven Arm		
D040	Calibrated Drive Arm	
D057	Adjustable drive arm for 1/2 inch output shaft of TA-1200 (Use with R008)	
Position Feedback		
F001	Potentiometer, 1000 ohms	
Heater		
H001	Anti-Condensation Heater (24 VDC) For units with enclosure	
H002	Anti-Condensation Heater (120 VAC) For units with enclosure	
H003	Anti-Condensation Heater (240 VAC) For units with enclosure	
Identification/Certifications		
J004	One year extended warranty	
J005	Two year extended warranty	
Shaft		
R008	1/2 inch Output Shaft	
Special Mechanical Options		
Y003	Increased rotation for TA-1200 Series actuators to 135°	

For a full description of options, go to the Complete Listing of Options starting on page 197.

TYPICAL WIRING DIAGRAMS



COMMAND SIGNAL OPTIONS



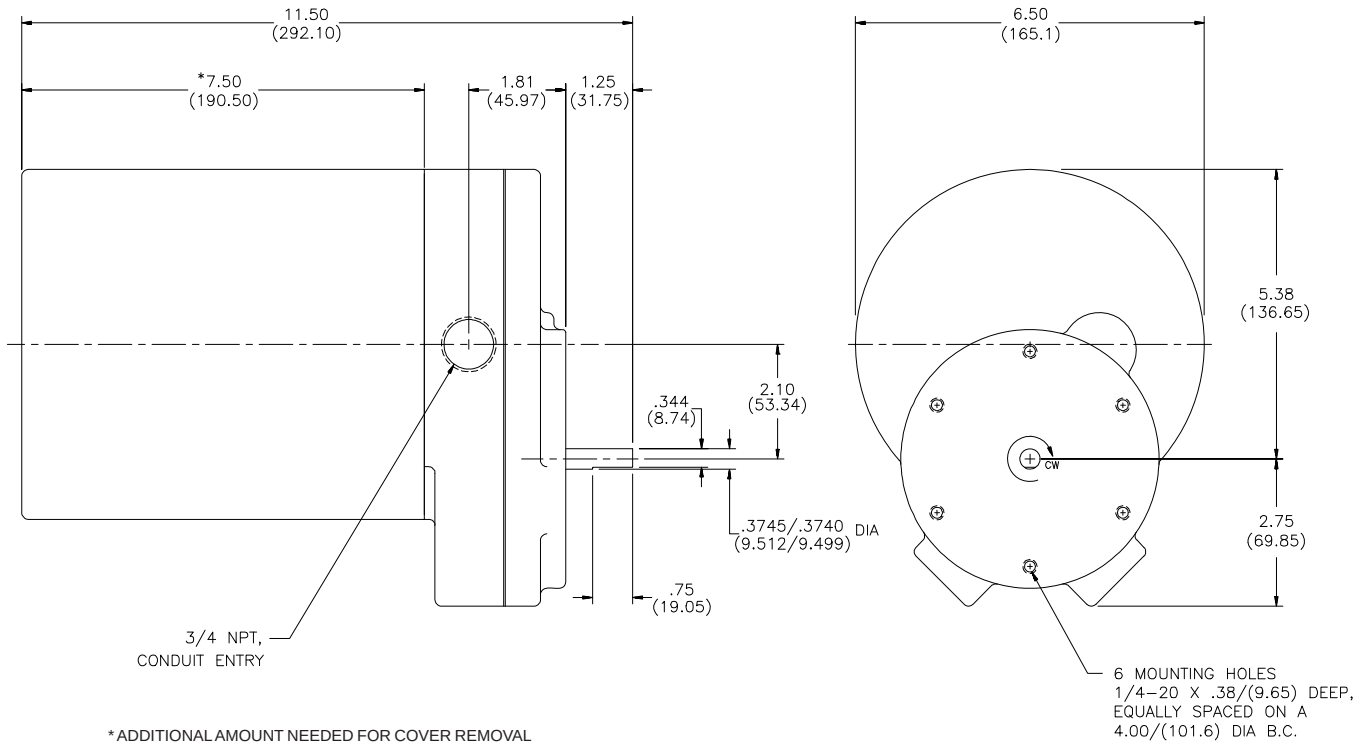
Notes:

1. An increase in command at terminal 1 of AD-7530 will result in a voltage increase at terminal 22 with respect to terminal 21 causing the actuator to rotate CW as shown and the feedback potentiometer wiper to move as shown by arrow.
2. *To reverse shaft direction*
   Change wires on the actuator as follows:
   Reverse wires to terminals 1 and 2 (Armature).
   Reverse wires to terminals 3 and 4 (Tachometer).
   Reverse wires to terminals 7 and 9 (Potentiometer).

These dimensions and specifications are subject to change without notice.
Current drawings and specifications are available upon request.

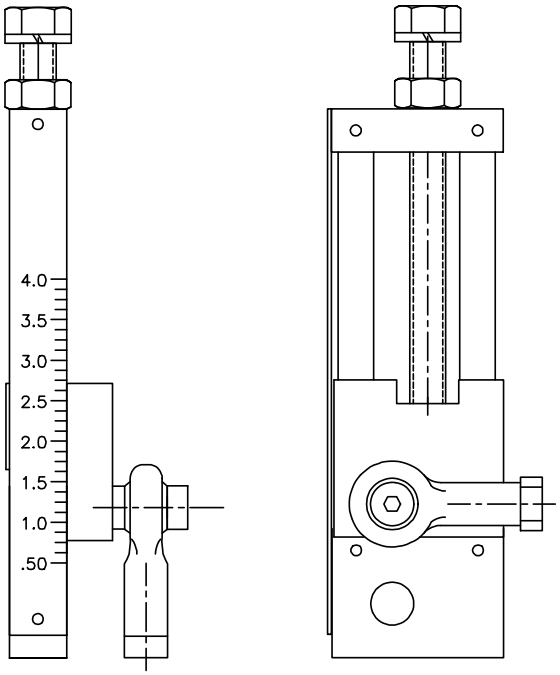
TA-1200 MAJOR DIMENSIONS

INCHES (MILLIMETERS)



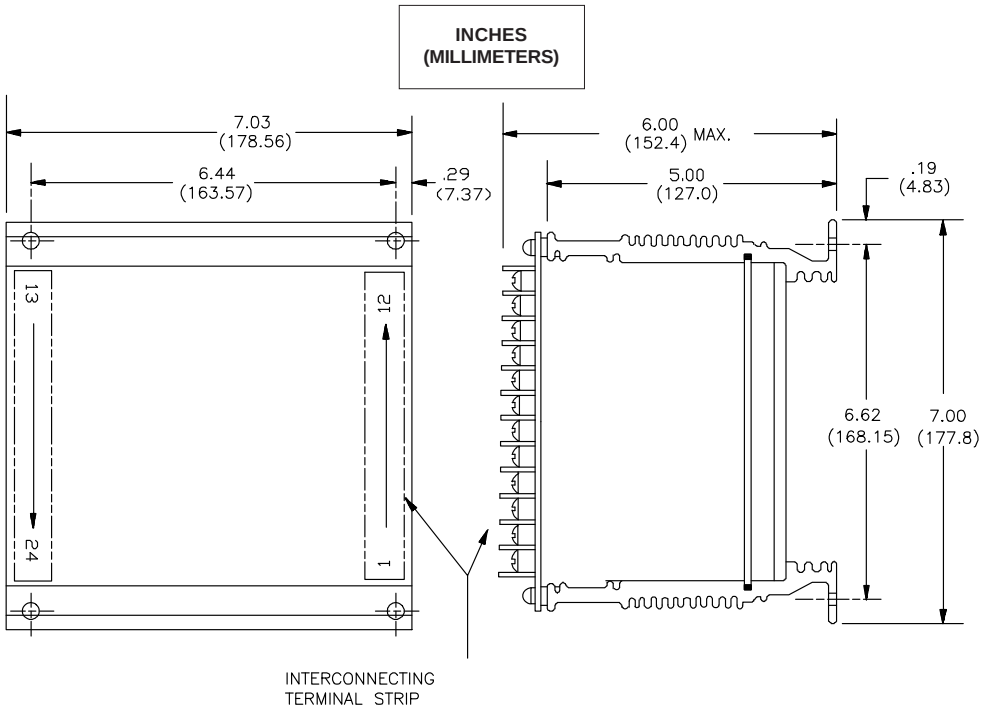
*ADDITIONAL AMOUNT NEEDED FOR COVER REMOVAL

Calibrated Drive Arm (D040)

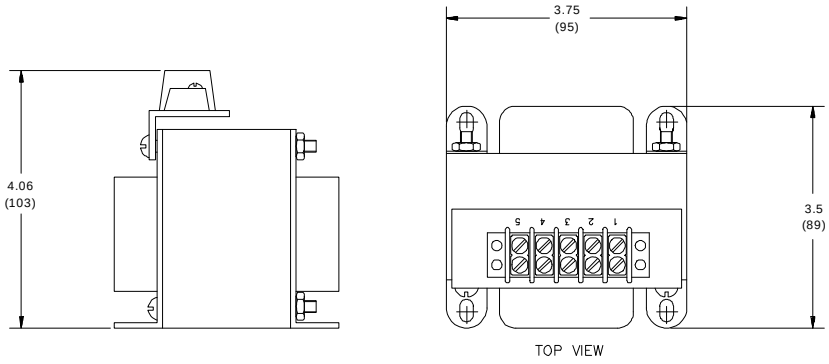


These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

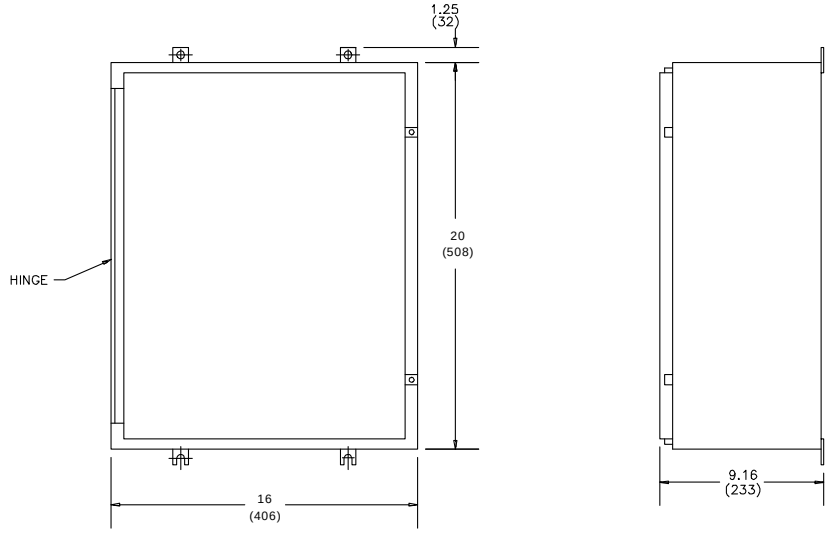
TA-1200-R MAJOR DIMENSIONS



INPUT TRANSFORMER



TA-1200-E ENCLOSURE



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

MC-1100 ELECTRIC TO PNEUMATIC CONVERTER

Jordan CONTROLS

GENERAL DESCRIPTION

These electric, actuator driven, precision pressure regulators are used for remote pressure adjustment. They can be controlled by switched line voltage inputs or operated from a voltage or current process command signal. A variety of pressure ranges and shift times are available to meet the most popular application requirements.

The MC-1100 is ideally suited for remote set-point control, remote brake and clutch control, remote web tension control, and pilot valve control.

All MC-1100 Series actuators are also available with an internal amplifier (excluding MC-1140). These amplifiers are all full-featured ac switching devices designed to work seamlessly with the unit for closed loop control.



FEATURES

- Explosion proof enclosure
- Permanently lubricated for any position mounting
- AC or DC input power versions
- Wide ambient temperature range
- Explosion proof enclosure
- High strength, corrosion resistant, cast aluminum enclosure rated for indoor/outdoor and hazardous dust installations
- End of travel position limit switches
- Variety of pressure ranges
- Maintains pressure setting when power is removed
- Regulator is virtually unaffected by changes in supply pressure
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Internal or remote-mounted amplifier
- Anti-condensation heater and thermostat
- Various enclosure coatings
- Battery back-up, RP-4000

AMPLIFIER FEATURES

- Isolated, two-wire, 4-20mA feedback
- Customer-adjustable zero and span
- Adjustable deadband
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place - Run to preset
- Field selectable dynamic brake
- AC triac switching of AC motors

ACTUATOR SPECIFICATIONS

Temperature Limits: -40°F to 150°F (-40°C to 65°C)

Field Wiring: To barrier terminal blocks

Modulation Rate: Modulating duty

Approximate Weight: 12 to 16 lbs.(5.4 to 7.3 kg)

REGULATOR SPECIFICATIONS

Pipe Connections: 1/4-NPT, 3/8-NPT

Supply Pressure: 500 psi (34.5 bar) maximum

Flow Capacity: (100 psi/6.9 bar supply, 20 psi/1.4 bar set) 40 SCFM (19 liters/sec.) maximum

Exhaust Capacity: (Downstream pressure 5 psi/0.34 bar above set pressure) 5.5 SCFM (2.6 liters/sec.)

Effect of Supply Pressure Variation: Less than 0.1 psi (0.007 bar) for a 100 psi (6.9 bar) change

SPECIFICATION CHART

Actuator Model	Input Power Volts/Phase/Hz	Current (Amps)		Command Input	Input Impedance in ohms	Feedback	Loss of Power	Loss of Command Signal
		Run	Stall					
MC-1110-N	120/1/50/60	0.25	0.26	N/A	N/A	N/A	Lock in Place	N/A
MC-1110-D	120/1/50/60	0.25	0.26	4-20 mA	200	1000 ohm pot or 4-20 mA	Lock in Place	Lock in Place or Run to Preset
				0-5 Vdc or 0-10 Vdc	100,000			
MC-1140-N	24 Vdc	1.2	4.8	N/A	N/A	N/A	Lock in Place	N/A
MC-1170-N	240/1/50/60	0.14	0.15	N/A	N/A	N/A	Lock in Place	N/A
MC-1170-D	240/1/50/60	0.14	0.15	4-20 mA	200	1000 ohm pot or 4-20 mA	Lock in Place	Lock in Place or Run to Preset
				0-5 Vdc or 0-10 Vdc	100,000			

N/A = Not Applicable

ACTUATOR/AMPLIFIER SPECIFICATIONS

	MC-1110-D & MC-1170-D
Position Accuracy	1.0 % of full travel
Repeatability	0.5 % of full travel
Hysteresis	0.5 % of full travel
Linearity	+/-1 % of full travel
Deadband	0.75 % of full travel
Resolution	0.75 % of full travel

REMOTE AMPLIFIERS

Actuator	Remote Mounted Amplifier	Catalog Page
MC-1110-N & MC-1170-N	AD-8240	173
MC-1140-N	AD-7530	170

MC-1100 SERIES SELECTION CHART

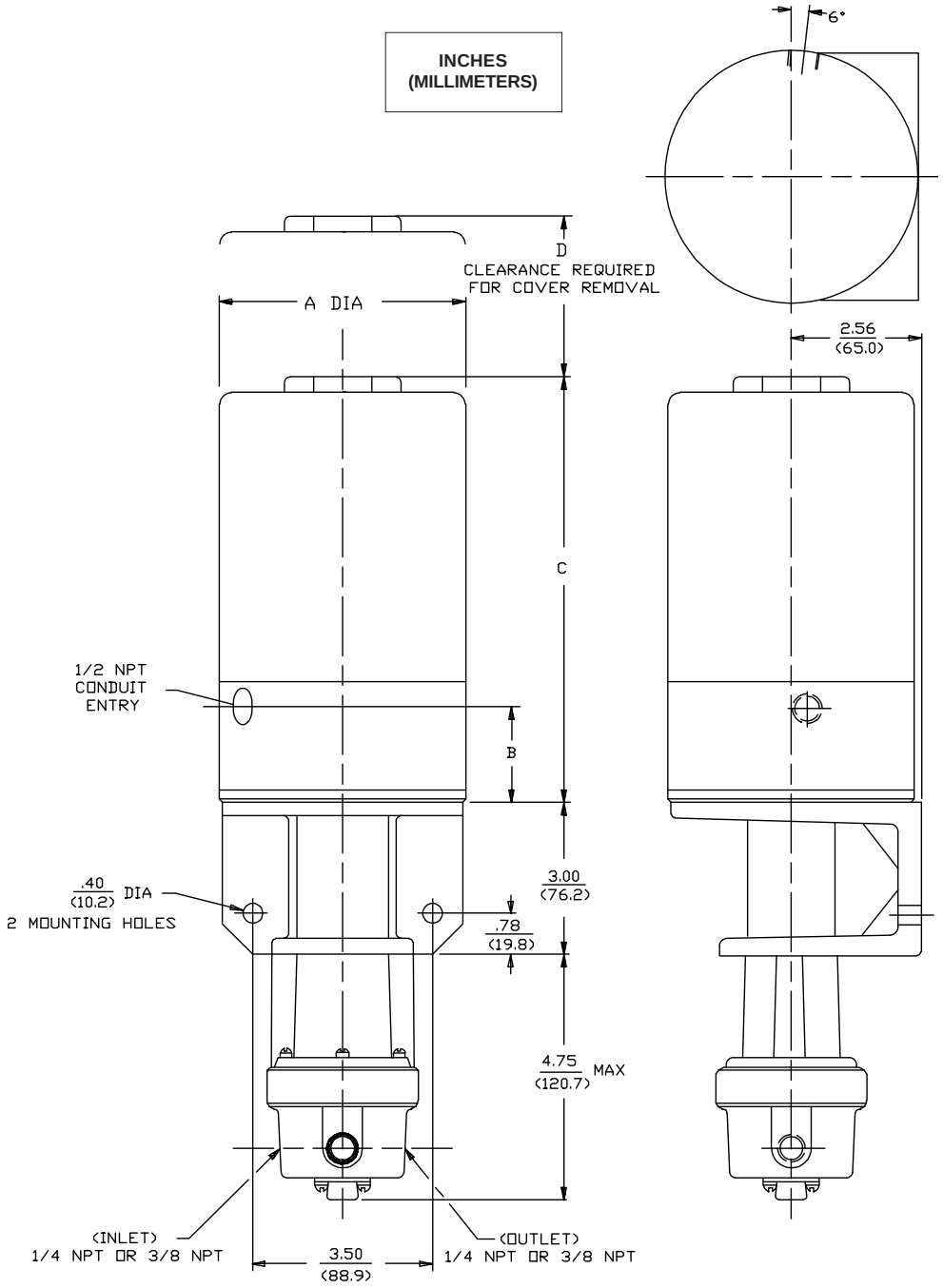
					Selection
1	Basic Model	**MC-1110-N**	1 phase, 120 Vac, modulating duty, without built-in amplifier		
		MC-1110-D	MC-1110 above with built-in amplifier		
		MC-1140-N	24 Vdc (If amplifier desired, see remote model AD-7530		
		MC-1170-N	1 phase, 240 Vac, modulating duty, without built-in amplifier		
		MC-1170-D	MC-1170 above with built-in amplifier		
2	Input Voltage	**1**: 120 Vac, (MC-1110)			
		2: 240 Vac, (MC-1170)			
		6: 24 Vdc, (MC-1140)			
3	Pressure Range and Shift Time (Nominal) *Must select pressure range and shift time for model selected.*	Pressure Range PSI (bar)	Shift Time (Minutes)		
			MC-1110 MC-1170	MC-1140	
		3 to 15 (0.21 to 1.03)	**7.5**	**6**	
			3.8	**3**	
			1.9	**1.5**	
			1	**0.8**	
			0.5	**0.4**	
			0.2	**0.2**	
			0.28		
			0.14		
		0 to 20 (0 to 1.38)	**25.2**	**20.1**	
			12.8	**10.2**	
			6.4	**5.1**	
			3.2	**2.6**	
			1.6	**1.3**	
			0.8	**0.66**	
		0.5 to 30 (0.034 to 2.07 or 1 to 60 (0.069 to 4.14)	**26**	**20.8**	
			13	**10.5**	
			6.6	**5.3**	
			3.3	**2.7**	
			1.7	**1.3**	
			0.9	**0.68**	
		2 to 150 (0.14 to 10.34 or 5 to 200 (0.34 to 13.79)	**21.2**	**17**	
			10.8	**8.6**	
			5.4	**4.3**	
			2.7	**2.2**	
			1.4	**1.1**	
				0.55	
4	Pipe Size	**1/4**: 1/4 NPT inlet and outlet pipe connections			
		3/8: 3/8 NPT inlet and outlet pipe connections			

MC-1100 STANDARD OPTIONS

Code	Description	Selection
	Toggles, Lights	
A008	Local Auto/Manual and INC/OFF/DEC (Close Coupled Enclosure)	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure)	
	Feedback	
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each	
F003	Potentiometer, 5000 ohms (Not for use with models with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (Not for use with models with internal amplifiers)	
F005	Transmitter, 4-20 mA	
F006	One Potentiometer, 1000 ohms plus Transmitter, 4-20 mA	
F011	Anti Backlash Feedback Gearing	
F999	Special Potentiometer (consult factory)	
	Heater	
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
	Identification/Certifications	
J001	CE Marking (AC input only)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
	Auxiliary Switches	
S007	Aluminum Feedback Frame (Multi Turn)	
S008	Two nonadjustable end of travel position switches, SPST, 10 Amp, 120/240 Vac	
S013	Two nonadjustable end of travel position switches, SPDT, 10 Amp, 120/240 Vac	
	Painting/Coating	
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	

For a full description of options, go to the Complete Listing of Options starting on page 197.

MC-1100 MAJOR DIMENSIONS



MC-1100-N	A	B	C	D
NEMA 4, X-proof	4.88 (124)	1.84 (46.7)	8.65 (219.7)	6.28 (159.5)
MC-1100-D	**A**	**B**	**C**	**D**
NEMA 4 Enclosure	4.88 (124)	1.84 (46.7)	10.21 (259.3)	7.81 (198.4)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

SM/LA-3300 SERIES SPRING RETURN ACTUATORS

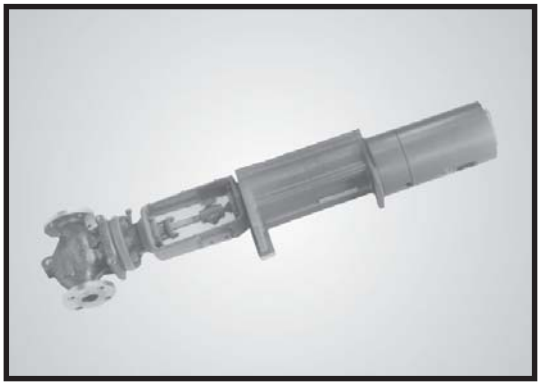
Jordan CONTROLS

GENERAL DESCRIPTION

The SM-3300 (rotary) and LA-3300 (linear) are spring return actuators designed to fail to either the full extend (full CW) or full retract (full CCW) position on loss of input power. Offering continuous modulating duty, these actuators may be used to position valves, dampers and other control devices, with spring-powered failsafe operation.

The SM/LA-3300 Series can provide closed-loop positioning from an analog input signal, or they may be operated by switched line voltage inputs for ON/OFF control.

All SM/LA-3300 Series actuators are also available with an internal amplifier. These amplifiers are all full-featured ac switching devices designed to work seamlessly with the actuator for closed loop control.



POPULAR OPTIONS *(See Pages 197-206)*

- Auxilliary position limit switches
- Custom mounting and interface hardware
- Local Auto/Manual toggle switches in close coupled enclosure
- Various enclosure coatings

FEATURES

- Permanently lubricated for any position mounting
- End of travel position limit switches
- Anti-condensation heater and thermostat
- Position indicator providing final output shaft position
- Internal or remote-mounted amplifier
- Wide ambient temperature range
- High strength, corrosion resistant, cast aluminum enclosure rated for indoor/outdoor and hazardous dust installations
- Full one year warranty

AMPLIFIER FEATURES

- Isolated, two-wire, 4-20mA feedback
- Customer-adjustable zero and span
- Adjustable deadband
- Accepts either voltage or current command inputs
- Selectable options for Loss-of-Signal action:
  - Lock in place
  - Run to preset
- AC triac switching of AC motors

SPECIFICATIONS

Speed:

Full rotation or stroke under AC power conditions: 12 sec.

Full rotation or stroke on loss of AC power: 8 sec.

Modulation Rate: 2000 starts/hr (minimum position change of 1%)

Temperature: -40° F to 150° F (-40° C to 65° C)
Maximum torque drops to 750 in. lbs. (85 Nm) when operating at temperature below 32° F (0° C).
Maximum thrust drops to 600 lbs. (2669 N) when operating at temperature below 32° F (0° C).

Environmental Rating: Hazardous locations for Class I, Div. 1, Groups C and D and Class II, Div. 1, Groups E, F, and G. Also suitable for NEMA Type 4 (IP65) indoor/outdoor use.

Field Wiring: To barrier terminal blocks

SM-3300 rotation: 90° maximum

SM-3300 Torque: 1000 in.lbs. (113 Nm)

Approximate Weight: 110 lbs. (50 kg) (without valve and mounting hardware)

Fail Mode on Loss of Power: Full CW or CCW

LA-3300 stroke: 2 in. (50.8 mm) maximum

LA-3300 thrust: 800 lbf (3558 N)

Approximate Weight: 85 lbs. (39 kg) (without valve and mounting hardware)

Fail Mode on Loss of Power:
LA-3300: Full extend (valve stem down)
LA-3300: Full retract (valve stem up)

ACTUATOR/AMPLIFIER SPECIFICATIONS

	SM-3330-D, SM-3350-D, LA-3330-D & LA-3350-D
Position Accuracy	1.0 % of full travel
Repeatability	0.5 % of full travel
Hysteresis	0.5 % of full travel
Linearity	+/-1 % of full travel
Deadband	0.75 % of full travel
Resolution	0.75 % of full travel

REMOTE AMPLIFIER

Actuator	Remote Mounted Amplifier	Catalog Page
SM-3330-N, SM-3350-N, LA-3330-N & LA-3350-N	AD-8240	173

SM-3300 SELECTION CHART

				Selection
1	Basic Model	**SM-3330-N**	90° rotary actuator, 1000 inch-pounds (113 Nm), spring return CW or CCW	
		SM-3330-D	SM-3330 above with built-in digital amplifier	
		SM-3350-N	90° rotary actuator, 1000 inch-pounds (113 Nm), spring return CW or CCW	
		SM-3350-D	SM-3350 above with built-in digital amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (SM-3330-N & SM-3330-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (SM-3350-N & SM-3350-D)		
3 4	Speed, sec. (a) / Torque, in. lbs. (Nm)	**12/1000** (113) *Maximum torque drops to 750 (85) when operating at temperature below 32° F (0° C)*		
5	Rotation	**Specify** output rotation - minimum 8.7° - maximum 90°		

(a) Speed is shift time for 90° rotation.

LA-3300 SELECTION CHART

				Selection
1	Basic Model	**LA-3330-N**	2-inch stroke linear actuator, 800 pounds thrust (363 N)	
		LA-3330-D	LA-3330 above with built-in digital amplifier	
		LA-3350-N	2-inch stroke linear actuator, 800 pounds thrust (363 N)	
		LA-3350-D	LA-3350 above with built-in digital amplifier	
2	Input Voltage	**1**: 120 Vac, 1 Phase, 50/60 Hz (LA-3330-N & LA-3330-D)		
		2: 240 Vac, 1 Phase, 50/60 Hz (LA-3350-N & LA-3350-D)		
3 4	Speed, sec. (a) / Thrust, lbs. (N)	**12/800** (3558) *Maximum thrust drops to 600 (2669) when operating at temperature below 32° F (0° C)*		
5	Linear Travel	**Specify** output stroke - minimum 0.19 inches (4.8 mm) - maximum 2.0 inches (50.8 mm)		
6	Mechanical Failsafe Direction	**E**: Spring to extend		
		R: Spring to retract		

(a) Speed is shift time for full 2 inch stroke.

SM/LA-3300 STANDARD OPTIONS

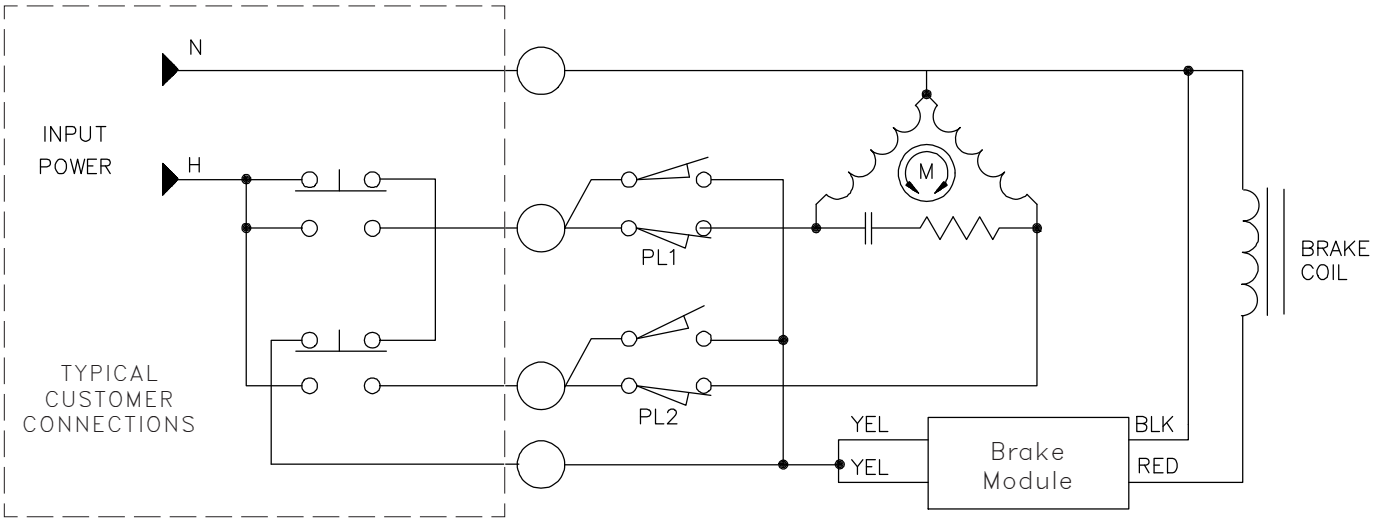
Code	Description	Selection
Toggles, Lights		
A008	Local Auto/Manual INC/OFF/DEC (Close Coupled Enclosure)	
A015	ON/OFF Toggle Power Switch (Close Coupled Enclosure)	
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each (not available for units with internal amplifiers)	
F003	Potentiometer, 5000 ohms (not available for units with internal amplifiers)	
F004	Potentiometer, 10,000 ohms (not available for units with internal amplifiers)	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (not available for units with internal amplifiers)	
F012	Tandem Potentiometers, 5000 ohms (not available for units with internal amplifiers)	
F013	Tandem Potentiometers, 10,000 ohms (not available for units with internal amplifiers)	
F999	Special Potentiometer (consult factory)	

Code	Description	Selection
Identification/Certifications		
J001	CE Marking (AC input only)	
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S005	Aluminum Cam	
S006	Aluminum Feedback Frame	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S013	Two nonadjustable end of travel position switches, SPDT, 00 Amp, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
Special Mechanical Options		
Y017	SM-3300 metric mounting holes, 6mm	
Y018	LA-3300 metric mounting holes, 8mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

TYPICAL WIRING DIAGRAM

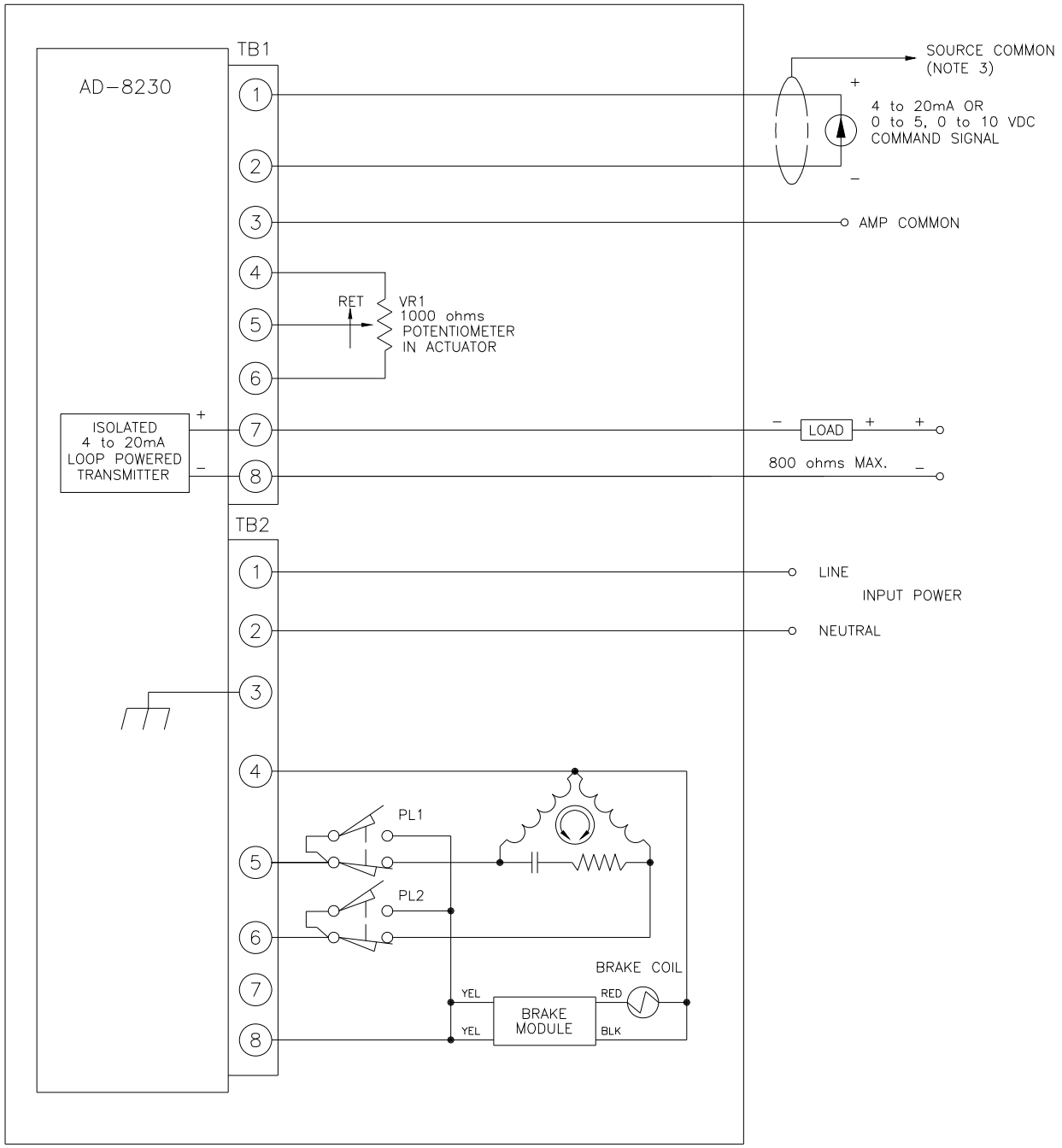
SM-3330-N, SM-3350-N, LA-3330-N & LA-3350-N



These dimensions and specifications are subject to change without notice.
Current drawings and specifications are available upon request.

TYPICAL WIRING DIAGRAM

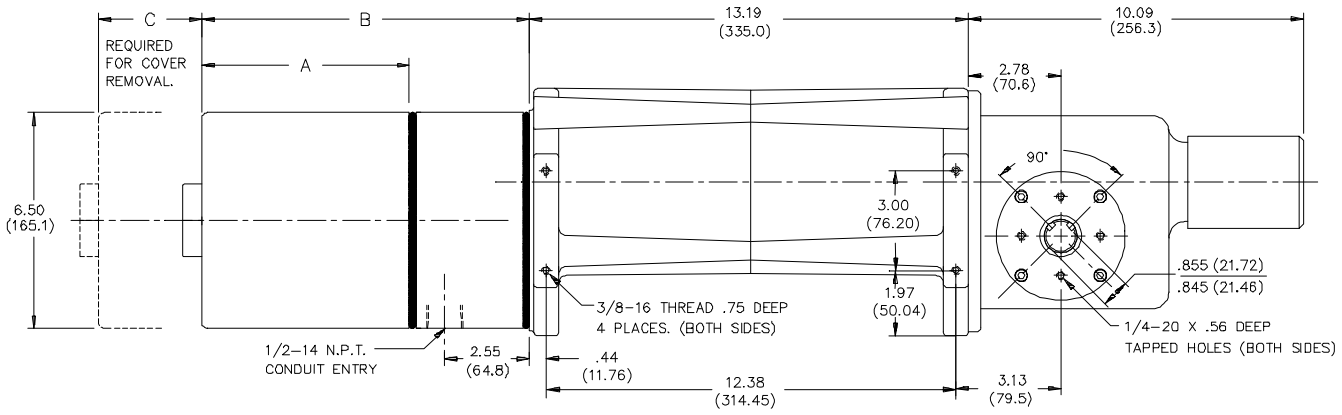
SM-3330-A, SM-3350-A, LA-3330-A & LA-3350-A



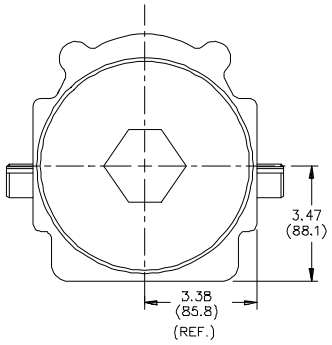
These dimensions and specifications are subject to change without notice.
Current drawings and specifications are available upon request.

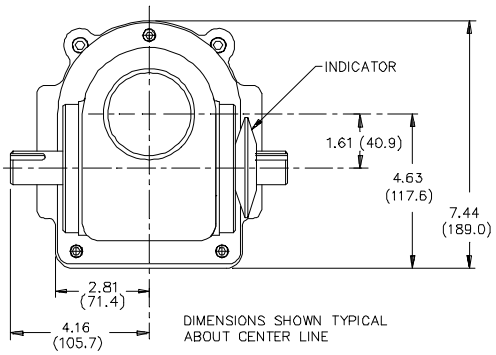
SM-3300 MAJOR DIMENSIONS

INCHES (MILLIMETERS)



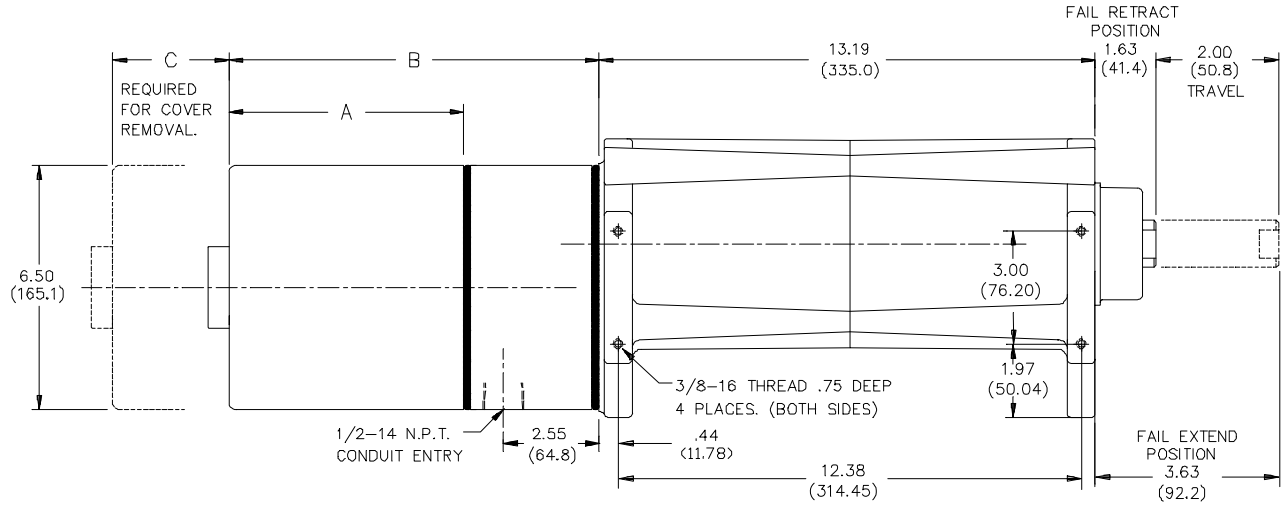
NOTE:
KEYWAY LOCATION CORRESPONDS TO FAIL CW POSITION.
OUTPUT SHAFT: 1.000/.998 (25.4/25.34) DIA.
KEYWAY: .251/.253 WIDE

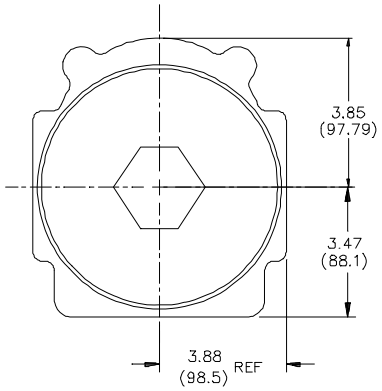


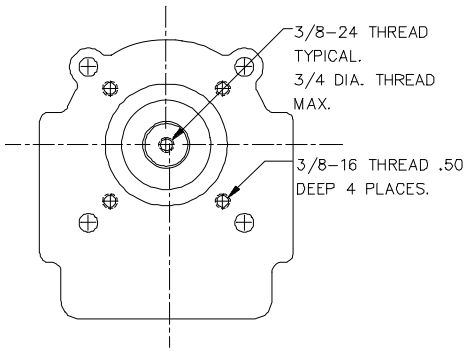


These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LA-3300 MAJOR DIMENSIONS







	A	B	C
Without Amplifier	6.88 (174.7)	9.86 (250.4)	7.00 (177.8)
With Amplifier	10.00 (254)	12.98 (329.7)	10.13 (257.3)

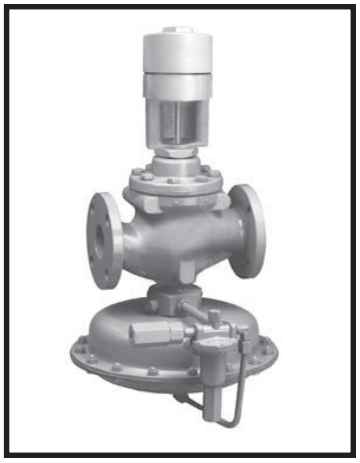
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

GENERAL DESCRIPTION

Jordan Control's Series of position transmitters are rugged, compact industrial components that couple a precision feedback component and/or limit switches to the input shaft through an enclosed gear train.

Typical applications include position feedback for valves, dampers, tension loops, winder/rewinder rolls and other process control equipment.

PT and PF models are rotary transmitters, LT and LF models are linear transmitters.



LT-1010

FEATURES

- NEMA 4 enclosure (IP65) & dust ignition-proof (NEMA 4 for PT-1330 & PT-1510)
- Permanently lubricated for any position mounting
- Wide ambient temperature range
- High strength, corrosion resistant, cast aluminum enclosure rated for indoor/outdoor and hazardous dust installations
- Wide range of input turn ratios
- Face or side servo mount (except LF-1210)
- Full one year warranty

POPULAR OPTIONS *(See Pages 197-206)*

- Anti-condensation heater and thermostat
- Auxilliary position limit switches
- Aluminum limit switch cams and frames
- Various enclosure coatings

$$\text{Maximum load (ohms)} = \frac{\text{DC power supply voltage - 8V}}{0.020\ \text{A}}$$

$$\text{Minimum load rating (watts)} = \text{Max.load} \times (0.020\ \text{A})$$

Figure 1

SPECIFICATIONS

- **Linearity**: ± 0.5%, ± 0.1% for PT-1330 & PT-1510
- **Ambient Operating Temperature Range**: -40° F to 150° F (-40° C to 65° C)
- **Maximum Radial Input Shaft Load**:
  PF/PT-1010 models: 400 lbs. (181 kg)
  PF-1210 models: 60 lbs. (27 kg) at midpoint of flat
  PT-1330 & PT-1510 models: 75 lbs. (34 kg)
- **Maximum Axial Input Shaft Load**:
  PF/PT-1010 models: 600 lbs. (272 kg)
  PF-1210 models: Not recommended
  PT-1330 & PT-1510 models: 320 lbs. (145.2 kg)
- **Standard Signal Range Output**:
  PF models: 60% to 90% (depending on gear ratio of 1000 ohm, 2W, ±0.25% linearity potentiometer
  LF models: 60% to 90% (depending on gear ratio of 1000 ohm, 2W, ±0.25% linearity potentiometer
  PT models: 4 to 20 mA, 2 wire, loop-powered, 800 ohm impedence load maximum, as calculated by the formula in *Figure 1*.
  LT models: 4 to 20 mA, 2 wire, loop-powered, 800 ohm impedence load maximum, as calculated by the formula in *Figure 1*.
- **Temperature Effect**:
  0.05% or less variation per °C
- **Weight**:
  1010 & 1330 models: 3.5 lbs. (1.5 kg)
  PF-1210 models: 6.5 lbs. (2.9 kg)
  LF-1210 models: 6.5 lbs. (2.9 kg) minimum, depends upon length of rack
  LT-1010 models: 6 lbs. (2.7 kg)
  PT-1510 models: 14.1 lbs. (6.4 kg)

BASIC SPECIFICATIONS FOR POSITION TRANSMITTERS

Model	Rotation/ Travel	Output Signal	Max. Loop Impedence	NEMA Rating	Limit Switches	1000 Ohm Pot	Catalog Page #
PF-1010	28° to 420 turns *	1000 ohm Pot and/or switches	NA	4/7/9	Option (See chart)	NA	160
PT-1010		4 to 20 mA	800 ohms		NA	Option (See chart)	161
PF-1210	0.25 to 60 turns	1000 ohm Pot and/or switches	NA	4/7/9	Option (See chart)	NA	162
LF-1210	3 in. to 18 in.				Option (See chart)	NA	163
PT-1330 & PT-1510	0° to 19° or 19° to 300°	4 to 20 mA	800 ohms	4/7/9	NA	Option (See chart)	164
LT-1010	0.38 in. to 2 in.	4 to 20 mA	800 ohms	4/7/9	NA	NA	165

* Rotation range is accomplished with 18 different gear ratios.
N/A = Not Applicable

PF-1010 SERIES SELECTION CHART

		Selection
Basic Model	**PF-1010, Position Sensor ***	
Input Shaft Turns	**Specify** specific input shaft turns required. ***(Not the Range)***	
	28° to 40 turns	
	41 to 420 turns	

** Select desired feedback options below.*

PF-1010 SERIES STANDARD OPTIONS

Code	Description	Selection
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each	
F003	Potentiometer, 5000 ohms	
F004	Potentiometer, 10,000 ohms	
F010	Anti Backlash Feedback Gearing *Up to 40 turns*	
F011	Anti Backlash Feedback Gearing *Up to 420 turns*	
F012	Tandem Potentiometers, 5000 ohms	
F013	Tandem Potentiometers, 10,000 ohms	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	

Code	Description	Selection
Auxiliary Switches		
S005	Aluminum Cam, Up to 40 turns	
S006	Aluminum Feedback Frame (Up to 40 turns)	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S011	Four Adjustable Position Switches, SPDT, 10 Amp, Up to 40 turns, 120/240 Vac	
S012	Four Adjustable Position Switches, DPDT, 10 Amp, Up to 40 turns, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
Special Mechanical Options		
Y019	Metric mounting holes, 6mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

PT-1010 SERIES SELECTION CHART

		Selection
Basic Model	**PT-1010, Two-wire, 4 to 20 mA output**	
Transmitter Type	**1**: 120 Vac line powered, 4 to 20 mA dc. Capable of powering loads of up to 600 ohms	
	2: 240 Vac line powered, 4 to 20 mA dc. Capable of powering loads of up to 600 ohms	
	12: Loop powered, two-wire, 4 to 20 mA dc. Customer-supplied, external dc power supply required	
Input Shaft Turns	**Specify** specific input shaft turns required. **(Not the Range)**	
	28° to 40 turns	
	41 to 420 turns	

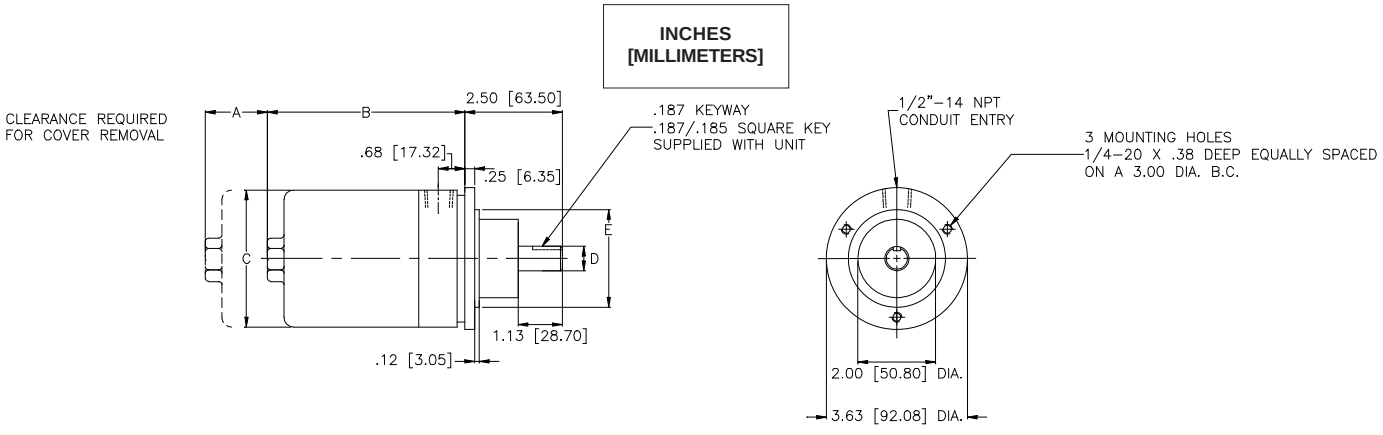
PT-1010 SERIES STANDARD OPTIONS

Code	Description	Selection
Position Feedback		
F010	Anti Backlash Feedback Gearing *Up to 40 turns*	
F011	Anti Backlash Feedback Gearing *Up to 420 turns*	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	

For a full description of options, go to the Complete Listing of Options starting on page 197.

Code	Description	Selection
Auxiliary Switches		
S005	Aluminum Cam (Up to 40 turns)	
S006	Aluminum Feedback Frame (Up to 40 turns)	
S007	Aluminum Feedback Frame (41 to 420 turns)	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S011	Four Adjustable Position Switches, SPDT, 10 Amp, Up to 40 turns, 120/240 Vac	
S012	Four Adjustable Position Switches, DPDT, 10 Amp, Up to 40 turns, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
Special Electrical Options		
X020	24 Vdc power supply for position transmitter, separately mounted, 120/240 Vac input. For use with DC transmitter	
Special Mechanical Options		
Y019	Metric mounting holes, 6mm	

PF-1010 & PT-1010 MAJOR DIMENSIONS



Model	A	B	C	D	E
NEMA Type 4 "X-Proof"	5.63 (143)	6.82 (172.23)	3.50 (88.9)	0.625 (18.875)	2.5 (63.5)
Short Cover	4.02 (102.11)	5.08 (129.03)	3.50 (88.9)	0.625 (18.875)	2.5 (63.5)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

PF-1210 SERIES SELECTION CHART

		Selection
Basic Model	**PF-1210, Position Sensor**	
Input Shaft Turns	**Specify** specific input shaft turns required. (***Not the Range***)	
	40° to 5.85 turns	
	5.85 to 60 turns	

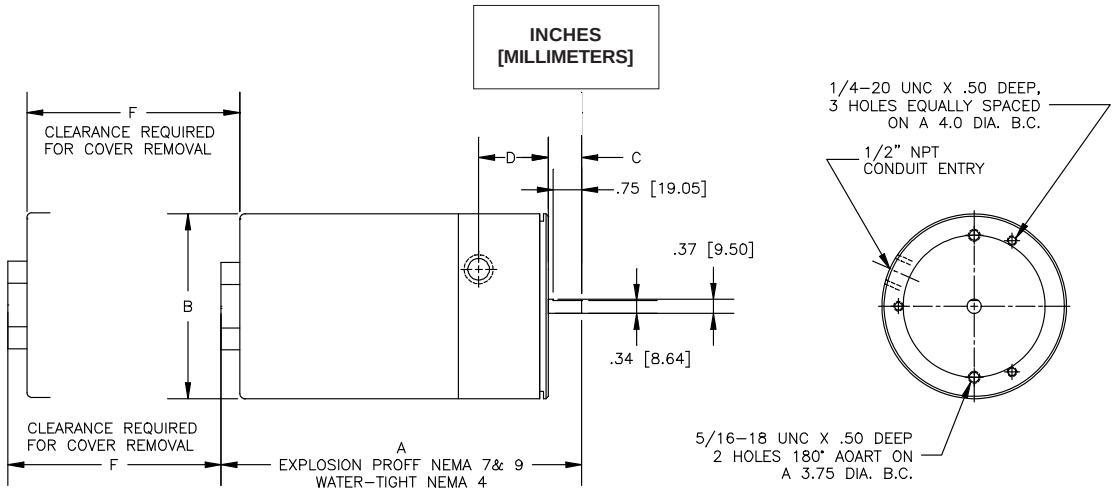
PF-1210 SERIES STANDARD OPTIONS

Code	Description	Selection
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each	
F003	Potentiometer, 5000 ohms	
F004	Potentiometer, 10,000 ohms	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter	
F010	Anti Backlash Feedback Gearing, *40° to 5.85 turns*	
F011	Anti Backlash Feedback Gearing, *5.85 turns to 60 turns*	
F012	Tandem Potentiometers, 5000 ohms	
F013	Tandem Potentiometers, 10,000 ohms	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	

For a full description of options, go to the Complete Listing of Options starting on page 197.

Code	Description	Selection
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S005	Aluminum Cam, 40° to 5.85 turns	
S006	Aluminum Feedback Frame, 40° to 5.85 turns	
S007	Aluminum Feedback Frame 5.85 turns to 60 turns	
S009	Two Adjustable Position Switches, SPDT, 10 Amp, 120/240 Vac	
S010	Two Adjustable Position Switches, DPDT, 10 Amp, 120/240 Vac	
S011	Four Adjustable Position Switches, SPDT, 10 Amp, 40° to 5.85 turns, 120/240 Vac	
S012	Four Adjustable Position Switches, DPDT, 10 Amp, 40° to 5.85 turns, 120/240 Vac	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
Special Mechanical Options		
Y020	Metric mounting holes, 6mm	

PF-1210 MAJOR DIMENSIONS



A	B	C	D	E	F
9.53 (190.6)	4.88 (123.95)	0.88 (22.35)	1.84 (46.74)	6.28 (159.51)	

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LF-1210 SERIES SELECTION CHART

		Selection
Basic Model	**LF-1210, Position Sensor**	
Input Shaft Turns	**Specify stroke (3 inch increments)** 0.75 inch (19mm to 18 inches (457mm)	

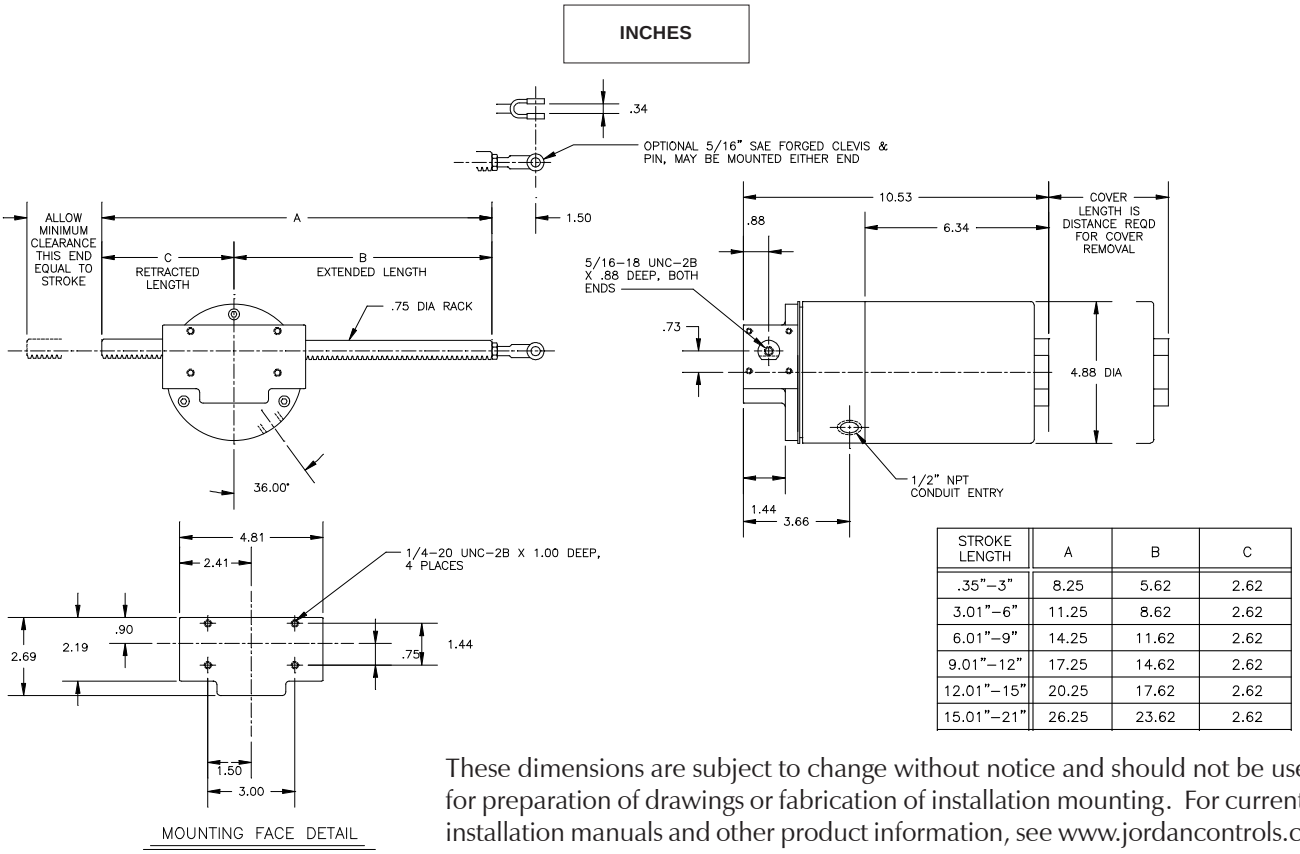
LF-1210 SERIES STANDARD OPTIONS

Code	Description	Selection
Drive Arm/Adapter Clevis/Driven Arm		
D036	Rack bellows	
Position Feedback		
F001	Potentiometer, 1000 ohms	
F002	Tandem Potentiometer, 1000 ohms each	
F003	Potentiometer, 5000 ohms	
F004	Potentiometer, 10,000 ohms	
F005	Transmitter, 4-20 mA output	
F006	One Potentiometer, 1000 ohms plus 4-20 mA Transmitter	
F012	Tandem Potentiometers, 5000 ohms	
F013	Tandem Potentiometers, 10,000 ohms	
F999	Special Potentiometer (consult factory)	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	

For a full description of options, go to the Complete Listing of Options starting on page 197.

Code	Description	Selection
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S005	Aluminum Cam	
S006	Aluminum Feedback Frame, *Up to 18 inches*	
S009	Two Adjustable Position Switches, SPDT, 10 Amp	
S010	Two Adjustable Position Switches, DPDT, 10 Amp	
S011	Four Adjustable Position Switches, SPDT, 10 Amp	
S012	Two nonadjustable end of travel position switches, SPST, 10 Amp	
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
Special Mechanical Options		
Y025	Metric mounting holes, 6mm	

LF-1210 MAJOR DIMENSIONS



STROKE LENGTH	A	B	C
.35"–3"	8.25	5.62	2.62
3.01"–6"	11.25	8.62	2.62
6.01"–9"	14.25	11.62	2.62
9.01"–12"	17.25	14.62	2.62
12.01"–15"	20.25	17.62	2.62
15.01"–21"	26.25	23.62	2.62

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

PT-1330 SERIES SELECTION CHART

		Selection
Basic Model	**PT-1330, Two-wire, 4 to 20 mA output**	
Transmitter Type	**1**: 120 Vac line powered, 4 to 20 mA dc. Capable of powering loads of up to 600 ohms	
	2: 240 Vac line powered, 4 to 20 mA dc. Capable of powering loads of up to 600 ohms	
	12: Loop powered, two-wire, 4 to 20 mA dc. Customer-supplied, external dc power supply required	

PT-1330 SERIES STANDARD OPTIONS

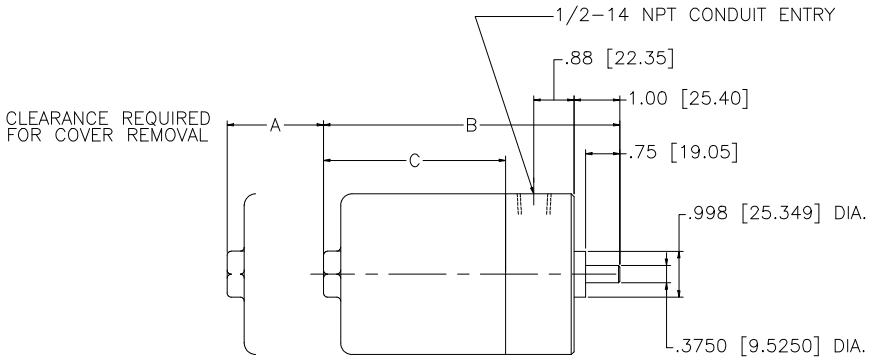
Code	Description	Selection
Drive Arm/Adapter Clevis/Driven Arm		
D029	PT-1300 Mounting Kit	
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	
Auxiliary Switches		
S009	Two Adjustable Position Switches, SPDT, 10 Amp	
S011	Four Adjustable Position Switches, SPDT, 10 Amp	

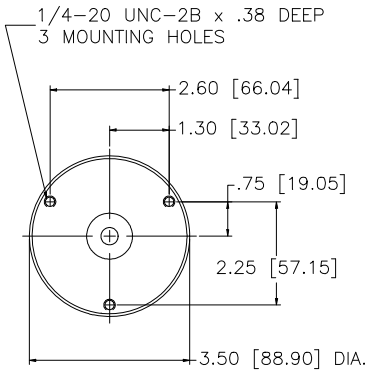
Code	Description	Selection
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	
W003	Food Grade Epoxy	
W004	Baked On Epoxy	
W005	Teflon	
Special Electrical Options		
X020	24 Vdc power supply for position transmitter, separately mounted, 120/240 Vac input. For use with DC transmitter	
Special Mechanical Options		
Y021	Metric mounting holes, 5mm	

For a full description of options, go to the Complete Listing of Options starting on page 197.

PT-1330 MAJOR DIMENSIONS

INCHES [MILLIMETERS]





STROKE LENGTH	A	B	C
PT–1330 SHORT COVER	4.01 [101.85]	6.47 [164.34]	3.98 [101.09]
PT–1330 LONG COVER	5.63 [144.00]	8.07 [204.98]	5.58 [141.73]

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

PT-1510 SERIES SELECTION CHART

		Selection
Basic Model	**PT-1510**, Two-wire, 4 to 20 mA output	
Transmitter Type	**1**: 120 Vac line powered, 4 to 20 mA dc. Capable of powering loads of up to 600 ohms	
	2: 240 Vac line powered, 4 to 20 mA dc. Capable of powering loads of up to 600 ohms	
	12: Loop powered, two-wire, 4 to 20 mA dc. Customer-supplied, external dc power supply required	

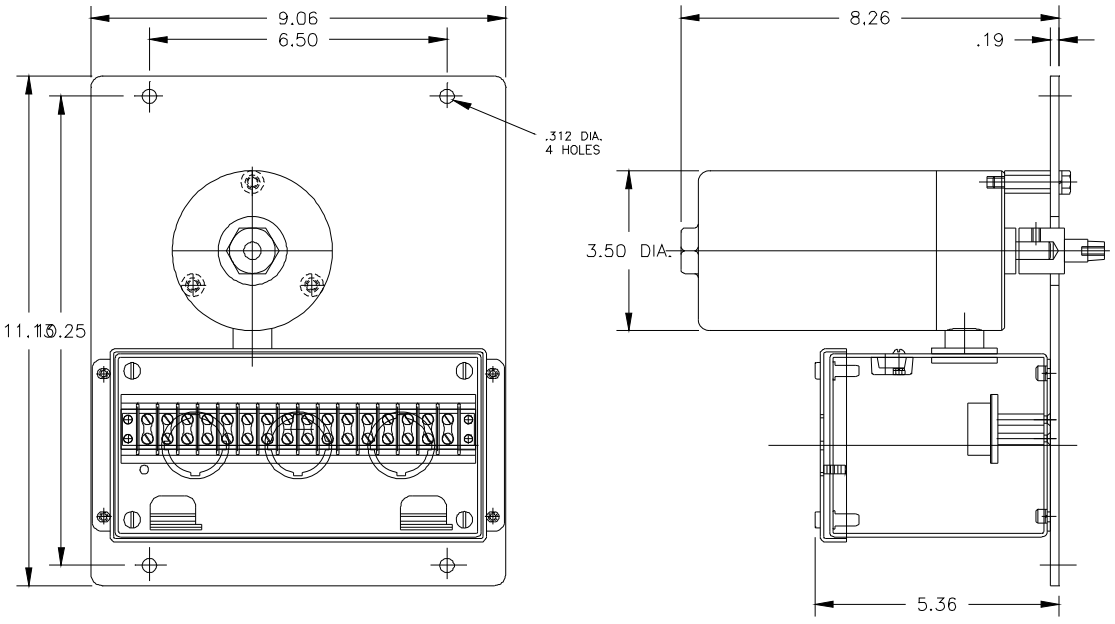
PT-1510 SERIES STANDARD OPTIONS

Code	Description	Selection
Heater		
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
Identification/Certifications		
J002	Stainless Steel Tags	
J004	One year extended warranty	
J005	Two year extended warranty	

Code	Description	Selection
Auxiliary Switches		
S009	Two Adjustable Position Switches, SPDT, 10 Amp	
S011	Four Adjustable Position Switches, SPDT, 10 Amp	
Special Electrical Options		
X020	24 Vdc power supply for position transmitter, separately mounted, 120/240 Vac input. For use with DC transmitter	

For a full description of options, go to the Complete Listing of Options starting on page 197.

PT-1510 MAJOR DIMENSIONS



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

LT-1010 SERIES SELECTION CHART

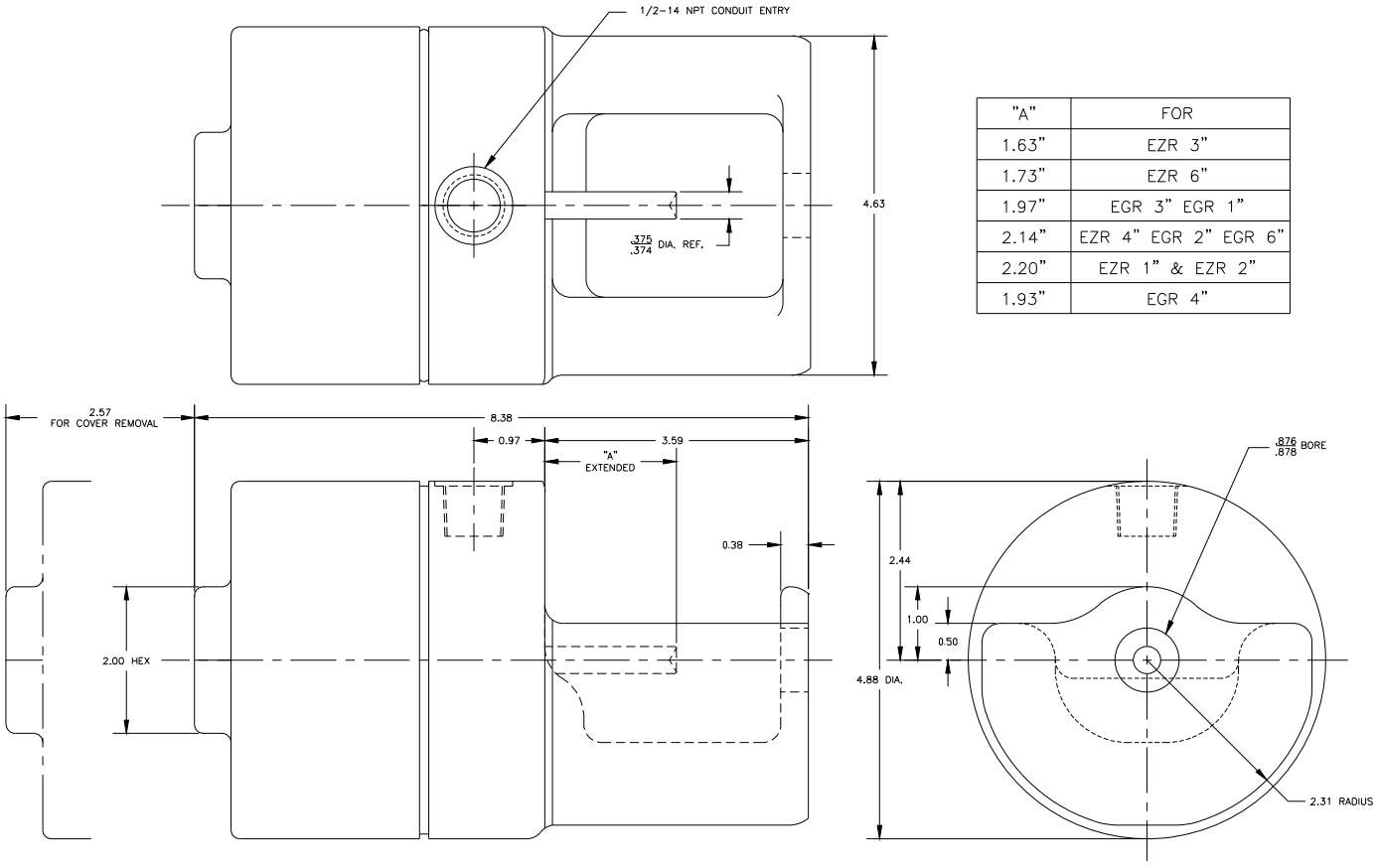
		Selection
Basic Model	**LT-1010**, Linear Position Transmitter	
Stroke	**1**: 0.38 - 1.5 in. (9.6 - 38.1 mm) stroke	
	2: 1.5 - 2 in. (38.1 - 50.8 mm) stroke	

LT-1010 SERIES STANDARD OPTIONS

Code	Description	Selection
Painting/Coating		
W001	JCI Standard Polyurethane Blue	
W002	Two Part Epoxy	

LT-1010 MAJOR DIMENSIONS

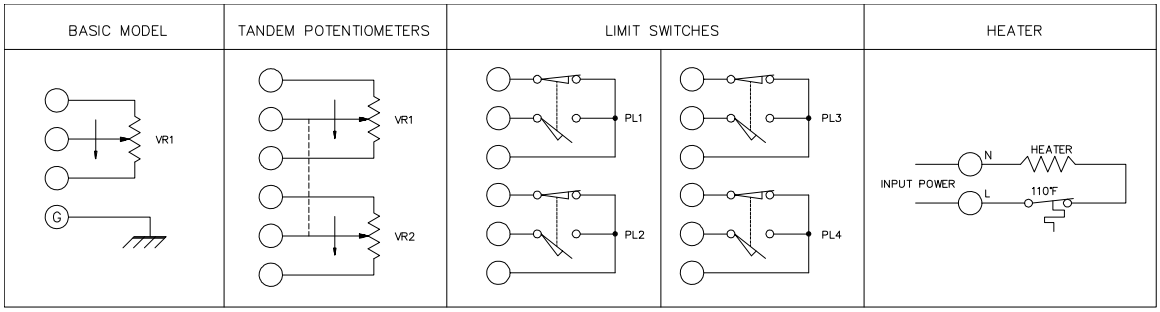
INCHES [MILLIMETERS]



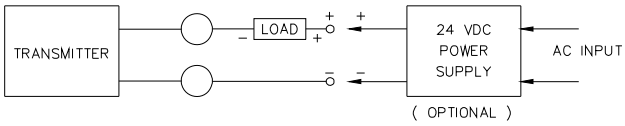
"A"	FOR
1.63"	EZR 3"
1.73"	EZR 6"
1.97"	EGR 3" EGR 1"
2.14"	EZR 4" EGR 2" EGR 6"
2.20"	EZR 1" & EZR 2"
1.93"	EGR 4"

TYPICAL WIRING DIAGRAMS

PF-1010, PF-1210 & LF-1210

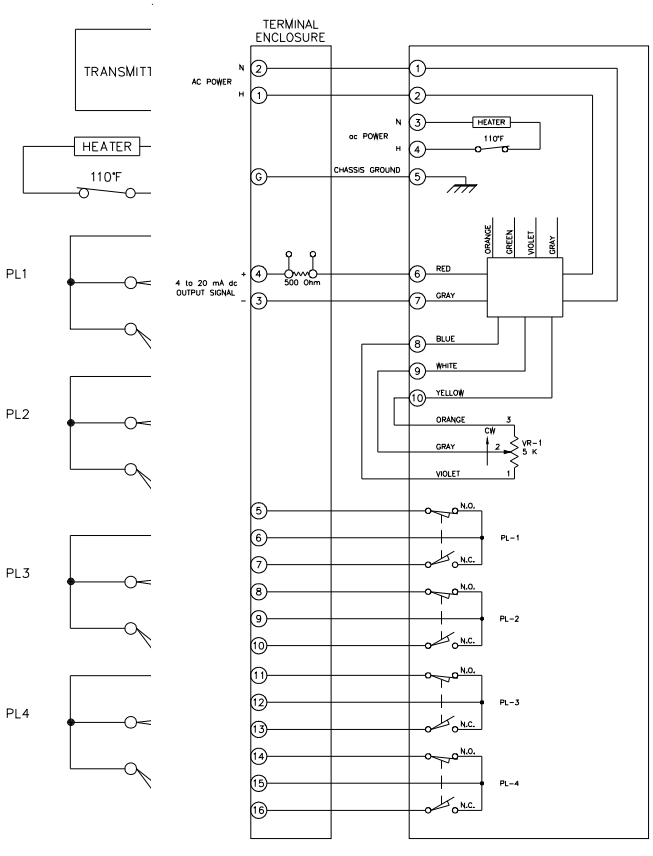


PT-1010 & LT-1010



PT-1330

PT-1510



These dimensions and specifications are subject to change without notice. Current drawings and specifications are available upon request.

REMOTE MOUNTED SERVO AMPLIFIERS

The basic amplifier is an electronic device used to control the response and output shaft positioning of an electric actuator in a closed-loop configuration.

All amplifiers require a minimum of three basic inputs and one output to function:

- Input Power
- Command Signal Input
- Feedback Signal Input
- Feedback Signal Output

TYPICAL AC AMPLIFIER FEATURES:

- Loss of signal detection: Monitors 4 to 20 mA command, when below 4 mA lock in place or run to position.
- Dynamic Braking: Energizes both motor directions at "null" to stall the motor.
- Feedback Output: Jordan Controls amplifiers (except AD7500) are provided with 2-wire, loop-powered, isolated, 4 to 20 mA signal output for customer monitoring of actuator position

TYPICAL DC AMPLIFIER FEATURES:

- Speed Adjust: Limits maximum motor armature voltage
- Current Limit: Sets maximum output current to motor
- Feedback Output: Jordan Controls amplifiers are provided with 2-wire, loop-powered, isolated, 4 to 20 mA signal output for customer monitoring of actuator position

GUIDE TO REMOTE AMPLIFIER SELECTIONS

Actuator	Remote Amplifier	Actuator	Remote Amplifier	Actuator	Remote Amplifier
SM-1110-N	AD-8240	SM-5190-N	AD-8240	LA-2910-N	AD-8900
SM-1120-N	AD-8240	SM-5210-N	AD-8900	LA-2980-N	AD-9120
SM-1140-N	AD-7530	SM-5220-N	AD-8240	LA-5115-N	AD-8900
SM-1150-N	AD-8240	SM-5260-N	AD-7830/AD-9120	LA-5120-N	AD-8240
SM-1170-N	AD-8240	SM-5310-N	AD-8900	LA-5140-N	AD-7540
SM-1180-N	AD-7530	SM-5360-N	AD-7830/AD-9120	LA-5160-N	AD-7830/AD-9120
SM-1520-N	AD-8240	SM-5430-N	AD-8900	LA-5190-N	AD-8240
SM-1530-N	AD-8240	SM-5480-N	AD-7830/AD-9120	LA-5210-N	AD-8900
SM-1540-N	AD-7530	LA-1110-N	AD-8240	LA-5220-N	AD-8240
SM-1550-N	AD-8240	LA-1140-N	AD-7530	LA-5260-N	AD-7830/AD-9120
SM-1580-N	AD-7540	LA-1170-N	AD-8240	LA-5310-N	AD-8900
SM-1590-N	AD-8240	LA-2320-N	AD-8240	LA-5360-N	AD-7830/AD-9120
SM-1630-N	AD-8240	LA-2390-N	AD-8240	MC-1110-N	AD-8240
SM-1650-N	AD-8240	LA-2410-N	AD-8240	MC-1140-N	AD-7530
SM-1680-N	AD-7540	LA-2415-N	AD-8900	MC-1170-N	AD-8240
SM-1715-N	AD-8900	LA-2420-N	AD-8240	SM-LA-3300-N	AD-8240
SM-1720-N	AD-8240	LA-2440-N	AD-7540	MV-1110-N	AD-8240
SM-1730-N	AD-8240	LA-2460-N	AD-7830/AD-9120	MV-1120-N	AD-8240
SM-1740-N	AD-7540	LA-2490-N	AD-8240	MV-1140-N	AD-7530
SM-1750-N	AD-8240	LA-2510-N	AD-8900	MV-1170-N	AD-8240
SM-1760-N	AD-7830/AD-9120	LA-2520-N	AD-8240	MV-1510-N	AD-8240
SM-1790-N	AD-8240	LA-2610-N	AD-8900	MV-1550-N	AD-8240
SM-5115-N	AD-8900	LA-2620-N	AD-8240	MV-1570-N	AD-8240
SM-5120-N	AD-8240	LA-2710-N	AD-8900	VA-1630-N	AD-8240
SM-5140-N	AD-7540	LA-2720-N	AD-8240	VA-1650-N	AD-8240
SM-5160-N	AD-7830/AD-9120			VA-1680-N	AD-7540

50 foot wire run maximum between amplifier and actuator. Consult factory for longer runs.

GENERAL DESCRIPTION

The AD-7500 Series Remote Servo Amplifiers are designed for bi-directional positioning of dc actuators. These amplifiers are normally used to control an actuator in closed-loop servo positioning, and are designed for remote mounting.

The output voltage is proportional to the difference between command and position feedback signals for optimum performance results.

FEATURES

- Optimum motor efficiency achieved
- Fast response
- Extremely broad speed range
- Anti-condensation heater and thermostat (for E & X models)
- Current limiting overload protection tailored to the actuator being controlled
- Peak current operation for optimum response
- Motor speed clamp circuit
- Limit switch circuit for positive motor stopping
- Null output
- EMF or tachometer rate input. Potentiometer, current or voltage command signal inputs

SPECIFICATIONS

Power Input: 120/240 Vac, 1 phase, 50/60 Hz into a 175 VA (AD-7530) or 350 VA (AD-7540) isolation transformer.

Power Output:
Voltage: 0 to 28 Vdc
Current: 5 or 10 amp peak
Null: 15 Vdc, 100 mA maximum de-energized at null
Reference: ± 15 Vdc, 200 mA maximum each

Signal Inputs:
Command: 0 to 10 Vdc, 4 to 20 mA
Position Feedback: 1000 ohm potentiometer
Tachometer: Integral to actuator
Limit Switch: One for each end of travel

Ambient Temperature Range:
32° F to 131° F (0° C to 55° C)

Approximate Weight: AD-7530: 15 lbs. (6.8 kg)
AD-7540: 20 lbs. (9.1 kg)
Enclosure: 35 lbs. (15.9 kg)

Remote Mounting Distance: 50 feet or less.
(Consult factory for longer runs.)

AD-7500 SERIES SELECTION CHART

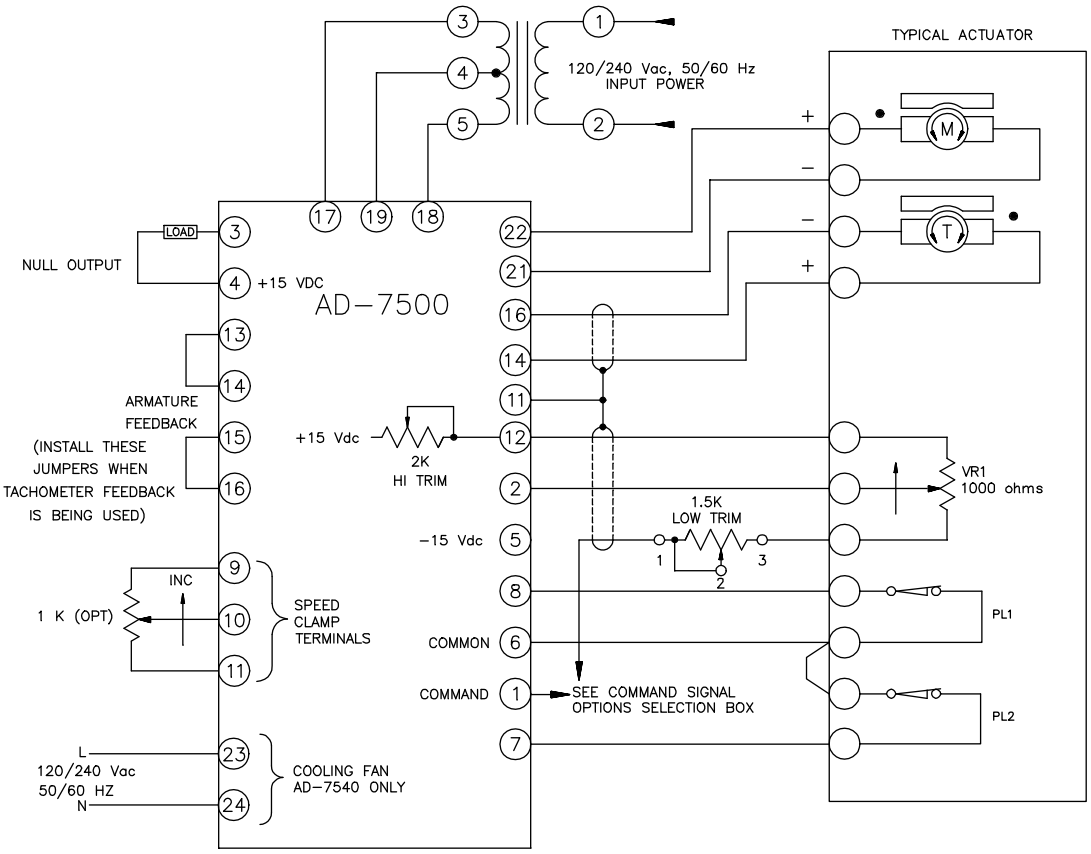
			Selection
1	Basic Model	**AD-7530-P**, 5A peak at 24 Vdc. 10A for 0.5 second peak	
		AD-7530-E, 5A peak at 24 Vdc. 10A for 0.5 second peak, mounted in a NEMA 4 enclosure	
		AD-7540-P, 10A peak at 24 Vdc. 20A for 0.5 second peak	
		AD-7540-E, 10A peak at 24 Vdc. 20A for 0.5 second peak, mounted in a NEMA 4 enclosure	
2	Voltage Input	**1**: 120 Vac, 1 Phase, 50/60 Hz	
		2: 240 Vac, 1 Phase, 50/60 Hz	
3	Actuator Controlled	**A**: The Jordan Controls 24 Vdc actuator to be used with this amplifier	
4	Input Signal	**P**: 1000 ohm potentiometer	
		V: 0 to 10 Vdc	
		M: 4 to 20 mA dc	

AD-7500 SERIES STANDARD OPTIONS

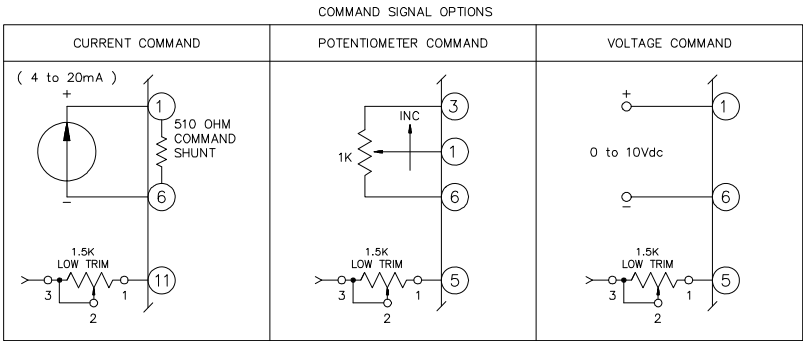
Code	Description	Selection
	Heater	
H001	Anti-Condensation Heater (24 VDC)	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	

For a full description of options, go to the Complete Listing of Options starting on page 197.

TYPICAL WIRING DIAGRAMS



COMMAND SIGNAL OPTIONS



Due to wide variations in the terminal numbering of actuator products, actual wiring should follow the print supplied with the actuator and amplifier.

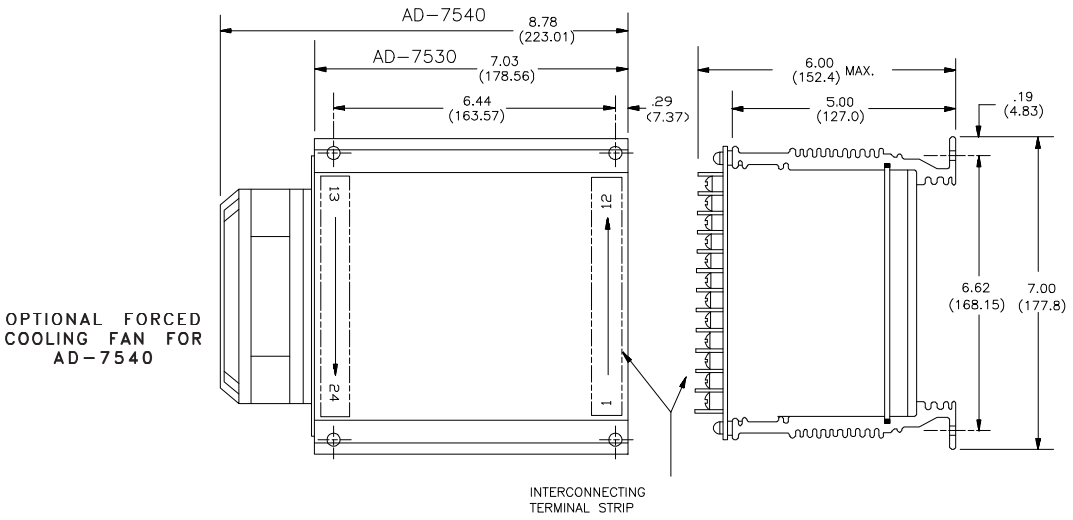
NOTES:

1. Maximum wire run from amplifier to actuator must not exceed 50 feet. Consult factory for longer runs.
2. Shielded wiring is required with the shield grounded at source common for all low level circuits. This includes command & feedback signals and position torque limit switches.
3. Wire size must allow for minimal voltage drop in wiring to the actuator motor, but not smaller than 14 AWG.

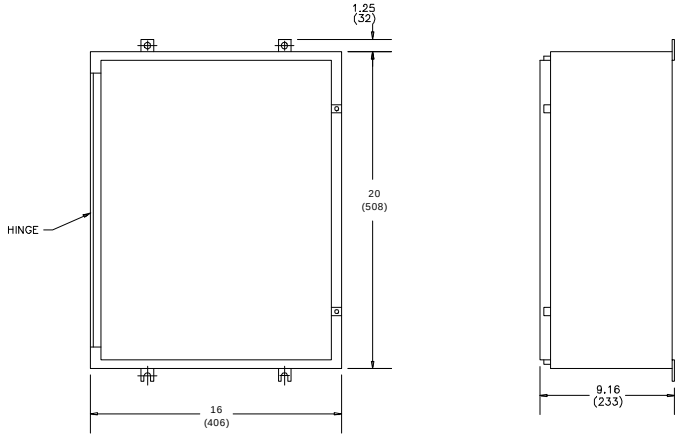
Actuator Model Prefix	Actuator Action on Increasing Command Signal
SM or TA	CW rotation as viewed facing output shaft
MC-1140-N	Increasing regulator outlet pressure
LA	Rack moves to the right when viewing rack block face with rack teeth down
MV	Actuator retracts (valve stem moves up)

AD-7530-P & AD-7540-P MAJOR DIMENSIONS

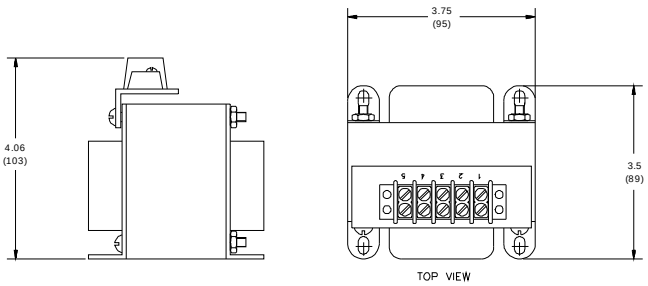
INCHES (MILLIMETERS)



AD-7530-E & AD-7540-E NEMA 4 ENCLOSURE



INPUT TRANSFORMER



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

AD-8240 SERIES REMOTE SERVO AMPLIFIERS

GENERAL DESCRIPTION

The AD-8240 Series Remote Servo Amplfiers are full-featured, single phase ac triac switching devices designed for remote mounting for new installations, or for direct replacement of existing AD-8800 Series remote mounted servo-amplifiers.

FEATURES

- Accepts analog current and voltage command signals that are DIP switch selectable
- Field selectable command signal monitor to cause actuator to lock-in-place or drive to a desired position on loss of current command signal
- Adjustable deadband
- Field selectable dynamic brake
- Adjustable SPAN and ZERO
- 120 or 240 Vac input, field selectable to match actuator motor voltage
- 4 to 20 mA position feedback signal
- Anti-condensation heater and thermostat (for E & X models)

POPULAR OPTIONS

- Versions are available to serve as direct replacement of all previous AD-8800 series remote mounted servo amplifiers
- On board signal conditioner for 4 to 20 mA position feedback signal (AD-8240/EC-10836 only)
- Watertight and explosion-proof enclosures
- Local controls for enclosed versions

BASIC MODELS

AD-8240: For use with all AC powered, single phase 1100, 1500, 1600, 1700, 3300, 2000 Series and 5000 Series actuators.

SPECIFICATIONS

Power:
Voltage Input: 120 or 240 Vac, 50/60 Hz, single phase. Switch selectable.
Voltage Output: Identical to voltage input
Current Output: 10 amps max. at 120 or 240 Vac
Null Output: 2 amps at 120 Vac or 240 Vac or half wave dc derivative. Jumper selectable
Fuse Protection: Customer supplied and sized based on actuator controlled

Command Signal Inputs: (field selectable)
4 to 20 mA current command into a 200 ohm impedence
0 to 5 or 0 to 10 Vdc voltage command into a 100,000 ohm impedence.
1000 ohm potentiometer command signal input is an option on the AD-8240/EC-10836 model only

Position Feedback Signal: 1000 ohm potentiometer

Position Signal Output: Loop powered, 2 wire 4 to 20 mA signal

Approximate Weights:
AD-8240-P: 2 lbs. (0.9 kg)
with enclosure "E": 25 lbs. (11 kg)
with enclosure "X": 40 lbs. (18 kg)

Remote Mounting Distance: 50 feet or less. (Consult factory for longer runs.)

AD-8240 SERIES SELECTION CHART

			Selection
1	Basic Model	**AD-8240-P**, 120/240 Vac, 1 phase, 50/60 Hz input; 10 amp maximum output.	
		AD-8240-E, 120/240 Vac, 1 phase, 50/60 Hz input; 10 amp maximum output, mounted in a NEMA 4 enclosure	
		AD-8240-X, 120/240 Vac, 1 phase, 50/60 Hz input; 10 amp maximum output, mounted in a NEMA 7 & 9 enclosure	
		AD-8240-R, replacement for AD-8813, AD-8823, AD-8833 and AD-8843	
2	Voltage Input	**1**: 120 Vac, 1 Phase, 50/60 Hz	
		2: 240 Vac, 1 Phase, 50/60 Hz	
3	Actuator Controlled	**A**: The Jordan Controls actuator to be used with this amplifier	
4	Feedback Signal	**P**: 1000 ohm potentiometer	
		V: 0 to 10 Vdc	
		M: 4 to 20 mA dc	
5	Command Signal	**O**: *Current* - 4 to 20 mA or *Voltage* - 0 to 5 Vdc or 0 to 10 Vdc	
		P: 1000 ohm potentiometer	

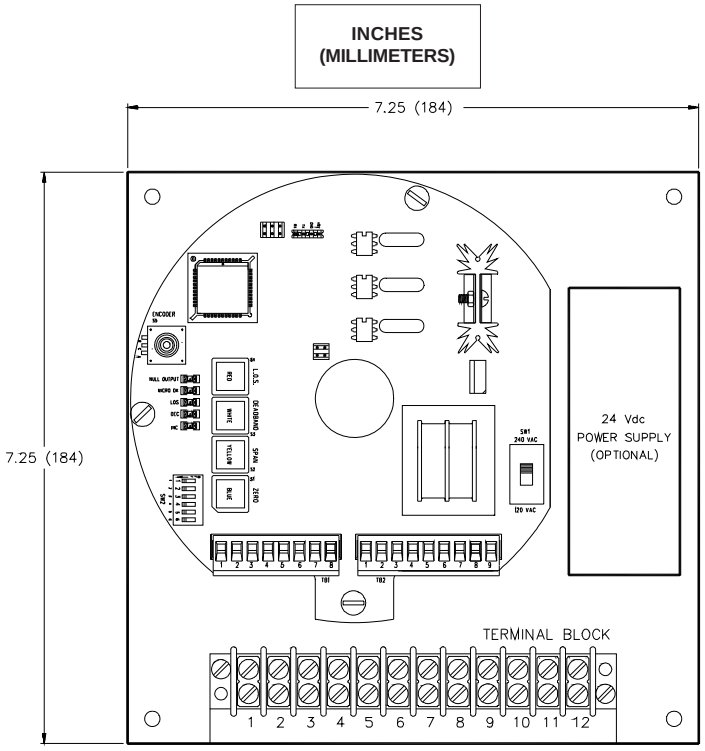
AD-8240 STANDARD OPTIONS

Code	Description	Selection
	Heater	
H002	Anti-Condensation Heater (120 VAC), only available with enclosure option	
H003	Anti-Condensation Heater (240 VAC), only available with enclosure option	
	Special Electrical Options	
X001	24 Vdc power supply, 320 mA, 100-240 Vac input (only available with AD-8240/EC-10836	
X021	EC-10842 (AD-8210 and AD-8220 replacement)	

For a full description of options, go to the Complete Listing of Options starting on page 197.

MAJOR DIMENSIONS

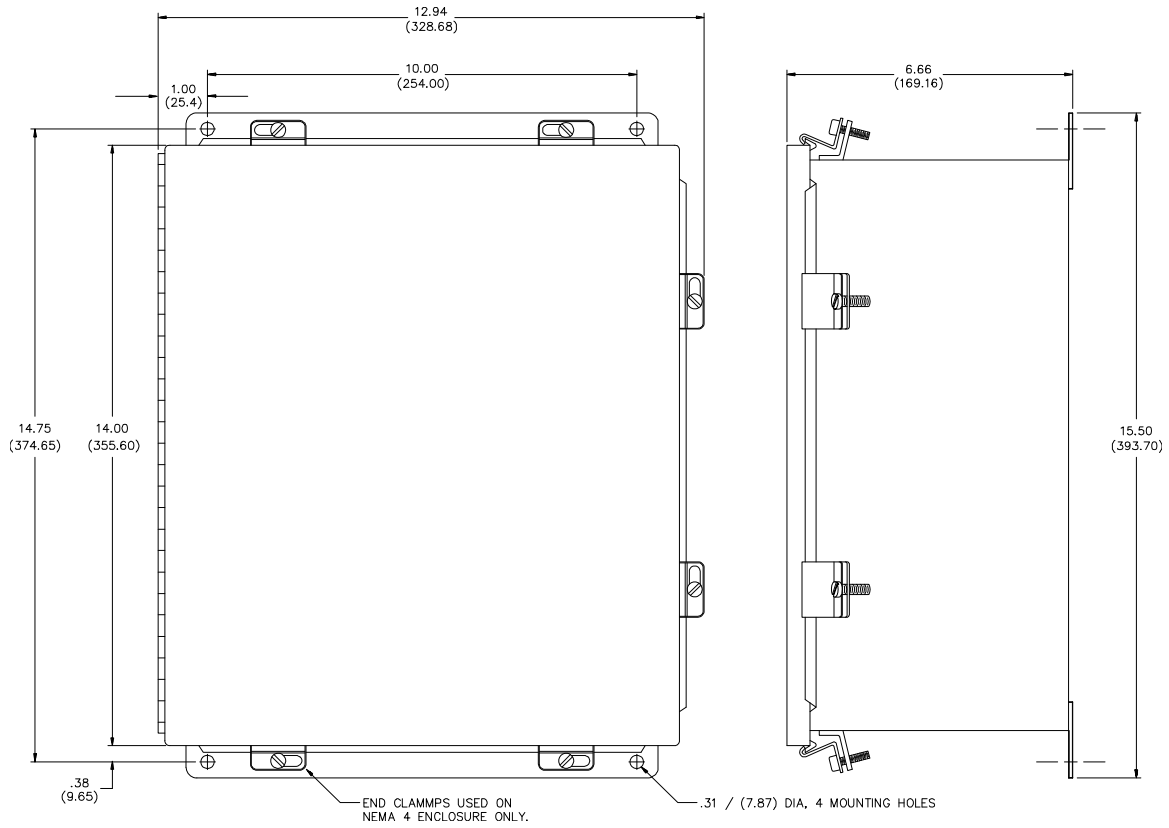
Panel Mount (AD-8240-P & AD-8240-R)



ENCLOSURE MAJOR DIMENSIONS

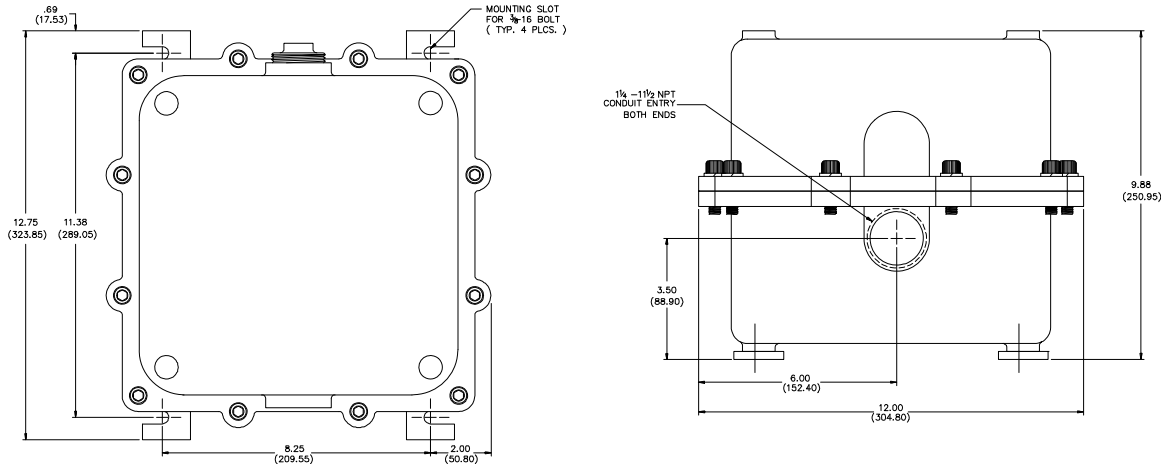
AD-8240-E

INCHES (MILLIMETERS)



AD-8240-X

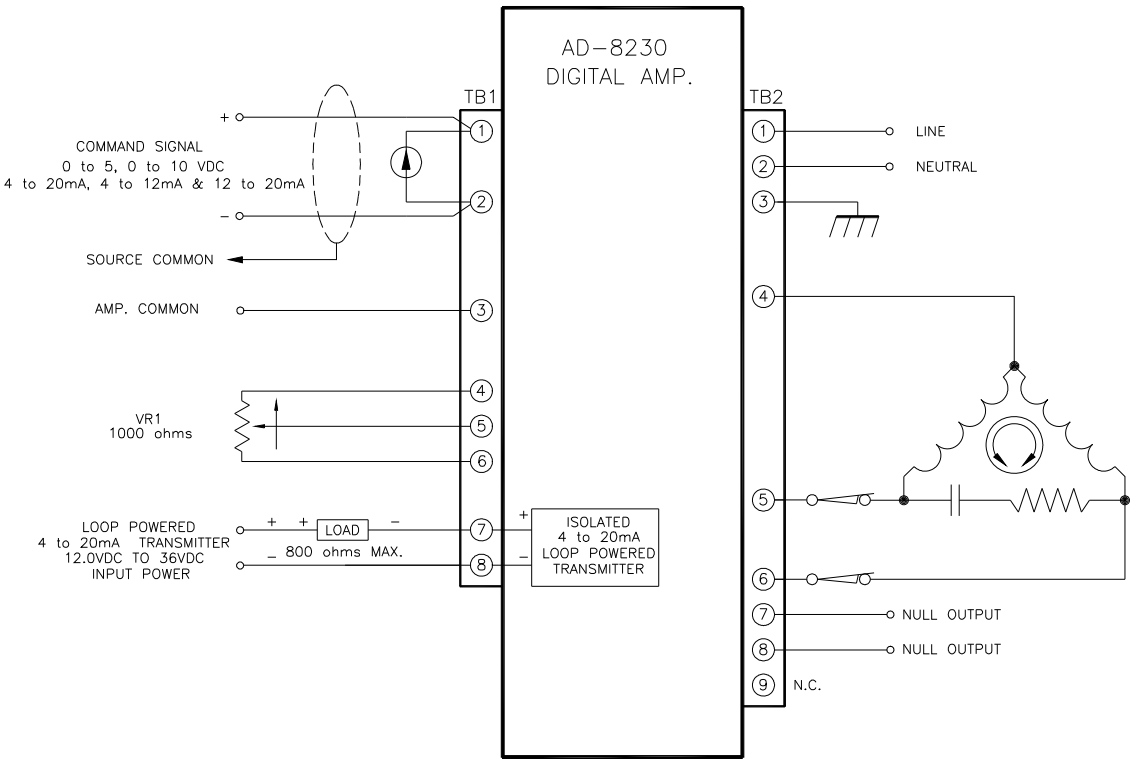
INCHES (MILLIMETERS)



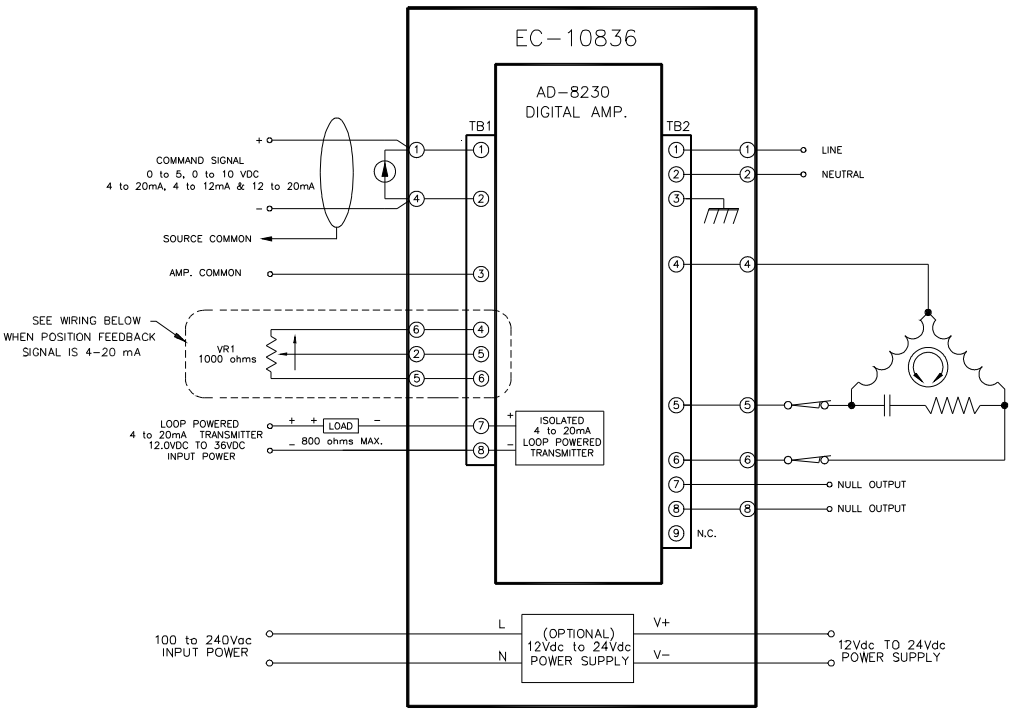
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

TYPICAL WIRING DIAGRAMS

AD-8240



AD-8240 Panel Mount



NOTES:

1. Maximum wire run from amplifier to actuator must not exceed 50 feet. Consult factory for longer runs.
2. Shielded wiring is required with the shield grounded at source common for all low level circuits. This includes command & feedback signals and position torque limit switches.
3. Wire size must allow for minimal voltage drop in wiring to the actuator motor, but not smaller than 14 AWG.

Due to wide variations in the terminal numbering of actuator products, actual wiring should follow the print supplied with the actuator and amplifier.

AD-8900 SERIES 3-PHASE AC REMOTE SERVO AMPLIFIERS

Jordan CONTROLS

GENERAL DESCRIPTION

The AD-8900 Series Remote Servo Amplifiers are full-featured, three-phase AC triac devices designed for remote mount to a variety of Jordan Controls electric actuators for closed loop control. There are no moving contacts or contact arcing; typical of three phase control equipment. The design incorporates solid state devices.

FEATURES

- Accepts analog current and voltage command signals that are DIP switch selectable
- Field selectable command signal monitor to cause actuator to lock-in-place or drive to a desired position on loss of current command signal
- Adjustable deadband
- Field adjustable dynamic brake
- Anti-condensation heater and thermostat (for E models)
- Adjustable zero and span
- 208, 240, 380 or 480 Vac, 3 phase input, to match actuator motor voltage
- Isolated 4 to 20 mA position feedback signal
- Automatic phase correction; user is able to connect incoming phases in any sequence.
- All solid-state with no moving contacts
- Cross firing protection providing short circuit protection
- Fault output on loss of any input or output phase or on loss of command signal
- On-Board diagnostic LEDs

POPULAR OPTIONS

- Watertight enclosure
- Local controls for enclosed versions

ACTUATOR COMPATIBILITY

Rotary: SM-1715-N, SM-5115-N, SM-5210-N, SM-5310-N, SM-5430-N.
Linear: LA-2415-N, LA-2510-N, LA-2610-N, LA-2710-N, LA-2910-N, LA-5115-N, LA-5210-N, LA-5310-N.

SPECIFICATIONS

Voltage & Current Input:
AD-8910 - 240 Vac, 50/60 Hz 3-phase
AD-8920 - 380 Vac, 50 Hz 3-phase
AD-8930 - 480 Vac, 50/60 Hz 3-phase
AD-8940 - 208 Vac, 50/60 Hz 3-phase

Voltage Output: Identical to voltage input

Fault Output: Triac going low on loss of input or output phase or loss of current command signal. Triac output rated 120/240 Vac, 2 amp

Fuse Protection: Customer supplied and sized based on actuator controlled

Command Signal Inputs: Field selectable
- 4 to 20 mA current command into a 200 ohm impedance
- 0 to 5 Vdc, 0 to 10 Vdc command into a 475 ohm impedance

Position Feedback Signal: 1000 ohm potentiometer or 4 to 20 mA, switch selectable

Position Output Signal: Loop powered, isolated 2 wire, 4 to 20 mA signal

Other Inputs:
End-of-travel limit switches (from actuator)
Overtorque limitswitches (from actuator)
Auto/Manual Jumper
- Selects between Automatic and Manual mode
  Automatic Mode, 4-20mA command input controls actuator position
  Manual Mode, Increase and Decrease inputs control actuator position
  Increase (INC)- Commands actuator to increase when in manual mode.
  Decrease (DEC)- Commands actuator to decrease when in manual mode.

Other Outputs: LED Indication:

Loss of Signal (LOS)	Phase A, B & C output
Decrease direction	Phase A, B & C input
Increase direction	Control running

Approximate Weight: 10 lbs. (4.5 kg)
With enclosure - 35 lbs. (16 kg)

Remote Mounting Distance: 50 feet or less. (Consult factory for longer runs.)

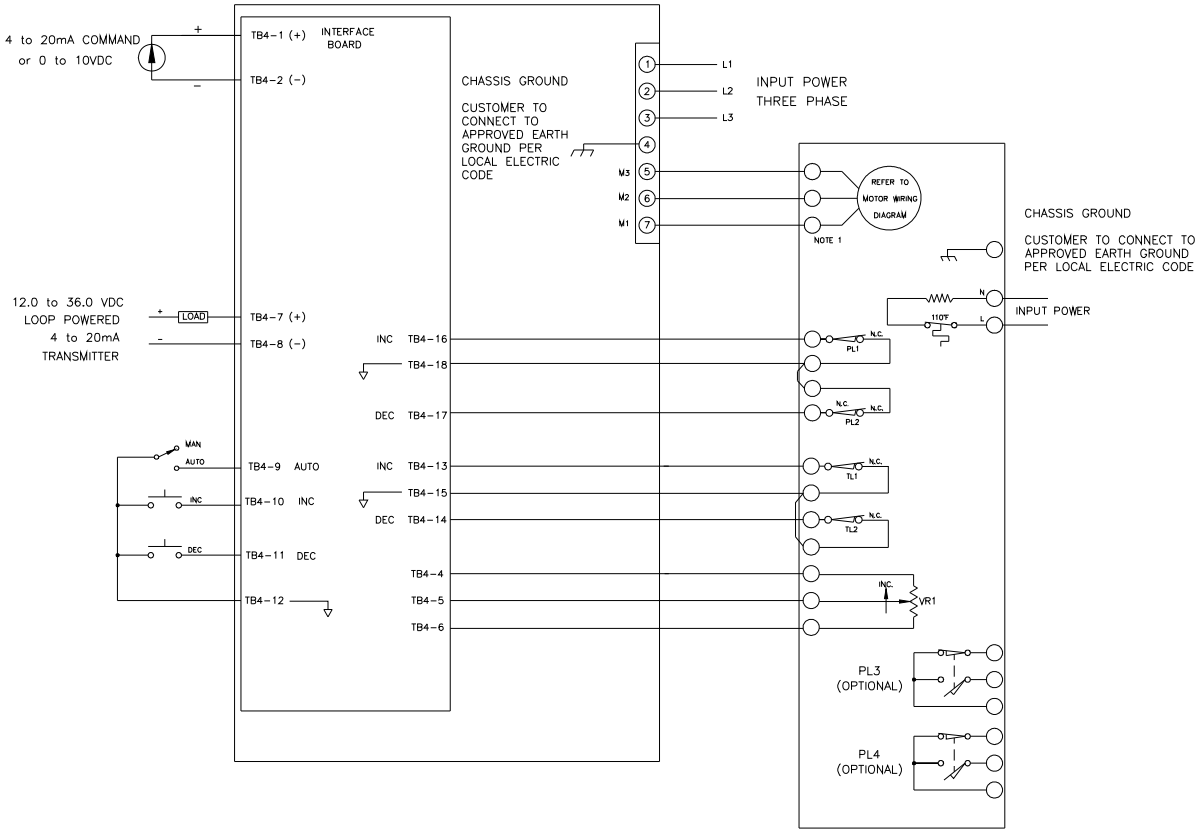
AD-8900 SERIES SELECTION CHART

				Selection
1	Basic Model	**AD-8910-P**	240 Vac, Panel Plate	
		AD-8920-P	380 Vac, Panel Plate	
		AD-8930-P	480 Vac, Panel Plate	
		AD-8940-P	208 Vac, Panel Plate	
		AD-8910-E	240 Vac, Panel Plate, mounted in a NEMA 4 enclosure	
		AD-8920-E	380 Vac, Panel Plate, mounted in a NEMA 4 enclosure	
		AD-8930-E	480 Vac, Panel Plate, mounted in a NEMA 4 enclosure	
		AD-8940-E	208 Vac, Panel Plate, mounted in a NEMA 4 enclosure	

AD-8900 STANDARD OPTIONS

Code	Description	Selection
	Heater	
H002	Anti-Condensation Heater (120 VAC), only available with enclosure option	
H003	Anti-Condensation Heater (240 VAC), only available with enclosure option	

TYPICAL WIRING DIAGRAM



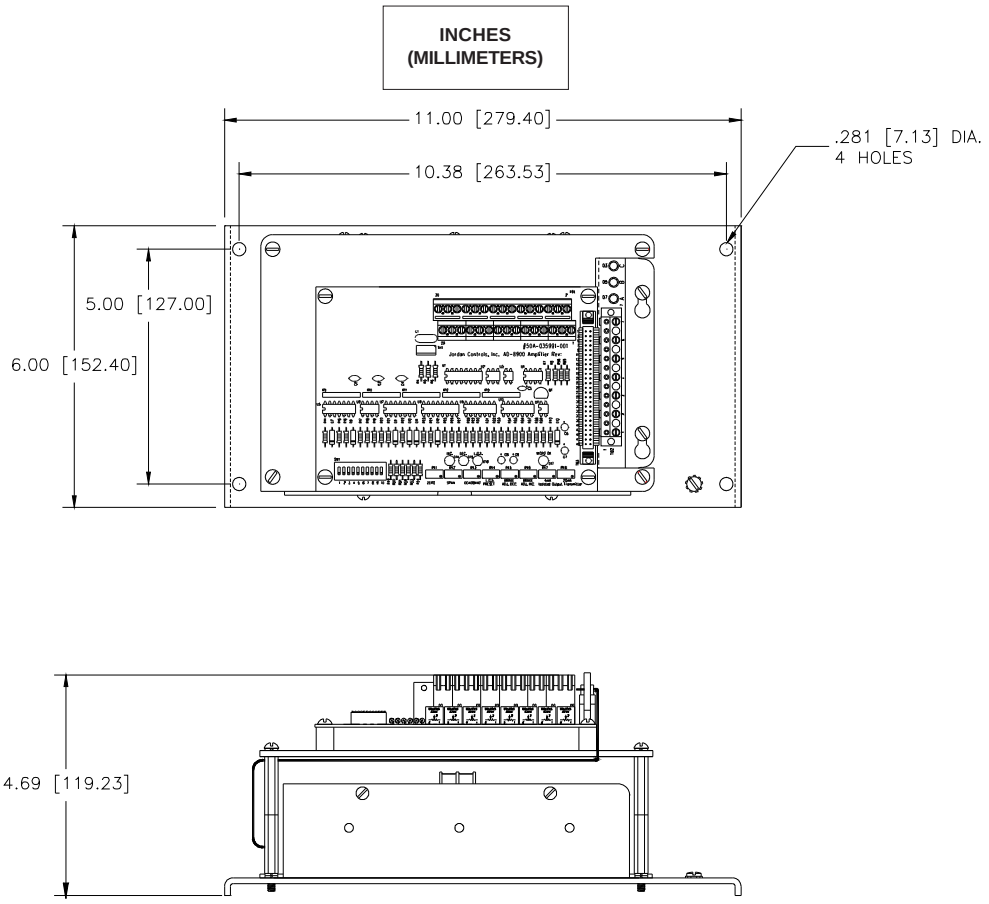
NOTES:

1. Maximum wire run from amplifier to actuator must not exceed 50 feet. Consult factory for longer runs.
2. Shielded wiring is required with the shield grounded at source common for all low level circuits. This includes command & feedback signals and position torque limit switches.
3. Wire size must allow for minimal voltage drop in wiring to the actuator motor, but not smaller than 14 AWG.

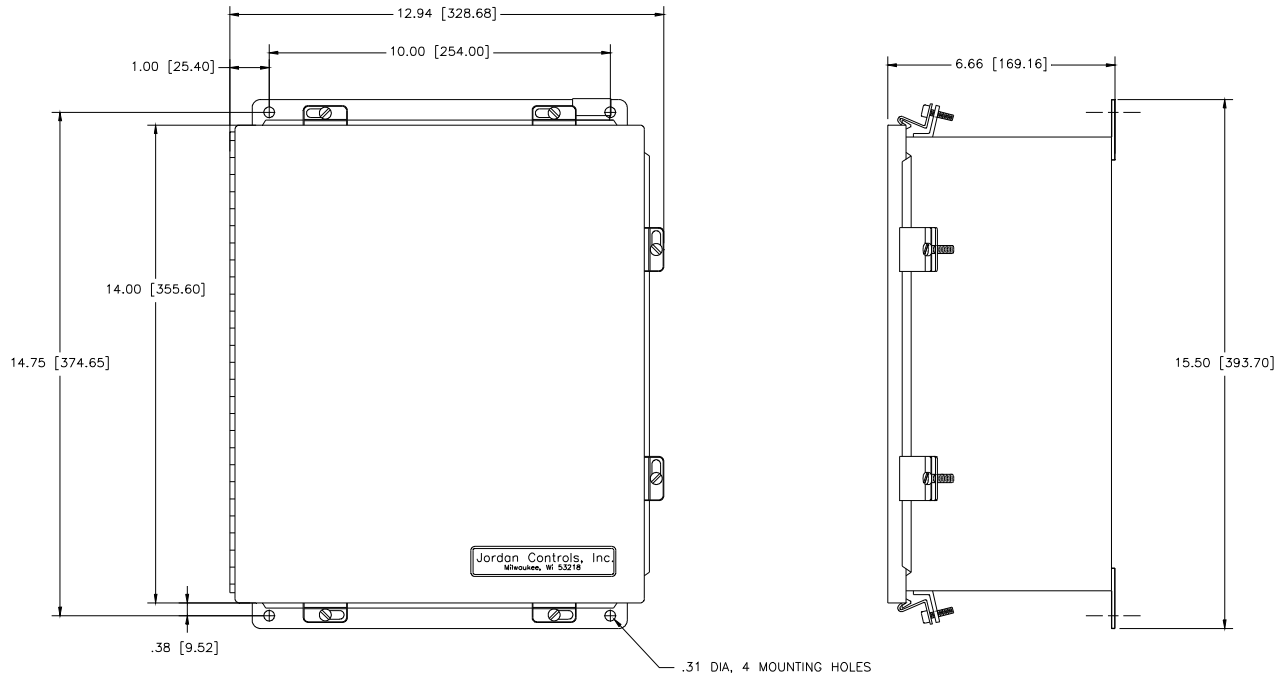
Due to wide variations in the terminal numbering of actuator products, actual wiring should follow the print supplied with the actuator and amplifier.

MAJOR DIMENSIONS

AD-8910-P, AD-8920-P, AD-8930-P, AD-8940-P



AD-8910-E, AD-8920-E, AD-8930-E, AD-8940-E



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

AD-7800 SERIES DC ANALOG REMOTE SERVO AMPLIFIERS

GENERAL DESCRIPTION

The AD-7800 Series Analog Remote Servo Amplifiers are designed for remote-mount bi-directional positioning of DC actuators. It uses a real-time hybrid operating system to drive an advanced, high-power Intelligent Power Module (IPM) output drive stage to the actuator motor. The conservative design allows operation with any of the larger Jordan actuators, and traditional potentiometer set-up requires no special training or equipment. Modular design allows for easy maintenance, or for upgrading to other Jordan amplifier products in the future.

FEATURES

- Drive stage control uses Intelligent Power Module (IPM) for pulse width modulation (PWM) power switching to the motor for longer motor life and cooler operation.
- Proportional Control - Compares the control signal against the feedback signal and accelerates quickly for a large control change and accelerates slowly for a small control change.
- Bi-Directional actuator positioning
- Over-current, under-voltage, short circuit and over temperature protection onboard
- Field selectable loss of signal control: Lock-in-place or go to pre-set position
- Field adjustable speed control
- Plugable terminal block connectors supplied for quick field wiring terminations
- Fast amplifier response time
- Isolated, two wire, 4-20mA output
- LED outputs for Loss of Signal (LOS) indication and Increase/Decrease driving indication
- Customer adjustable gain, zero and span and deadband
- Interchangable powerstage with the AD-9120
- Potentiometer/Dip switch Set-up
- Balance speed control for different load dynamics like overhung loads or back driving type units
- Increase/decrease pushbuttons
- DIP or dry contact auto/manual control

ACTUATOR COMPATIBILITY

Rotary: SM-1760-N, SM-5160-N, SM-5260-N, SM-5360-N, SM-5480-N.

Linear: LA-2460-N, LA-2560-N, LA-2660-N, LA-2660-N, LA-2980-N, LA-5160-N, LA-5260-N.

SPECIFICATIONS

Power Input: 120/240Vac, +/- 10%, 1 Ph., 50/60 Hz

Power Output: 90Vdc/180Vdc, 20 Amperes peak

Command Inputs: 4-20mA into 200 ohm shunt
0-10 Vdc into 100K minimum impedance
0-5 Vdc into 100K minimum impedance

Position Feedback Signal: 1000 ohm potentiometer
4-20mA

Position Output Signal: Loop powered, isolated, 2 wire 4-20mA signal

Other Inputs:
End-of-travel limit switches (from actuator)
Overtorque limit switches (from actuator)

Auto/Manual Jumper

Selects between Automatic and Manual mode
Automatic Mode, 4-20mA command input controls actuator position
Manual Mode, Increase and Decrease inputs control actuator position
Increase (INC)- Commands actuator to increase when in manual mode.
Decrease (DEC)- Commands actuator to decrease when in manual mode.

Other Outputs: LED Indication:
Loss of Signal (LOS)
Increase direction
Decrease direction

Temperature Range: -40°F to 150°F (-40°C to 65°C)

Approximate Weight: 10 lbs. (4.5 kg)
With enclosure - 35 lbs. (16 kg)

Remote Mounting Distance: 50 feet or less. (Consult factory for longer runs.)

AD-7800 SERIES SELECTION CHART

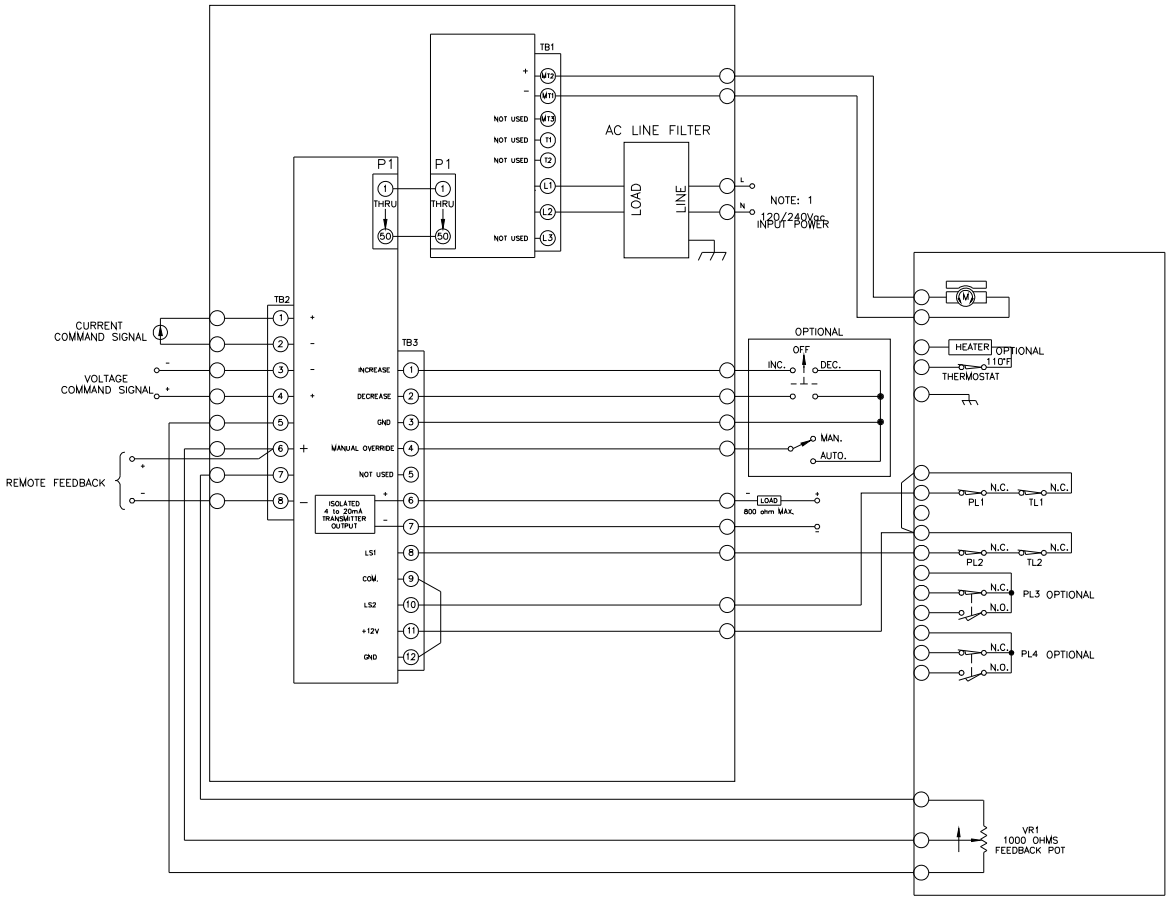
				Selection
1	Basic Model	**AD-7830-P**	Panel Plate, 120/240 Vac, 1 Phase, 50/60 Hz	
		AD-7830-E	120/240 Vac, 1 Phase, 50/60 Hz, mounted in a NEMA 4 enclosure	
		AD-7830-R	AD-7300 Replacement Panel, 120/240 Vac, 1 Phase, 50/60 Hz	

LA-2980 and SM-5480 require 240 Vac input power to the amplifier.

AD-7800 STANDARD OPTIONS

Code	Description	Selection
	Heater	
H002	Anti-Condensation Heater (120 VAC), only available with enclosure option	
H003	Anti-Condensation Heater (240 VAC), only available with enclosure option	

TYPICAL WIRING DIAGRAM



Notes:

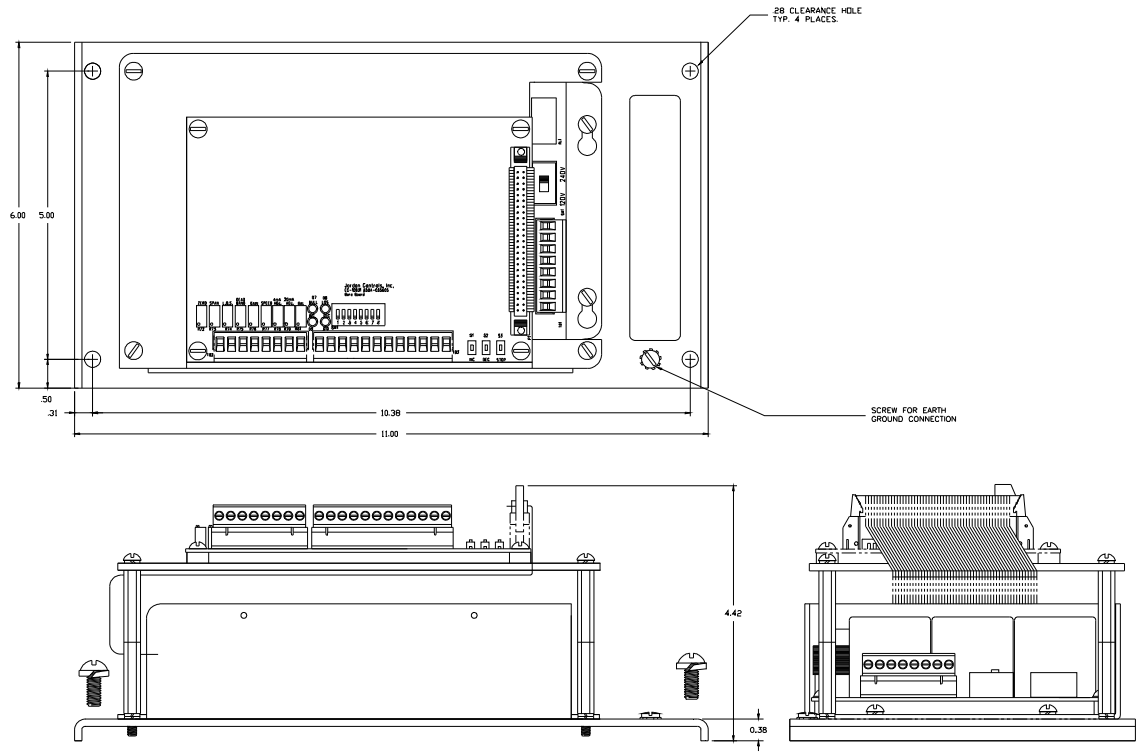
1. 240 Vac input is required where actuator nameplate voltage is 180 Vdc.
2. 24 Vdc power supply is required.
3. Maximum wire run from amplifier to actuator must not exceed 50 feet. Consult factory for longer runs.
4. Shielded wiring is required with the shield grounded at source common for all low level circuits. This includes command & feedback signals and position torque limit switches.
5. Wire size must allow for minimal voltage drop in wiring to the actuator motor, but not smaller than 14 AWG.

Due to wide variations in the terminal numbering of actuator products, actual wiring should follow the print supplied with the actuator and amplifier.

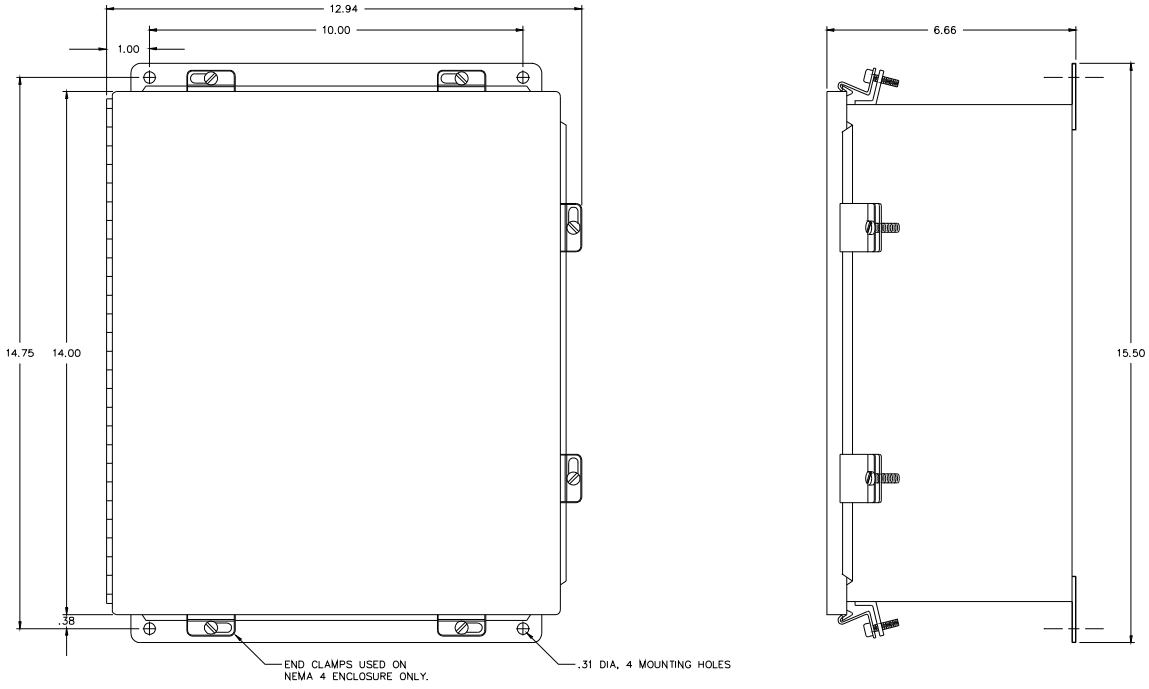
MAJOR DIMENSIONS

AD-7830-P

INCHES



AD-7830-E

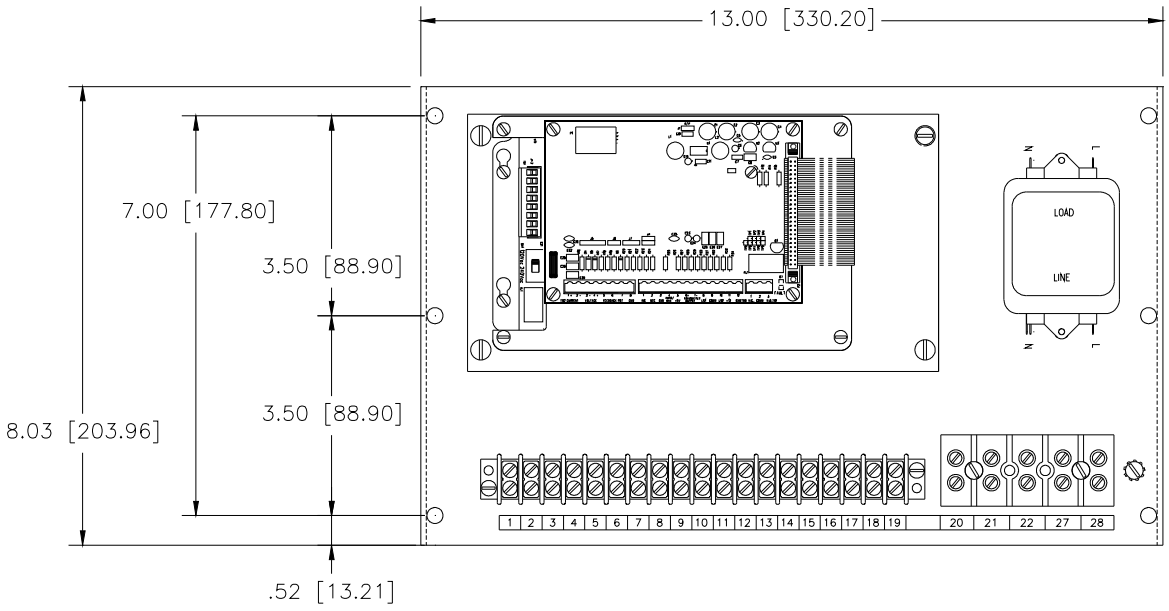


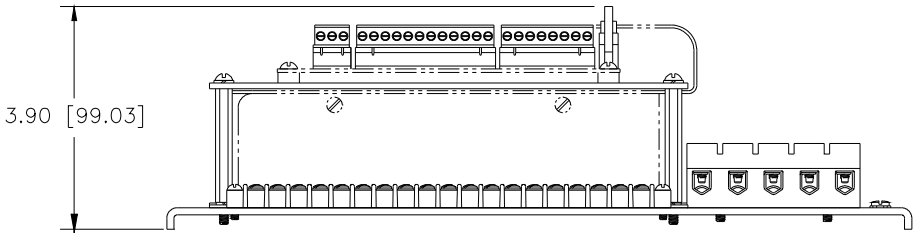
These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

MAJOR DIMENSIONS

AD-7830-R

INCHES (MILLIMETERS)





These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

GENERAL DESCRIPTION

The AD-9120 Digital Remote Servo-Amplifiers are designed for to control appropriate Jordan Controls Actuators. A microprocessor-based control and an IGBT-based intelligent power module is used to drive the actuator to perform highly accurate, bidirectional positioning. The "Rapid Setup" feature allows actuator setup using push-button switches (no potentiometers or jumpers). Self diagnostics and prompts are available through an on-board visual display. Modular design allows for easy maintenance, or for upgrading to other Jordan amplifier products in the future.

FEATURES

- Quick pushbutton setup
- Vacuum fluorescent display
- LED outputs for Loss of Signal (LOS) indication and Increase/Decrease driving indication
- Form C fault relay output
- Drive stage control uses Intelligent Power Module (IPM) for pulse width modulation (PWM) power switching to the motor for longer motor life and cooler operation.
- Proportional Control - Compares the control signal against the feedback signal and accelerates quickly for a large control change and accelerates slowly for a small control change.
- Bi-Directional actuator positioning
- Over-current, under-voltage, short circuit and over temperature protection onboard
- Field selectable loss of signal control: Lock-in-place or go to pre-set position
- Field adjustable speed control
- Plugable terminal block connectors supplied for quick field wiring terminations
- Fast amplifier response time
- Isolated, two wire, 4-20mA output
- Customer adjustable GAIN, ZERO, SPAN, deadband and speed
- Interchangable powerstage with the AD-7830

POPULAR OPTIONS

- 208 Vac input voltage
- Anti-condensation heater

ACTUATOR COMPATIBILITY

Rotary: SM-1760-N, SM-5160-N, SM-5260-N, SM-5360-N, SM-5480-N.

Linear: LA-2460-N, LA-2560-N, LA-2660-N, LA-2660-N, LA-2980-N, LA-5160-N, LA-5260-N.

SPECIFICATIONS

Power Input: 120/240Vac, +/- 10%, 1 Ph., 50/60 Hz

Power Output: 90Vdc/180Vdc, 20 Amperes peak

Command Inputs: 4-20mA into 200 ohm shunt
0-10 Vdc into 100K minimum impedance
0-5 Vdc into 100K minimum impedance

Position Feedback Signal: 1000 ohm potentiometer
4-20mA

Position Output Signal: Loop powered, isolated, 2 wire 4-20mA signal

Other Inputs:
End-of-travel limit switches (from actuator)
Overtorque limit switches (from actuator)
Auto/Manual Jumper

Selects between Automatic and Manual mode
Automatic Mode, 4-20mA command input controls actuator position
Manual Mode, Increase and Decrease inputs control actuator position
Increase (INC)- Commands actuator to increase when in manual mode.
Decrease (DEC)- Commands actuator to decrease when in manual mode.

Other Outputs: Form C fault relay output
LED Indication:
Power
Fault

Approximate Weight: 10 lbs. (4.5 kg)
With enclosure - 35 lbs. (16 kg)

Remote Mounting Distance: 50 feet or less. (Consult factory for longer runs.)

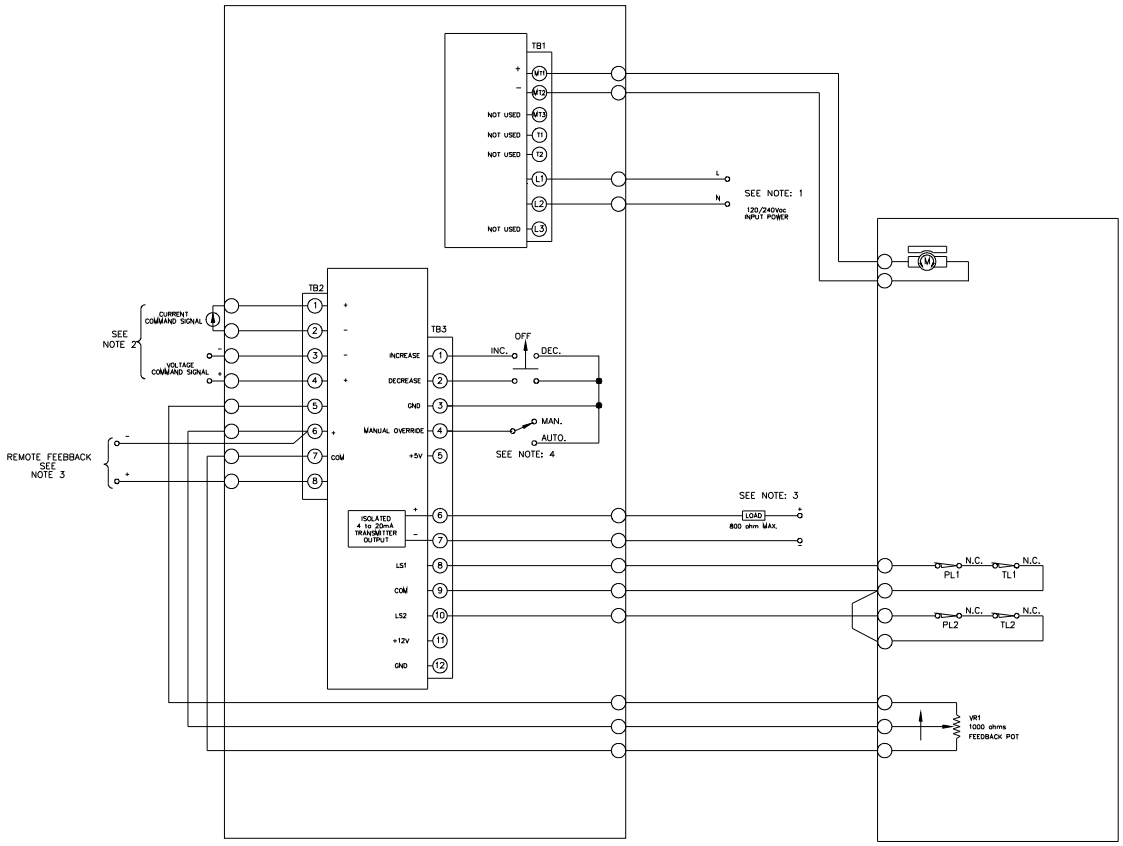
AD-9120 SERIES SELECTION CHART

				Selection
1	Basic Model	**AD-9120-P**	Panel Plate, 120/240 Vac, 1 Phase, 50/60 Hz	
		AD-9120-E	120/240 Vac, 1 Phase, 50/60 Hz, mounted in a NEMA 4 enclosure	
		AD-9120-R	AD-7300 Replacement Panel, 120/240 Vac, 1 Phase, 50/60 Hz	

AD-9120 STANDARD OPTIONS

Code	Description	Selection
	Heater	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	
	Special Electrical Options	
X020	Palm Pilot Setup	
X028	208 Vac input power for AD-9120	

TYPICAL WIRING DIAGRAM



Notes:

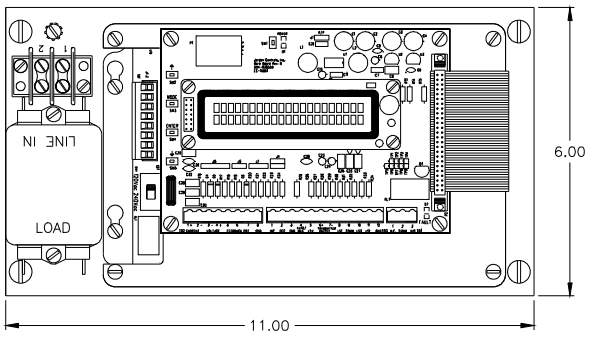
1. 240 Vac input is required where actuator nameplate voltage is 180 Vdc.
2. 24 Vdc power supply is required.
3. For 4-20mA signal to actuator. Maximum wire run from amplifier to actuator must not exceed 50 feet. Consult factory for longer runs.
4. Shielded wiring is required with the shield grounded at source common for all low level circuits. This includes command & feedback signals and limit switches.
5. Wire size must allow for minimal voltage drop in wiring to the actuator motor, but not smaller than 14 AWG.

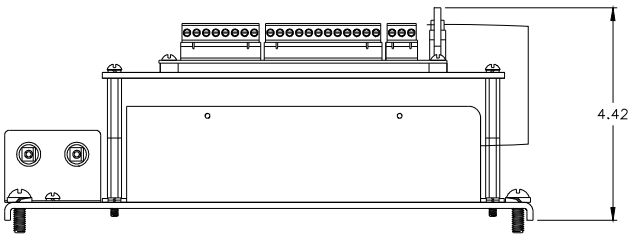
Due to wide variations in the terminal numbering of actuator products, actual wiring should follow the print supplied with the actuator and amplifier.

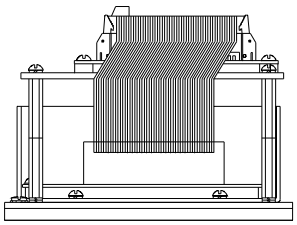
MAJOR DIMENSIONS

AD-9120-P

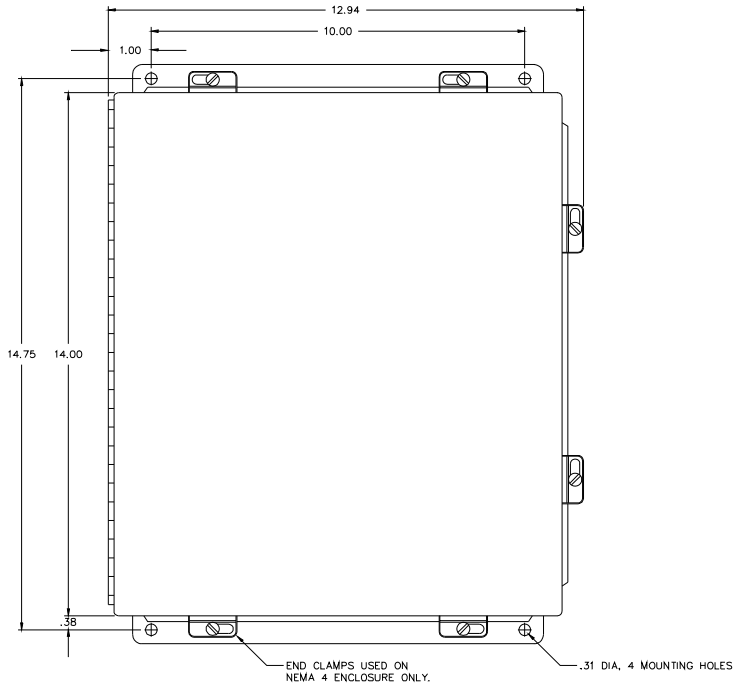
INCHES

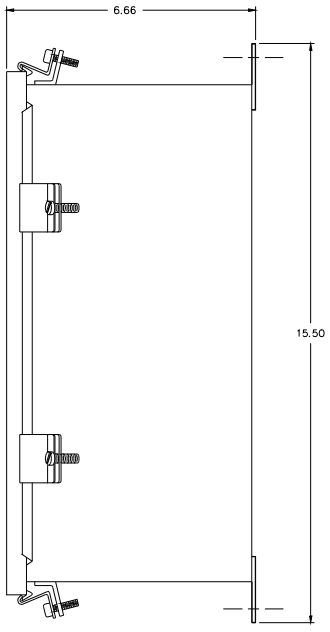






AD-9120-E



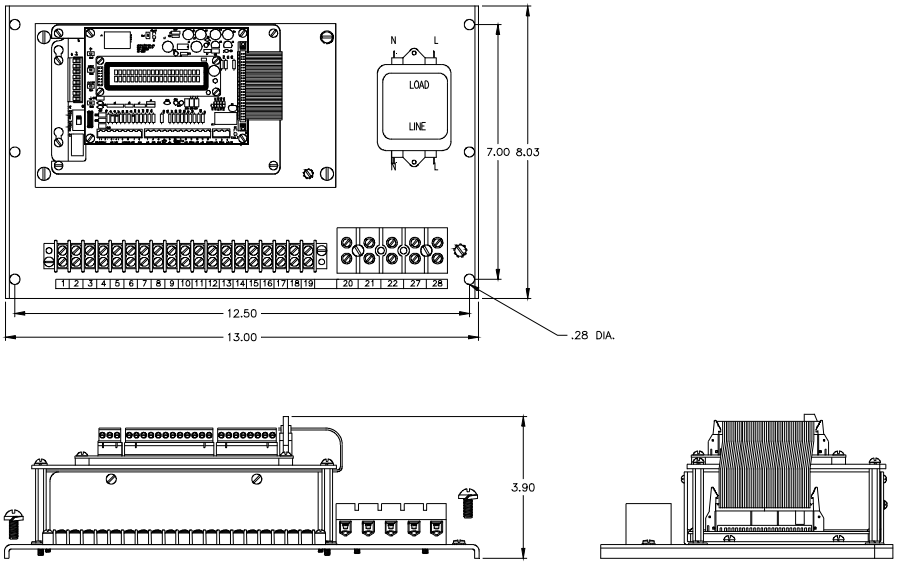


These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

MAJOR DIMENSIONS

AD-9120-R

INCHES



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

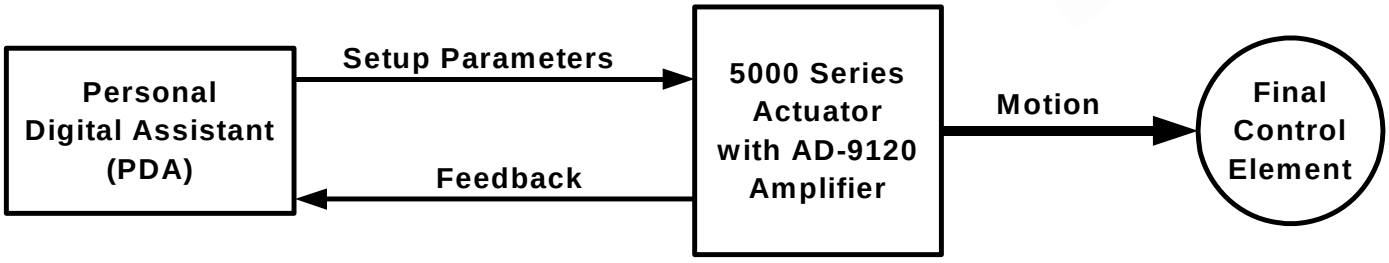
PALM OS® SETUP OPTION FOR THE AD-9120

(Option X020)

The Palm OS® setup option is designed for fast and easy infrared setup and feedback of most Jordan Controls actuators using the AD-9120 digital servo amplifier.

Most Personal Digital Assistants (PDA) using PALM OS® 3.5 or higher can be used to transmit and receive information through a transparent window mounted in the actuator's enclosure, thereby simplifying setup while maintaining enclosure integrity. The maximum infrared transmission distance from the PDA is approximately 5 feet.





Palm OS® and Palm™ are registered trademarks of Palm, Inc. or its subsidiaries.

RP-4000 RESERVE POWER CONTROL

GENERAL DESCRIPTION

The RP-4000 Series is designed to provide stand-by power to operate 120 and 240 Vac single phase Jordan Controls actuators in the event line power is interrupted. Actuators with amplifiers can be configured to move to a preset position or provide a controlled shut down to the actuator. Actuators without amplifiers can be configured to move to either a full open or full closed position. Also, operation can continue as normal up to the time specified for the specific model.



FEATURES

- Remote mounted from actuator
- High capacity, long life, sealed power supply
- Automatic reset on return of AC power
- ON/OFF power switch for actuator
- Battery / Actuator test operation switch
- Battery trickle recharge during normal running
- Auto sensing 50/60 Hz
- Standard NEMA 4 enclosure
- Anti-condensation heater
- Shipped fully charged from the factory
- Audible Alarms: battery discharge, low battery, overload, battery replacement needed.
- Indicators: battery status, battery discharge level, load level
- Full two year warranty

POPULAR OPTIONS

- Run to fail position with or without amplifier.
- Preset positioning for units with 4-20mA control on loss of power.

BASIC MODELS

RP-4010: Provides up to 750VA / 450 W for a maximum full load run time of between 24 and 80 minutes depending on actuator selected.

RP-4020: Provides up to 1425VA / 950 W for a maximum full load run time of between 13 and 180 minutes depending on actuator selected.

RP-4030: Provides up to 2250VA / 1600 W for a maximum full load run time of between 14 and 200 minutes depending on actuator selected.

SPECIFICATIONS

Input Voltage: 120 Vac, single phase, 50/60 Hz or 240Vac, single phase, 50/60 Hz

Output Voltage: 120 Vac, single phase, 50/60 Hz or 240Vac, single phase, 50/60 Hz

Full Recharge Time: 3 to 7 hours, depending on model and load on the RP-4000.

Maximum Run Time: See Cross Reference Chart on next page for specific values.

Enclosure: NEMA 4

Test Switch: This test switch provides a means to simulate a loss of power condition without physically removing main power to the system.

Power Switch: The power switch allows power to be removed from the actuator.

Battery Life: Approximately 5 years

Ambient Operating Temperature:
5° to 122° F (-15° to 50° C)

Approximate Weight:
RP-4010: 65 lbs (30 kg)
RP-4020: 80 lbs (36 kg)
RP-4030: 120 lbs (54 kg)

Enclosure Dimensions: Height x Width x Depth
RP-4010: 24 x 20 x 10 in (610 x 508 x 254 mm)
RP-4020: 24 x 20 x 10 in (610 x 508 x 254 mm)
RP-4030: 36 x 30 x 16 in (914 x 762 x 406 mm)

RP-4000 SERIES SELECTION CHART

				Selection
1	Basic Model	**RP-4010**	See chart below for actuator selection (Includes NEMA 4 Enclosure)	
		RP-4020	See chart below for actuator selection (Includes NEMA 4 Enclosure)	
		RP-4030	See chart below for actuator selection (Includes NEMA 4 Enclosure)	
2	Input Voltage	**1**: 120 Vac (Input and output voltage)		
		2: 240 Vac (Input and output voltage)		
3	Command Signal	**0**: For ON / OFF actuators (For units without built in amplifiers)		
		1: For modulating actuators with built in amplifiers for run to fail position		
		2: For modulating actuators with built in amplifiers for controlled shut down		

AD-4000 STANDARD OPTIONS

Code	Description	Selection
	Heater	
H002	Anti-Condensation Heater (120 VAC)	
H003	Anti-Condensation Heater (240 VAC)	

ACTUATOR CROSS REFERENCE CHART

Actuator Models	Input Voltage (AC)	Maximum Run Time (min)		
		RP-4010	**RP-4020**	**RP-4030**
SM/LA/MC/MV-1110	120	80	180	200
SM/LA-1120	120	80	180	200
SM/LA-1150	240	80	180	200
SM/LA/MC-1170	240	80	180	200
MV-1510	120	80	180	200
SM/MV-1530	120	52	113	121
SM/MV-1550	240	52	113	121
MV-1570	240	52	113	121
SM-1520	120	24	43	54
SM-1590	240	24	43	54
SM/VA-1630	120	52	113	121
SM/VA-1650	240	52	113	121
SM-1730	120	80	180	200
SM-1720	120	52	113	121
SM-1750	240	80	180	200
SM/LA-5120	120	24	43	54
SM/LA-5190	240	24	43	54
SM/LA-5220	120 / 240	NA	13	19
SM/LA-5320	120 / 240	NA	NA	14
LA-2410	120	52	113	121
LA-2420	120	24	43	54
LA-2450	240	52	113	121
LA-2450	240	24	43	54
LA-2520	120 / 240	NA	13	19
LA-2620 (2500 thrust)	120 / 240	NA	13	19
LA-2620 (3600 thrust)	120 / 240	NA	NA	14
LA-2720 (2500 thrust)	120 / 240	NA	NA	14
LA-2720 (4700 thrust)	120 / 240	NA	NA	14
SM-6000	120 / 240	NA	NA	3
J Series (Up to JOM4)	120 / 240	52	113	121
J Series (JOM5 to JOM8)	120 / 240	NA	NA	14

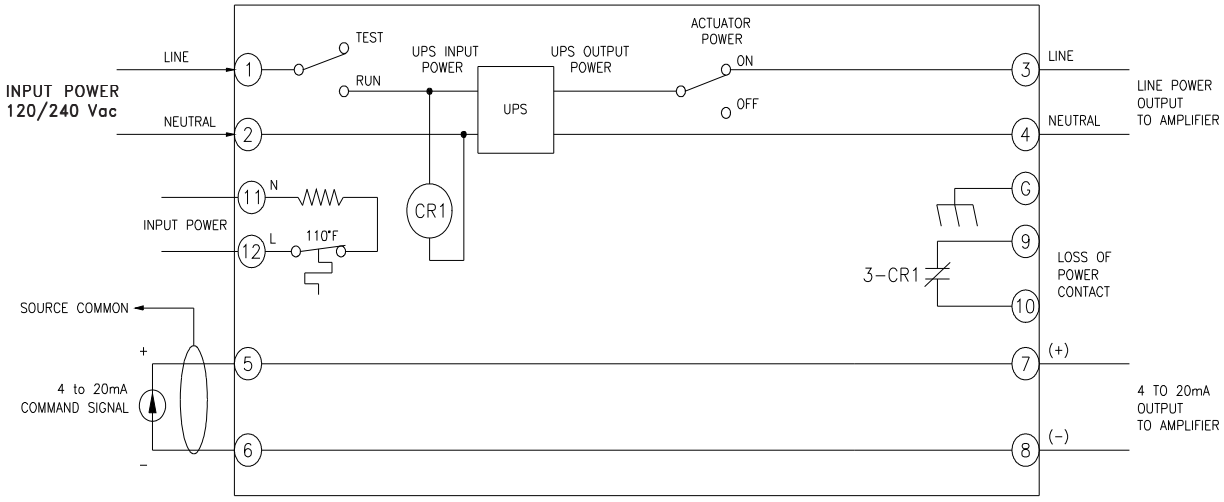
NOTES:

Not compatible with three phase actuators.

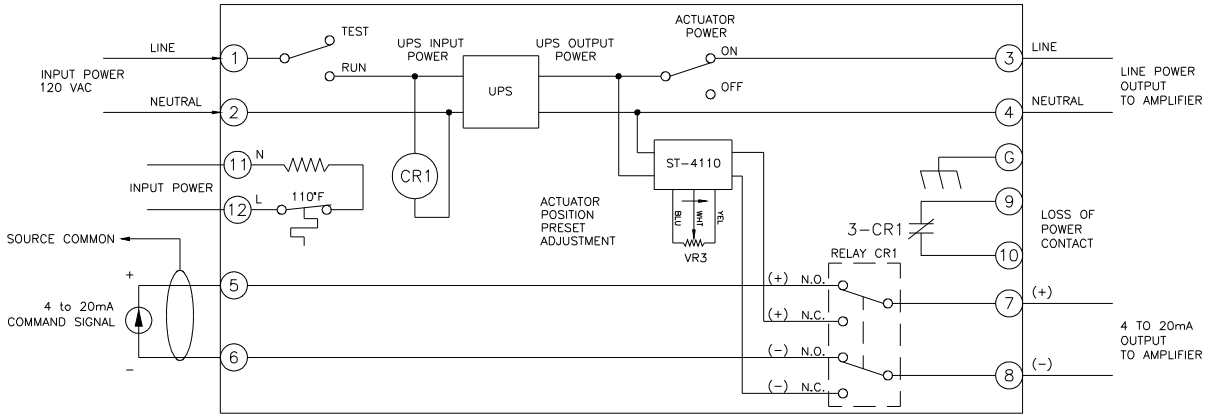
Maximum run time is nominally approximate and is dependent on battery condition, charge state, actuator load and actuator idle time. For most applications under normal conditions, the maximum run time will not be less than 50% of the chart values.

TYPICAL WIRING DIAGRAMS

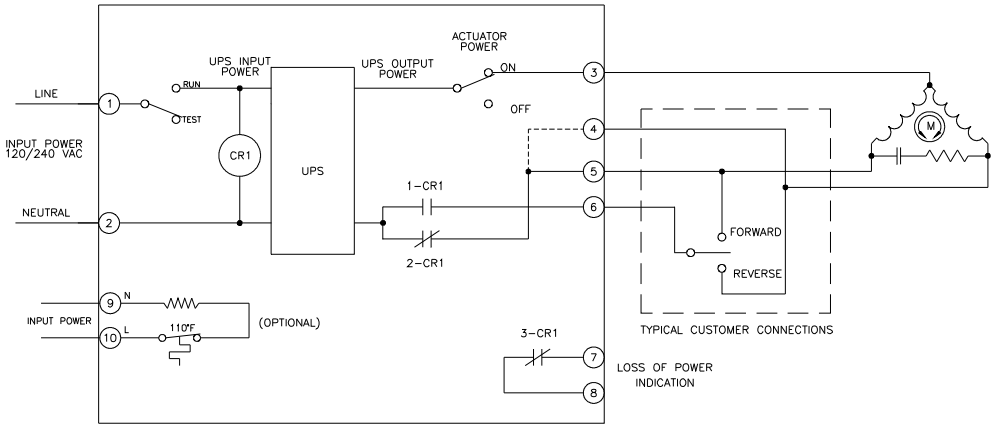
4 to 20 mA Control Shut Down or continued operation on Loss of line Power



4 to 20 mA Run to Preset on Loss of line Power

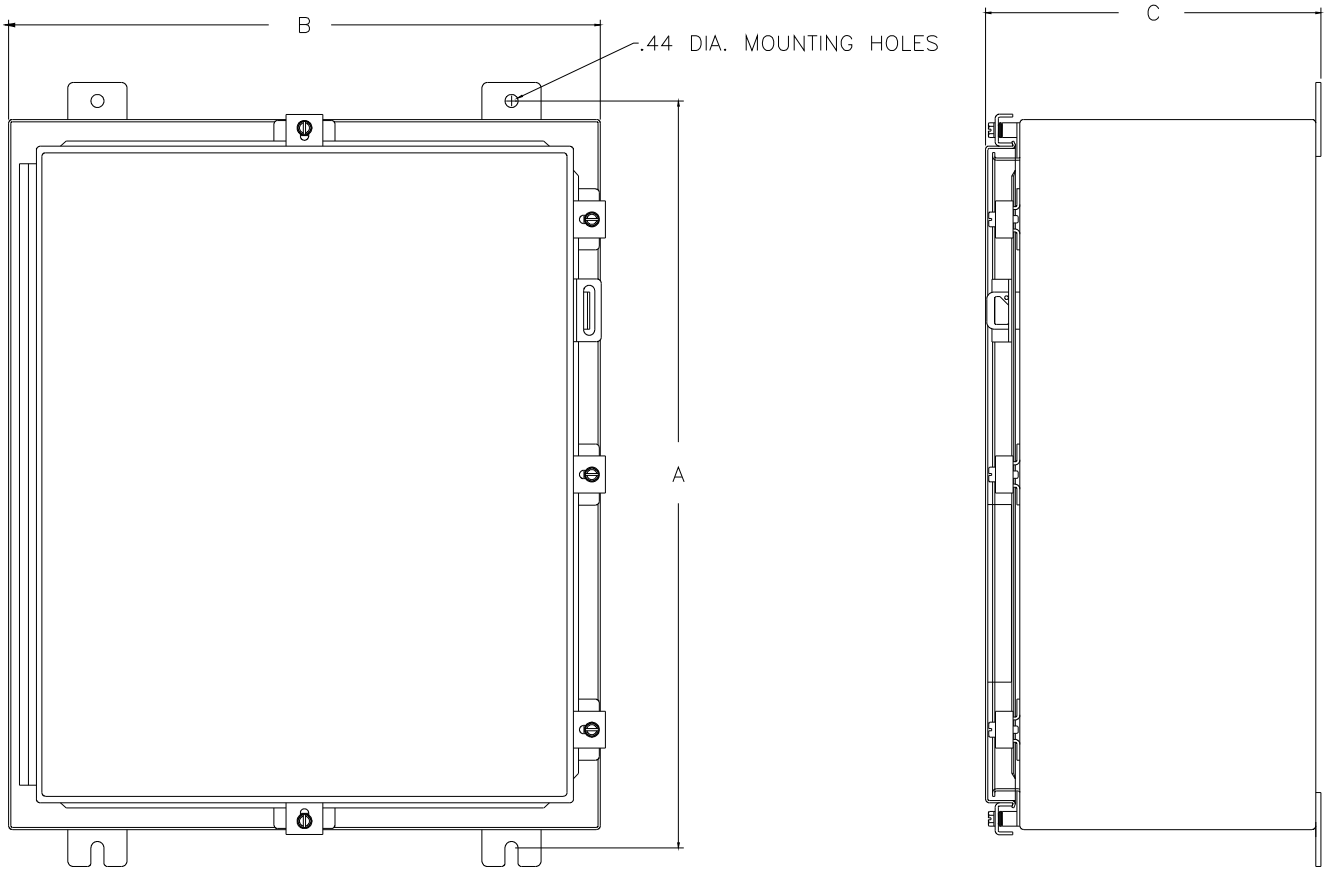


Loss of Power Run to End of Travel Position



Due to wide variations in the terminal numbering of actuator products, actual wiring should follow the print supplied with the actuator and reserve power unit.

MAJOR DIMENSIONS - ENCLOSURE



NOTE: Conduit openings are customer provided.

	A	B	C
RP-4010 RP-4020	24 in. (610 mm)	20 in. (508 mm)	10 in. (254 mm)
RP-4030	36 in. (914 mm)	30 in. (762 mm)	16 in. (406 mm)

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

GENERAL DESCRIPTION

The MT-6400 Remote Readout provides position indication of Jordan Controls actuators, position feedback transmitters or any 1000 ohm potentiometer feedback device.



FEATURES

- Large digital display
- Calibrated to display 0 - 100, or process range
- Universal input: 90 - 240 Vac, 50/60 Hz
- Built-in 24 Vdc supply provides reference voltage for 1000 ohm feedback potentiometer normally used in actuators for position indication, or powers 4-20 mA process input.

SPECIFICATIONS

Signal input: 1000 ohm potentiometer or 4-20 mA

Meter: Standard ±2% meter calibrated 0 to 100

Calibration: Front panel bushbutton calibration

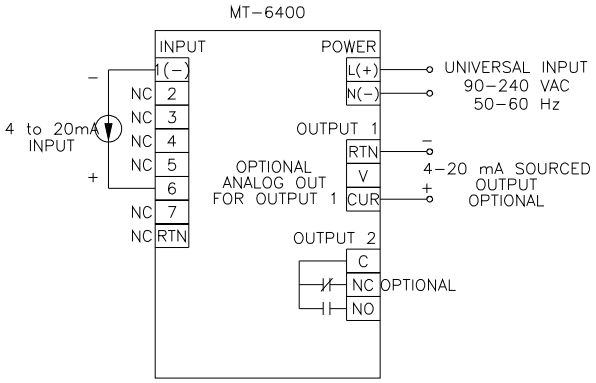
MT-6400 SERIES SELECTION CHART

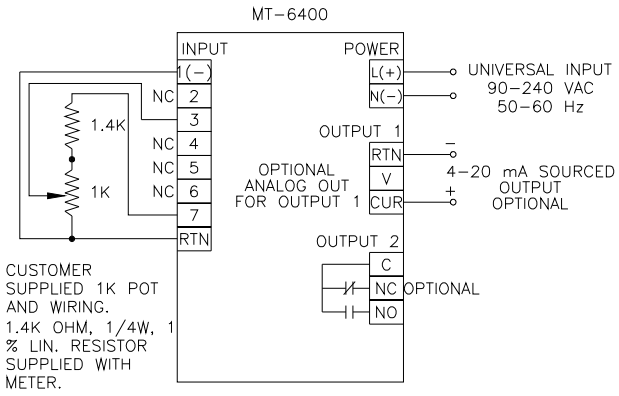
			Selection
Basic Model	**MT-6410**	Position indicator, panel mount, NEMA 4 front bezel	
	MT-6420	Position indicator, panel mount with actuator control switch panel mounted	
Signal Input	**P**: 1000 ohm potentiometer (by user) using 24 Vdc ecitation from meter		
	M: 4 to 20 mA signal		
Meter Readout	**D**: Display of process input, 4 to 20 or 0-10 (for pot input)		
	P: Percentage of process input 0-100		

MT-6400 STANDARD OPTIONS

Code	Description	Selection
Special Electrical Options		
X022	Form C relay contact and powered retransmission of process input for MT-6400	
X023	Special control outputs for MT-6400	

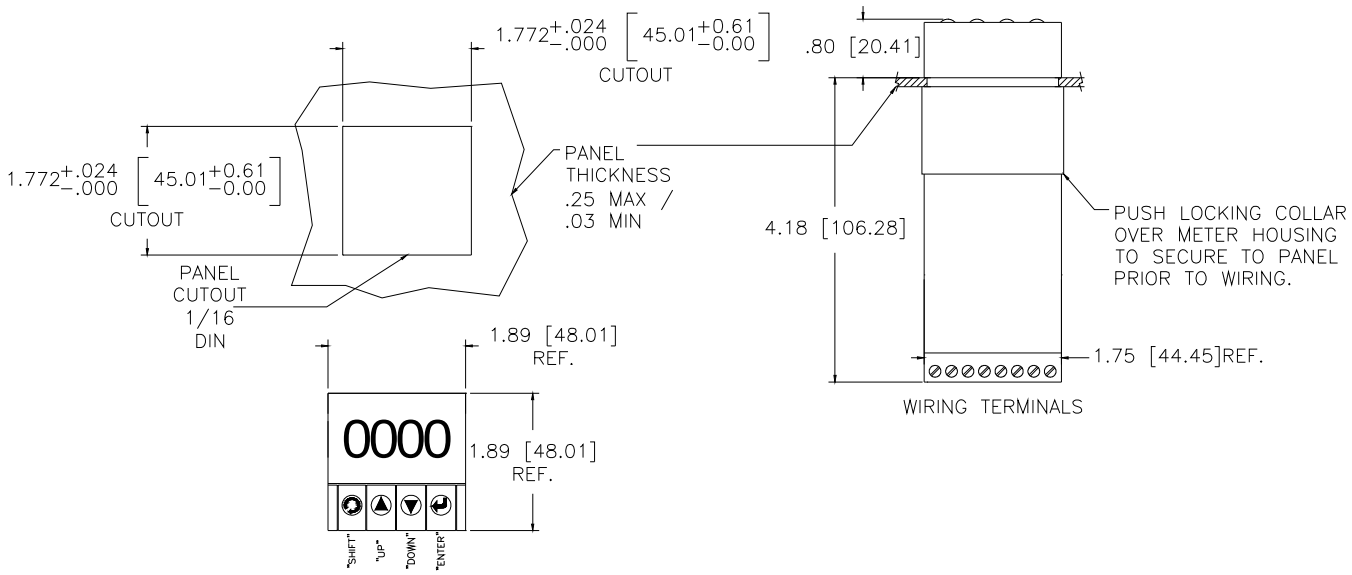
TYPICAL WIRING DIAGRAMS





MAJOR DIMENSIONS

INCHES (MILLIMETERS)



These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

GENERAL DESCRIPTION

The CS-7200 Series control stations are designed for position indication and control of Jordan Controls actuators. They may be used as a manual loading station with or without a separate, remotely located process controller or servo amplifier.

The CS-7200 Series are the ideal addition to any control loop for easy signal interfacing, LOCAL/REMOTE control and status indication of command and feedback signals. Any number of these control stations can be chained together to provide control and/or indication at various plant locations.



FEATURES

- 120 or 240 Vac input models
- Built-in 24 Vdc, 60 mA dc power supply
- Easy to read 3½ digit LCD display with 0.35 inch (8.9 mm) character height
- Display accuracy: 2% of span, ± digit (maximum), over full operating temperature range
- 4 to 20 mA output signal into a 600 ohm maximum load
- Accepts a variety of process control signals
- Easy installation and calibration
- Terminal strip for field wiring connections
- Temperature Ranges: Operating: 32° to 122° F (0° to 50° C) Storage: 14° to 140° F (-10° to 60° C)
- Humidity to 95% non-condensing
- 4 to 20 mA dc input, 250 ohm load; 0 to 24 Vdc input; 1K to 10K ohms input, powered from built-in power supply
- Full one year warranty

BASIC MODELS

***CS-7210**: Local Control Station*

Local, manual control of output signal via a front panel thumbwheel setting. Meter displays the output signal as a percent of full range. Used for manually setting the command signal in a closed-loop servo or process control system.

***CS-7240**: Local/Remote Control Station with display selector*

Local, manual control of the output signal via a front panel thumbwheel setting or remote control of the output signal by a separate, automatic or manual process controller or servo amplifier. Meter displays the output signal or the feedback signal as a percent of full range.

CS-7200 SERIES SELECTION CHART

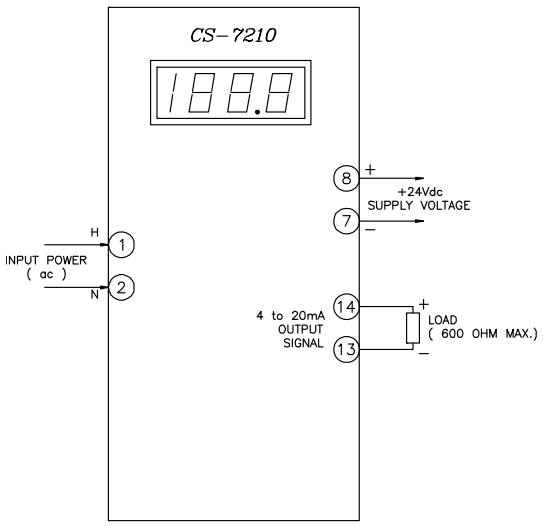
			Selection
Basic Model	**CS-7210**	Local control station	
	CS-7240	Local/remote control station with display selector	
Input Power	**1**: 120 Vac		
	2: 240 Vac		

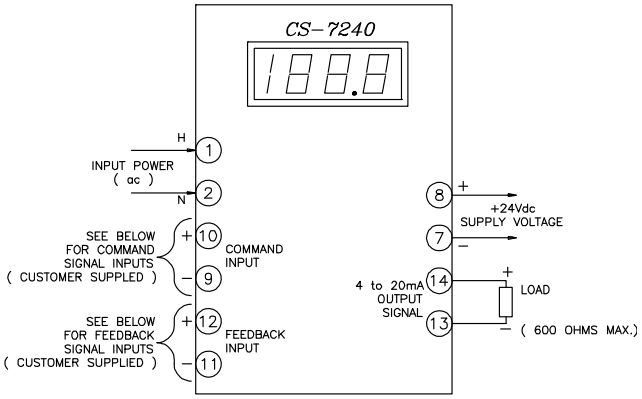
CS-7200 STANDARD OPTIONS

Code	Description	Selection
Enclosure		
E006	Clear, hinged door with keylock	

For a full description of options, go to the Complete Listing of Options starting on page 197.

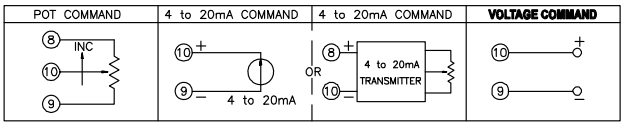
TYPICAL WIRING DIAGRAMS



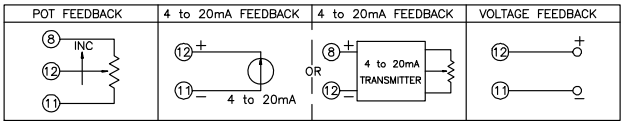


Due to wide variations in the terminal numbering of actuator products, actual wiring should follow the print supplied with the control station.

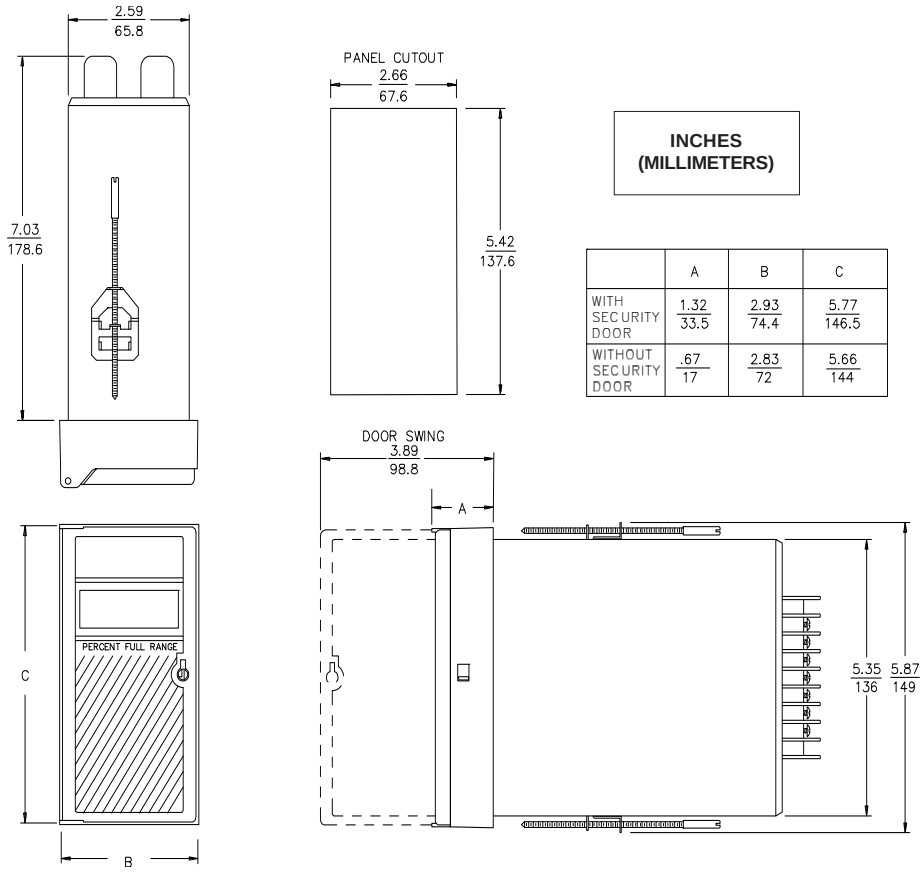
COMMAND SIGNAL INPUTS



FEEDBACK SIGNAL INPUTS



MAJOR DIMENSIONS



	A	B	C
WITH SECURITY DOOR	1.32 / 33.5	2.93 / 74.4	5.77 / 146.5
WITHOUT SECURITY DOOR	.67 / 17	2.83 / 72	5.66 / 144

These dimensions are subject to change without notice and should not be used for preparation of drawings or fabrication of installation mounting. For current installation manuals and other product information, see www.jordancontrols.com.

Toggles, Lights

A001 Local Auto/Manual Toggle Switch - NEMA 4. Switch for Auto (i.e. 4-20 mA from control room), or Manual (local control at actuator). Switch is maintain type.



A002 INCREASE/OFF/DECREASE Toggle Switch - NEMA 4. Momentary toggle type switch for local control of actuator.



A003 Local Auto/Manual Toggle Switch & INCREASE/OFF/DECREASE Toggle Switch - NEMA 4. (A001 & A002) (See photos for A001 & A002)

A004 Open-Close-Stop Push Buttons - NEMA 4. Three pushbutton control. Open & Close momentary type pushbutton with maintain type Stop button (pull to release)

A005 Two Pilot Lights - NEMA 4. Used for indication of ends of travel or intermediate position.





A006 Two Position Selector Switch, NEMA 4 (Auto/Manual). Switch for Auto (i.e. 4-20 mA from control room), or Manual (local control at actuator). Switch is maintain type.



A007 INCREASE/OFF/DECREASE Three Position Selector Switch. NEMA 4 momentary type switch for local control of actuator.



A008 Local Auto/Manual, INCREASE/OFF/DECREASE (Close Coupled Enclosure). Option A003 mounted in a NEMA 4 enclosure.

A009 ON/OFF Toggle Power Switch - NEMA 4. Removes main power to the actuator for single phase input.

A010 Local Auto/Manual Switch & (INCREASE/OFF/DECREASE) Switch - NEMA 4 (See photos for A006 & A007)

A011 Key lock for rotary switches - NEMA 4 (Auto/Manual). Manual operation only after unlocked.

A012 ON/OFF NEMA 4 Toggle Power Switch. Removes main power to the actuator for single phase input.

A013 Local Auto/Manual (Close Coupled Enclosure) Option A001 mounted in a NEMA 4 enclosure.

A014 INCREASE/OFF/DECREASE (Close Coupled Enclosure) Option A002 mounted in a NEMA 4 enclosure.

A015 ON/OFF Toggle Switch (Power) (Close Coupled Enclosure) Option A009 mounted in a NEMA 4 enclosure.

A016 Fault Indicator Light (SM-6000 Series)

A017 Power Indicator Light (SM-6000 Series)

A018 Auto/Manual Indicator Light (SM-6000 Series)

A031 Auto/Manual Switches (JOM2-N, JOM3-N), 1 Ph.

A032 Auto/Manual Switches (JOM4-N, JOM5-N, JOM6-N), 1 Ph.

A033 Auto/Manual Switches (JOM7-N, JOM8-N), 1 Ph.

A034 Auto/Manual Switches (JOM2-N, JOM3-N), 3 Ph.

A035 Auto/Manual Switches (JOM4-N, JOM5-N, JOM6-N), 3 Ph.

A036 Auto/Manual Switches (JOM7-N, JOM8-N), 3 Ph.

Brakes

B001 120 Vac Electrically Operated Brake. Electronically released brake, HOLDS when there is no power to HOLD a load, or to minimize coasting of the actuator. (Used on units with a high RPM rating). See drawing on page 186.

B002 240 Vac Electrically Operated Brake. Electronically released brake, HOLDS when there is no power to HOLD a load, or to minimize coasting of the actuator. (Used on units with a high RPM rating).. See drawing on page 186.

B003 Drag Brake (will decrease torque rating, contact factory for rating). Mechanical drag brake mounted on motor shaft to minimize coasting and to HOLD a load. (Used on units with a high RPM rating).

B004 Drag Brake (800 lbs. maximum thrust). Mechanical drag brake mounted on motor shaft to minimize coasting.

Connectors

C001 Quick disconnect NEMA 4 connector on enclosure for power and control.

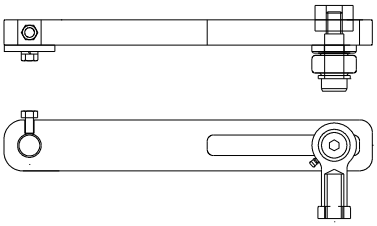
C002 Mating quick disconnect NEMA 4 connector on enclosure for power and control. (For use with C001).

Drive Arm/Adapter Clevis/Driven Arm

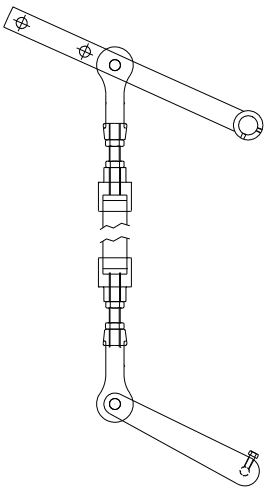
D001 Fixed (10") drive arm for splined shaft of SM-1700 and SM-5100. Pinhole diameter of 0.75 in. and thickness of 1.0 in.



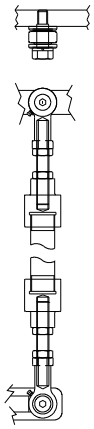
D002 Adjustable drive arm for splined shaft of SM-1700 and SM-5100. Adjustable range of 6 in. - 10 in., with a ¾-16 R.H. mating rod end.



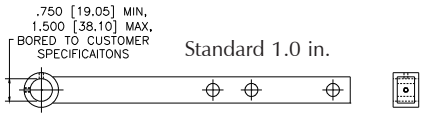
D003 Adapter clevis kit used with D001 & D005 for splined shaft of SM-1700 and SM-5100. Female clevis ends with a pinhole diameter of 0.75 in. and width of 1.0 in.



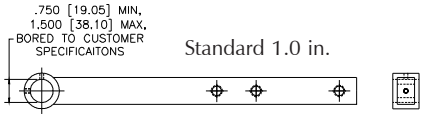
D004 Adapter clevis kit used with D002 & D006 for splined shaft of SM-1700 and SM-5100. Threaded ends of ¾-16 to meet with rod ends.



D005 Driven arm used with D001 and D003. 1.0 in. width driven arm with 1.0 in. diameter pin holes at 10, 12.25 and 17. End bored to 1.0 in. for damper shaft.



D006 Driven arm used with D002 and D004. 1.0 in. width driven arm with 5/8-11 in. taps at 10, 12.25 and 17 in. End bored to 1.0 in. for damper shaft.



D007 Fixed (10") drive arm for splined shaft of SM-5200. Pinhole diameter of 1.0 in. and thickness of 1.50 in. *See drawing for D001.*

D008 Adjustable drive arm for splined shaft of SM-5200. Adjustable range of 6 in. - 10 in., with a 1¼-12 R.H. mating rod end. *See drawing for D002.*

D009 Adapter clevis kit used with D007 and D011 for splined shaft of SM-5200. Female clevis ends with a pinhole diameter of 1.0 in. and width of 1.50 in. *See drawing for D003.*

D010 Adapter clevis kit used with D008 and D012 for splined shaft of SM-5200. Threaded ends of 1¼-16 to meet with rod ends. *See drawing for D004.*

D011 Driven arm used with D007 and D009. 1.50 in. width driven arm with 1.0 in. diameter pin holes at 10, 12.25 and 17 in. End bored to 0.25 pilot for damper shaft. *See drawing for D005.*

D012 Driven arm used with D008 and D010. 1.5 in. width driven arm with ¾-10 taps at 10, 12.25 and 17 in. End bored to 0.25 pilot for damper shaft. *See drawing for D006.*

D013 Fixed (12") drive arm for splined shaft of SM-5300. Pinhole diameter of 1.0 in. and thickness of 1.50 in. *See drawing for D001.*

D014 Adapter clevis kit used with D013 and D015 for splined shaft of SM-5300. Female clevis ends with a pinhole diameter of 1.0 in. and width of 1.50 in. *See drawing for D003.*

D015 Driven arm used with D013 and D014. 1.50 in. width driven arm with 1.0 in. diameter pin holes at 12, 14.7 and 20.4 in. End bored to 1.938/1.940 in. for damper shaft. *See drawing for D005.*

Drive Arm/Adapter Clevis/Driven Arm

D016 Fixed (15") drive arm for splined shaft of SM-5400. Pinhole diameter of 1.0 in. and thickness of 1.50 in. *See drawing for D001.*

D017 Adapter clevis kit used with D016 and D018 for splined shaft of SM-5400. Female clevis ends with a pinhole diameter of 1.0 in. and width of 1.50 in. *See drawing for D003.*

D018 Driven arm used with D016 and D017. 1.50 in. width driven arm with 1.0 in. diameter pin holes at 15, 18.38 and 25.5. End bored to 0.25 pilot for damper shaft. *See drawing for D005.*

D019 Hagan Bailey Pneumatic Replacement (Pedestal Drive Arm) for SM-1700. Retrofit pedestal and drive arm to match Hagan and Bailey pneumatic drives up to 200 ft. lbs. of torque. Allows use of existing linkage and mounting. See D022 photo.

D020 Hagan Bailey Pneumatic Replacement (Pedestal Drive Arm) for SM-5100. Same as D019, but up to 300 ft. lbs. of torque. See D022 photo.

D021 Hagan and Bailey Pneumatic Replacement (Pedestal Drive Arm) for SM-5200. Same as D019, but up to 1,000 ft. lbs. of torque. See D022 photo.

D022 Hagan and Bailey Pneumatic Replacement (Pedestal Drive Arm) for SM-5300. Same as D019, but up to 3,200 ft. lbs. of torque.



D023 Hagan and Bailey Pneumatic Replacement (Pedestal Drive Arm) for SM-5400. Same as D019, but up to 12,500 ft. lbs. of torque. See photo for D022.

D024 Wedgelock Coupling for the SM-1700. Three piece coupling that allows adjustment of actuator and valve/damper without removal of either from the mounting bracket.



D025 Wedgelock Coupling for SM-5100. *See description and photo of D024.*

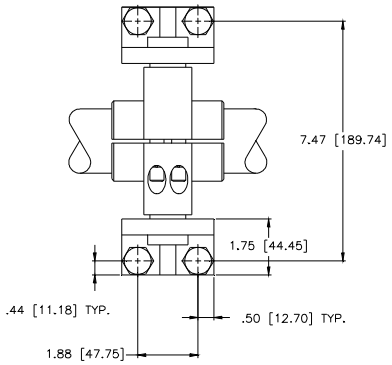
D026 Wedgelock Coupling for SM-5200. *See description and photo of D024.*

D027 Wedgelock Coupling for SM-5300. *See description and photo of D024.*

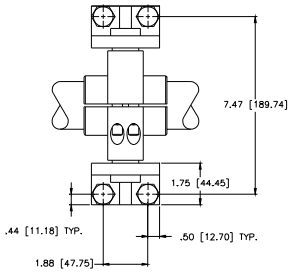
D028 Wedgelock Coupling for SM-5400. *See description and photo of D024.*

D029 PT-1300 Mounting Kit. Universal mounting kit for sliding stem valves.

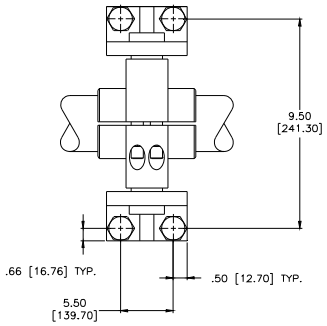
D030 Trunnion Mount for LA-2300. Allows actuator to be foot mounted. (Not available for 6 in. stroke).



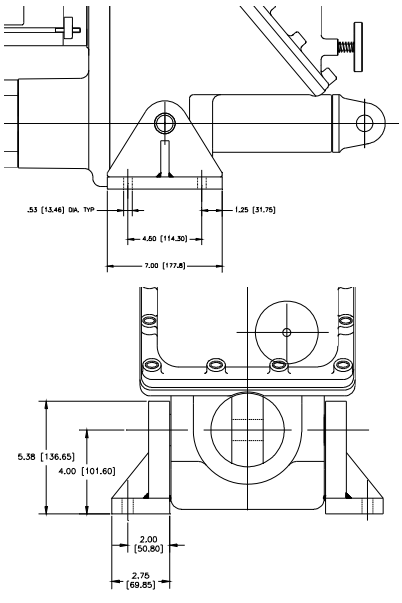
D031 Trunnion Mount for LA-2400. Allows actuator to be foot mounted. (Not available for 6 in. stroke).



D032 Trunnion Mount for LA-2500. Allows actuator to be foot mounted. (Not available for 6 in. stroke).

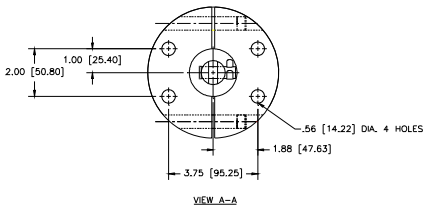


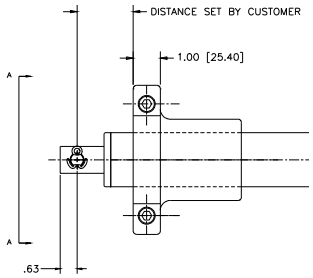
D033 Trunnion Mount for LA-2600. Allows actuator to be foot mounted. (Available up to 3,600 lbf)



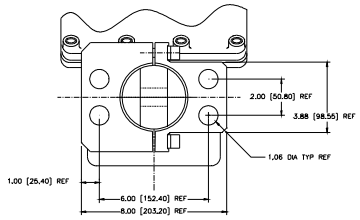
Drive Arm/Adapter Clevis/Driven Arm

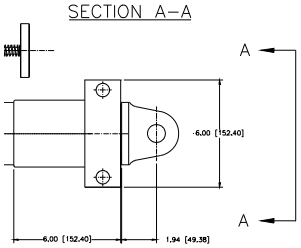
D034 Flange Mount for LA-2300. Mounting bracket which clamps to the tube of the actuator with a four bolt mounting face. (Not available for 6 in. stroke).



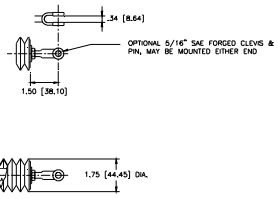


D035 Flange Mount for LA-2600 and LA-2700. Mounting bracket which clamps to the tube of the actuator with a four bolt mounting face. (Available up to 3,600 lbf).

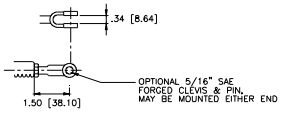




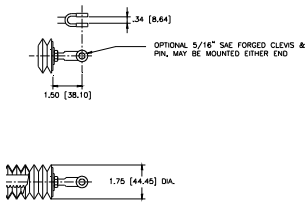
D036 Rack Bellows for LF-1210. Boot that protects the rack from dirt and moisture.



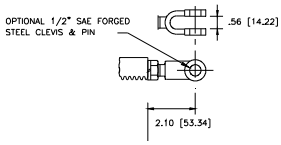
D037 Female clevis with pin for LA-1100.



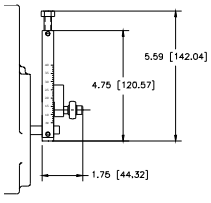
D038 Rack Bellows for LA-1500. Boot that protects the rack from dirt and moisture.

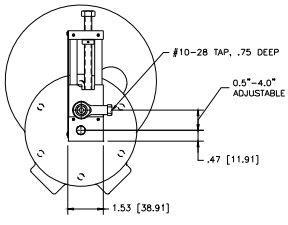


D039 Female clevis with pin for LA-1500.

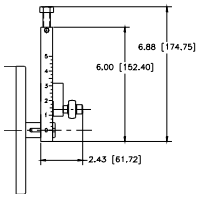


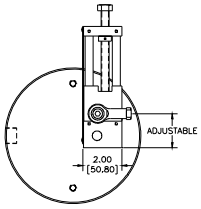
D040 Adjustable Drive Arm for TA-1200. Adjustable range of 0.5 in. to 4.0 in., with a #10-28 tap, 0.75 deep.



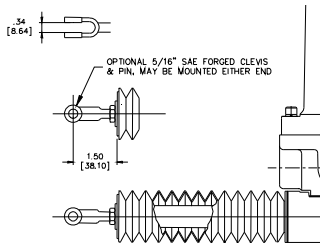


D041 Adjustable Drive Arm for SM-1500. Adjustable range of 0.5 in. to 4.0 in., with a #10-28 tap, 0.75 deep.





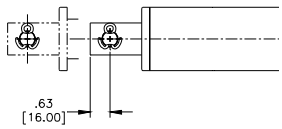
D042 Clevis, one end for LA-1000 & LA-1100.

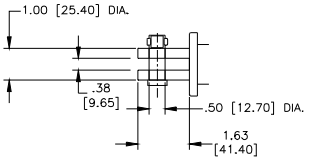


D043 Clevis, both ends for LA-1000 & LA-1100.

D044 Flange Mount for LA-2400. Mounting bracket which clamps to the tube of the actuator with a four bolt mounting face. (Not available for 6 in. stroke). *See drawing for D034.*

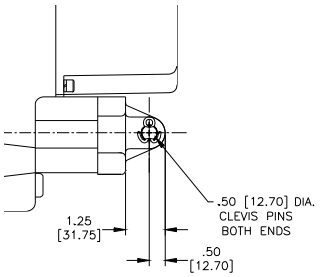
D045 Female front clevis for LA-2400.





Drive Arm/Adapter Clevis/Driven Arm

D046 Female rear clevis for LA-2400.



D047 Special front clevis for LA-2300 & LA-2400 (consult factory, customer must provide dimensions).

D048 Special rear clevis for LA-2300 & LA-2400 (consult factory, customer must provide dimensions).

D049 Special front clevis for LA-2500 (consult factory, customer must provide dimensions).

D050 Special rear clevis for LA-2500 (consult factory, customer must provide dimensions).

D051 Special front clevis for LA-2600 & LA-2700 (consult factory, customer must provide dimensions).

D052 Special rear clevis for LA-2600 & LA-2700 (consult factory, customer must provide dimensions).

D053 Special front clevis for LA-2900 (consult factory, customer must provide dimensions).

D054 Special rear clevis for LA-2900 (consult factory, customer must provide dimensions).

D055 Bellows for Linear Actuators (LA-2400/2500). Protects against dust and moisture when ram extends up.

D056 Vertical Moisture Shield (LA-2300/2400). Protects against rain when ram extends up.

D057 Adjustable drive arm for 1/2 inch output shaft of TA-1200. Used with R008.

D058 Rotating clevis for LA-2400

D059 Keyed drive arm for 370 ft. lb. SM-6000. *See drawing on page 53.*

D060 Keyed drive arm for 800 ft. lb. SM-6000. *See drawing on page 53.*

D061 Keyed drive arm for 1400 ft. lb. SM-6000. *See drawing on page 53.*

D062 Keyed drive arm for 2500 ft. lb. SM-6000. *See drawing on page 53.*

D063 Keyed drive arm for 4400 ft. lb. SM-6000. *See drawing on page 53.*

D064 Keyed drive arm for 6200 ft. lb. SM-6000. *See drawing on page 53.*

D065 Keyed drive arm for 8000 & 11000 ft. lb. SM-6000. *See drawing on page 53.*

D066 Keyed drive arm for 16500 ft. lb. SM-6000. *See drawing on page 53.*

D067 Keyed drive arm for 26000 ft. lb. SM-6000. *See drawing on page 53.*

D068 Adapter assembly kit for 370 ft. lb. SM-6000

D069 Adapter assembly kit for 800 & 1400 ft. lb. SM-6000

D070 Adapter assembly kit for 2500 & 4400 ft. lb. SM-6000

D071 Adapter assembly kit for 6200, 8000, 11000 & 16500 ft. lb. SM-6000

D072 Adapter assembly kit for 26000 ft. lb. SM-6000

D073 Driven arm for 370 ft. lb. SM-6000

D074 Driven arm for 800 & 1400 ft. lb. SM-6000

D075 Driven arm for 2500 & 4400 ft. lb. SM-6000

D076 Driven arm for 6200, 8000, 11000 & 16500 ft. lb. SM-6000

D077 Driven arm for 26000 ft. lb. SM-6000

D078 Pedestal replacement & drive arm to replace SM-5100

D079 Pedestal replacement & drive arm to replace SM-5200

D080 Pedestal replacement & drive arm to replace SM-5300

D081 Pedestal replacement & drive arm to replace SM-5400

D082 Pedestal replacement & drive arm to replace Hagan Bailey

D083 Adjustable length drive arm for 370 ft. lb. SM-6000

D084 Adjustable length drive arm for 800 ft. lb. SM-6000

D085 Adapter clevis kit for 370 ft. lb. SM-6000 (Used with D083)

D086 Adapter clevis kit for 800 ft. lb. SM-6000 (Used with D084)

D124 Adjustable length drive arm for 550 ft. lb. SM-6000

D125 Adapter clevis kit for 550 ft. lb. SM-6000 (Used with D124)

D126 Driven arm for 550 ft. lb. SM-6000

OPTIONS

Enclosure

E001 Watertight, NEMA 4 (IP65).

E002 NEMA 12 (IP52).

E003 Explosion Proof - Class 1, Div. 1, Groups C & D, indoor/outdoor (NEMA 7).

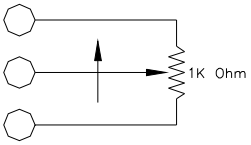
E004 Dust-Ignition Proof - Class II, Div. 1, Groups E, F & G, indoor/outdoor (NEMA 9).

E005 Group B enclosure for use in Hydrogen atmosphere (AC input only).

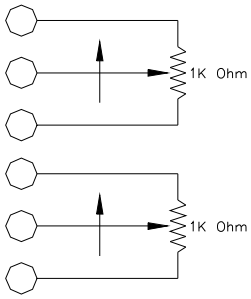
E006 Clear, hinged door with keylock.

Position Feedback

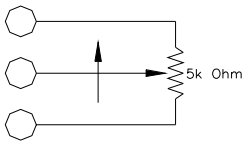
F001 1000 ohm, wire wound potentiometer.



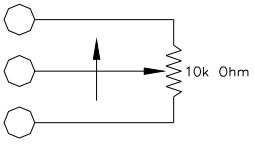
F002 Tandem Potentiometer, 1000 ohms each, wire wound, (not available for units with internal amplifiers).



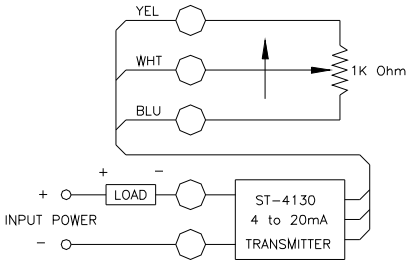
F003 5,000 ohm, wire wound potentiometer, (not avail. for units with internal amplifiers).



F004 10,000 ohm, wire wound potentiometer, (not available for units with internal amplifiers).



F005 Transmitter, 4-20 mA output



F006 One Potentiometer, 1000 ohms plus 4-20 mA Transmitter (not available for units with internal amplifiers)

F007 Hall Effect Contactless 4-20 mA Feedback.

F008 Contactless 4-20 mA

F009 Contactless 0-10Vdc

F010 Anti Backlash Feedback Gearing (Single Turn)

F011 Anti Backlash Feedback Gearing (Multi Turn)

F012 Tandem Potentiometers, 5000 ohms (not available for units with internal amplifiers). See drawing for F002.

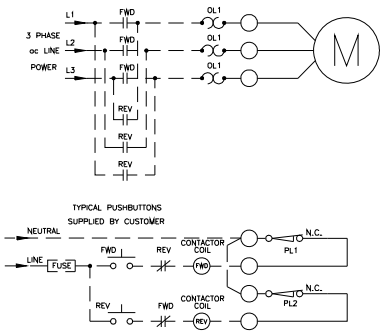
F013 Tandem Potentiometers, 10,000 ohms (not avail. for units with internal amplifiers). See drawing for F002.

F014 Absolute Encoder Contactless Feedback.

F036 4-20 mA output transmitter

3-Phase Controls

G001 3-Phase Contactor with Overloads

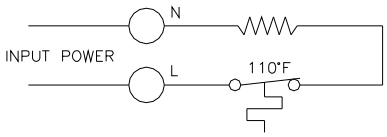


G002 120 Vac Control Transformer.

G003 240 Vac Control Transformer.

Heater

H001 Anti-Condensation Heater (24 VDC)



H002 Anti-Condensation Heater (120 VAC). See drawing for H001.

H003 Anti-Condensation Heater (240 VAC). See drawing for H001.

OPTIONS

Identification/ Certifications

J001 CE Marking (AC input only)

J002 Stainless Steel Tags

J003 FM approved (Factory Mutual)

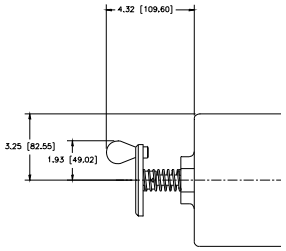
J004 One year extended warranty.

J005 Two year extended warranty.

J006 ATEX Approval.

Handcrank

K001 Manual Handcrank for SM-1500/1600 Series.



Valve Mounting

L001 Mounting of actuator to valve by customer. Compatible hardware supplied by Jordan Controls per customer specifications. Complete, certified valve drawings showing topworks detail must be submitted with order.

L002 Valve provided by customer for mounting and set-up with actuator by Jordan Controls.

L003 Valve provided by Jordan Controls, mounted and set-up with actuator per customer specifications.

L004 No regulator

L005 Rotate unit 180 degrees CW viewing actuator from top so conduit entry is 174 degrees from current.

Position Indicator

P001 Local Position Indicator



Motor

M001 575 Motor - 3 Phase, for SM-5000 and LA-5000 Series actuators.

Electro-Hydraulic

N004 Failsafe, Spring return on loss of power

N006 Quick Failsafe (see Table 2 for speed select)

N007 Manual override (336 in. lbs. or less)

N008 Manual override (460 to 1035 in. lbs.)

N009 Manual override (1858 to 4513 in. lbs.)

N011 0-10 Vdc Command

N013 Failsafe - Spring return on loss of power

N015 Mounting direction, if not above valve

N016 Silicone oil for low temperature operation,-40° F (-40° C)

N017 Valve mounting kit

N018 Manual override hand pump, NEMA 4

N019 Thermal jacket for low, -40° F (-40° C) and high, 482° F (250°C) (for two hours) temperature applications

N021 Geared manual override (speed/torque A-C)

N022 Geared manual override (speed/torque D-E)

N023 Geared manual override (speed/torque F-G)

N024 Explosion-proof, Class 1, Zone 1 Aex md IIBT3Ta

N025 M20 threaded cable entry

N026 Bio-degradable fluid for food/ beverage

N027 Quick speed ESD failsafe - explosion proof

N028 Overthrust protection using a pressure switch - NEMA 4 & Explosion-proof

N030 Geared manual overdrive, Code A

N031 Geared manual overdrive, Code B

N032 Overthrust protection using a pressure switch

N034 Split range command input

N035 Manual override

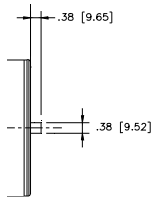
N036 Geared manual overdrive, Code C

N039 Manual override hand pump, Explosion-proof

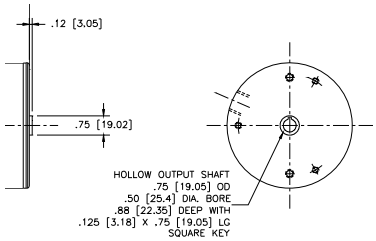
N040 Remote mounted amplifier for EH-1000 Series

Shaft

R001 3/8" Lg Shaft for SM-1100.

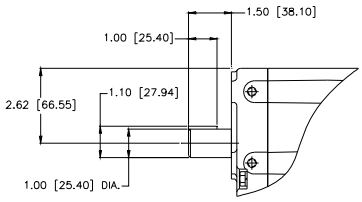


R002 Hollow Shaft for SM-1100.

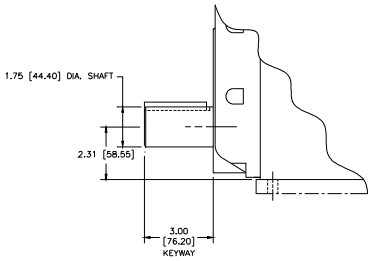


R003 Hollow Shaft for SM-1500.

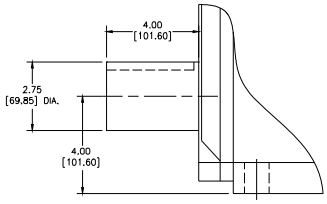
R004 Keyed Shaft for SM-1700 & SM-5100.



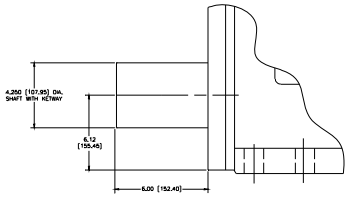
R005 Keyed Shaft for SM-5200.



R006 Keyed Shaft for SM-5300.

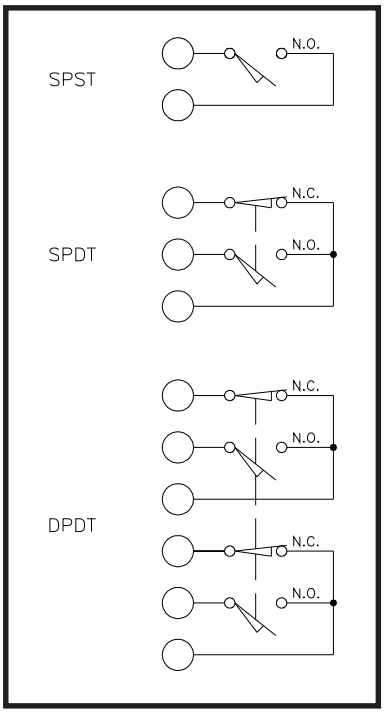


R007 Keyed Shaft for SM-5400.



R008 1/2 inch Output Shaft for TA-1200 with woodruf key.

Auxiliary Position Switches



Switch options are multiples of above configuration.

S001 Two Adjustable Position Switches, SPDT, 20 Amp.

S002 Two Adjustable Position Switches, DPDT, 20 Amp.

S003 Four Adjustable Position Switches, SPDT, 20 Amp.

S004 Four Adjustable Position Switches, DPDT, 20 Amp.

S005 Aluminum Cam (*Up to 5.85 turns)*

S006 Aluminum Feedback Frame *(Up to 5.85 turns)*

S007 Aluminum Feedback Frame *(5.85 turns and greater)*

S008 Two nonadjustable end of travel position switches, SPST, 10 Amp

S009 Two Adjustable Position Switches, SPDT, 10 Amp.

S010 Two Adjustable Position Switches, DPDT, 10 Amp.

S011 Four Adjustable Position Switches, SPDT, 10 Amp.

S012 Four Adjustable Position Switches, DPDT, 10 Amp.

S013 Two nonadjustable end of travel position switches, SPDT, 10 Amp

S014 Two nonadjustable end of travel position switches, SPDT, 20 Amp

S015 Two nonadjustable torque limit switches (CW & CCW), SPST, 10 Amp

S016 Two nonadjustable torque limit switches (CW & CCW), SPDT, 10 Amp

S017 Two nonadjustable thrust limit switches (EXT & RET), SPDT, 10 Amp

S018 Two nonadjustable thrust limit switches (EXT & RET), SPST, 10 Amp

S019 Two nonadjustable end of travel position switches, SPST, 20 Amp

S020 Two nonadjustable torque limit switches (CW & CCW), SPST, 20 Amp

S021 Two nonadjustable thrust limit switches (EXT & RET), SPST, 20 Amp

S042 Adjustable torque switches (J Series)

Temperature Rating

T001 Extended operating temperature of -40° F to 185° F (-40° C to 85° C).

T002 Extended operating temperature of -40° F to 225° F (-40° C to 107° C).

T003 Extended operating temperature of -40° F to 150° F (-40° C to 66° C). *Electro-Hydraulic Only.*

Conduit Entries

V001 1/2 inch Conduit Entry Additional (1000 Series, LA2500).

Painting/Coating

W001 JCI Standard Polyurethane Blue

W002 Two Part Epoxy

W003 Food Grade Epoxy

W004 Baked On Epoxy

W005 Teflon

Special Electrical Options

X001 24 Vdc power supply, 320 mA, 100-240 Vac input.

X002 Loss of signal high or low.

X003 Loss of signal low.

X004 Voltage to current converter.

X005 Loop powered 4 to 20 isolator, M-system HSN-IAA.

X006 EC-10726 ± 15 Vdc power supply.

X007 MOVs for AD-8000 (Required for output to solenoids or starters).

X008 Fail open at loss of power and loss of command signal.

X009 Fail close at loss of power and loss of command signal.

X010 Command and feedback, 0-10 Vdc or 0-5 Vdc instead of 4-20 mA.

X012 Signal conversion module, installed and wired to actuator.

X013 2 wire incremental dc input.

X014 3 wire incremental 120/240 Vac input.

X015 Battery Backup, only for 120/240 Vac, not availabe with X012.

X016 HART communication protocol.

X017 Process variable. (See page 7)

X018 Low current at null. (See page 7)

X019 AC Power Sensing with Output.

X020 Palm Pilot Setup for AD-9120.

X021 AD-8120 and AD-8220 replacement for use with AD-8240.

X022 Form C relay contact and powered retransmission of process input for MT-6400

X023 Special control outputs for MT-6400

X025 Relay contact to detect overtorque (stall) for SM-6000

X026 Remote Amplifier for SM-6000 (185° F)

X027 Remote Amplifier for SM-6000 (225° F)

X028 208 Vac input power for AD-9120

X033 Loss of line power relay for SM-6000

X034 Loss of signal relay for SM-6000

X055 Step down transformer for 480 to 240 Vac. Close coupled, 4400 ft. lbs. and higher.

X057 Step down transformer for 480 to 240 Vac. Remote-mounted, up to 2500 ft. lbs.

X070 Torque overload indicator for SM-6000.

X088 75% Duty cycle for JOM1 & JOMB (AC only).

X089 75% Duty cycle for JOM2 - JOM8 (AC only).

X095 75% Duty cycle (All Vdc, 12 Vac & 24 Vac).

Special Mechanical Options

Y001 Increased rotation for SM-1000 Series actuators to 33 turns.

Y002 Helicoil Screw Thread Inserts (specify screw thread).

Y003 Increased rotation for TA-1200 Series actuators to 135°.

Y004 Metric mounting holes, 6mm (LF-1210)

Y005 Metric mounting holes, 6mm (SM-1000)

Y006 Metric mounting holes, 6mm (SM-1100)

Y008 Metric mounting holes, 12mm (SM-1700)

Y009 Metric mounting holes, 12mm (SM-5100)

Y010 Metric mounting holes, 20mm (SM-5200)

Y011 Metric mounting holes, 20mm (SM-5300)

Y012 Metric mounting holes, 26mm (SM-5400)

Y013 Metric mounting holes, 8mm (SM-1600)

Y014 Metric mounting holes, 6mm (LA-1000)

Y015 Metric mounting holes, 6mm (LA-1100)

Y016 Metric mounting holes, 8mm (LA-1500)

Y017 Metric mounting holes, 8mm (SM-3300)

Y018 Metric mounting holes, 8mm (LA-3300)

Y019 Metric mounting holes, 6mm (PT-1010 & PF-1010)

Y020 Metric mounting holes, 6mm (PF-1210)

Y021 Metric mounting holes, 6mm (PT-1330)

Y022 Faster speeds (SM-6000)

Y023 Extended rotation beyond 300° (SM-6000)

Y024 Mechanical stop for 370 ft. lb. SM-6000

Y025 Mechanical stop for 800 ft. lb. SM-6000

Y026 Mechanical stop for 1400 ft. lb. SM-6000

Y027 Mechanical stop for 2500 ft. lb. SM-6000

Y028 Mechanical stop for 4400 ft. lb. SM-6000

Y029 Mechanical stop for 6200 & 8000 ft. lb. SM-6000

Y030 Mechanical stop for 11000 ft. lb. SM-6000

Y031 Mechanical stop for 16500 ft. lb. SM-6000

Y032 Mechanical stop for 26000 ft. lb. SM-6000

Y033 M20 conduit entries for J Series

CSA APPROVED PRODUCTS

Actuator	Comments	Approved Ratings
SM-1015	Not available with battery backup, 3 wire incremental input, or signal conversion modules.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1020	Not available with HART Communication Protocol, battery backup, 3 wire incremental input, or signal conversion modules.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
	Hydrogen "heavy wall enclosure"	Class I, Group B, C & D Class II, Group E, F & G Ex d IIB + Hydrogen CSA Enclosure 4. Max. ambient temperature: 65°C
SM-1110	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1120	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1140	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1150	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1170	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1520	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1530	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1540	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1550	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1590	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1630	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1650	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1730	Not available with local position indicator, contactless feedback or Auto/Manual or INC/OFF/DEC switches	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1740	Not available with local position indicator, contactless feedback or Auto/Manual or INC/OFF/DEC switches	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-1750	Not available with local position indicator, contactless feedback or Auto/Manual or INC/OFF/DEC switches	Class II, Div. 1, Group E, F & G CSA Enclosure 4.

Actuator	Comments	Approved Ratings
LA-1020	Not available with HART Communication Protocol, battery backup, 3 wire incremental input, or signal conversion modules.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
	Hydrogen "heavy wall enclosure"	Class I, Group B, C & D Class II, Group E, F & G Ex d IIB + Hydrogen CSA Enclosure 4. Max. ambient temperature: 65°C
LA-1110	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-1140	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-1170	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2320	Maximum of four position switches. Not available with contactless feedback.	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2390	Maximum of four position switches. Not available with contactless feedback.	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2410	Maximum of four position switches. Not available with contactless feedback.	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2420	Maximum of four position switches. Not available with contactless feedback.	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2440	Maximum of four position switches. Not available with contactless feedback.	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2450	Maximum of four position switches. Not available with contactless feedback.	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2490	Maximum of four position switches. Not available with contactless feedback.	Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2510	Not available with drag brake, ball screw or contactless feedback.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
LA-2520	Not available with drag brake, ball screw or contactless feedback.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MC-1110	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MC-1140	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MC-1170	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-LA-3330	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
SM-LA-3350	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.

CSA APPROVED PRODUCTS

Actuator	Comments	Approved Ratings
MV-1005	Not available with battery backup, 3 wire incremental input, yoke bases or signal conversion modules.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MV-1010	Not available with battery backup, 3 wire incremental input, yoke bases or signal conversion modules.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MV-1020	Not available with HART Communication Protocol, battery backup, 3 wire incremental input, yoke bases or signal conversion modules.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
	Hydrogen "heavy wall enclosure"	Ex d IIB + Hydrogen Max. ambient temperature: 60°C
MV-1110	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MV-1120	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MV-1140	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MV-1170	Maximum of four position switches. Not available with battery backup or 12-36 Vdc input.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MV-1510	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MV-1550	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
MV-1570	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
VA-1010	Not available with battery backup, 3 wire incremental input, or signal conversion modules.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
VA-1020	Not available with HART Communication Protocol, battery backup, 3 wire incremental input, or signal conversion modules.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
	Hydrogen "heavy wall enclosure"	Class I, Group B, C & D; Class II, Group E, F & G; Ex d IIB + Hydrogen; CSA Enclosure 4. Max. ambient temperature: 65°C
VA-1630	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
VA-1650	Maximum of four position switches.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
PF/PT-1010	Not available with 120 or 240 Vac power supply.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.
PT-1330	Not available with 120 or 240 Vac power supply, limit switches or heater.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G CSA Enclosure 4.

FM APPROVED PRODUCTS

Actuator	Comments	Approved Ratings
SM-1015 SM-1020	Not available with process variable option, battery backup, 3 wire incremental input, or signal conversion modules.	Class I, Div. 1, Group C & D Class II/III, Div. 1, Group E, F & G NEMA 4
SM-1680	Solar wellhead versions only.	Class I, Div. 1, Group C & D Class II, Div. 1, Group E, F & G NEMA 4, T6, Maximum Ambient Temperature: 60° C
LA-1020	Not available with battery backup, 3 wire incremental input, or signal conversion modules.	Class I, Div. 1, Group C & D Class II/III, Div. 1, Group E, F & G NEMA 4
MV-1005 MV-1010 MV-1020	Not available with battery backup, 3 wire incremental input, signal conversion modules, or yoke base.	Class I, Div. 1, Group C & D Class II/III, Div. 1, Group E, F & G NEMA 4
VA-1010 VA-1020	Not available with battery backup, 3 wire incremental input or signal conversion modules.	Class I, Div. 1, Group C & D Class II/III, Div. 1, Group E, F & G NEMA 4

CE APPROVED PRODUCTS

Actuator	Comments	Approved Ratings
SM-1015	Not available with battery backup or signal conversion modules.	CE
SM-1020	Not available with battery backup or signal conversion modules.	CE
	Hydrogen "heavy wall enclosure".	CE
SM-1530	Requires additional filtering.	CE
SM-1550	Requires additional filtering.	CE
SM-1590	SM-1590-N requires additional filtering.	CE
SM-1630	Requires additional filtering.	CE
SM-1650	Requires additional filtering.	CE
SM-1720	Requires additional filtering. Maximum of six position switches. Not available with contacless feedback.	CE
SM-1730	Requires additional filtering. Maximum of six position switches. Not available with contacless feedback.	CE
SM-1750	Requires additional filtering. Maximum of six position switches. Not available with contacless feedback.	CE
SM-5150	Requires additional filtering. Not available with contacless feedback.	CE
SM-5190	Requires additional filtering. Not available with contacless feedback.	CE
LA-1020	Not available with battery backup or signal conversion modules.	CE
	Hydrogen "heavy wall enclosure".	CE
LA-2410	Requires additional filtering. Not available with contacless feedback.	CE
LA-2420	Requires additional filtering. Not available with contacless feedback.	CE
LA-2450	Requires additional filtering. Not available with contacless feedback.	CE
LA-2490	Requires additional filtering. Not available with contacless feedback.	CE
LA-5150	Requires additional filtering. Not available with contacless feedback.	CE
LA-5190	Requires additional filtering. Not available with contacless feedback.	CE
MV-1010	Not available with battery backup or signal conversion modules.	CE
MV-1020	Not available with battery backup or signal conversion modules.	CE
	Hydrogen "heavy wall enclosure".	CE
VA-1010	Not available with battery backup or signal conversion modules.	CE
VA-1020	Hydrogen "heavy wall enclosure".	CE

ATEX APPROVED PRODUCTS

Actuator	Comments	Approved Ratings
SM-1015, SM-1020, MV-1010, MV-1020, LA-1020, VA-1015 & VA-1020	Available for input voltages 1 , 2 & 6. Not available with process variable option, battery backup, 3 wire incremental input, or signal conversion modules.	ATEX Approval
SM-1590-N & SM-1590-D	240 Vac input voltage	ATEX Approval

CERTIFICATE OF COMPLIANCE

This is to certify that the Quality Management System of:

JORDAN CONTROLS, INC.
5607 W. DOUGLAS AVENUE
MILWAUKEE, WI 53218

has been assessed by ABS Quality Evaluations, Inc. and found to be in compliance with the requirements set forth by:

ISO 9001:2000

The Quality Management System is applicable to:

DESIGN AND MANUFACTURE OF POSITION CONTROL PRODUCTS

Further clarifications regarding the scope of this certificate and the applicability of ISO 9001:2000 requirements may be obtained by consulting the organization

Certificate No.:	30866
Original Certification Date:	08 January 1996
Effective Date:	19 August 2003
Expiration Date:	14 December 2003
Issue Date:	20 August 2003

Robert W. Gilman, President

ANSI-RAB QMS ACCREDITED

ABS Quality Evaluations, Inc. ★ Management System Certification ★

Validity of this certificate is based on periodic audits of the management system defined by the above scope and is contingent upon prompt, written notification to ABS Quality Evaluations, Inc. of significant changes to the management system or components thereof.

ABS Quality Evaluations, Inc. 16855 Northchase Drive, Houston, Texas 77060 U.S.A.

CSA INTERNATIONAL

Certificate of Compliance

Certificate: 1032440 (LR 38878-23) **Master Contract:** 157504

Project: 1258595 **Date Issued:** February 06, 2002

Issued To: **Jordan Controls, Inc.**
5607 W. Douglas Ave.
Milwaukee, WI 53218
USA
Attention: Mr. Bob Yach

The products listed below are eligible to bear the CSA Mark shown

Issued by: Sorin Tat
Hazardous Location Team

Authorized by: Nick Arfano
Operations Manager

Example

PRODUCTS

CLASS 3218 06 - INDUSTRIAL CONTROL EQUIPMENT - Miscellaneous Apparatus - For Hazardous Locations

Class I, Groups C and D; Class II, Groups E, F and G; Encl Type 4:

- Rotary or Linear 1000 Series Actuators, Models SM-1010, SM-1020, MV-1010, MV-1020, LA-1010, LA-1020, VA-1010, VA-1020, MV-1005, SM-1015, LA-1015, VA-1015 and MC-10796, rated 120/240V ac, 50-60Hz, 23W or 65W max, operation ambient 65C. May be used with an optional signal conversion unit Model EC-10797. May also be used with an external gear box mounted to the base of the SM-1020.

CLASS 3218 06 - INDUSTRIAL CONTROL EQUIPMENT - Miscellaneous Apparatus - For Hazardous Locations

Class I, Groups BCD; Class II, Groups EFG, CSA Enclosure Type 4, Max Ambient 40C:
Class I, Zone I, Ex d IIB + Hydrogen:

- Linear and Bi-Directional Actuator Model MV-1020, rated 120/240V ac, 50-60Hz, 23W or 65W maximum, Max Ambient 60C.
- Rotary Actuator Model SM-1020 (with GB option), VA-1020 and LA-1020, rated 120/240V ac, 50-60Hz, 23W or 65W max, Max Ambient 65C.

QD 507WP 2001/07/20 Page 1

Jordan CONTROLS

CERTIFICATE OF CONFORMANCE

5607 West Douglas Avenue
Milwaukee, WI 53218
USA

Customer:_____ Jordan Ref #:_____

Customer P.O. #:_____ Date Shipped: ____/____/____

☐ ** PREVIOUSLY SHIPPED UNIT*

AMPLIFIER INFORMATION

Model #:_____ IM:_____

ACTUATOR INFORMATION

Sales Code:_____

Model #:_____ Serial #:_____

Installation Dwg:_____ Wiring Di_____

IM:_____ Enclosure:_____ Input Volta_____

Full Load Current Rating:_____ Rotation/Stroke:_____

Control Signal:_____ Option For Inc Signal:_____

Feedback Device(s):_____ Spring Return Direction:_____

Heater Rating:_____ # of Limit Switches:_____

Torque/Thrust Trip Protection: yes____ no____ Rated load:_____

Torque/Thrust Trip Load: Extend/CW:_____ Retract/CCW:_____

No Load Shift Time/Rate: Extend/CW:_____ Retract/CCW:_____

VALVE INFORMATION

Mfg:_____ Size:_____ Model#:_____

Serial #:_____ Tag #:_____ Orientation Per Specs:_____

Valve Supplied By:_____

* We certify that the part listed above was inspected for conformance to the data as shown at the time of shipment. Raw test data is no longer available. Data was obtained from our files based upon original customer specifications.

NAME:_____ DATE:____/____/____

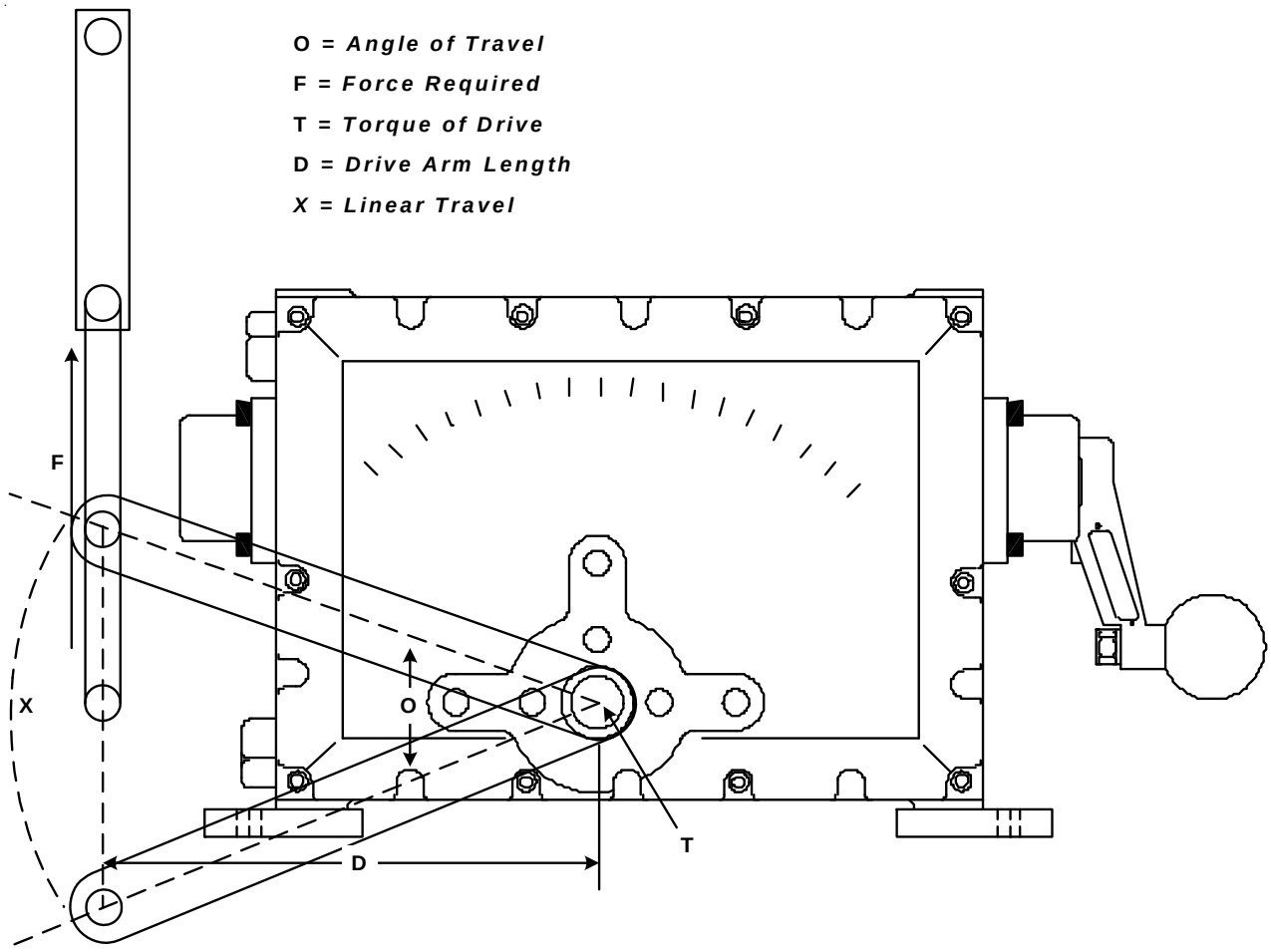
TITLE:_____

Example

CONVERSION CHART

To Convert From	Into	Multiply By
Thrust		
oz. Force	Newton	0.278
lb. Force	Newton	4.4482
Newton	oz. Force	3.596
Newton	lb. Force	0.2248
Torque		
in. lbf.	Nm	0.1129
ft. lbf.	Nm	1.355
Nm	in. lbf.	8.85
Nm	ft. lbf.	0.7375
Length		
Inch	Centimeter	2.54
Inch	Meter	0.0254
Inch	Millimeter	25.4
Foot	Centimeter	30.48
Foot	Meter	0.3048
Millimeter	Inch	0.0393
Centimeter	Inch	0.3937
Centimeter	Foot	0.0328
Meter	Inch	39.36
Meter	Foot	3.28
Mass		
Ounce	Kilogram	0.0283
Pound	Kilogram	0.4535
Kilogram	Ounce	35.27
Kilogram	Pound	2.204
Temperature		
°F	°C	(F - 32) x .555
°C	°F	(C x 1.8) + 32
Flow		
Gallon/hour	Liter/hour	3.7854
Gallon/minute	Liter/minute	3.785
Liter/hour	Gallon/hour	0.2641
Liter/minute	Gallon/minute	0.2641
Pressure		
PSI	Bar	0.0689
Bar	PSI	14.5
Feet of water (at 4° C)	PSI	0.4335
PSI	Feet of water (at 4° C)	2.3067

Using a 90° Actuator for Linear Applications



$$\text{Drive Arm Length} = \frac{\text{Linear Travel}}{1.42}$$

Torque of Drive = Force Required x Drive Arm Length

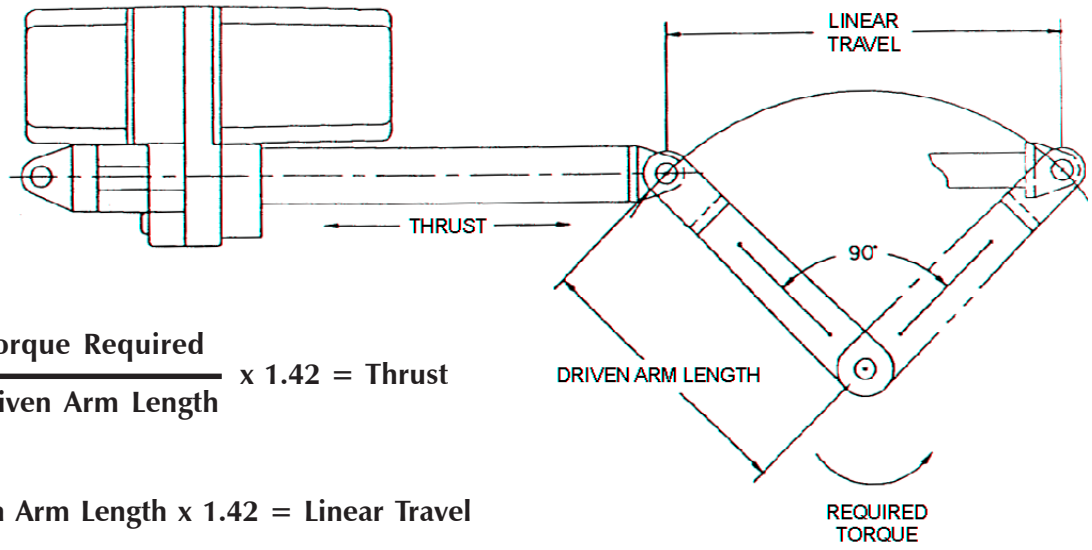
***Example*:**

- **Angle of Travel = *90°***
- **Force required = *500 N (112 lbf.)***
- **Linear Travel = *0.4 m (15.7 in.)***

$$\text{Drive Arm Length} = \frac{0.4\text{ m (15.7 in.)}}{1.42} = \textbf{0.281 m (11.1 in)}$$

Torque of Drive = 500 N x 0.281 m (11.1 in.) = 140.5 Nm (103.6 ft.lb)

Using a Linear Actuator for 90° Applications



$$\frac{\text{Torque Required}}{\text{Driven Arm Length}} \times 1.42 = \text{Thrust}$$

Driven Arm Length x 1.42 = Linear Travel

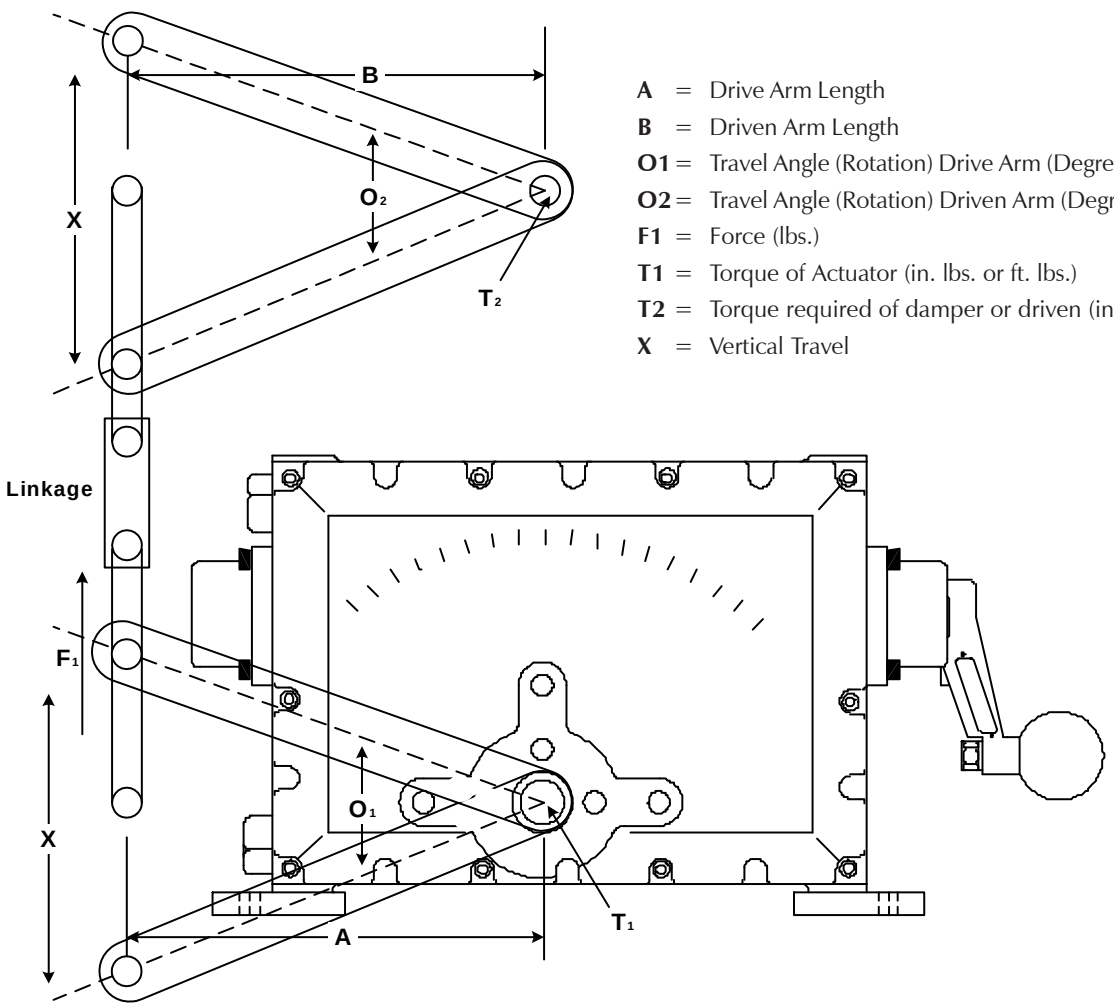
***Example*:**

- **Driven Arm = *0.3 m (11.8 in.)***
- **Torque required = *200 Nm (1772 in. lb.)***

$$\text{Thrust} = \frac{200\ \text{Nm (1772 in. lb.)}}{0.3\ \text{m (11.8 in.)}} \times 1.42 = \textbf{947 N (213 lbf.)}$$

Linear Travel = *0.3m (11.8 in.)* x *1.42* = **0.426 m (16.77 in.)**

Using a Rotary Actuator for Rotary Applications



A = Drive Arm Length
B = Driven Arm Length
O1 = Travel Angle (Rotation) Drive Arm (Degrees)
O2 = Travel Angle (Rotation) Driven Arm (Degrees)
F1 = Force (lbs.)
T1 = Torque of Actuator (in. lbs. or ft. lbs.)
T2 = Torque required of damper or driven (in. lbs. or ft. lbs.)
X = Vertical Travel

***Example*:**

- **Drive Arm Rotation = *90°***
- **Driven Arm Rotation = *120°***
- **Driven Arm Length = *12 in.***
- **Driven Arm Torque Required = *1000 in. lb.***

$$F1 = \frac{T2}{B} = \frac{1000}{12} = 83.33 \text{ lb.}$$

To get Vertical Travel

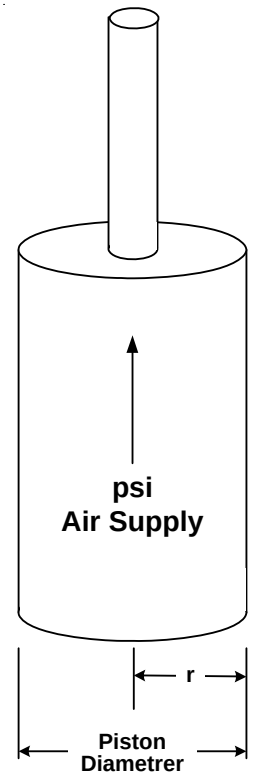
$$\frac{X}{\text{Sin}120} = \frac{12}{\text{Sin}30} \qquad X = \frac{\text{Sin}120\ (12)}{\text{Sin}30} \qquad X = 20.78 \text{ in.}$$

To get Drive Arm Length

$$\frac{20.78}{\text{Sin}90} = \frac{A}{\text{Sin}45} \qquad A = \frac{\text{Sin}45\ (20.78)}{\text{Sin}90} \qquad A = 14.7 \text{ in.}$$

Required Torque of Actuator*: T1 = F1 x A = 83.33 x 14.7 = *1,225 in. lb.

Replacement of Pneumatic or Hydraulic Cylinders



The linear force (lbs.) produced by a pneumatic cylinder can be calculated if the pressure supplied to the cylinder and the piston diameter of the cylinder are known.

***Example*:**

- **Piston Diameter = *3 inches***
- **Pressure Supplied = *120 psi***

Area of Piston = $(3.14)(r)^2 = (3.14)(1.5)^2 = 7.07$ in.^{2}

Force = (Area)(Pressure) = (7.07 in.^{2})(120 psi) = 848 lbs.

A cylinder with a 3 inch diameter piston pressurized to 120 psi will produce 848 lbs. of axial force when extending. The force produced while retracting will be slightly less due to the diameter of the rod (the rod reduces the effective area that the pressure works over).

To Convert RPM to Seconds for 90° of Travel

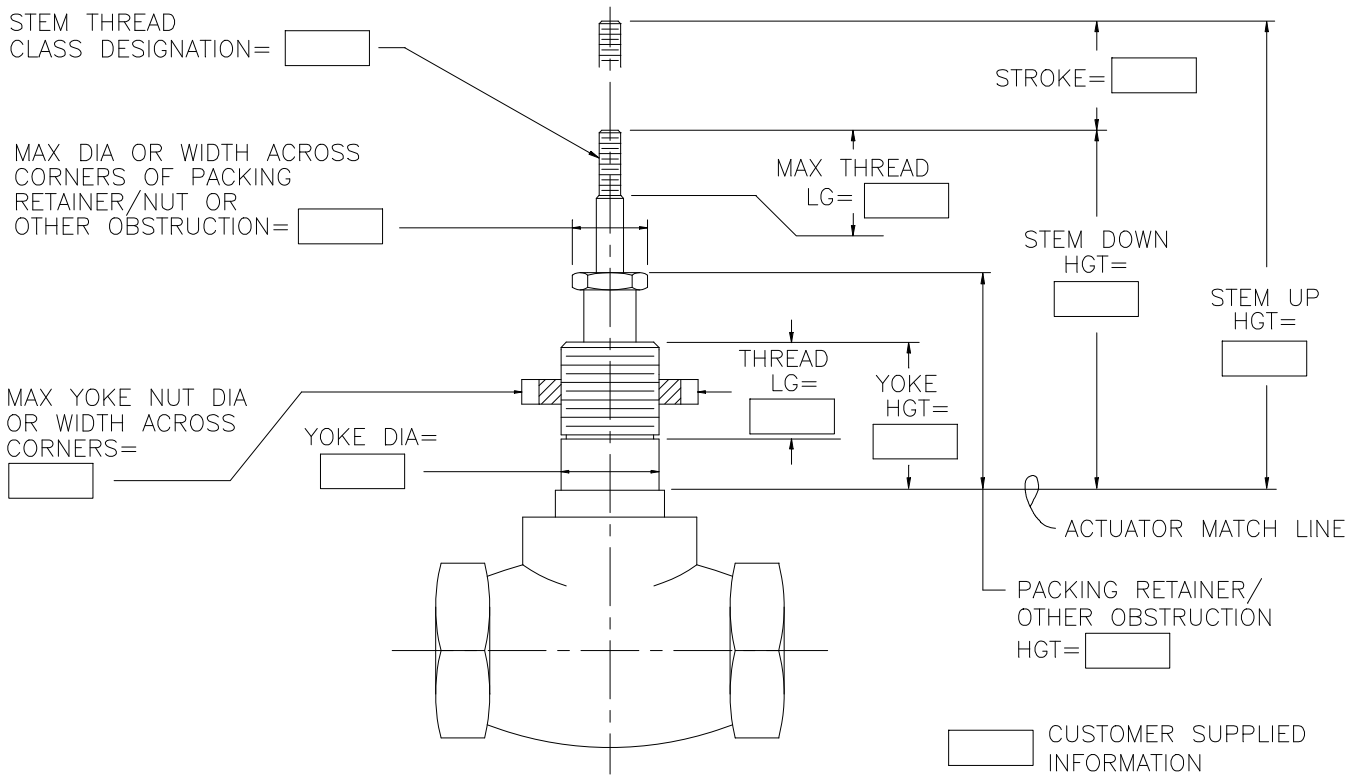
***To convert RPM to sec./90°*:**

X = *RPM*
Y = *Sec. for 90°*

$$\frac{1}{X} \times 15 = Y$$

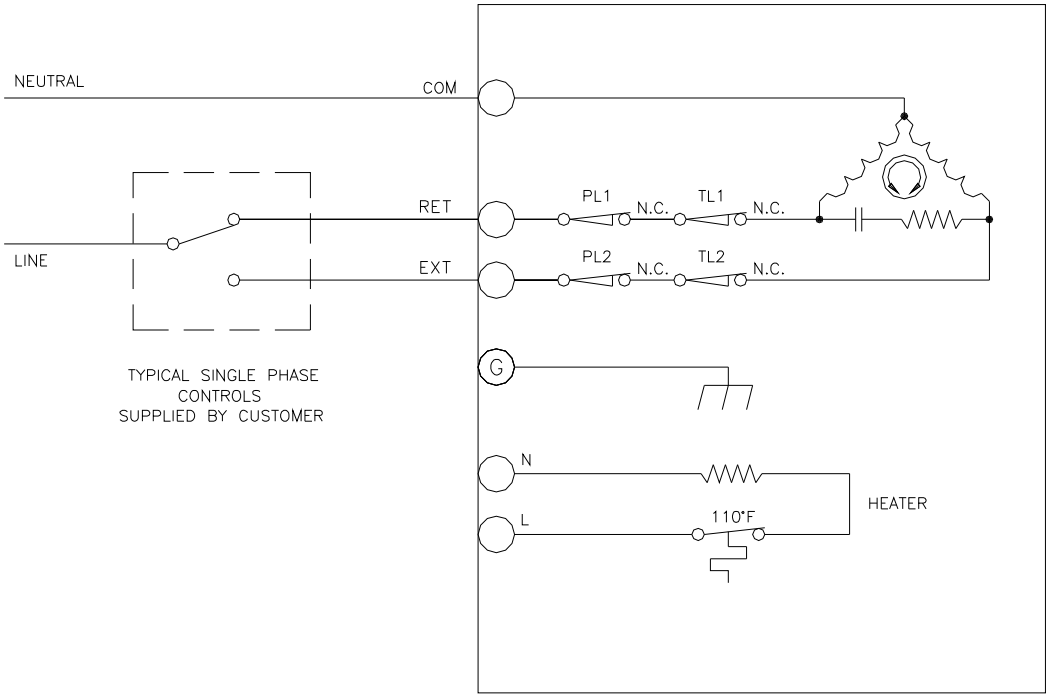
Required Valve Mounting Information

When mounting of the actuator to the valve is done by the customer, complete "top-works" of valve dimensions for the drawing below must be furnished to Jordan Controls. This is necessary to properly machine the actuator yoke, and to provide appropriate coupling hardware.

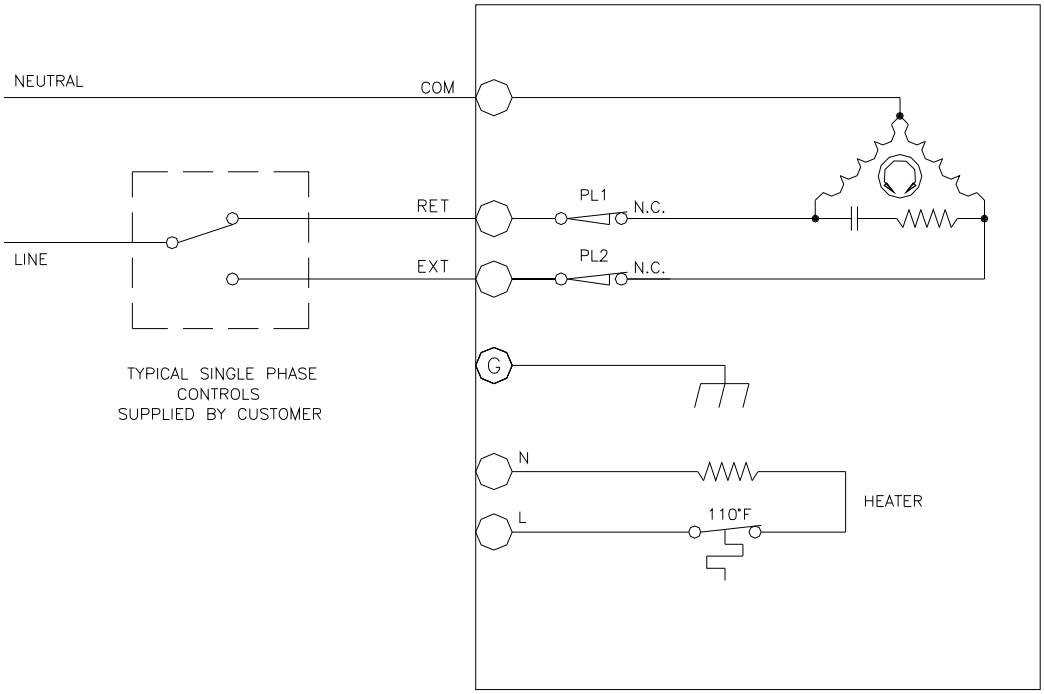


If valve is to be attached to yoke by any other method or has a unique mounting configuration, please provide necessary drawings.

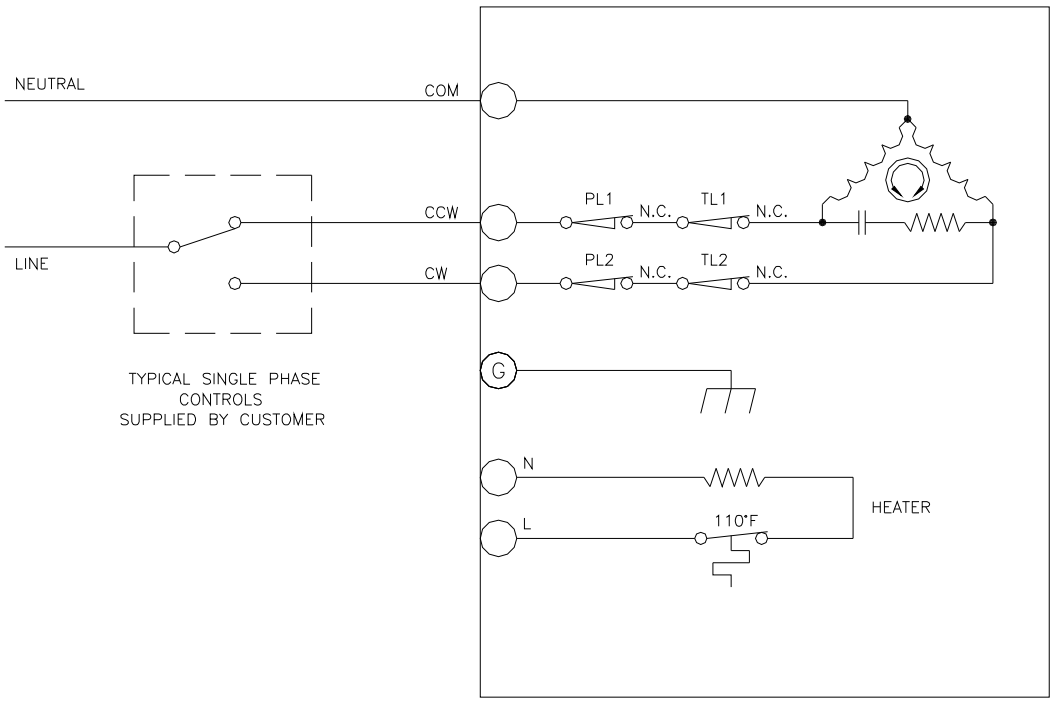
Single Phase AC Linear Actuators with Thrust Limiting



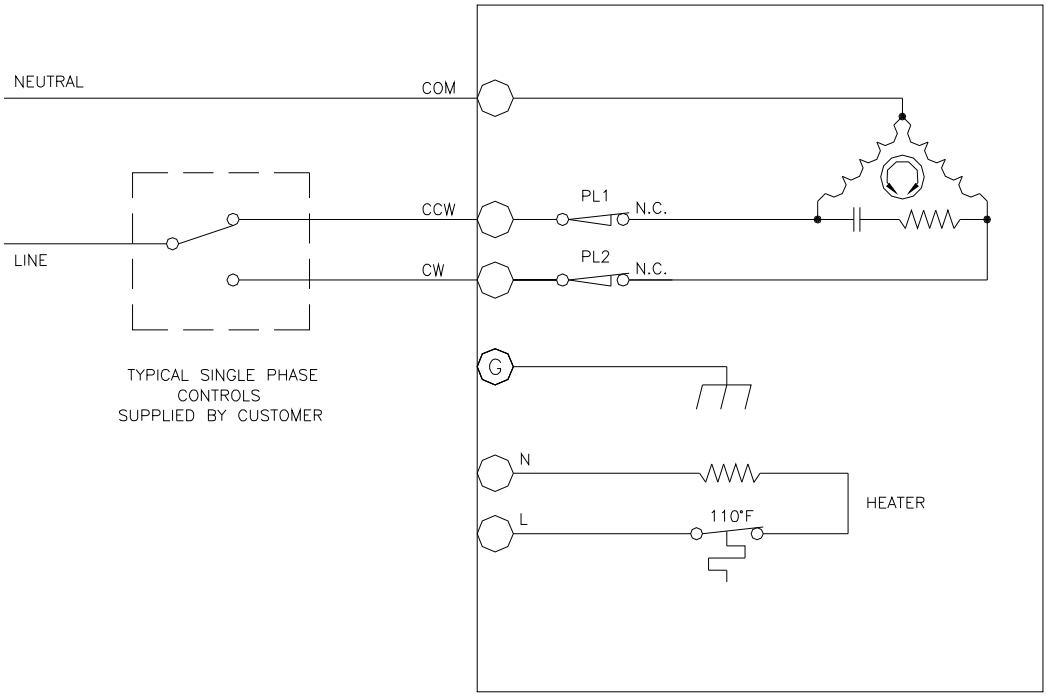
Single Phase AC Linear Actuators without Thrust Limiting



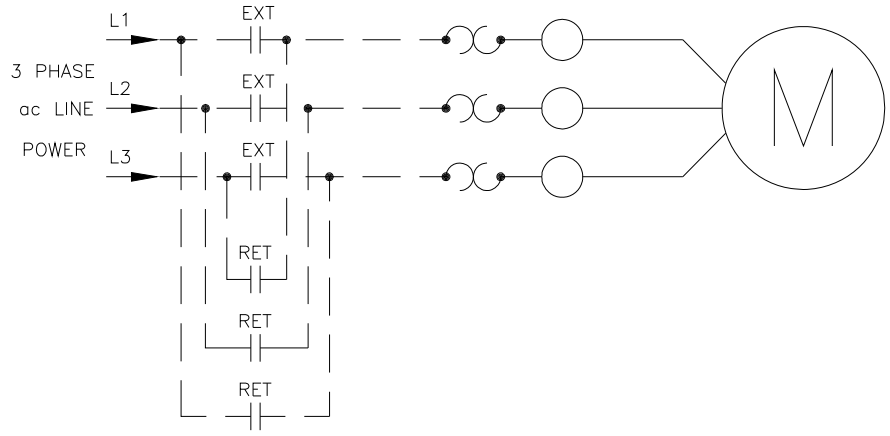
Single Phase AC Rotary Actuators with Torque Limiting

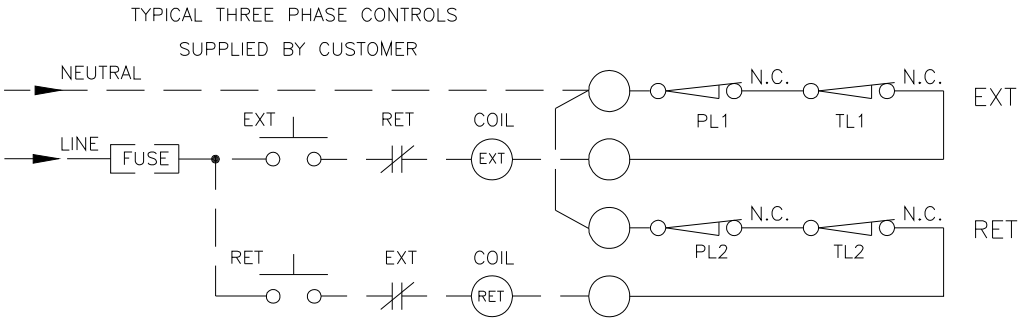


Single Phase AC Rotary Actuators without Torque Limiting

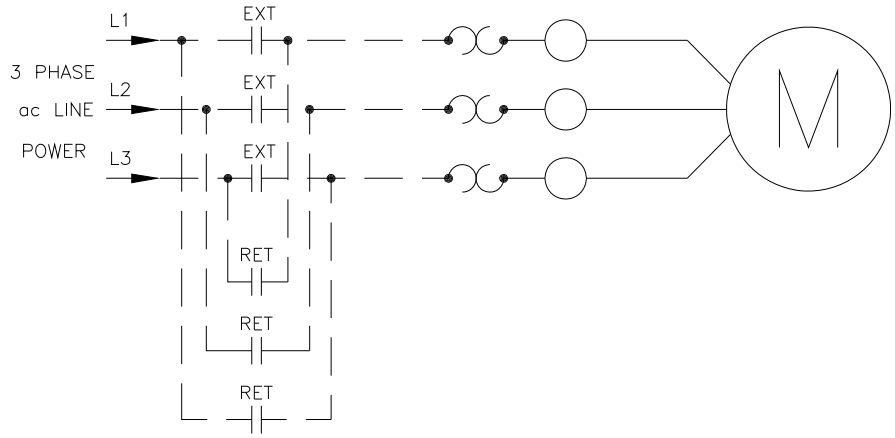


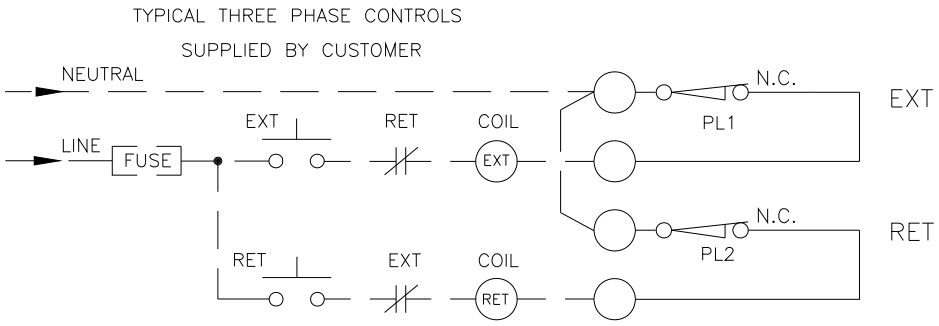
Three Phase AC Linear Actuators with Thrust Limiting



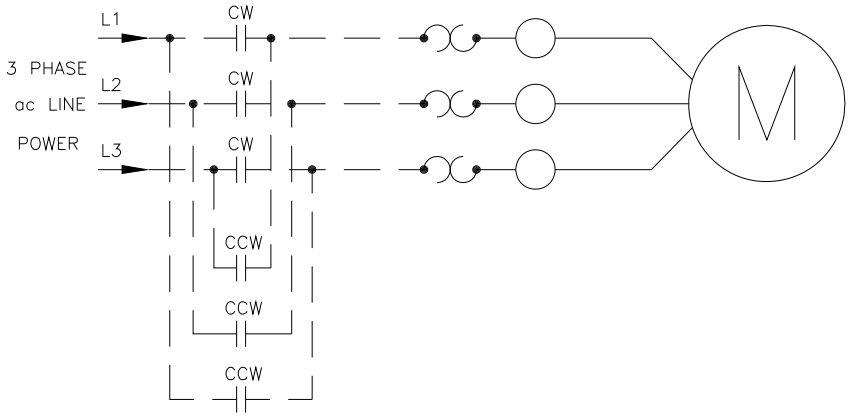


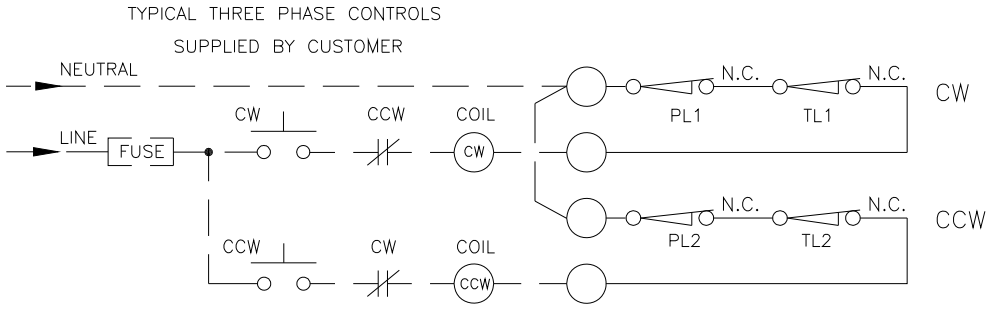
Three Phase AC Linear Actuators without Thrust Limiting



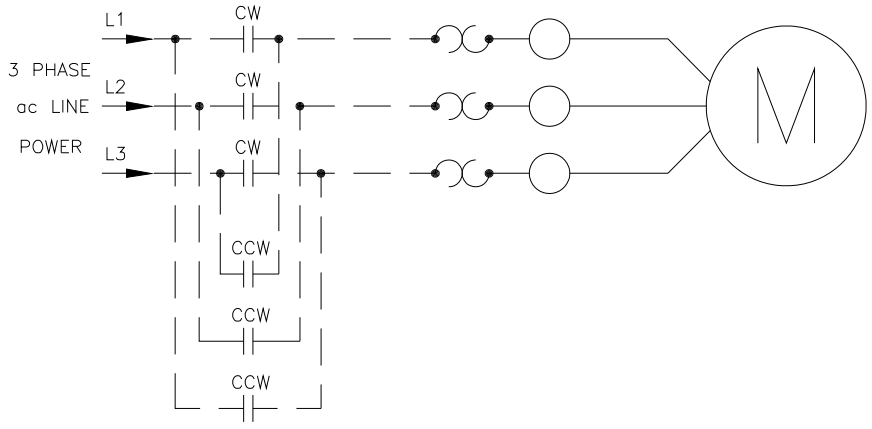


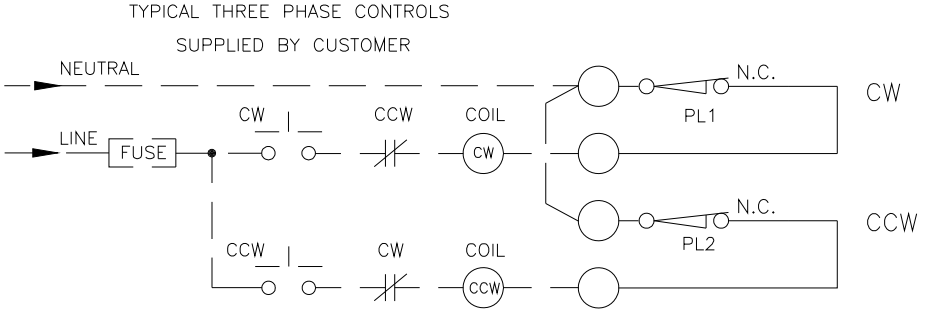
Three Phase AC Rotary Actuators with Torque Limiting



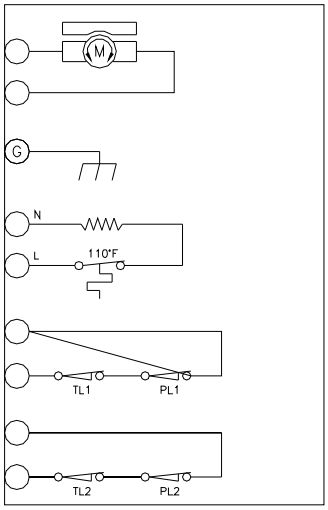


Three Phase AC Rotary Actuators without Torque Limiting

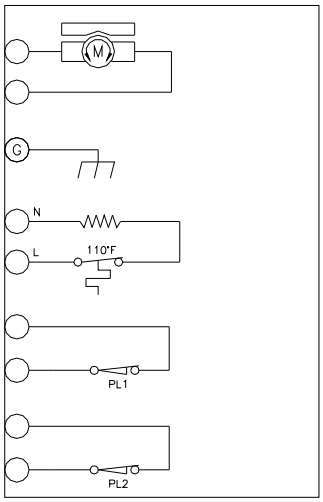




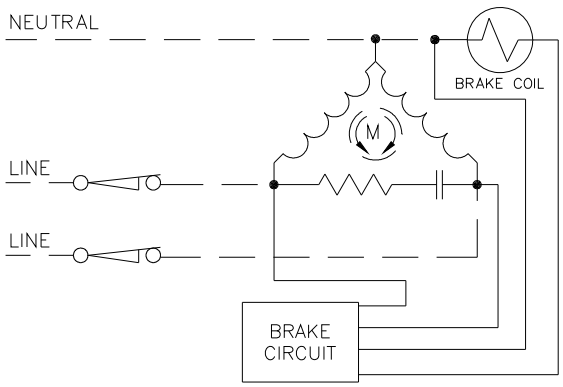
DC Actuators with Thrust Limiting



DC Actuators without Thrust Limiting



120 Vac & 240 Vac Electronically Controlled Brake



GLOSSARY OF TERMS

AC - Alternating current or voltage which changes direction and polarity at a rate of 50 or 60 Hz. The most common alternating current used in industry is sinusoidal.

Accumulator - A gas pressurized storage vessel used to store hydraulic fluid under pressure so it may be used as a source of potential energy.

Accuracy - Measured in percent of span, indicates how close the actual movement was compared to command set point.

Actuator - A device that transforms a source of energy into another upon command. In Jordan actuators, electrical motor energy provides rotary or linear motion upon command.

Acme Screw - A special rolled thread on a straight shaft designed to have contact with a nut on the faces of the thread, and to transmit rotary motion into linear motion.

Ampere (Amp) - Unit of measurement of electron flow or current.

Amplifier - See servo amplifier.

Analog - Measurement system where the units and signals bear a direct linear relationship to functions. Measured in milliamps (mA).

Auxiliary Switches - Auxiliary circuit switches fitted and mechanically coupled to switching devices for the purpose of indicating the state of the main contacts.

Backdrive - The ability of a device to be driven "backwards". Force applied to the output shaft can cause the input shaft to rotate.

Bellows – A protective "accordion" type mechanical covering used to protect the rod of a cylinder or actuator from the environment.

Brake - *Dynamic, Motor, Electric, Drag.*

Breakaway (unseating) Torque or Thrust - Torque or thrust initially used at start-up for a very brief period of time, usually expressed in m. sec.

Clevis - A device on the ends of a push-pull linear actuator or linkage for mounting and connecting.. Either forked or spade.

Close-Coupled - An enclosure mounted next to an actuator.

Closed Loop – A control system that will automatically respond to changes without the need for an operator.

Command - A DC control signal supplied by the customer to the servo amplifier, i.e.: of 4-20 mA current or 0-5 or 0-10 Vdc.

Continuous Modulating Duty - An actuator that is constantly in the process of reaching a commanded position. Typically the actuator will constantly be moving at a very slow speed, in an attempt to reach a constantly moving command position.

Control - The method by with the actuator is caused to move in response to a process change i.e.: incrementally pulsed, DC signal input.

Control Switches - Power circuit switches for the direct control of electrical equipment, or auxiliary circuit switches for indirect control.

Current - Flow of electrons through an electric circuit.

DC - Direct current or voltage which has a constant value related to zero value. Can be stipulated as a positive or negative value.

Deadband - An adjustable window around the desired set point of the actuator in which there is no movement of the actuator.

Digital – Typically using a microprocessor type device to compare and act upon input signals to it.

Drive Arm - A lever that is attached to a rotary actuator to provide linear motion through an arc.

Driven Arm - A lever that is attached to a device which produces rotary motion from a drive arm.

Duty Cycle - Maximum "On" time of a motor in a specific time period expressed as a percent of a time period.

Dynamic Brake - An electronic method used to stop a motor by applying voltage to both windings of the motor at the same time for a controlled period of time.

Efficiency - Ratio of useful ouput power to total input power.

Electric - Self Contained electric motor power package with or without closed loop feedback.

Electro-Hydraulic - Self contained, motorized, hydraulic package. Typically designed with closed loop feed back.

Enclosure Rating – Resistance to intrusion by either liquids or gasses, as specified by a standards organization, typically NEMA.

Explosion-proof - Characteristic of an enclosure to contain or prevent an explosion.

Fail safe - Characteristic of an actuator to move to a safe condition upon the failure of electric power, power source, etc.

Failure mode - Upon electric power failure to the actuator, the actuator moves to a predetermined position (fail open, fail closed, fail in last position).

Feedback - A signal from an actuator feedback element that relates to 0 to 100% of the actuator output shaft movement.

- Internal feedback to the servo amplifier from a pot or contactless device.
- Output position signal from the actuator transmitted to the DCS.

Feed Back Element - Component(s) that measure or indicate the position of the output shaft, i.e.: position limit switches, potentiometer, Hall Effect sensor, LVDT (linear variable displace transducer) or RVDT.

Gain – The rate at which the speed of an actuator will begin to slow as the actuator approaches the desired position.

Gear Ratio - Ratio of number of turns the input device takes to rotate the output device one turn.

Gearbox - A mechanical device that typically transforms high speed, low torque rotary motion into low speed, high torque rotary motion.

Hazardous environment - Any combination of combustible or potentially explosive mixtures existing in the surrounding atmosphere.

Hertz Unit of Frequency - Number of times per second that alternating current flows in one direction, reverses, and flows in the other direction. Power companies in the United States hold this figure constant at 60 cycles per second (hertz).

Hydraulic - Pressurized fluid (oil) used for power transmission incorporating a hydraulic power unit, hoses, and a cylinder or motor. Typically designed with out closed loop feed back.

Hysteresis - Measured effect of the delay in action due to compressive or elastic effects.

IGBT – Insulated Gate Bipolar Transistor

Incremental Control – Control that uses ON and OFF pulses, either automatically generated, or by an operator.

In-rush Current - Current draw of a motor on starting.

Limit Switch - Set of electric contacts that are activated mechanically.

Linear Travel - Movement in a straight line.

Linearity - Maximum deviation from best straight line representing ratio of output amplitude to input amplitude.

Lock-in-Place (LIP) – Refers to stopping when an event occurs.

Loop Control - A process management system designed to maintain a process variable at a desired set point.

Loss of Signal (LOS) – An absence of a command, or demand, signal.

Manual Override - Capability provided to actuators that allows the manual positioning of the actuator. Also known as handcrank.

Milliamp (mA) – A measure of electric current, one thousandth of an ampere.

Modulating - Regulating or adjusting a control element proportional to a process change.

Modulation Rate - Number of actuator starts (cycles) per hour. Minimum off-time of actuator is one-half of a cycle. (Example: If actuator is rated at 1200 starts/hr, this is 3 seconds per cycle, therefore minimum off-time is 1.5 sec.)

Moment Arm - Linear distance from the center of rotation upon which a force may act to produce torque.

Multi-Turn/Rising Stem - Describes a valve that requires multiple turns of the actuator output shaft to raise and lower the stem of a valve using a threaded portion of the stem.

NEMA Classification - Code established for the construction of electrical components by the National Electrical Manufacturers Association.

Null - Output of servo amplifier at rest or off when feedback signal matches command signal within deadband window.

Planetary Gears - A type of gear reduction where spur gears rotate around the axis of a central spur gear.

Plug Reversal – Immediate reversing of a motor without allowing the shaft to stop.

Position/Limit Switch - A mechanically tripped switch used to limit end travel movement, or to trip at a desired position.

Potentiometer - A three terminal, variable resistance device, that when turning the potentiometer shaft will result in a varying resistance.

Power Supply - There are two types of power supply: direct current and alternating current. The latter is the most widely used power supply.

Pneumatic - Compressed air used for power or control of devices utilizing clean dry instrument air.

Position Accuracy - See Accuracy.

Power - Input voltage required to drive the actuator i.e.: 120vAC/1ph/60hz.

Process Control - Act of maintaining or correcting a process within preset limits using a measurement of the actual process conditions in comparison to an established, desired condition.

Proportional Controller - Provides a response from an actuator that is directly related to the feedback from a remote sensor. Used to control a process automatically.

Rack - Linear gear that operates in conjunction with a pinion gear to provide rotary motion.

Repeatability - Measured accuracy in percent of span of a control device to match a set point when approaching from either direction.

Resolution - Smallest possible movement that can be measured.

Rotary Actuator - Device that provides rotating movement to a final control element.

Scotch Yoke - Method of converting linear motion to rotary motion by using a slotted lever arm.

Seat - That portion of the seat ring or valve body which a valve closure member contacts for closure.

Seating Torque - Value depicting the turning force required to seat a valve or damper into its closed position.

Self-Locking - The ability of a device to counteract the effects of gravity. An actuator that is self-locking will hold a load from lowering.

Servo Amplifier - An electronic device used to control actuator output shaft position based on the difference between command and feedback signal.

Shift Time - Actuator travel time in seconds for 0 - 100% travel.

Span - The high end adjustment for a signal input to a servo amplifier. aka: Range or High Trim.

Splined Shaft - A shaft with raised areas to positively transmit torque. Could be thought of as multiple keys/ keyway.

Split Command Input – A system whereby multiple actuators move in sequence with each other.

Spring-return Actuator - Device that contains a spring element having the capability of returning a valve-actuator system to its normal position in the absence of electric power.

Spur Gearing - Method of transferring torque from one shaft to another. Usually results in a decrease in speed with an increase in torque.

Stability - Ability of an actuator to not show a change in position when subjected to extraneous input like electronic noise, temperature or vibration.

Throttling - Action of an actuator in modulating motion.

Thrust - A linear force acting on an object, causing motion, measured in lbs., or Newtons

Thrust Limiting – A mechanical device inside the actuator which stops the actuator if the maximum thrust rating is exceeded.

Torque - A twisting force acting on a shaft, resulting in rotation, measured in in-lbs., ft-lbs.,or Nm.

Torque Limiting - A mechanical device inside the actuator which stops the actuator if the maximum torque rating is exceeded.

Triac – An electronic device which serves as an AC switch.

Unrestricted Modulating Duty - See modulating duty.

Valves - Butterfly, globe, plug, ball, etc.

Zero - Low end adjustment for signal input, aka: Low Trim.

PREVENTATIVE MAINTENANCE PROGRAM

Jordan Controls actuators are designed and manufactured for a long life span. Our "customer-first" approach to preventative maintenance can significantly reduce the cost of ownership.

Our Preventative Maintenance Program ensures that your actuators are always in top working condition.

Once a year, all units are fully tested for proper operation. Gaskets will be replaced and gears will be re-greased. Other mechanical and electrical parts will be repaired or replaced, as necessary.

ON-SITE REPAIR

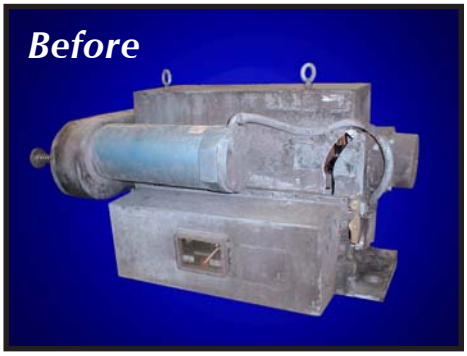
Occasionally it becomes necessary to have repairs done on-site, either due to an emergency or because it's not feasible to remove the unit. In these situations, Jordan Controls has technicians available to travel to your location.

Our skilled technicians have the specialized tools and knowledge to repair actuators at your facility, anywhere in the world. All of our technicians have years of factory experience to quickly diagnose and repair equipment, with a minimum of downtime and/or disruption to normal activity.



IN-HOUSE REPAIR

Our plant has complete facilities for repair or complete overhaul of equipment. Depending on your needs, we can tear a unit down to the basic castings, sandblast, repaint, re-assemble and test it as a new unit. Rush jobs often can be turned around in 5 days or less. Call us to coordinate the repair.





START-UP ASSISTANCE

Sometimes when a new product is purchased, on-site assistance is desired to ensure a correct and timely start-up. We at Jordan Controls recognize this need and offer start-up assistance for all products.

IN-HOUSE & ON-SITE TRAINING

Jordan Controls offers complete on-site training services. These "hands-on" sessions include a complete review of all electrical and mechanical options.

If it is more convenient, we also offer training classes at our facility in Milwaukee.

For hands-on training at your site, we ask that one installed unit be made available for classroom training.

Call, FAX or e-mail Jordan Controls for a quote on any of these services